

(45) **Date of Patent:** Sep. 9, 2025

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(21) Appl. No.: 17/681,902

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(22) Filed: **Feb. 28, 2022**

17681902_2024-11-05_CN_108023418_A_H.pdf (Year: 2024).*
Nov. 10, 2020 Search Report issued in International Patent Application No. PCT/JP2020/031877.

(65) **Prior Publication Data**

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Assistant Examiner — Ahmed F Seck

(74) *Attorney, Agent, or Firm* — Oliff PLC

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2020/031877, filed on Aug. 24, 2020.

(30) **Foreign Application Priority Data**

Aug. 26, 2019 (JP) 2019-153901

(51) **Int. Cl.**
H02K 1/276 (2022.01)
H02K 3/47 (2006.01)

(Continued)

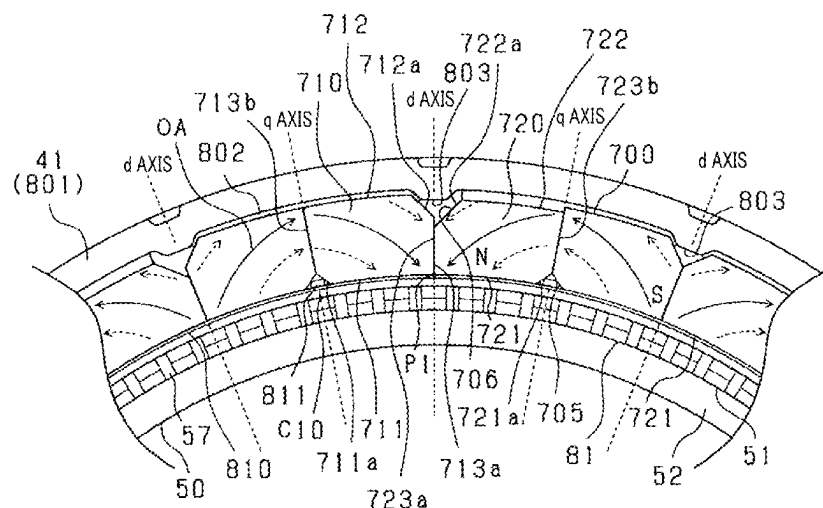
(52) **U.S. Cl.**
CPC ***H02K 1/276*** (2013.01); ***H02K 3/47***
(2013.01); ***H02K 15/02*** (2013.01); ***H02K***
21/12 (2013.01)

(58) **Field of Classification Search**
CPC .. H02K 1/04; H02K 1/02; H02K 1/06; H02K
1/08; H02K 1/14; H02K 1/146;
(Continued)

(57) **ABSTRACT**

A rotating electric machine is provided with a field magnet having a magnet unit including a plurality of magnetic poles having alternating polarities in a circumferential direction and a magnet supporting section having a cylindrical shape to which the magnet unit is fixed, and an armature having a multi-phase armature winding, disposed facing the magnet unit, either one of the field magnet or the armature being configured as a rotor. The magnet unit is configured annularly along the circumferential direction. The magnet unit is fixed to a fixed surface which is either an outer peripheral surface or an inner peripheral surface of the magnet supporting section via a foamable resin, and a thickness dimension of the foamable resin in a radial direction is set to be differentiated depending on a distance from the fixed surface of the magnet supporting section to an anti-armature side circumferential surface of the magnet.

8 Claims, 61 Drawing Sheets



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- (51) **Int. Cl.** USPC 310/156.01, 156.21, 156.37, 156.46,
H02K 15/02 (2025.01) 310/156.53
H02K 21/12 (2006.01) See application file for complete search history.
- (58) **Field of Classification Search** (56) **References Cited**
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H02K 1/27; H02K 1/2786; H02K 1/2787;
H02K 1/2788; H02K 1/2789; H02K
1/2791; H02K 1/27915; H02K 1/2792;
H02K 21/022; H02K 21/023; H02K
21/025; H02K 21/042; H02K 21/10;
H02K 21/12; H02K 21/22; H02K 21/14;
H02K 21/26; H02K 21/28; H02K 15/02;
H02K 15/12; H02K 15/03; H02K 3/47;
H02K 1/17; H02K 1/187; H02K 37/06;
H02K 37/10; H02K 37/12; H02K 37/14;
H02K 1/279; H02K 1/28; H02K 23/04;
H02K 23/42; H02K 35/02; H02K 37/02;
H02K 1/2706; H02K 1/272; H02K
1/2733; H02K 1/274; H02K 1/2753;
H02K 1/276
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- * cited by examiner

FIG. 1

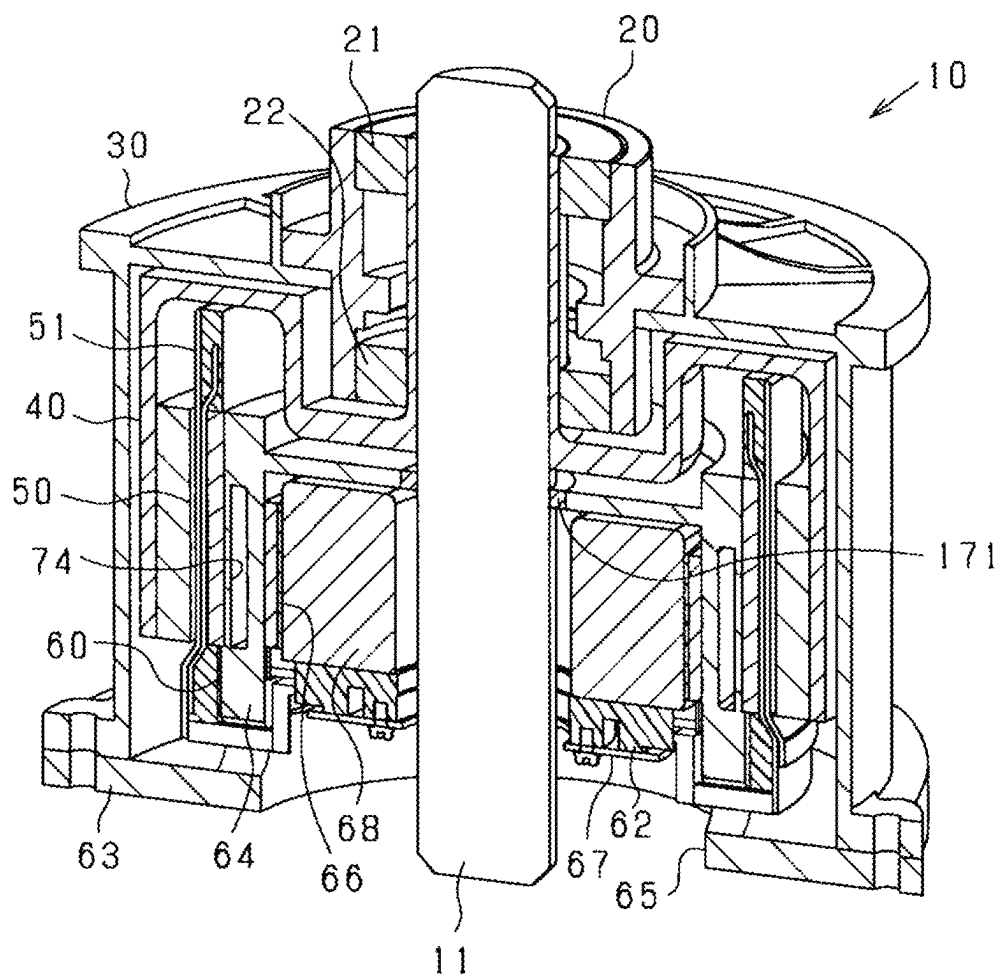


FIG.2

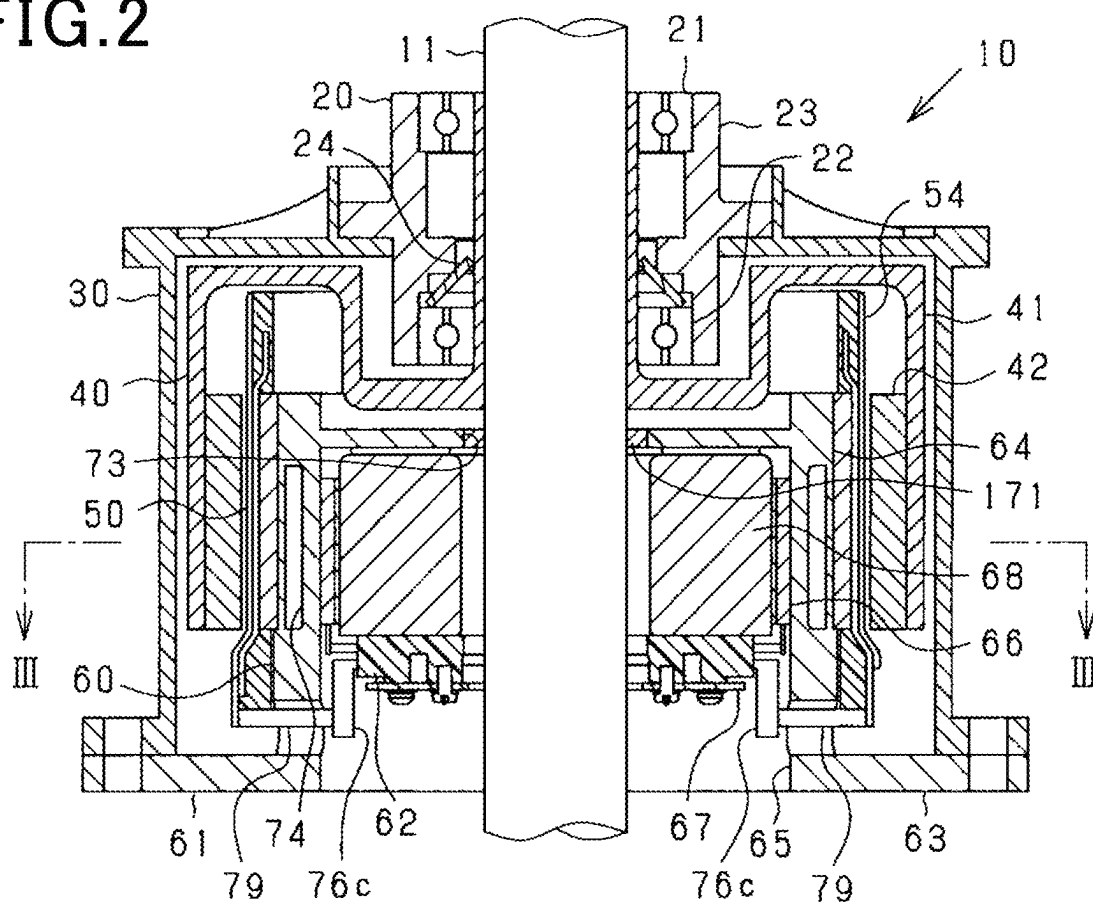


FIG.3

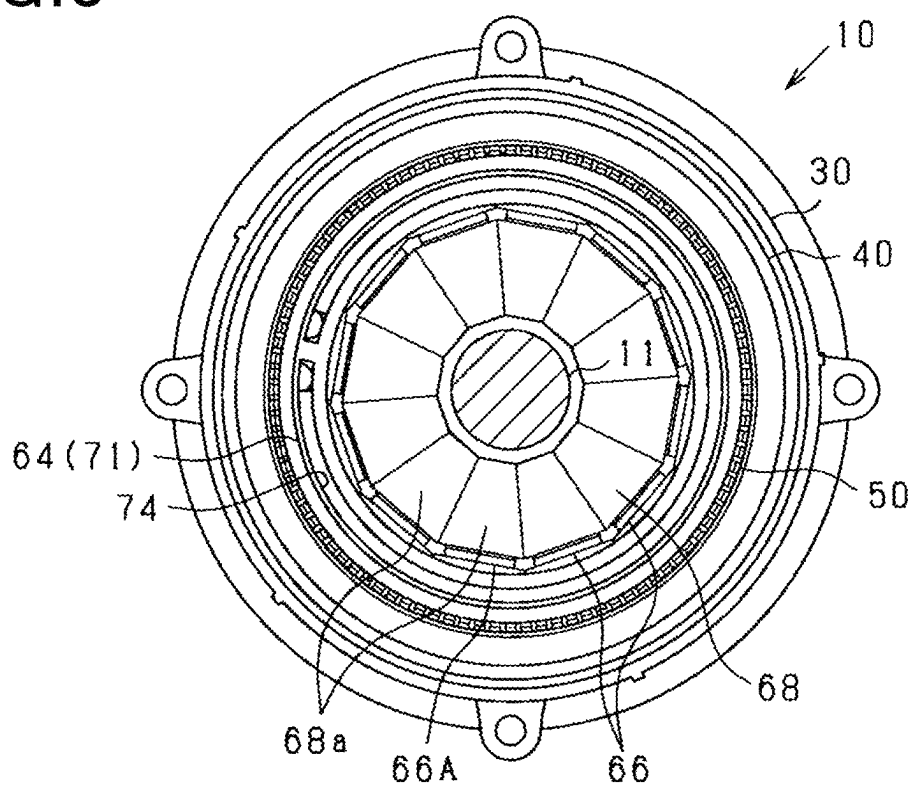


FIG. 4

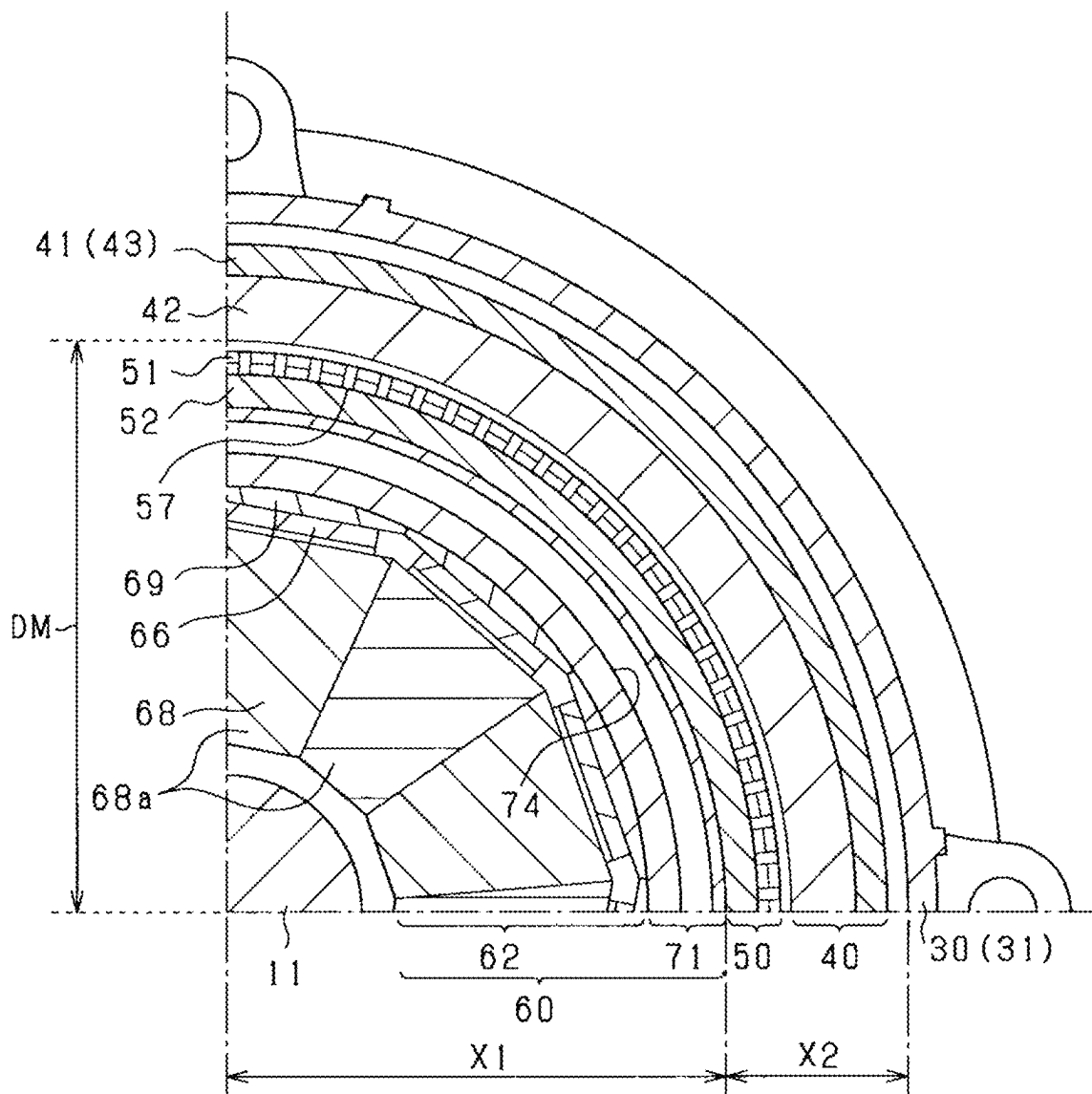
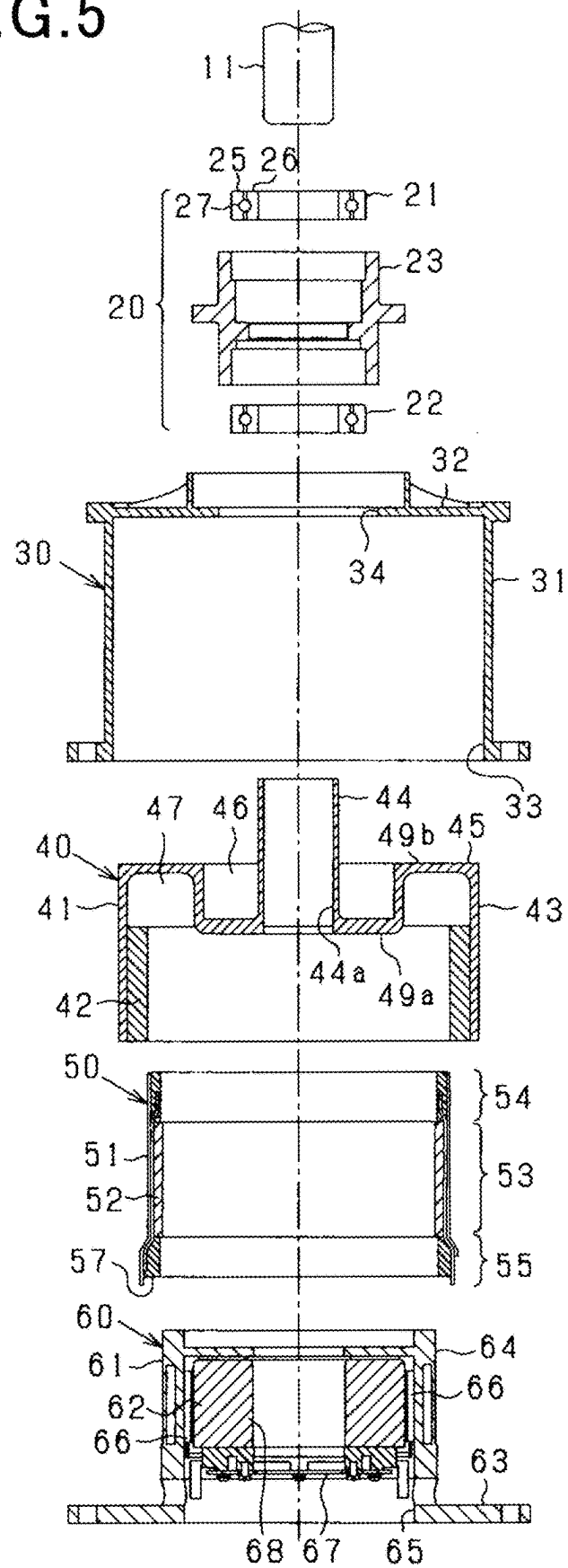


FIG. 5



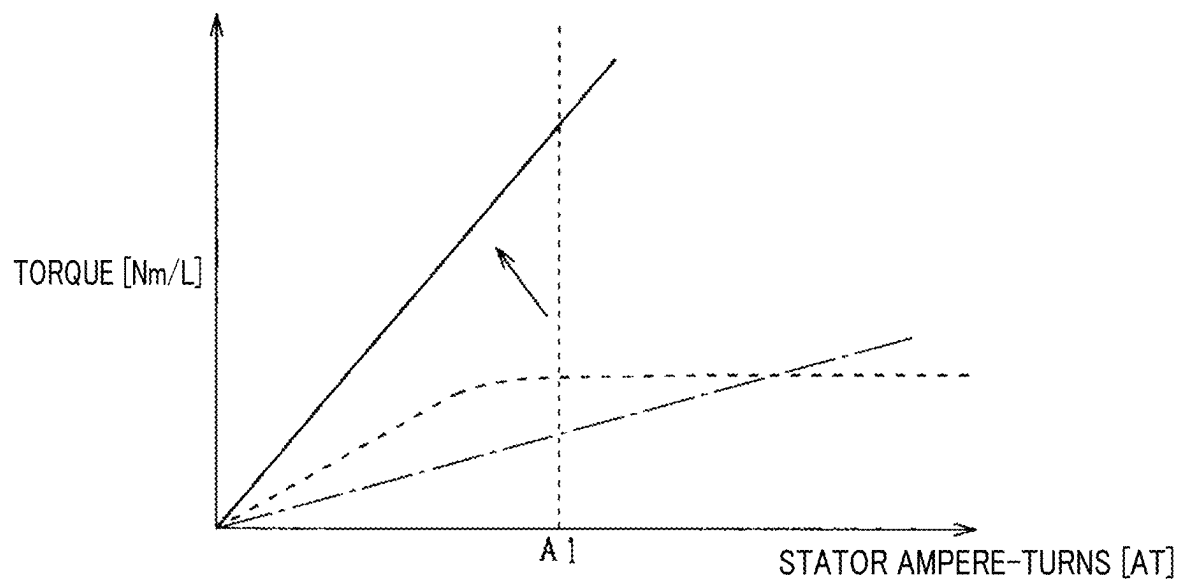


FIG. 8

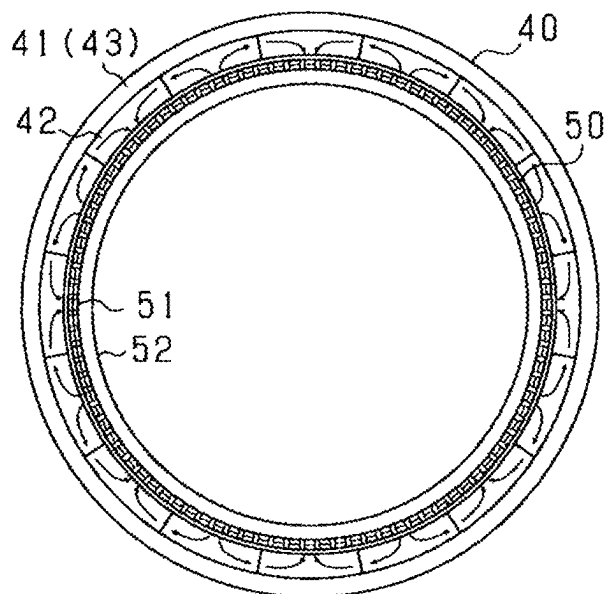
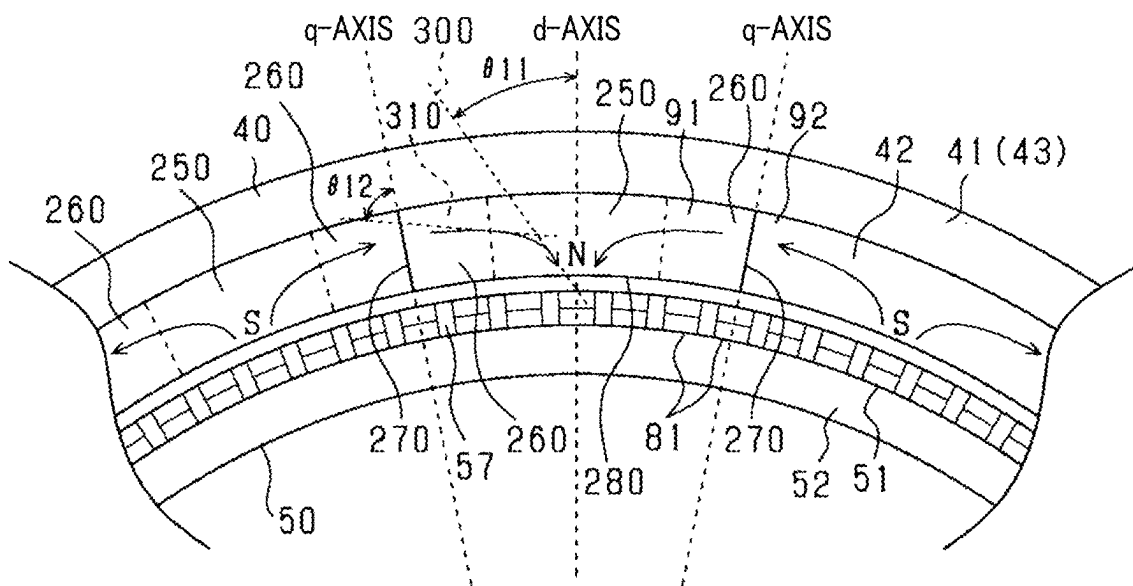


FIG. 9



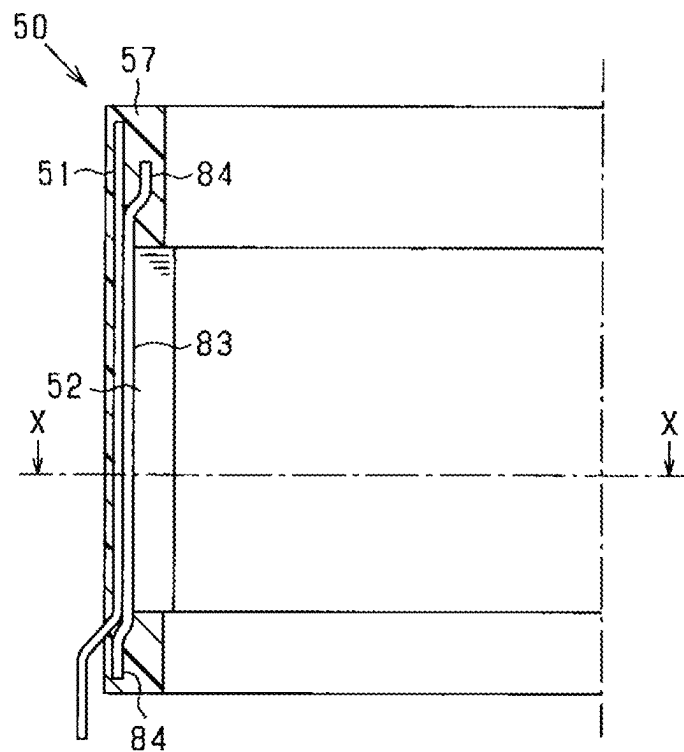


FIG. 12

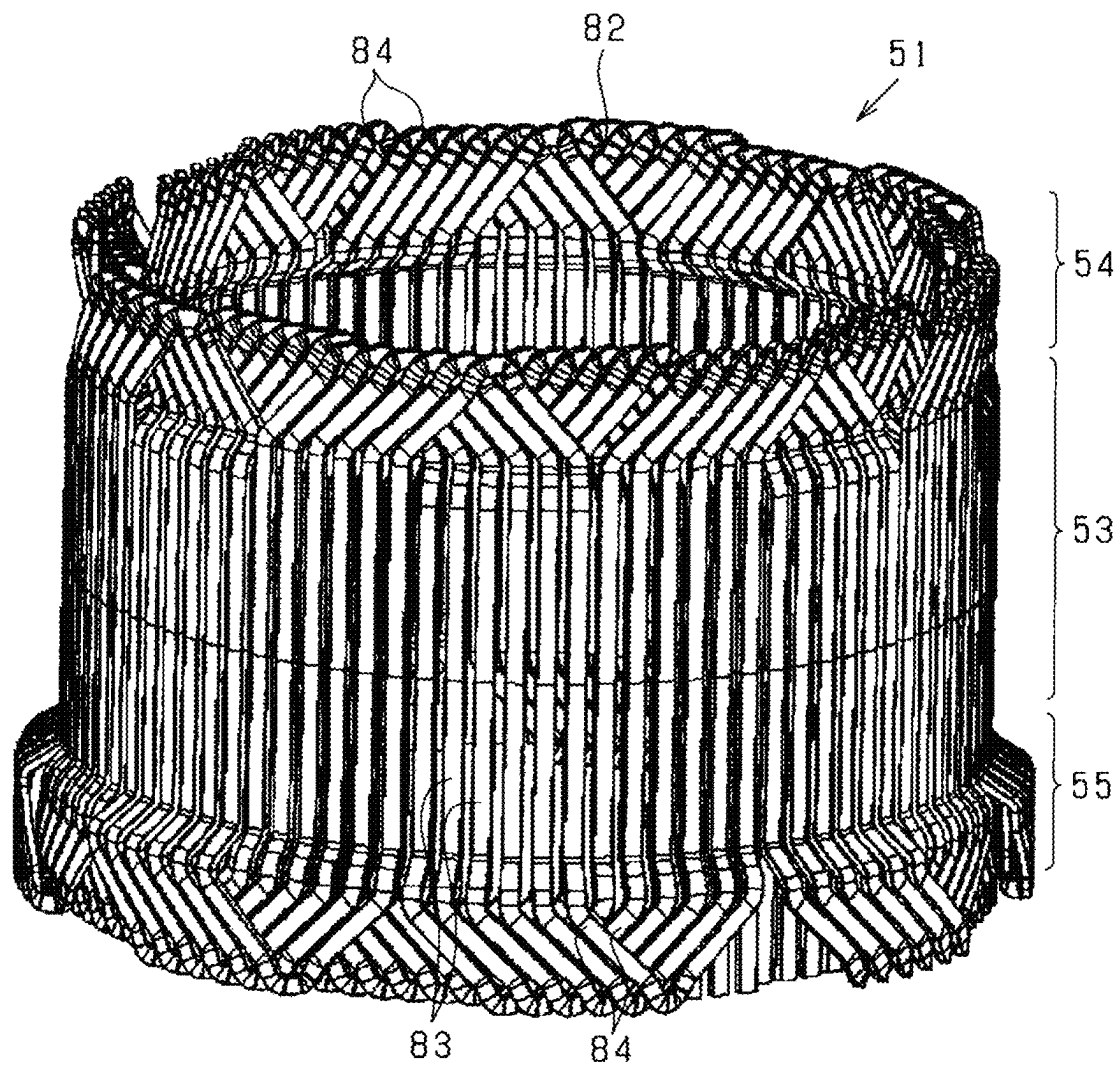


FIG. 13

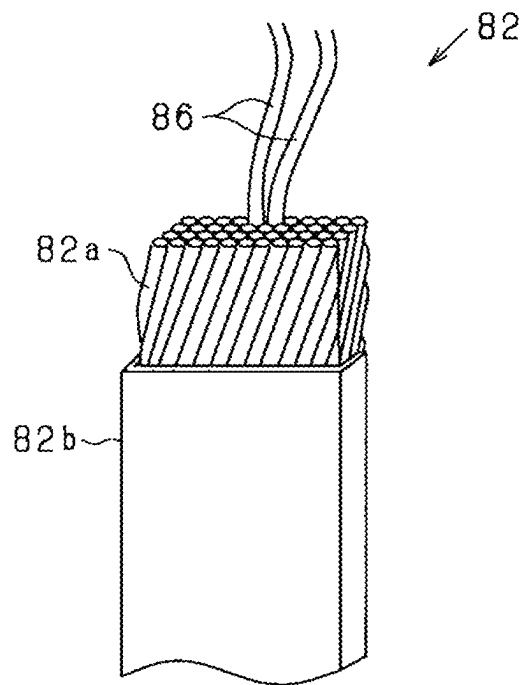


FIG. 14

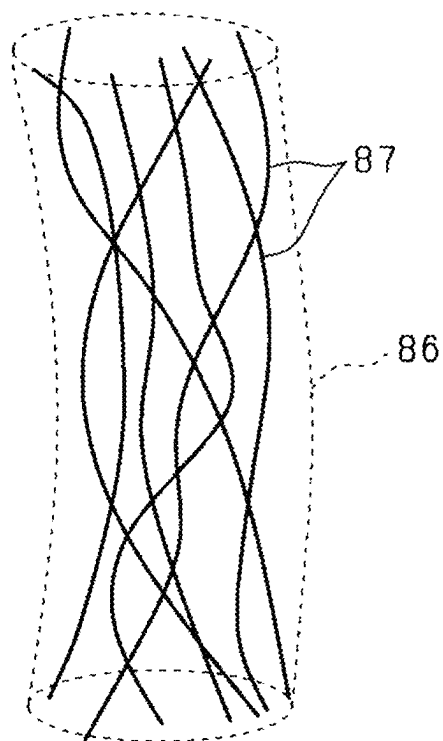


FIG. 15A

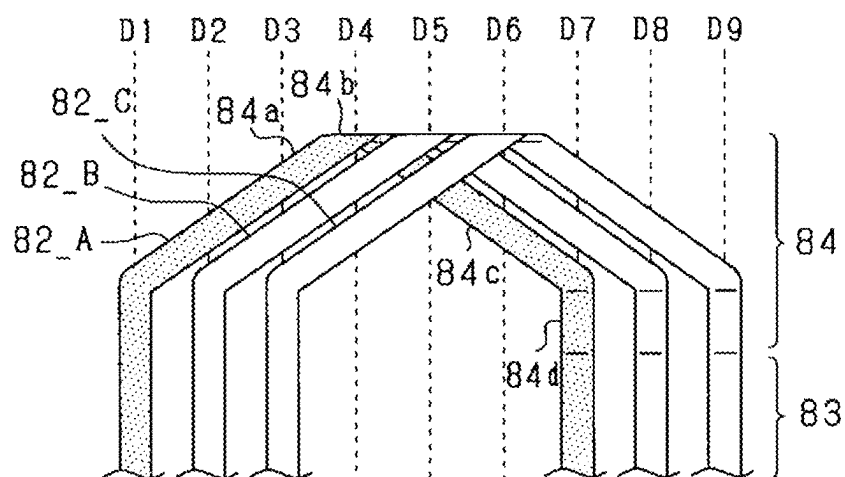


FIG. 15B

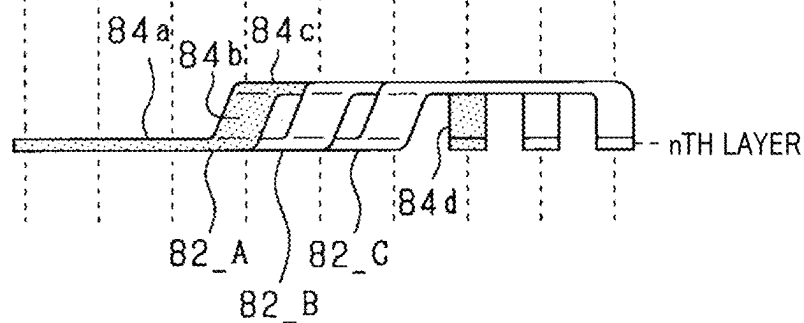


FIG. 16

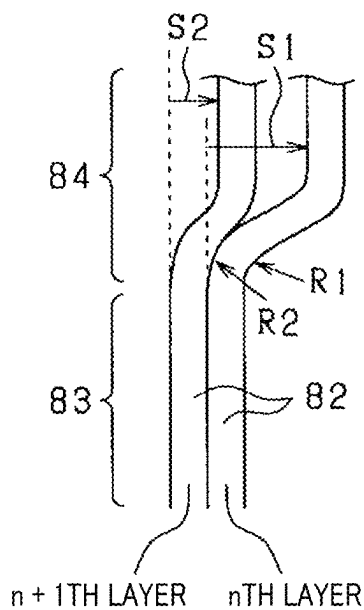


FIG.17

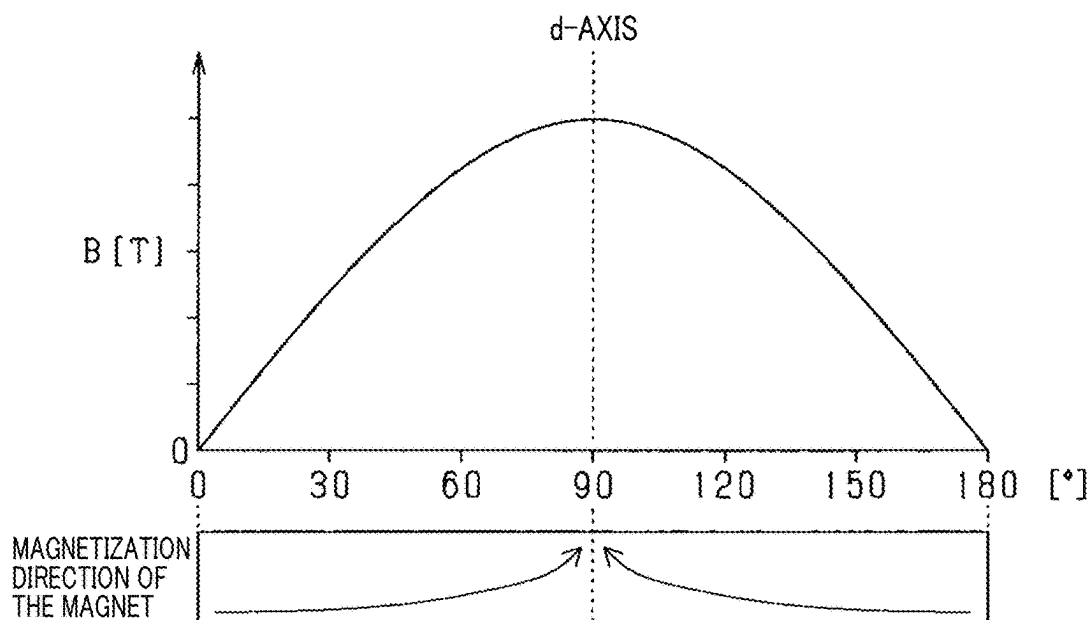


FIG.18

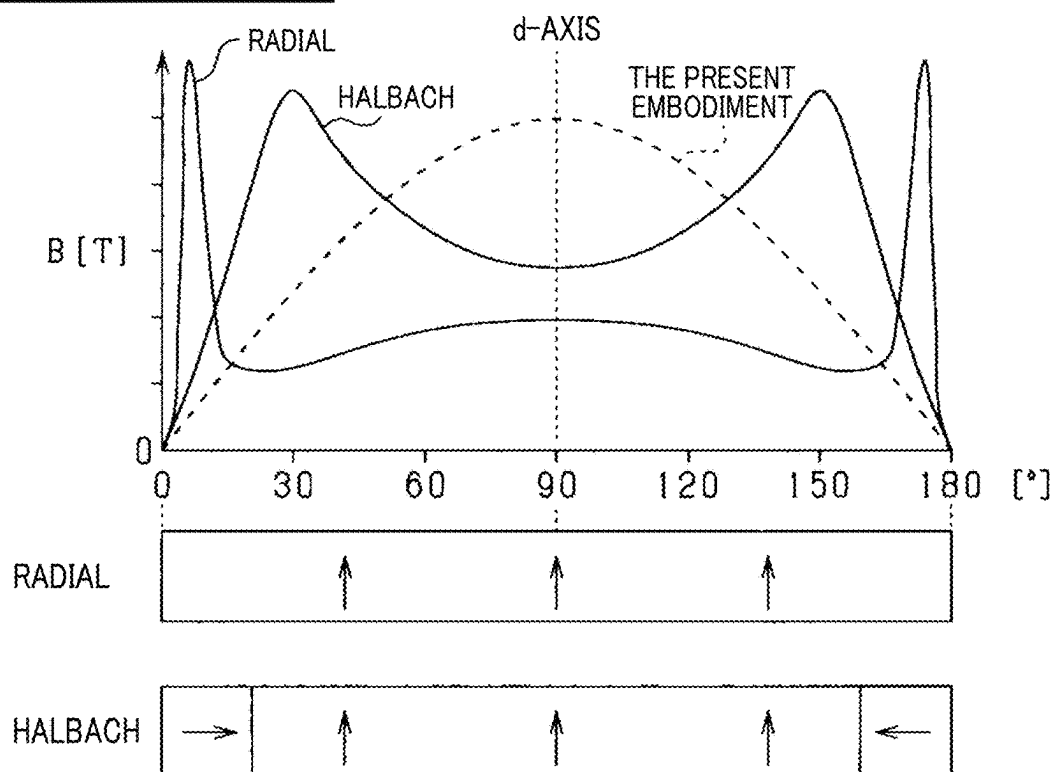
COMPARATIVE EXAMPLE

FIG. 19

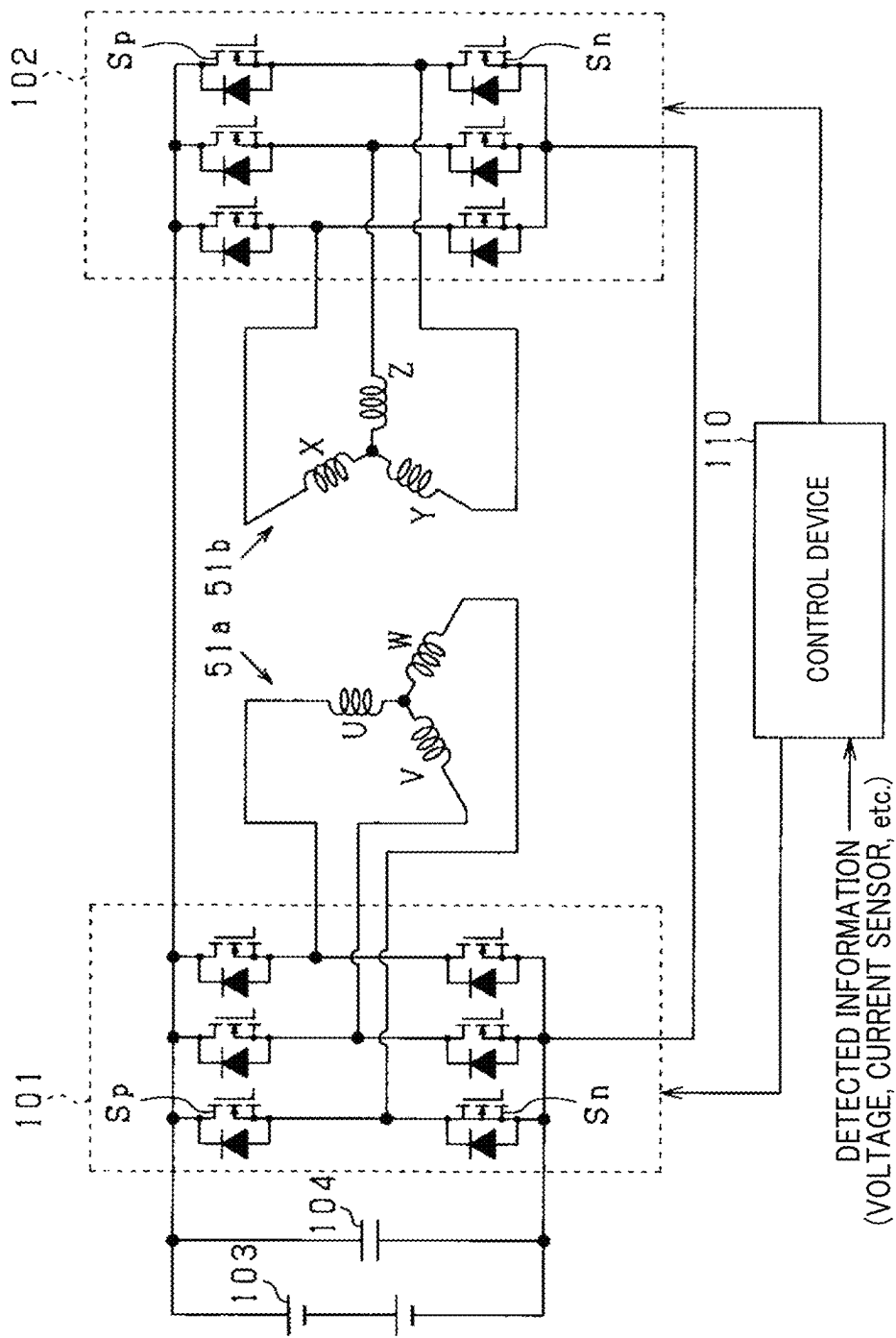


FIG. 20

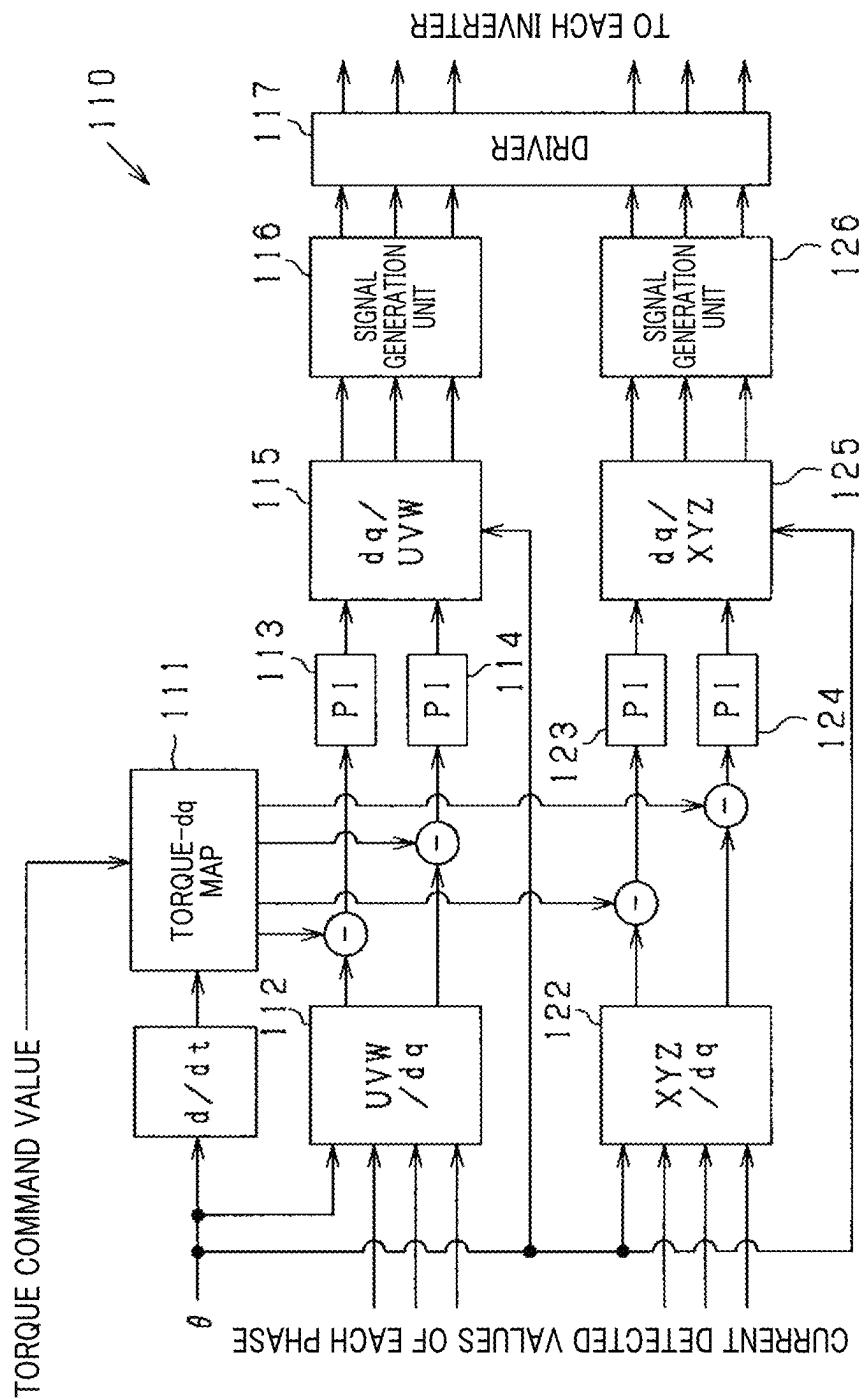


FIG. 21

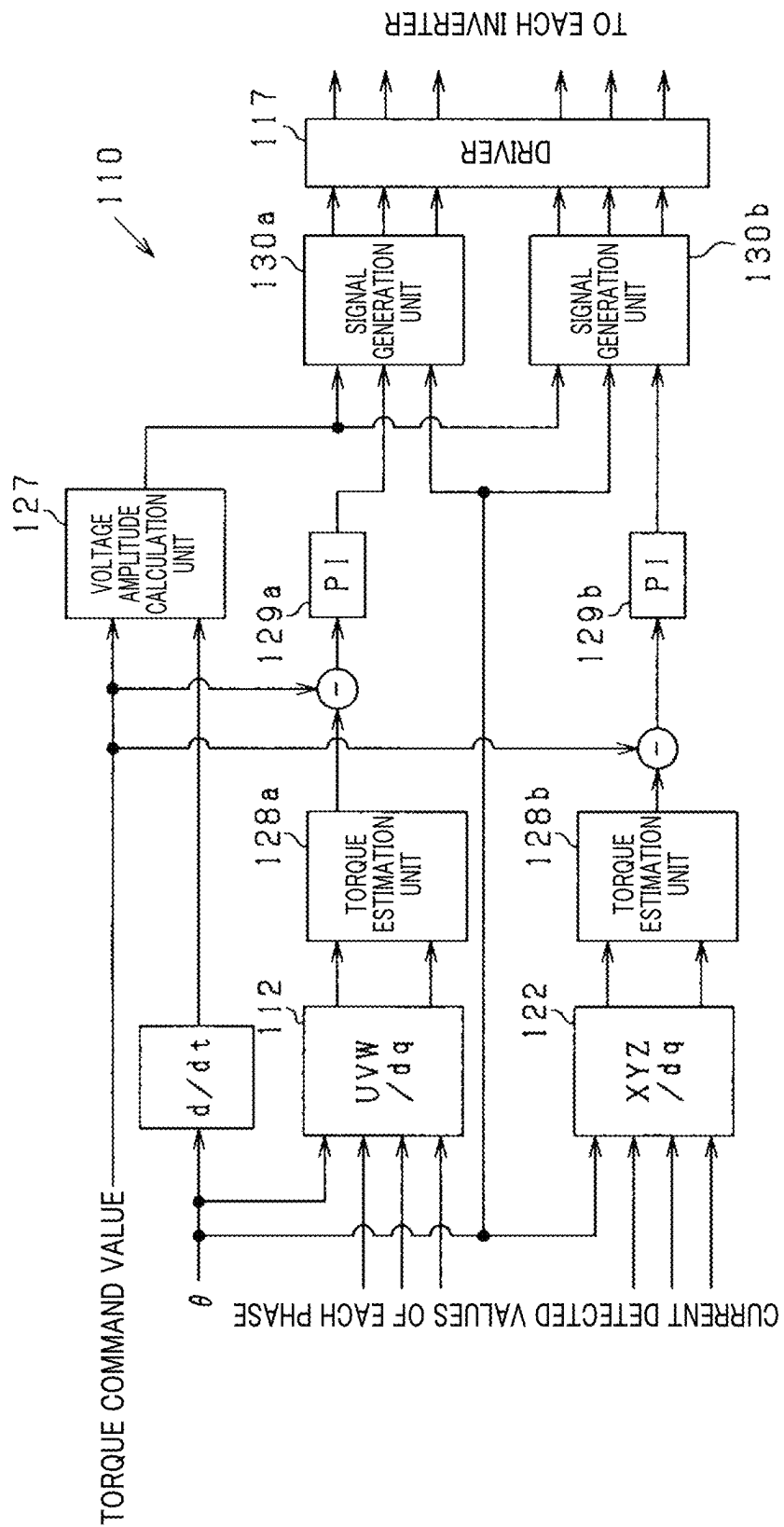


FIG.22

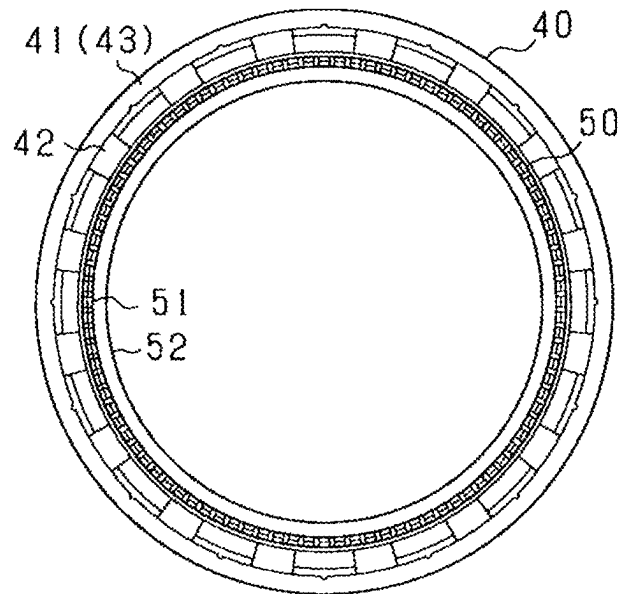


FIG.23

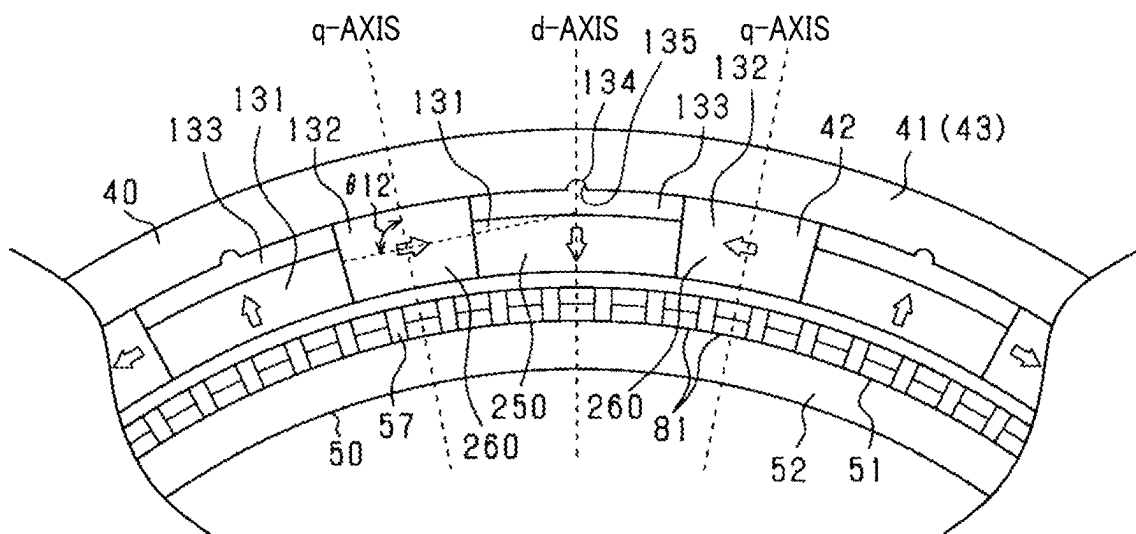


FIG. 24A

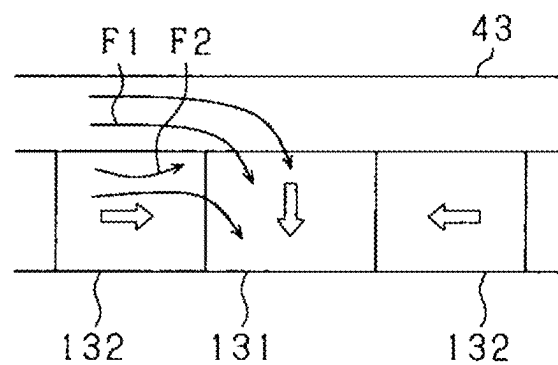


FIG. 24B

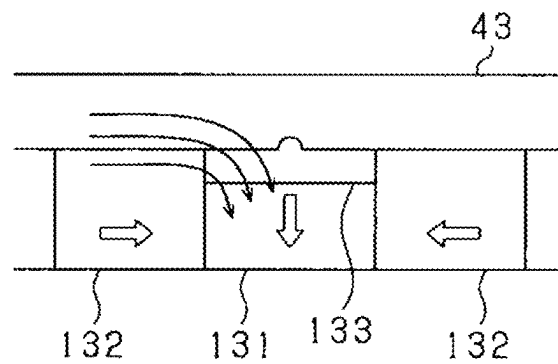


FIG. 25

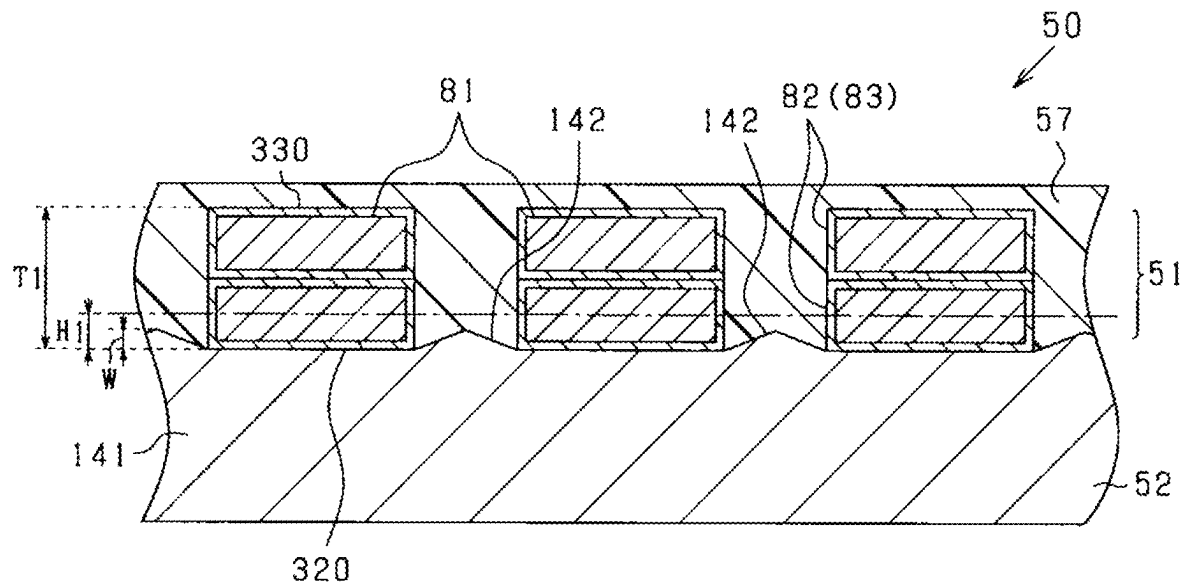


FIG. 26

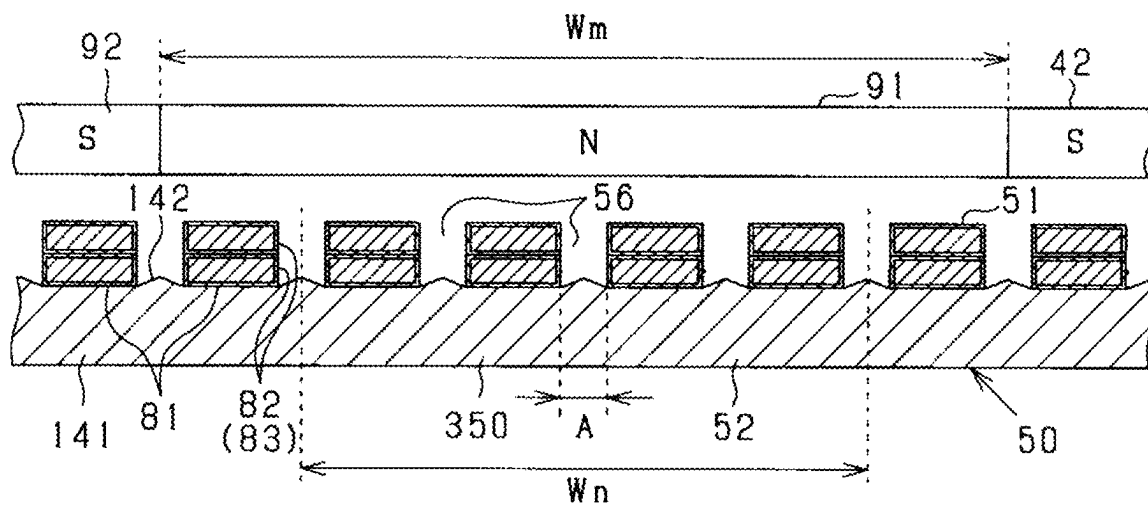


FIG.27

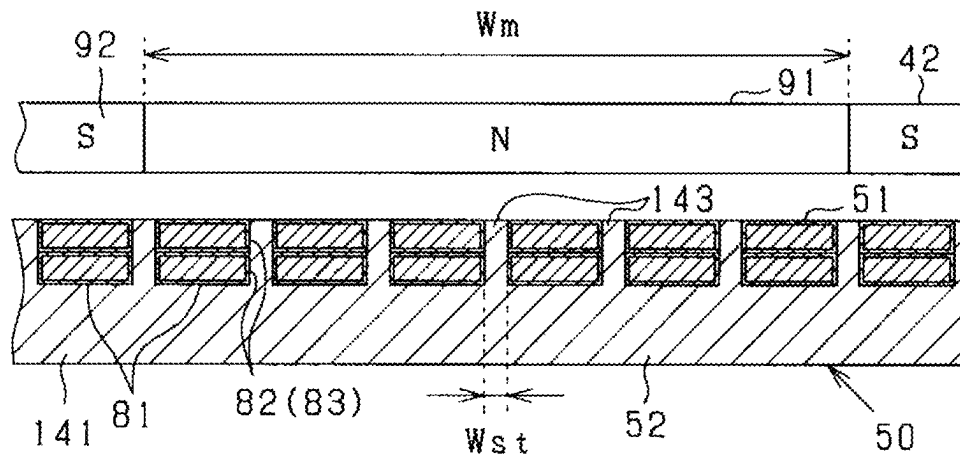


FIG.28

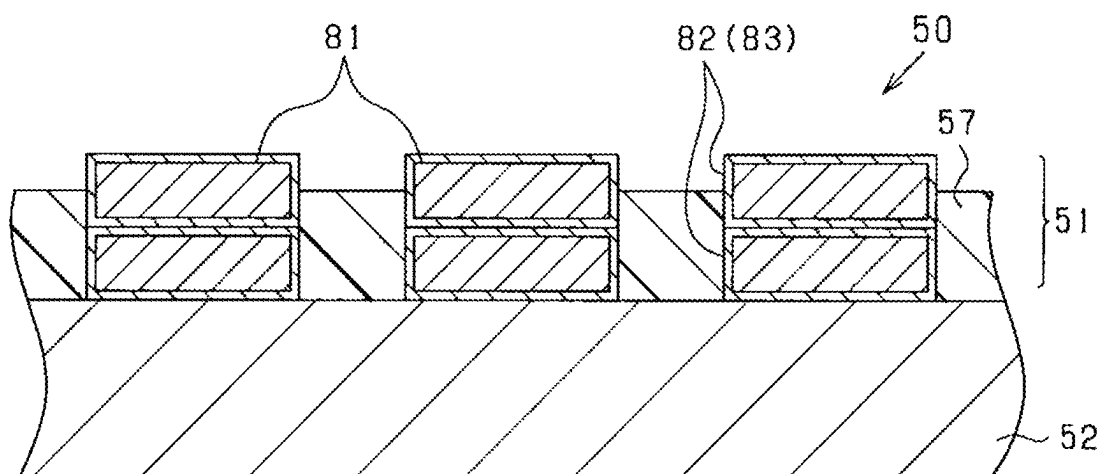


FIG.29

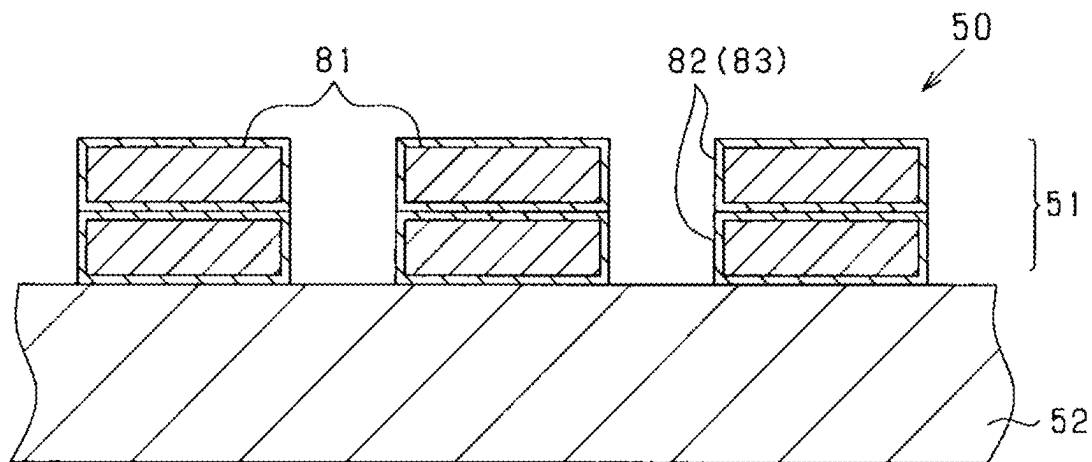


FIG.30

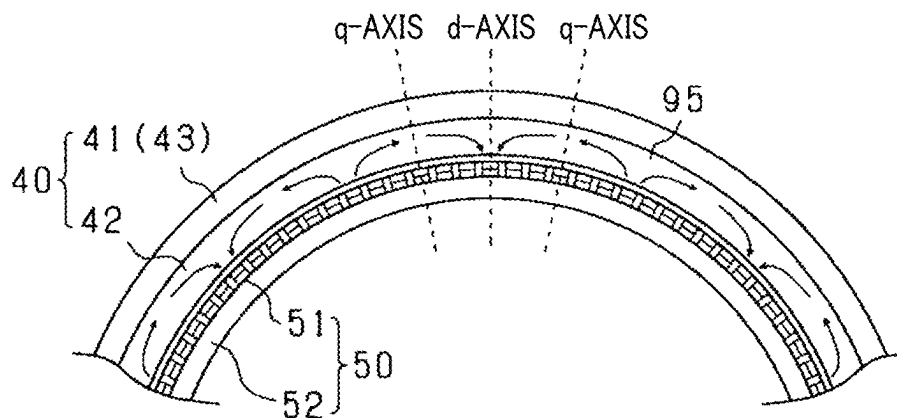


FIG.31

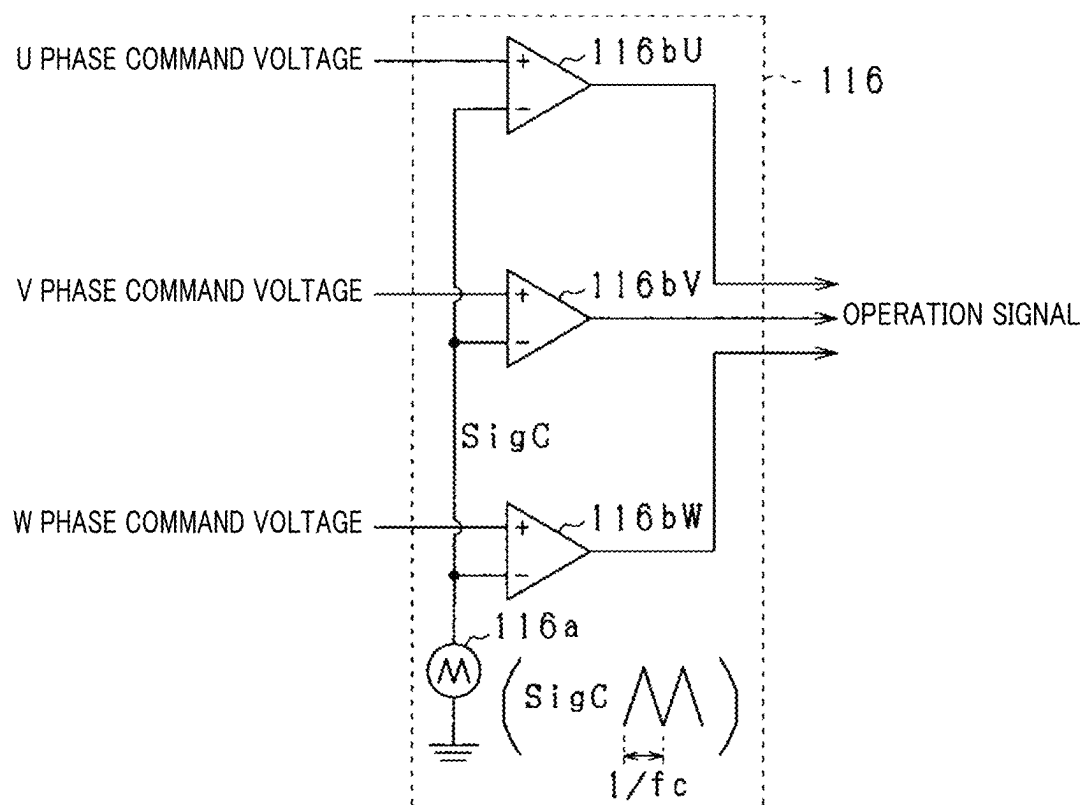


FIG.32

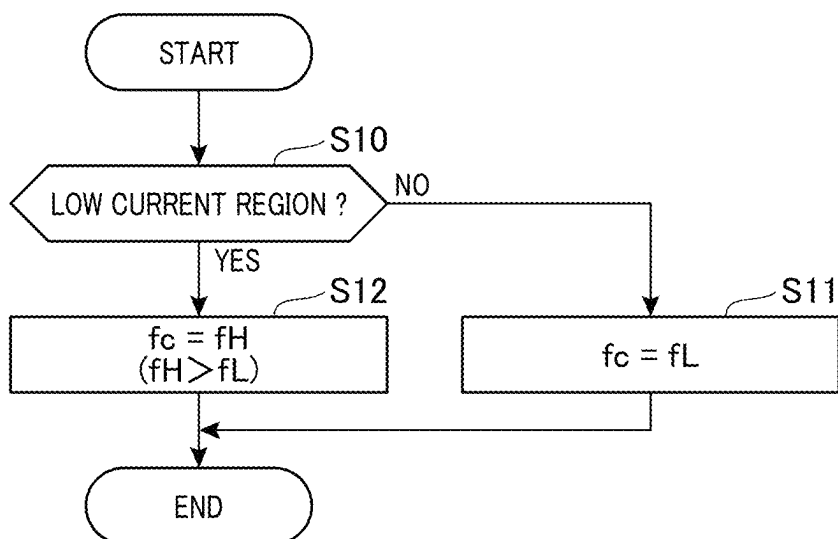


FIG. 33A

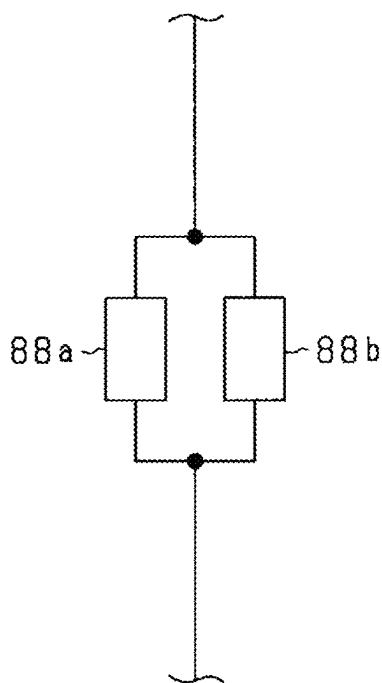


FIG. 33B

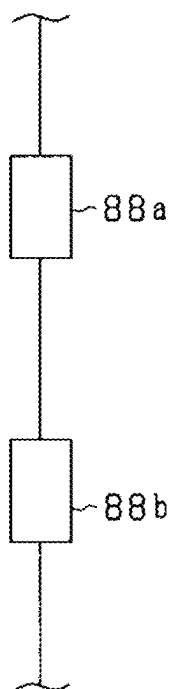


FIG. 33C

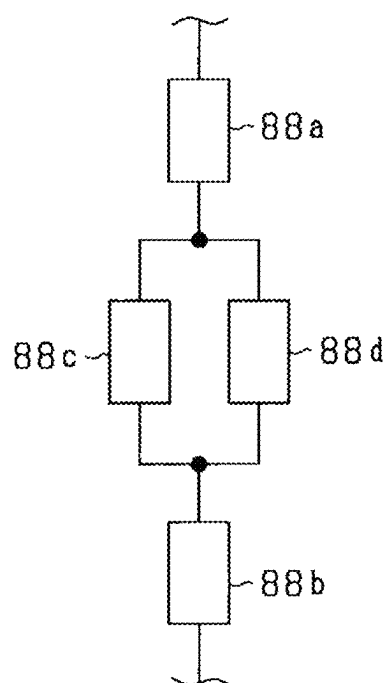
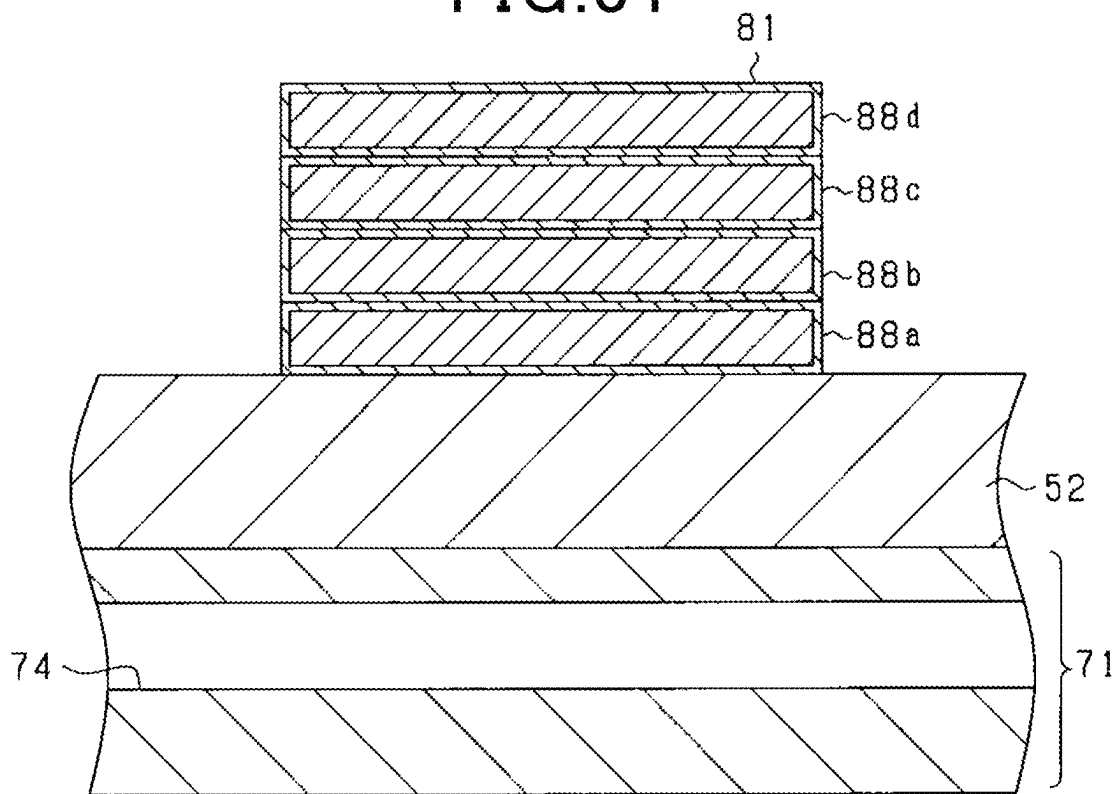


FIG. 34



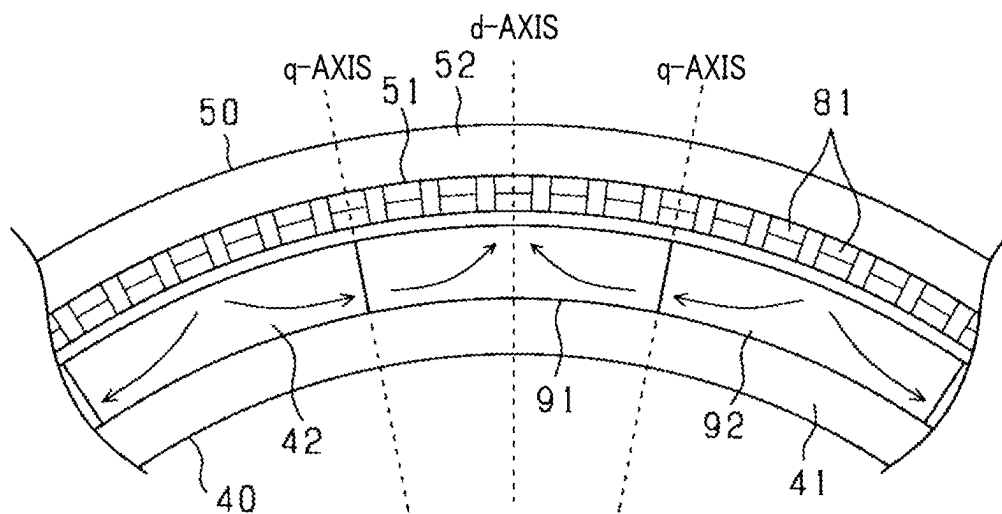


FIG. 37

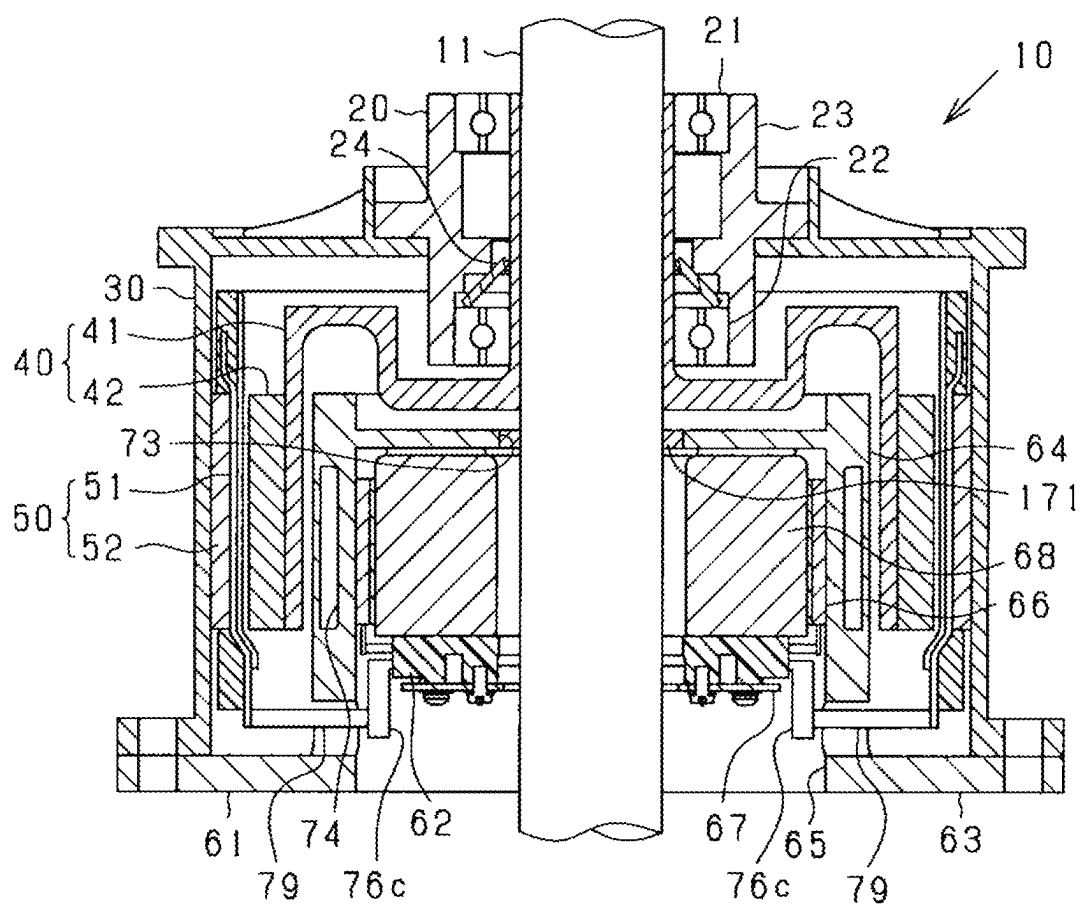


FIG. 38

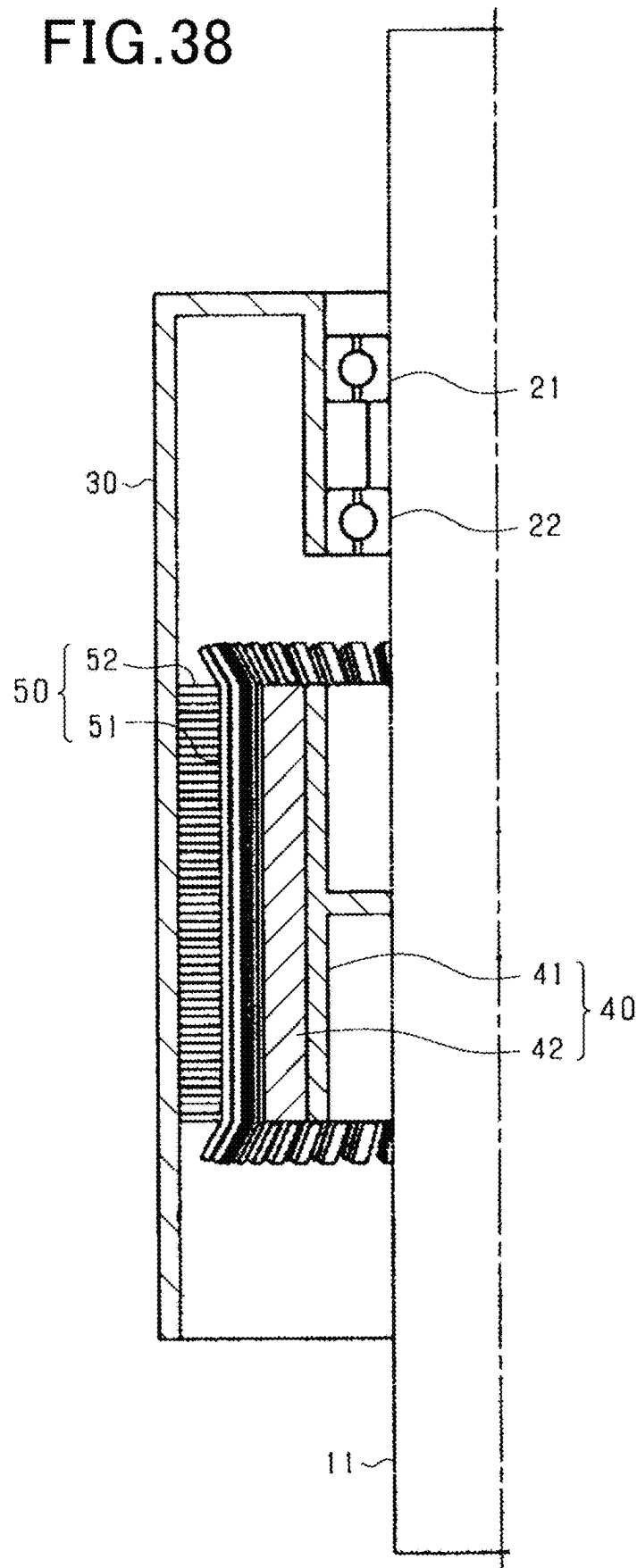


FIG. 39

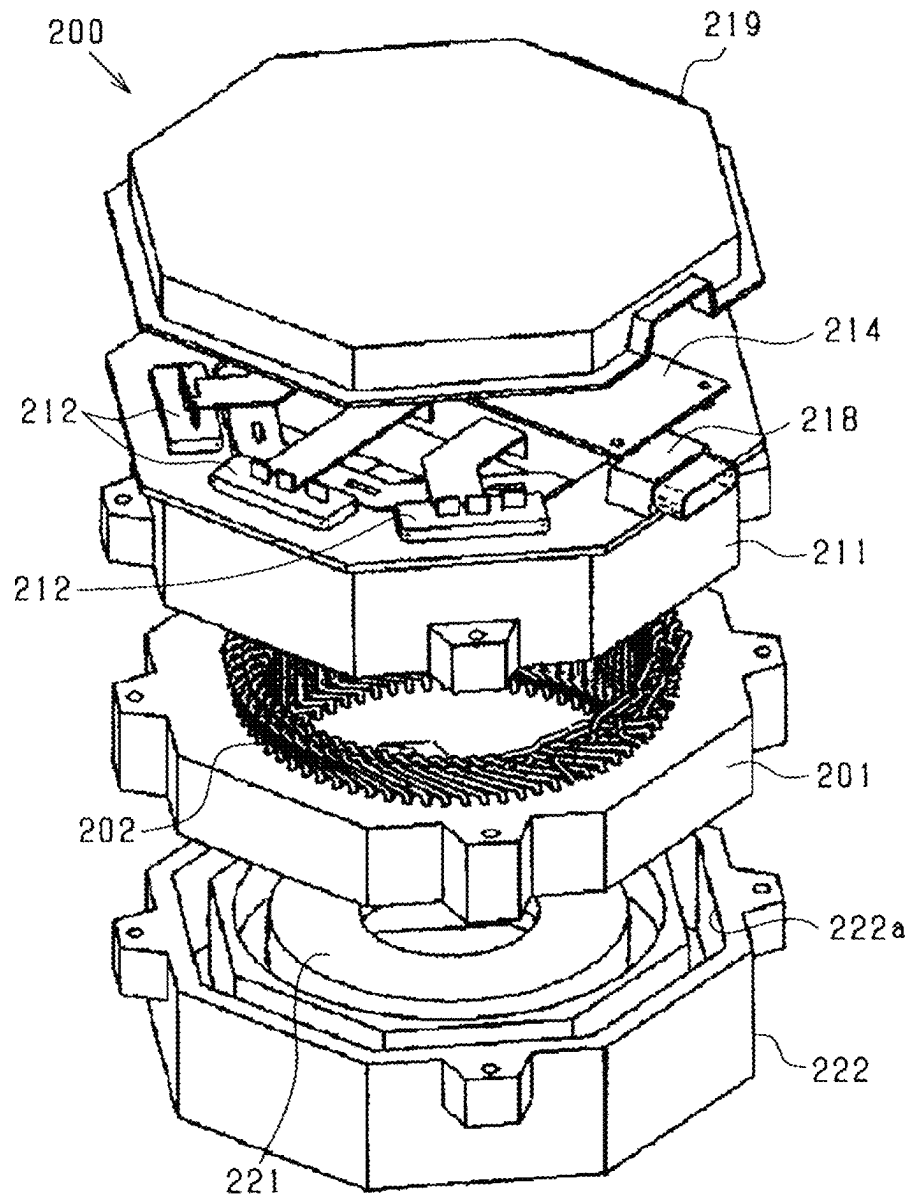


FIG. 40

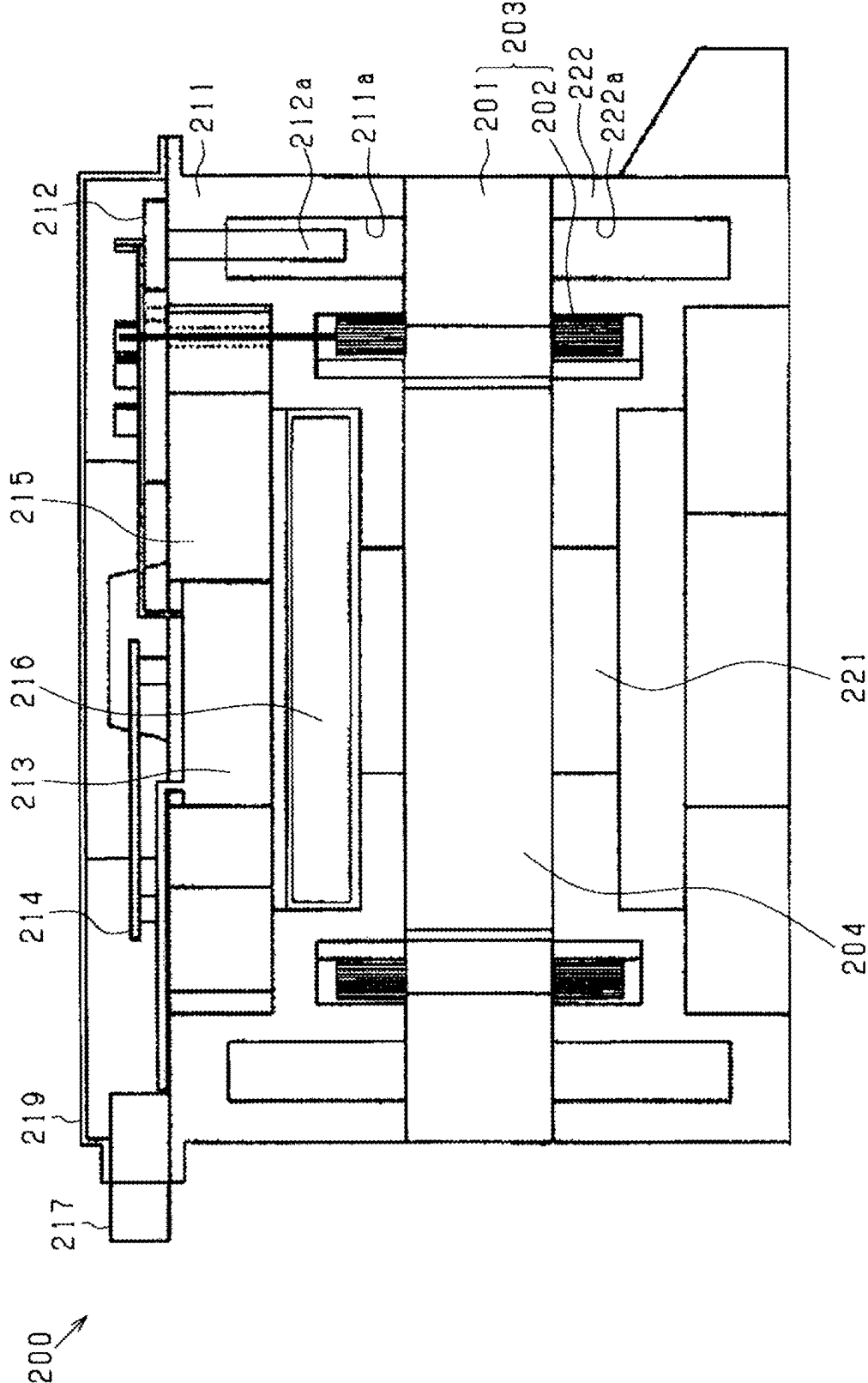


FIG. 41

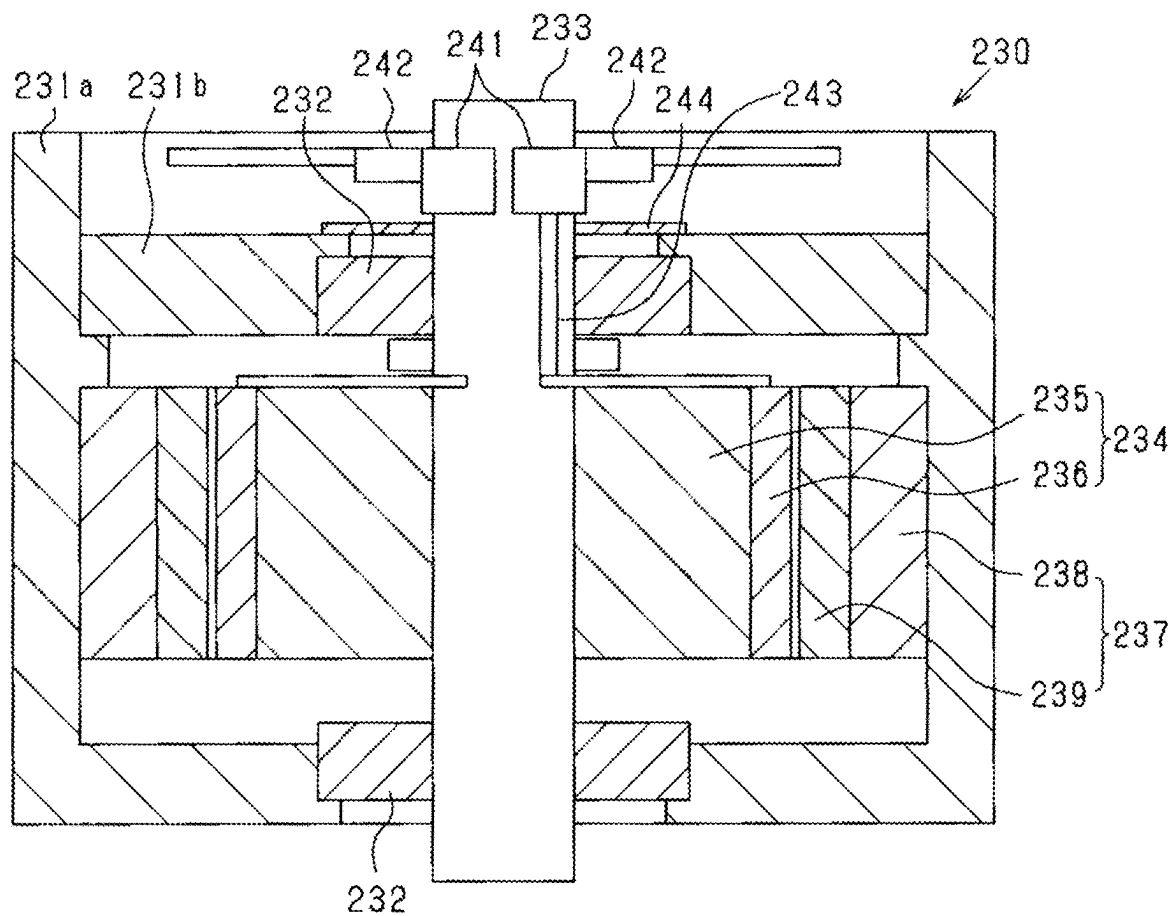


FIG. 42

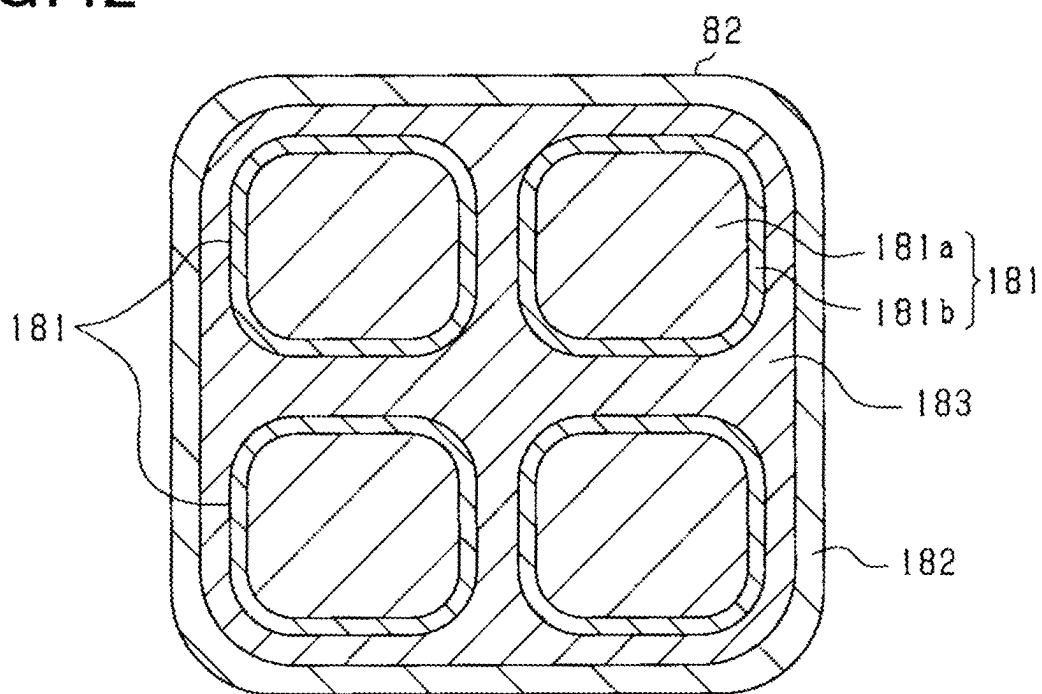


FIG. 43

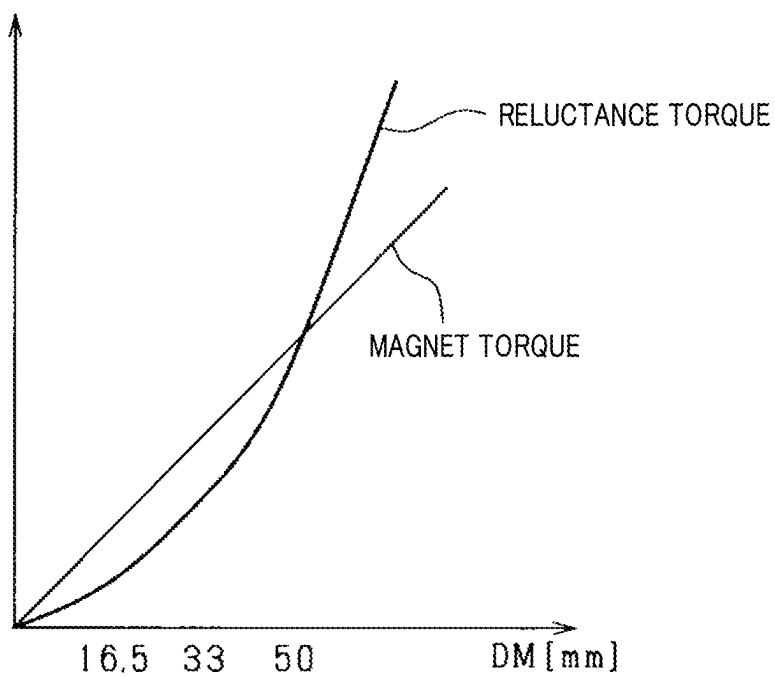


FIG. 44

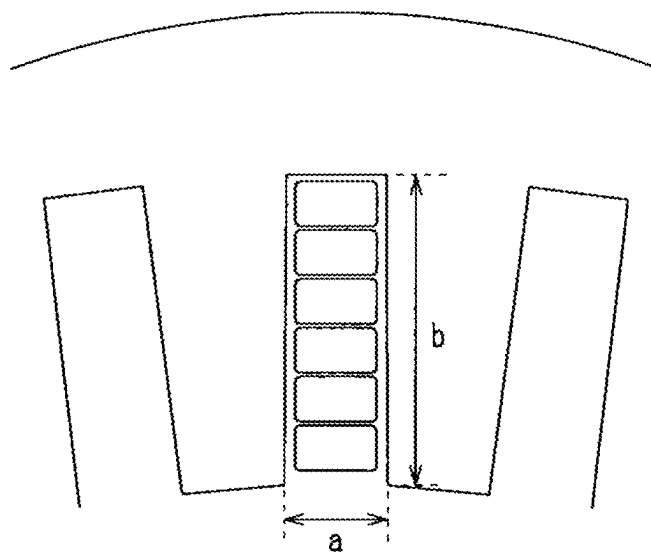


FIG.45

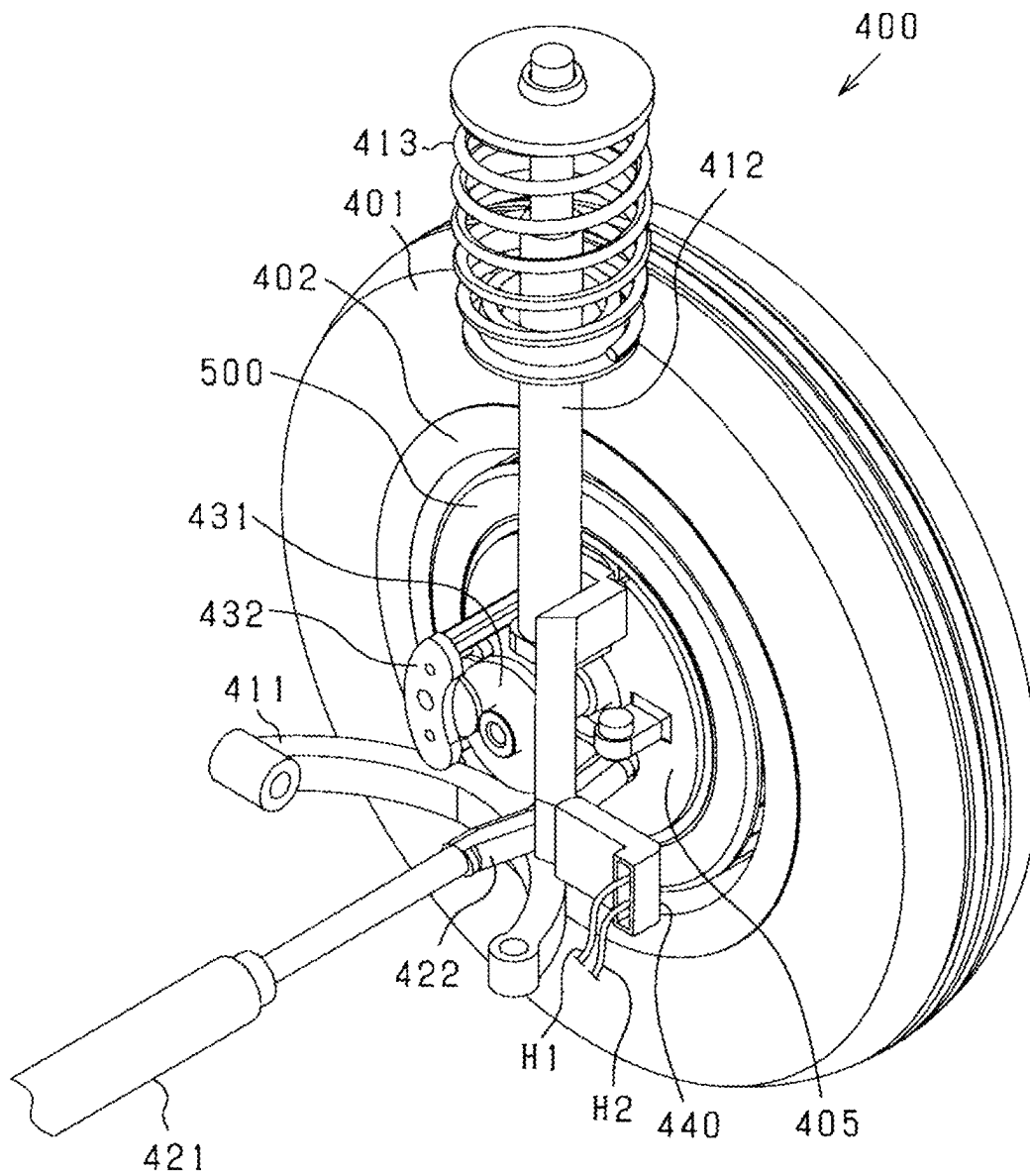


FIG. 46

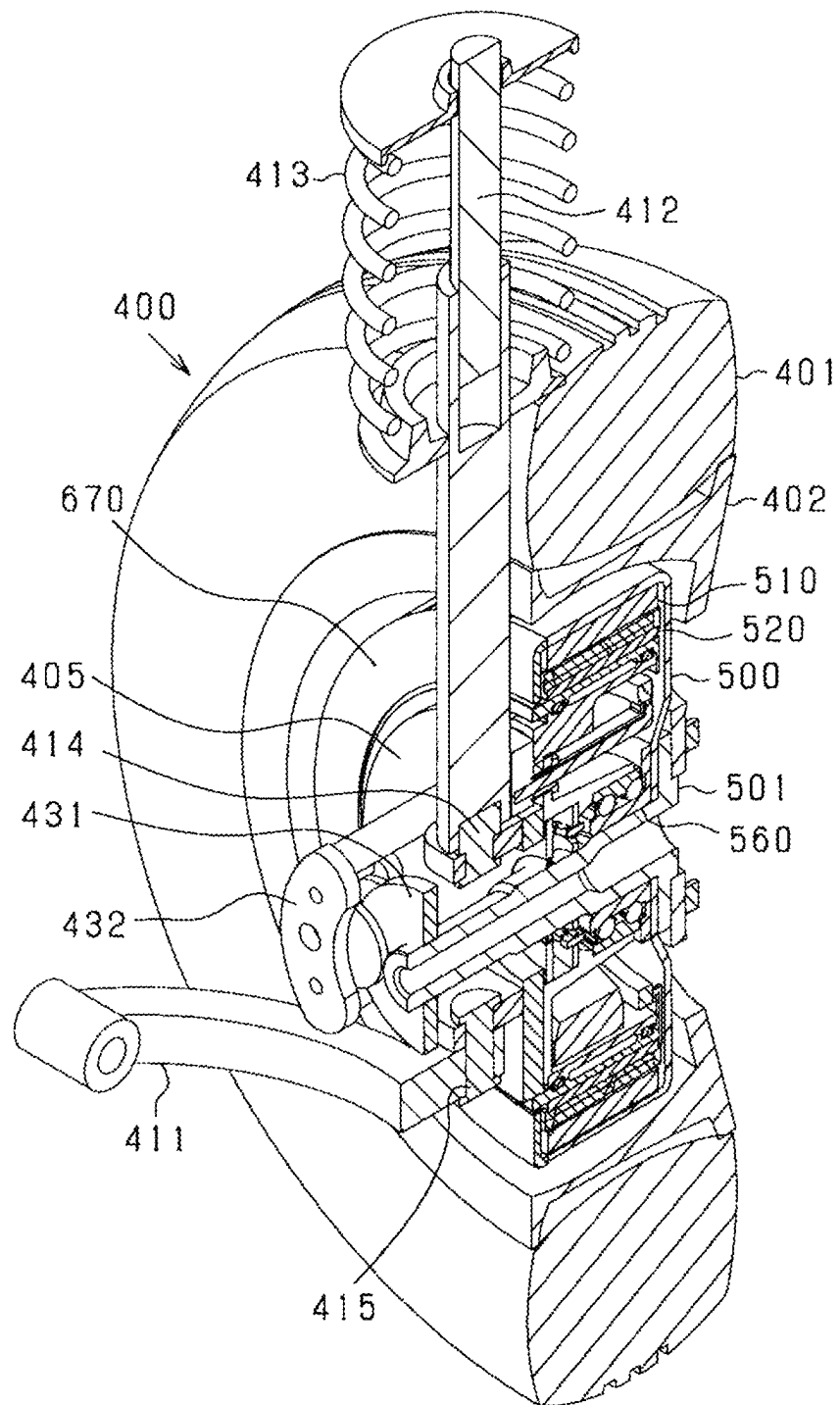


FIG. 47

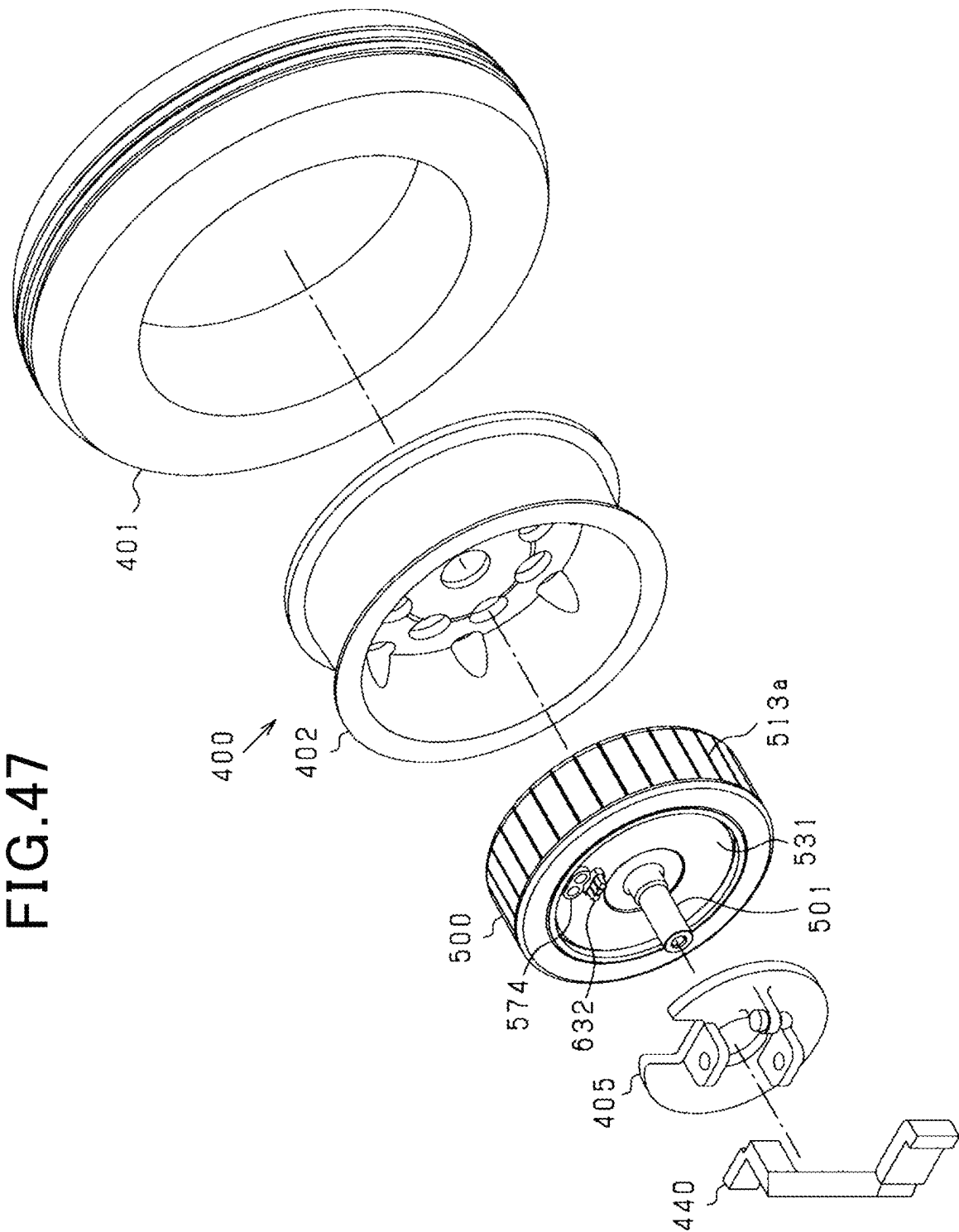


FIG. 48

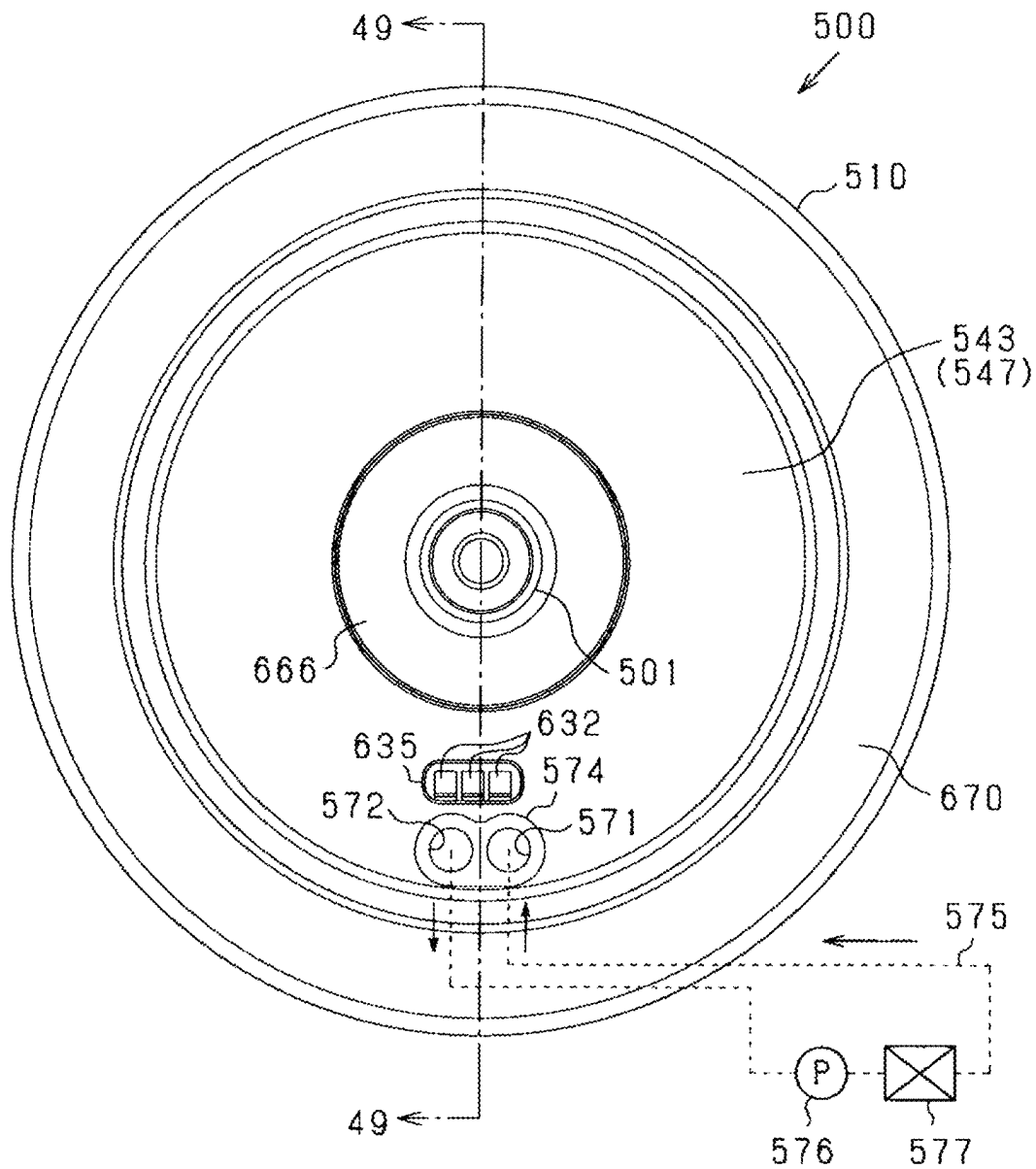


FIG. 49

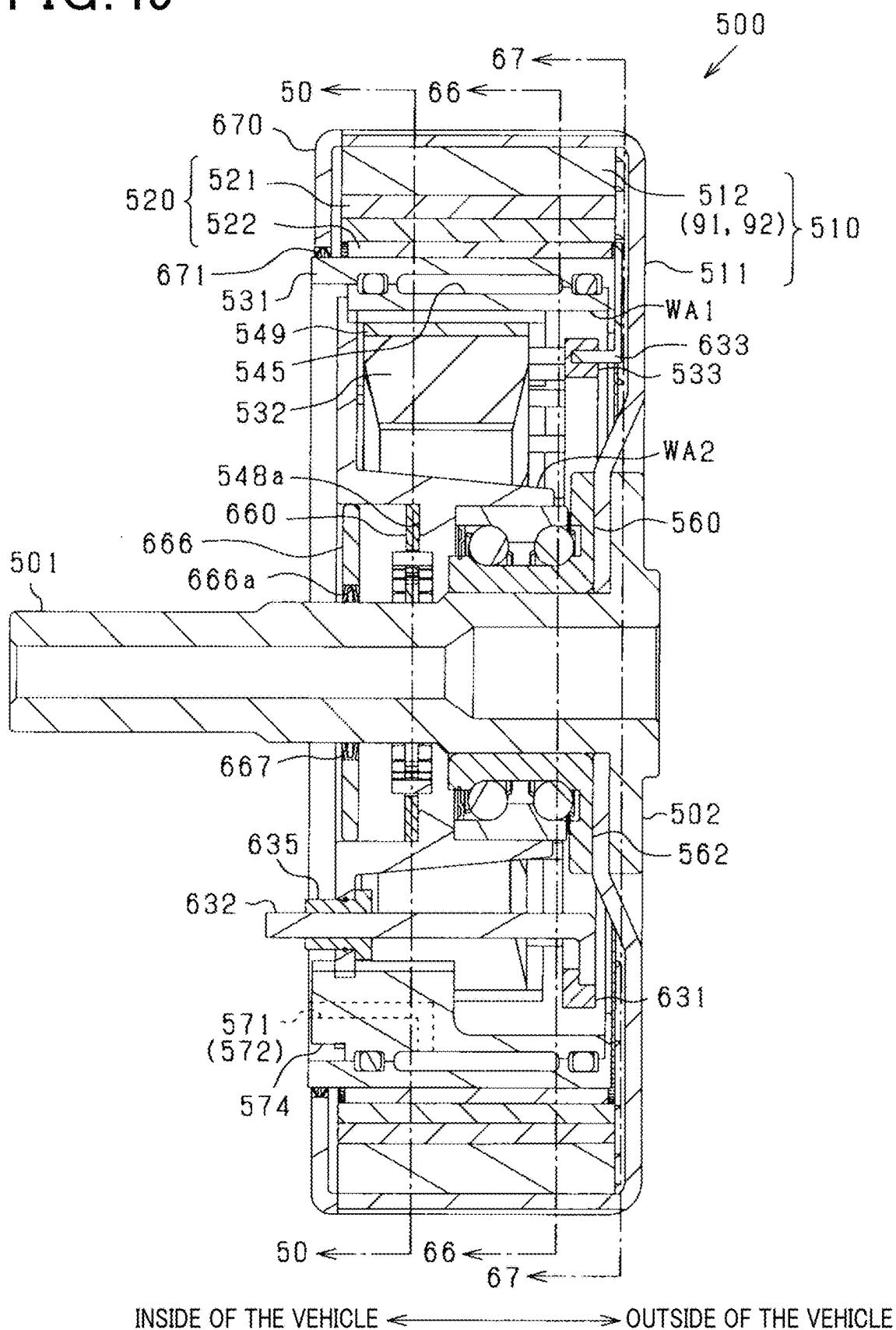


FIG. 50

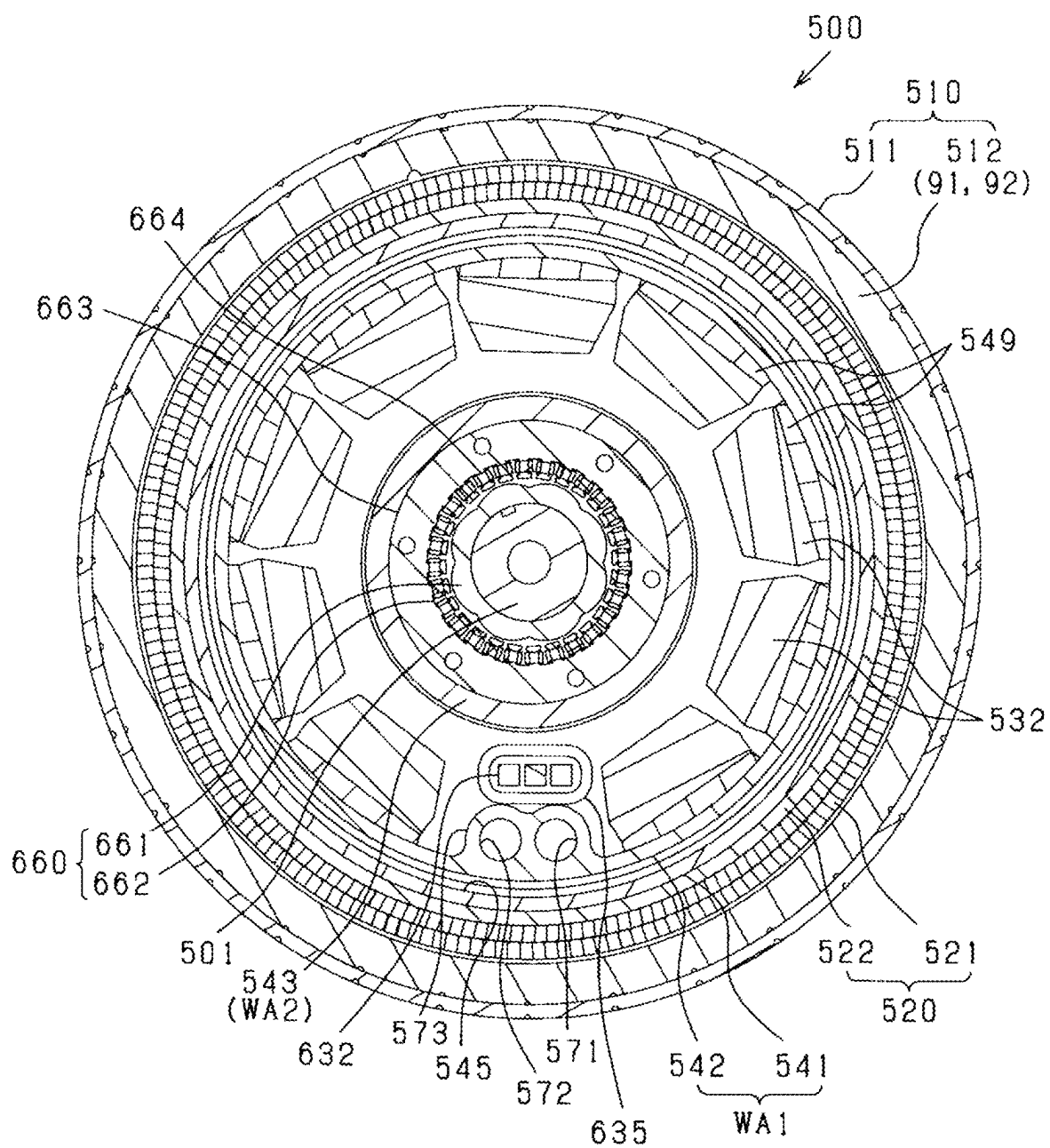


FIG. 51

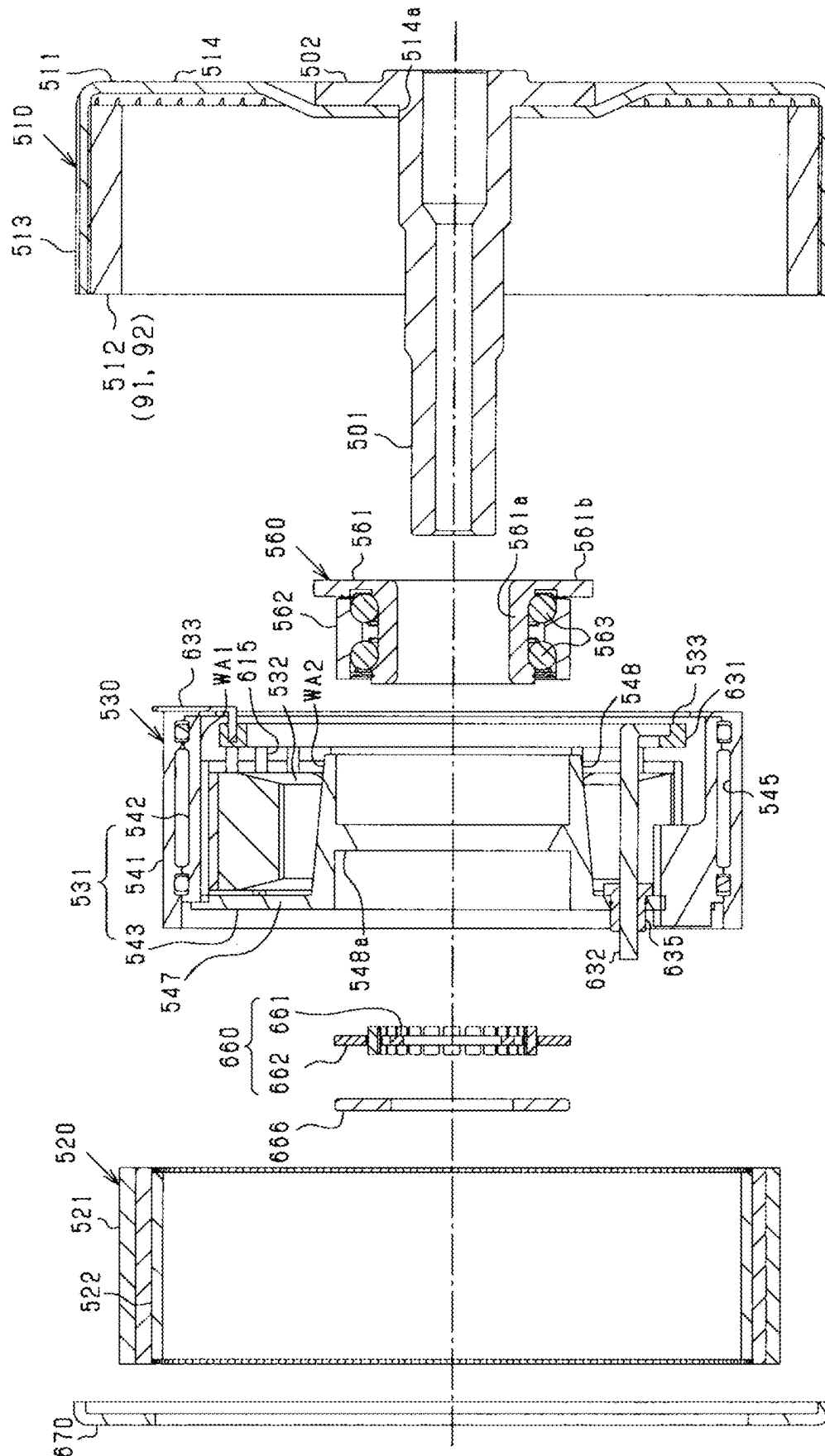


FIG. 52

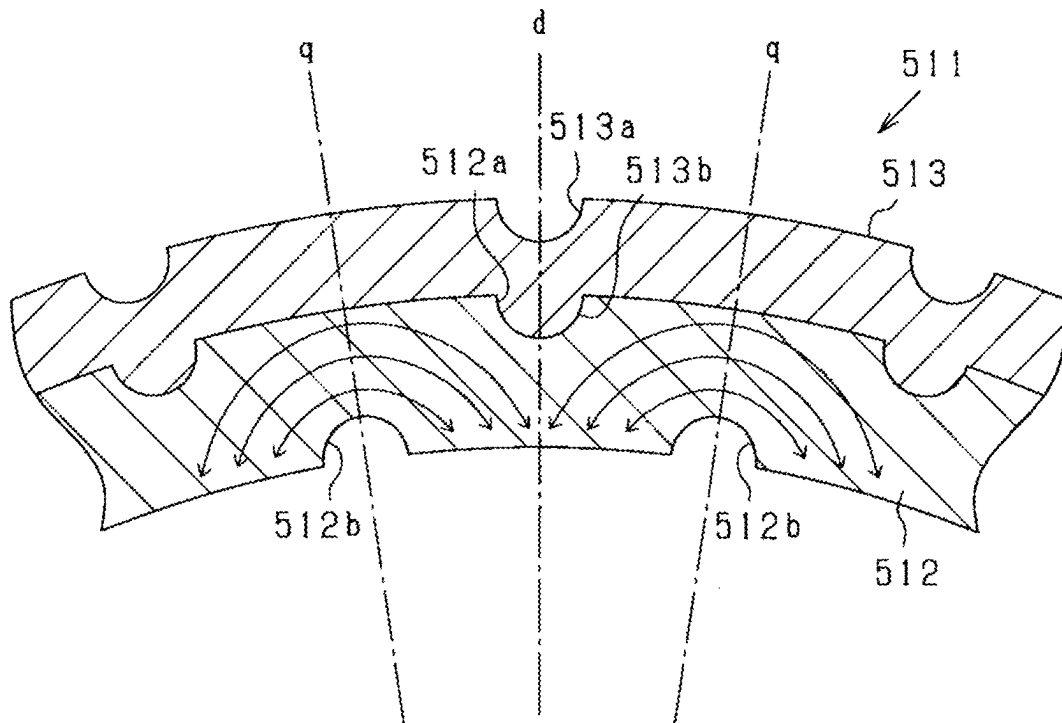


FIG. 53

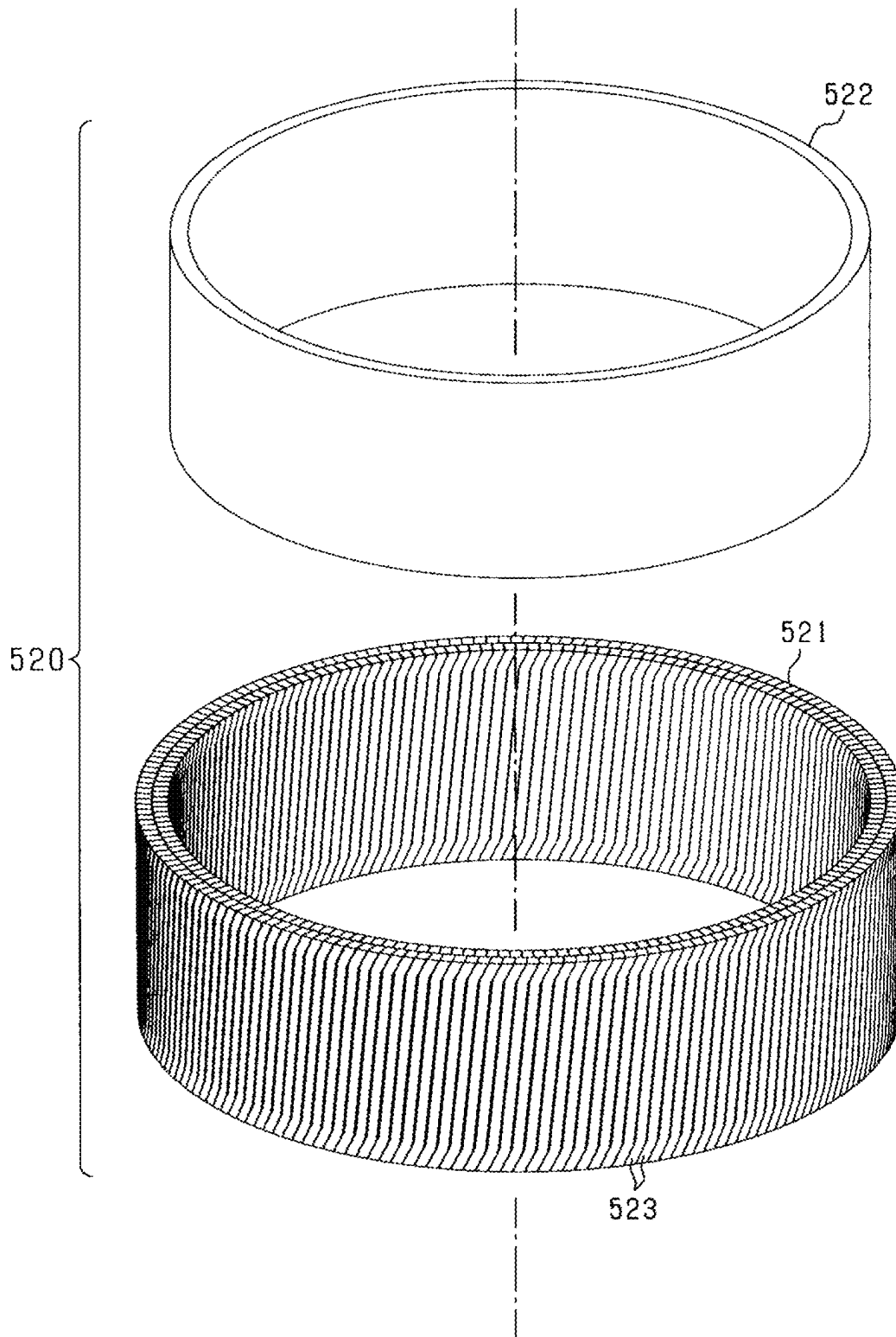


FIG.54A

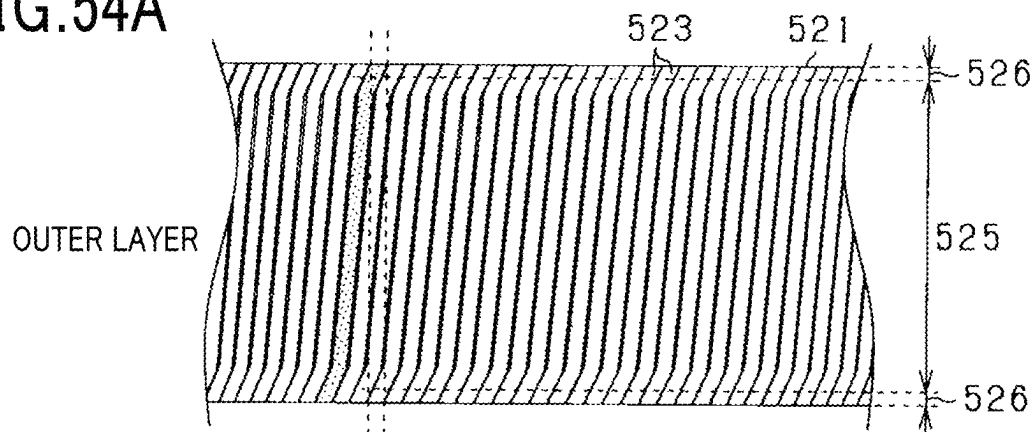


FIG.54B

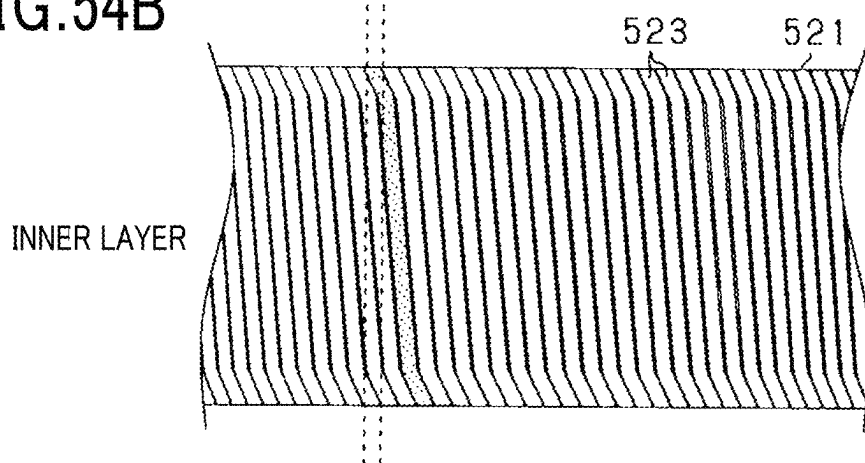


FIG.55

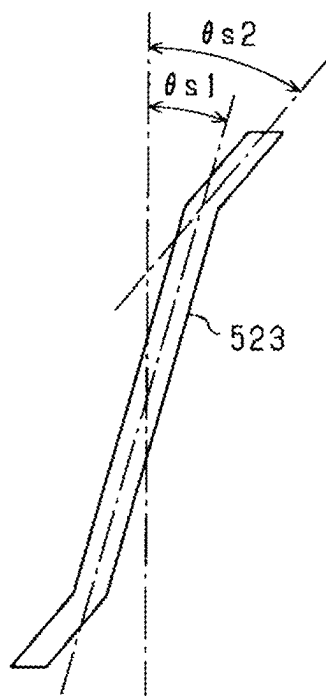


FIG. 56

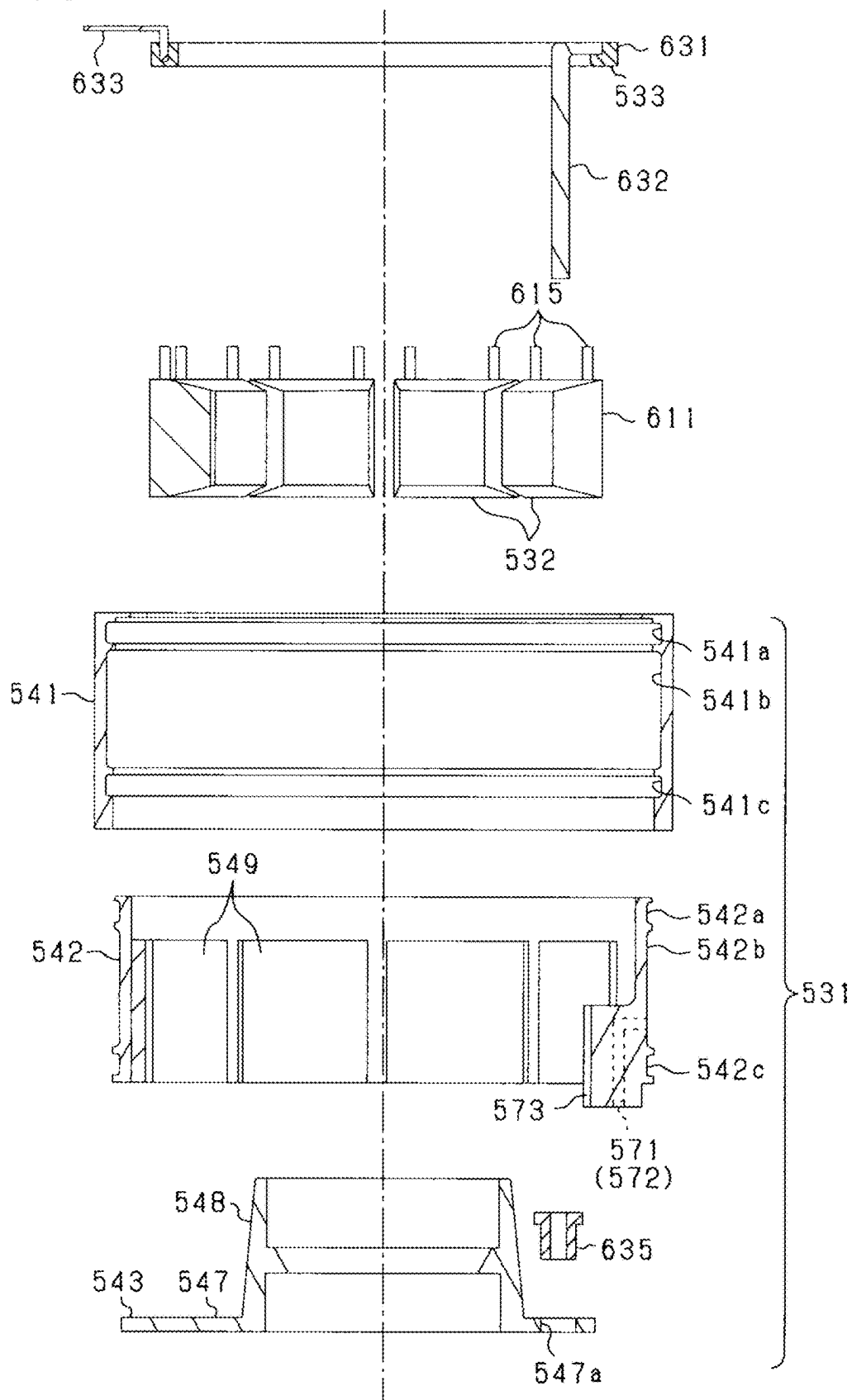


FIG. 57

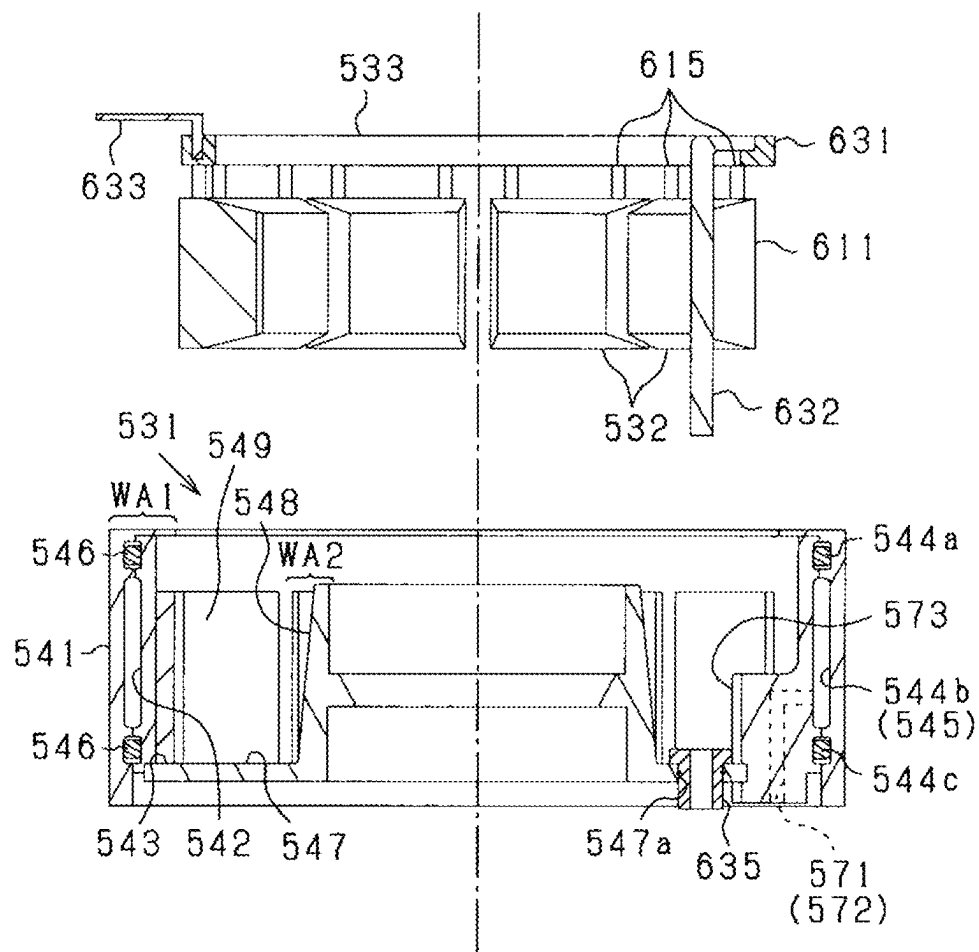


FIG. 58

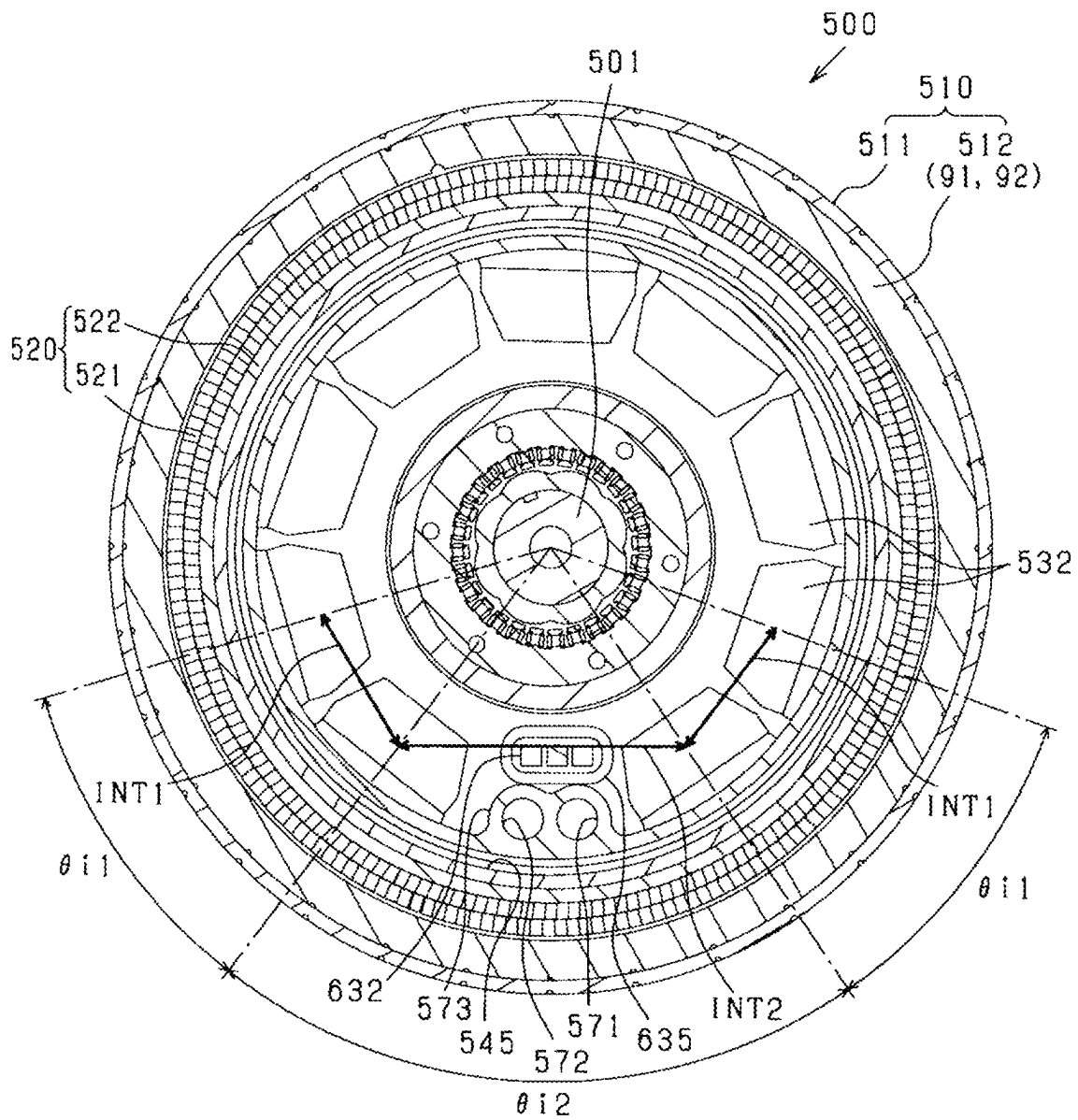


FIG. 59

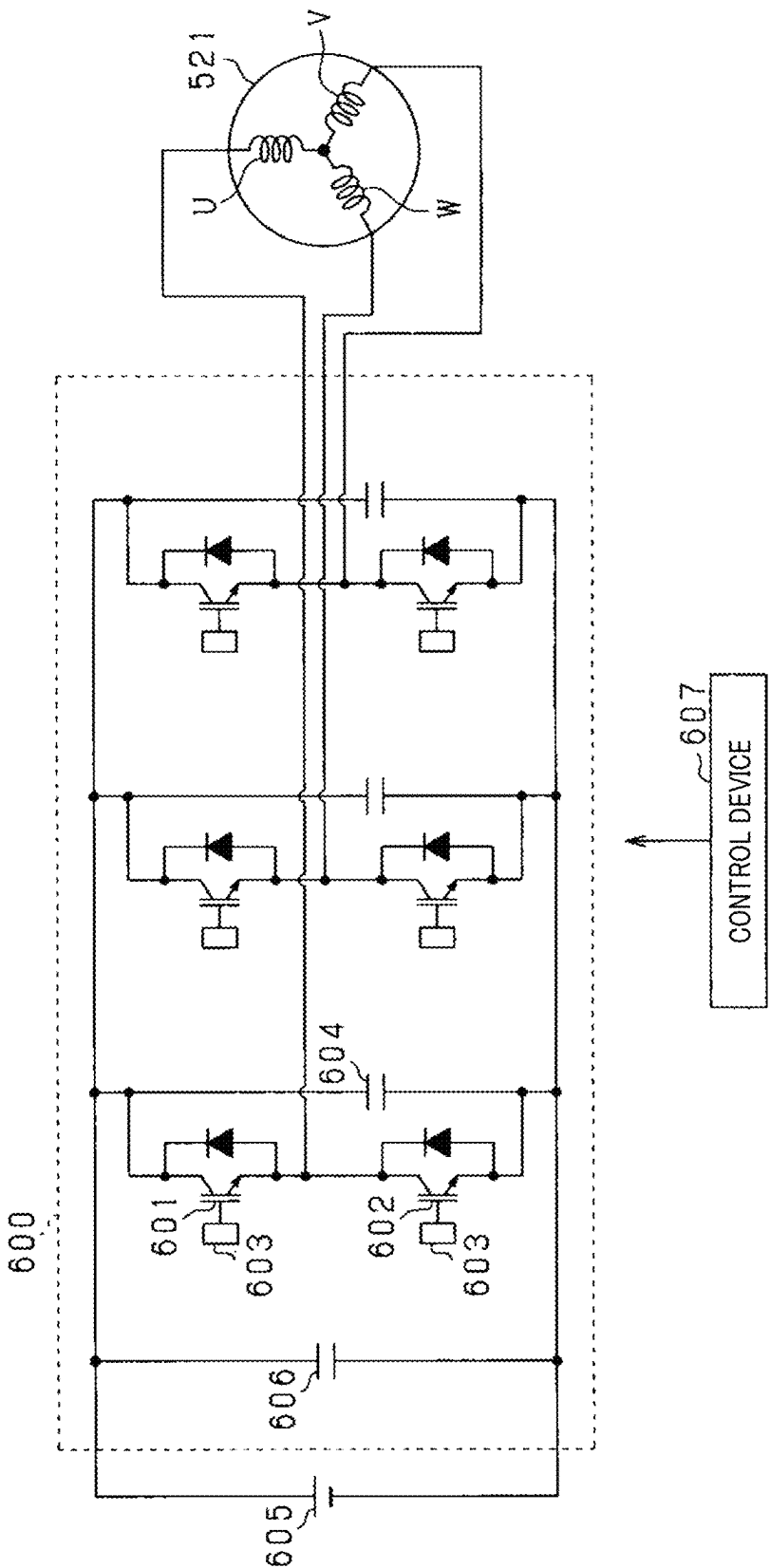


FIG. 60

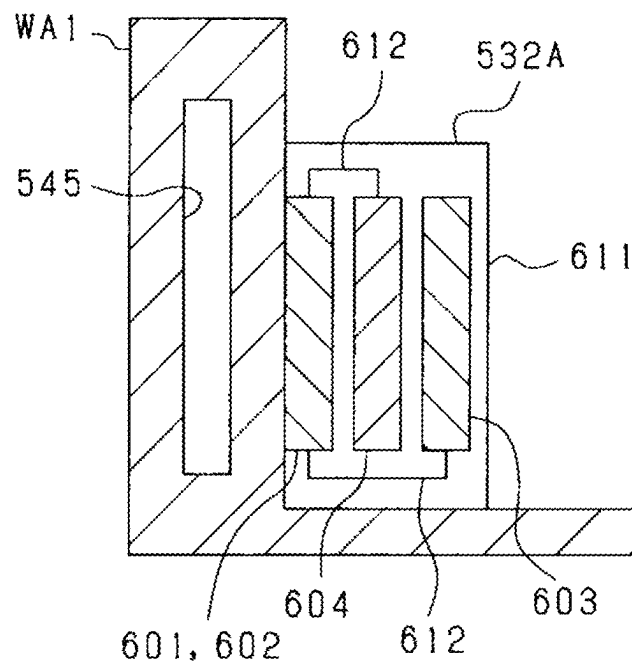


FIG. 61A

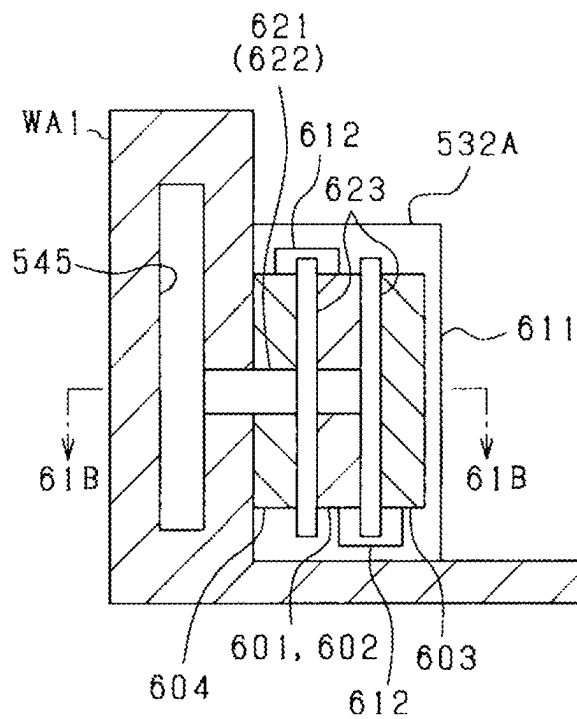


FIG. 61B

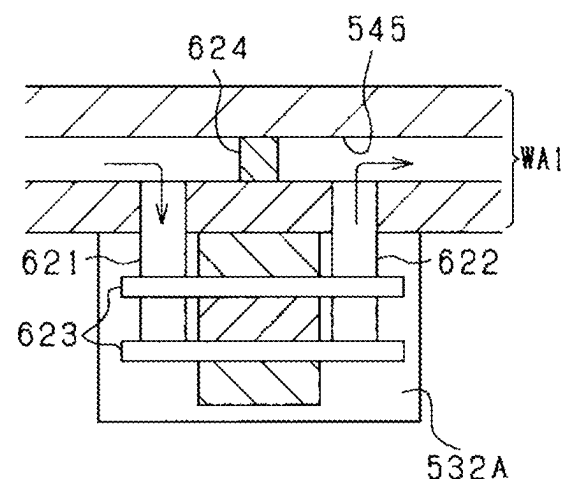


FIG. 62A

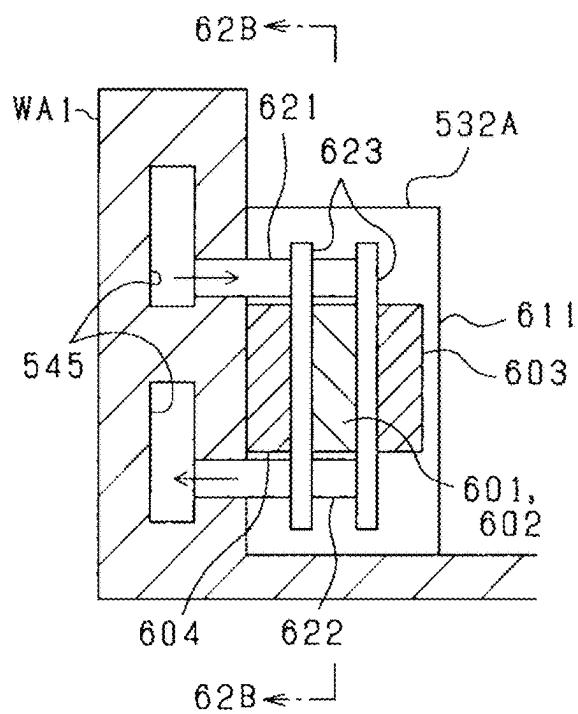


FIG. 62B

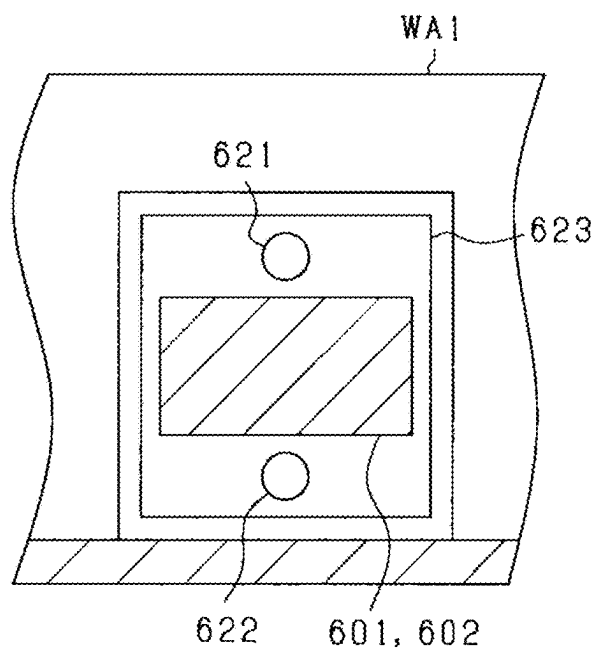


FIG. 62C

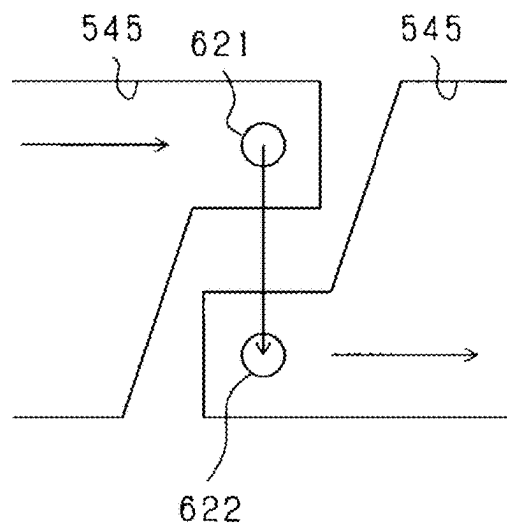


FIG. 63A

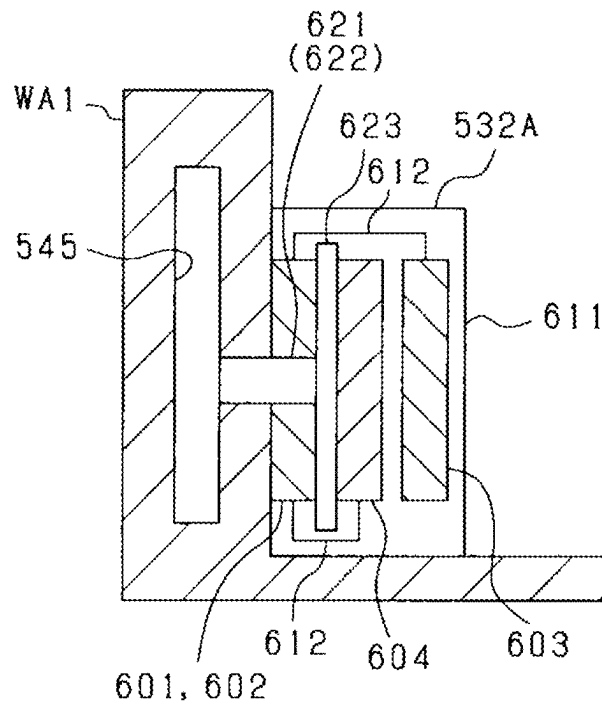


FIG. 63B

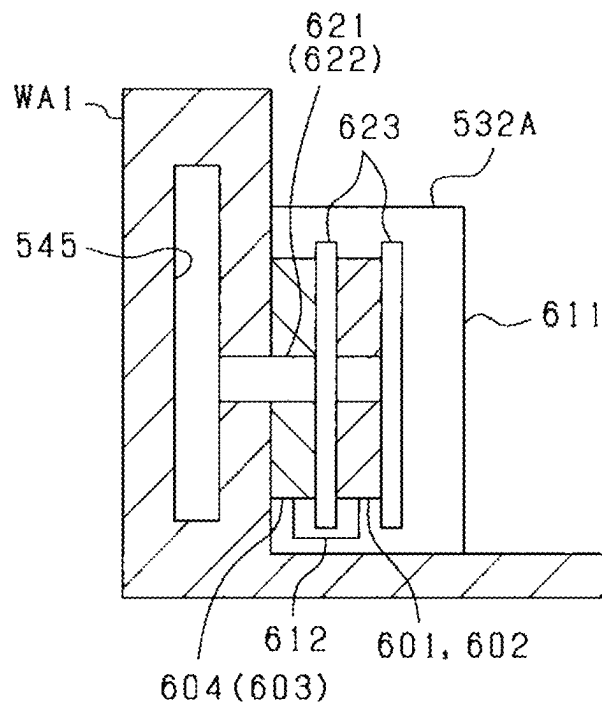


FIG.64

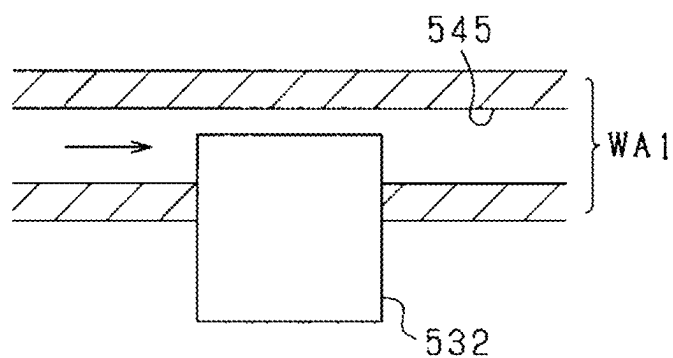


FIG.65

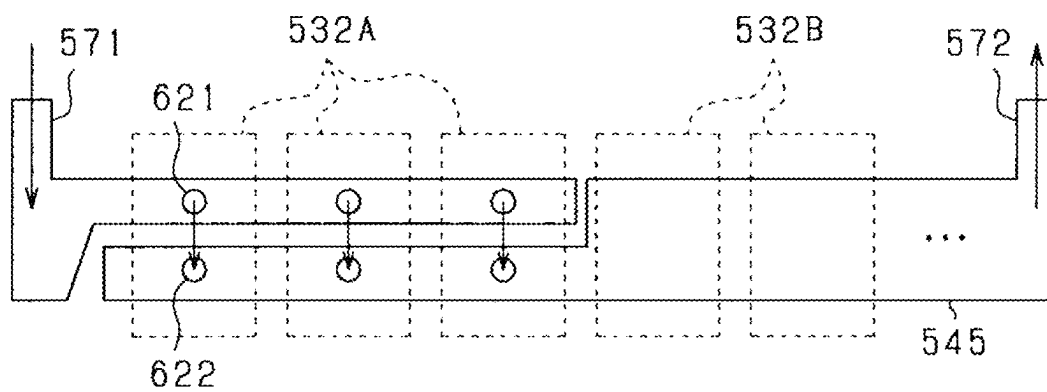


FIG. 66

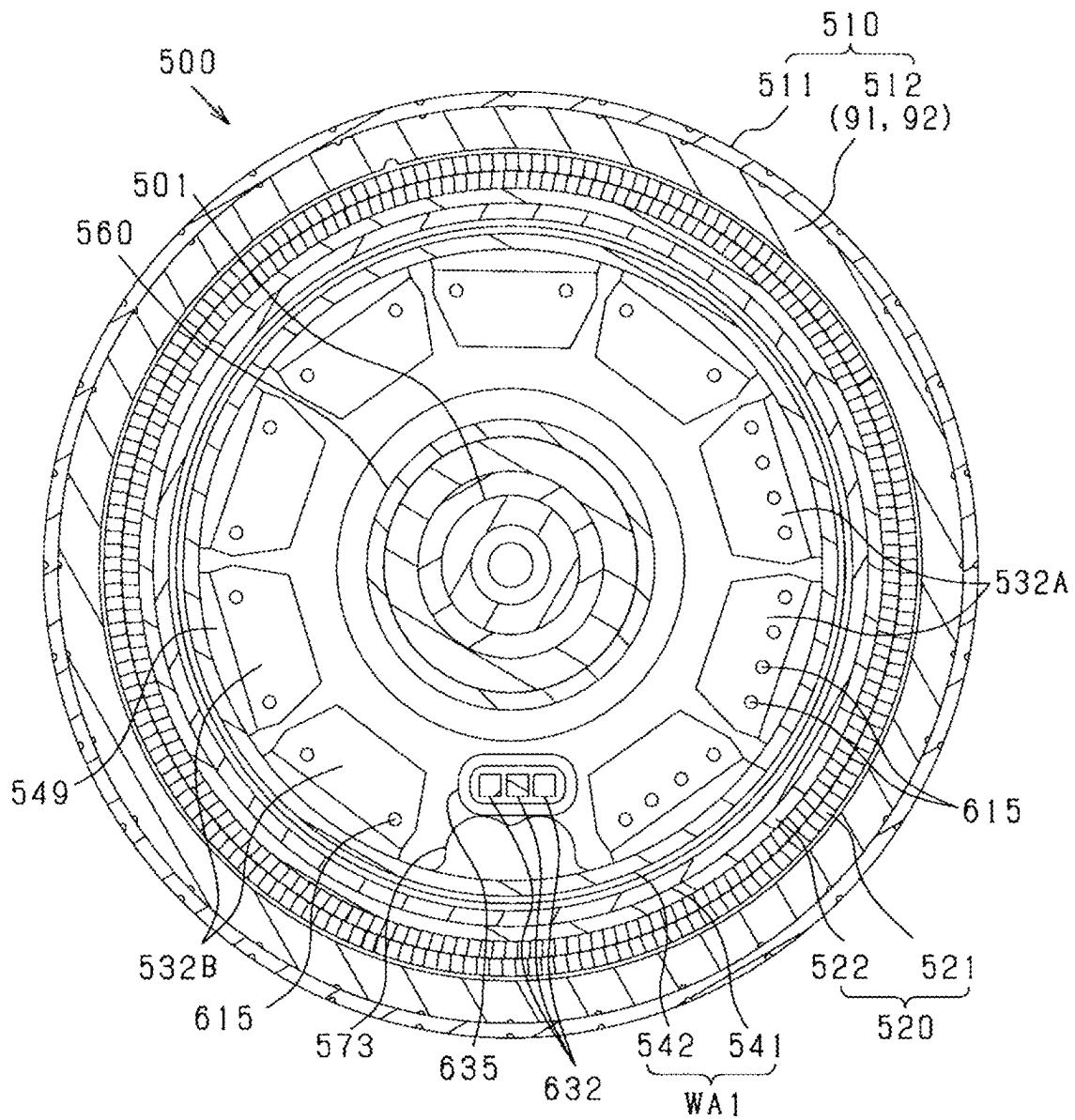


FIG. 67

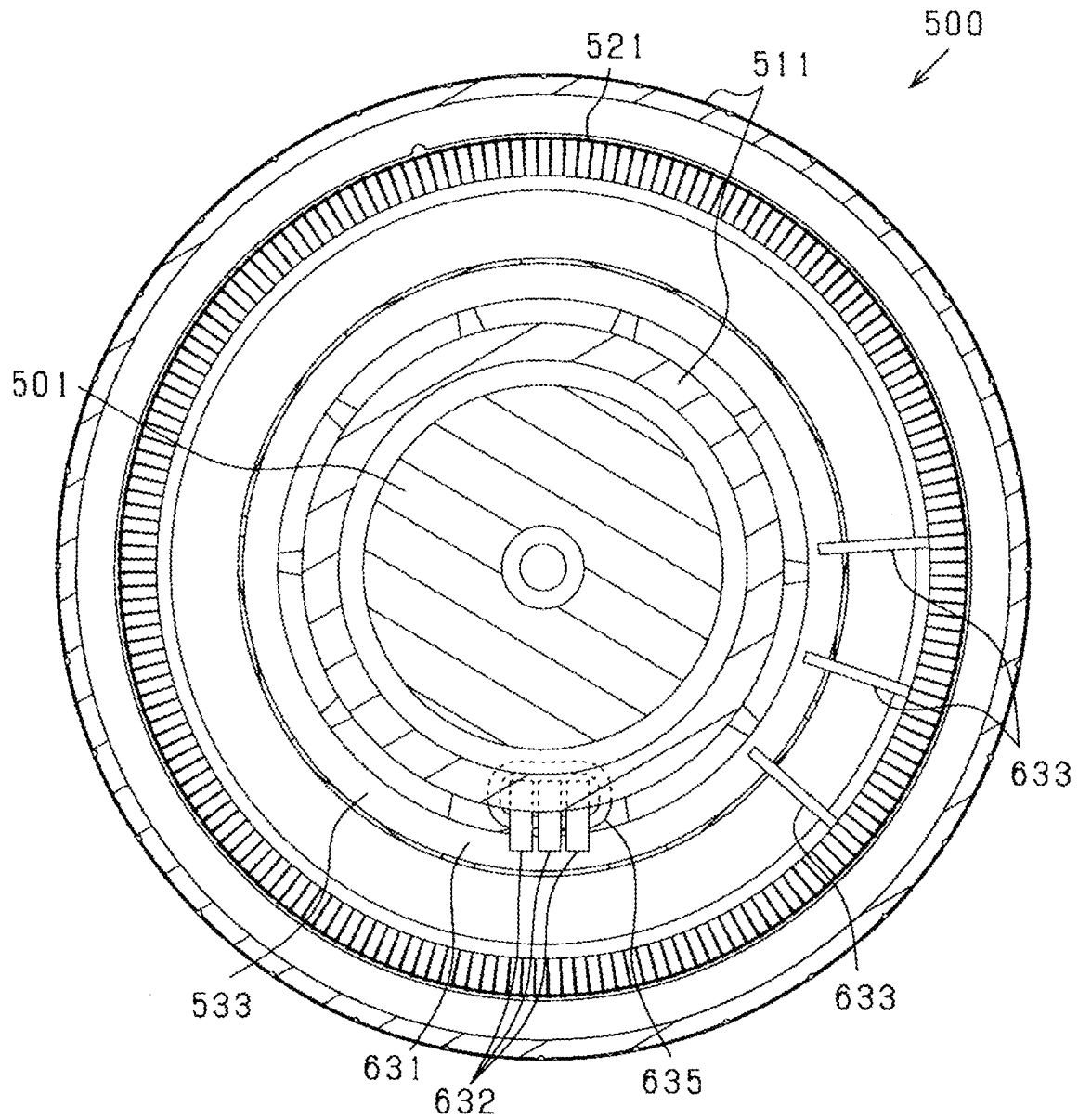


FIG. 68

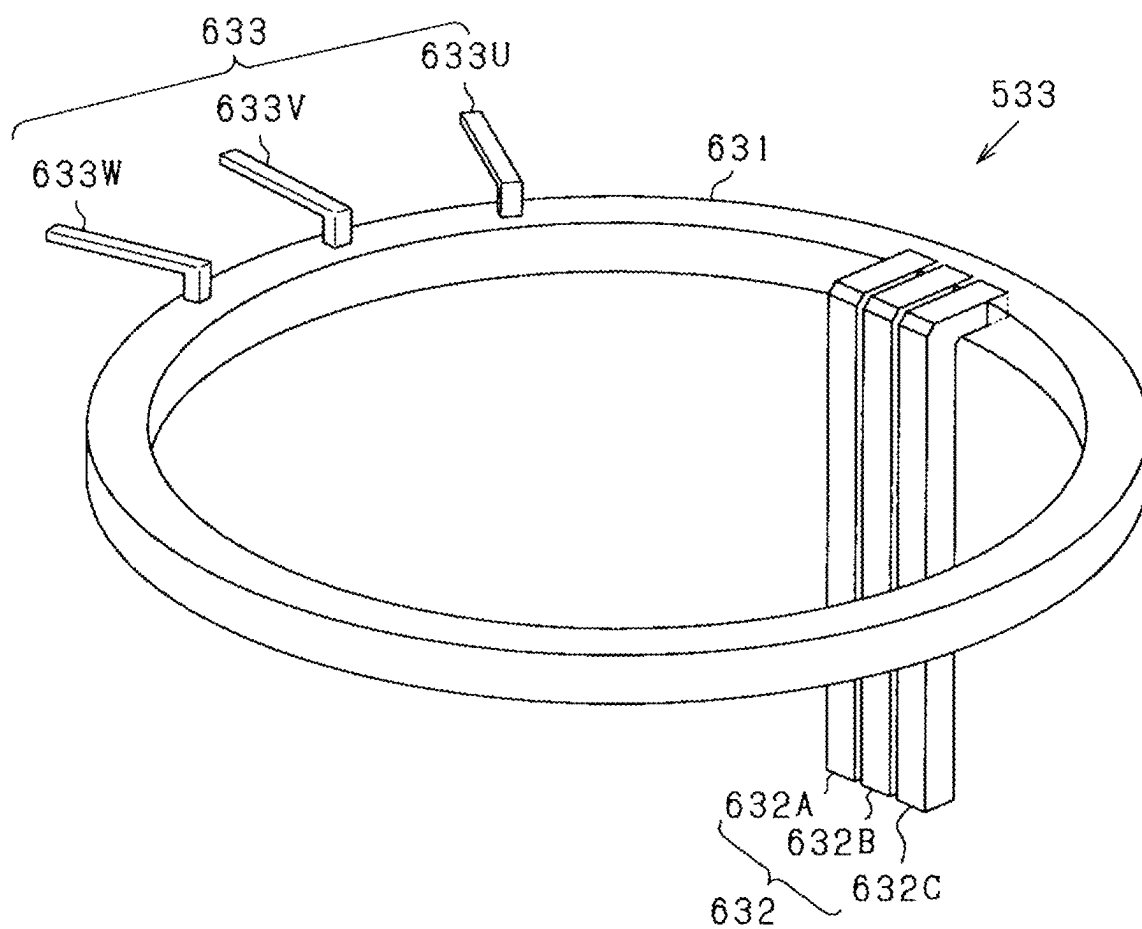


FIG. 69

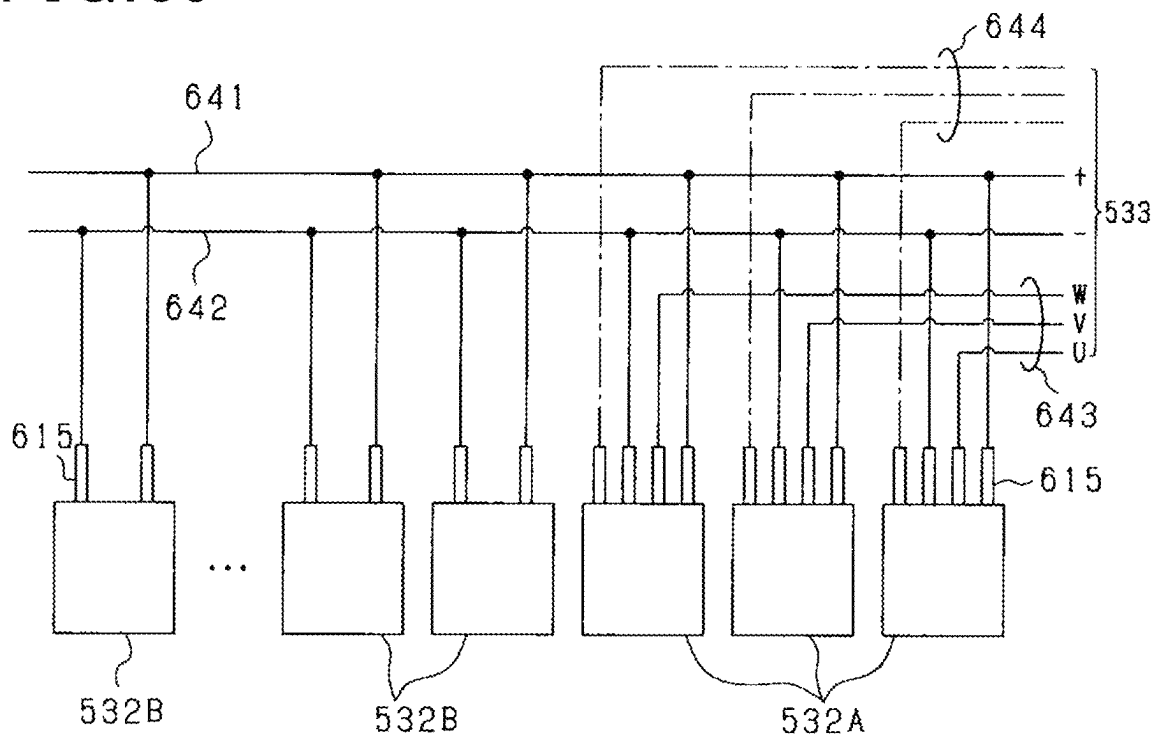


FIG. 70

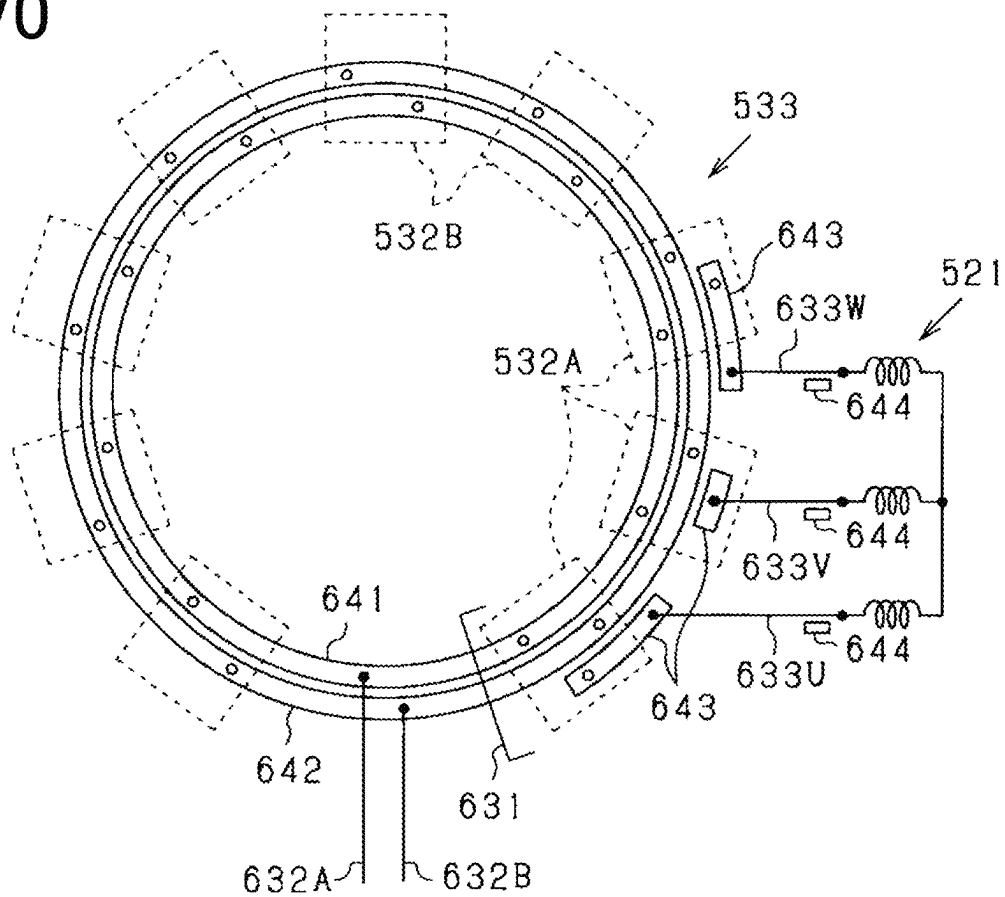


FIG. 71

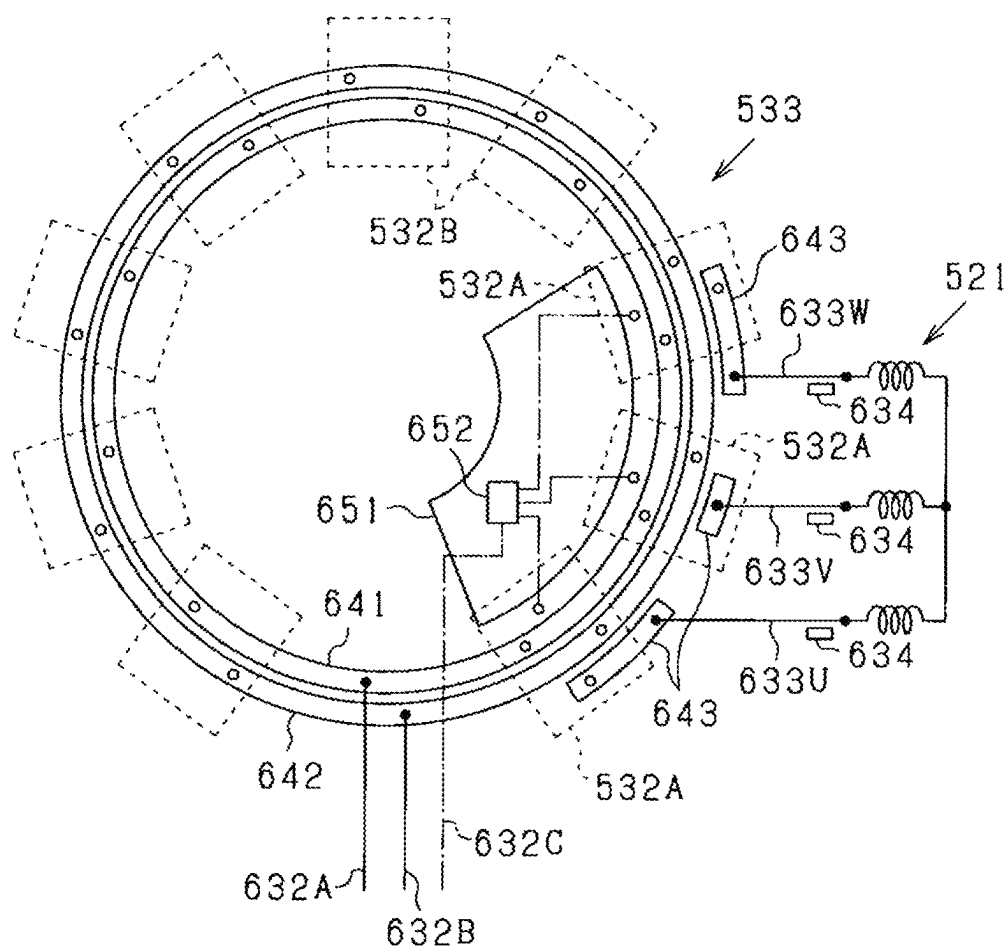


FIG. 72A

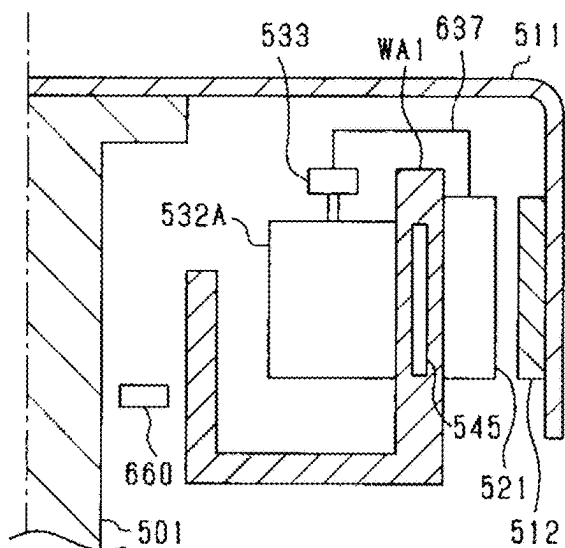


FIG. 72B

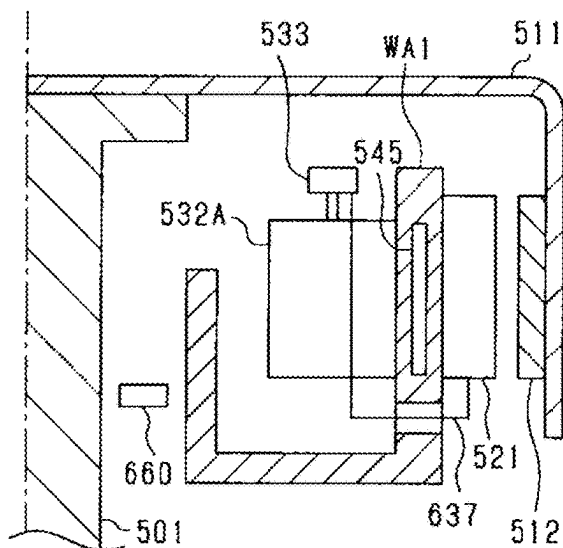


FIG. 72C

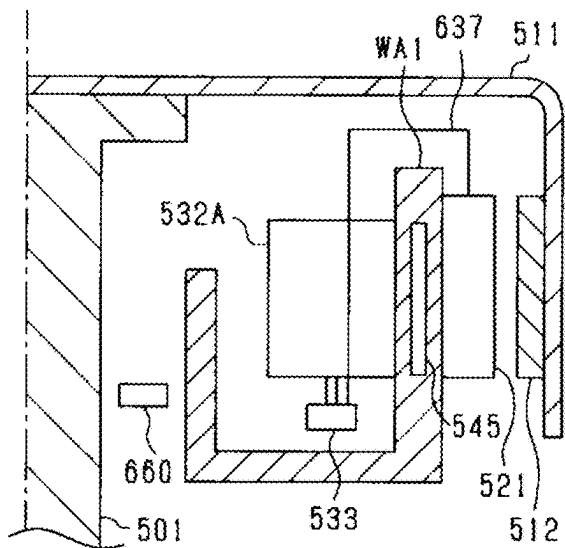


FIG. 72D

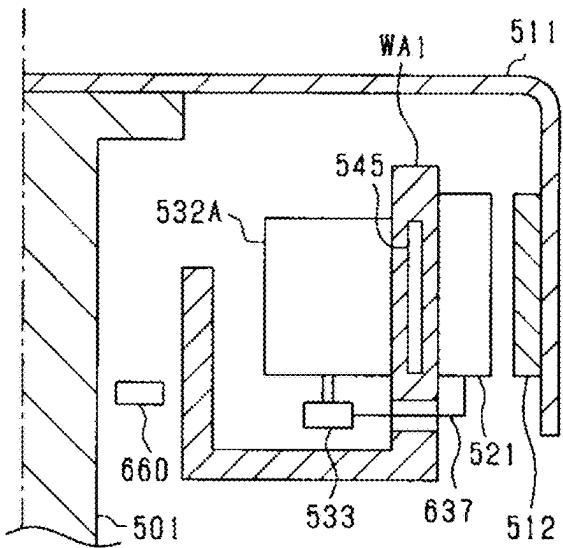


FIG.73A

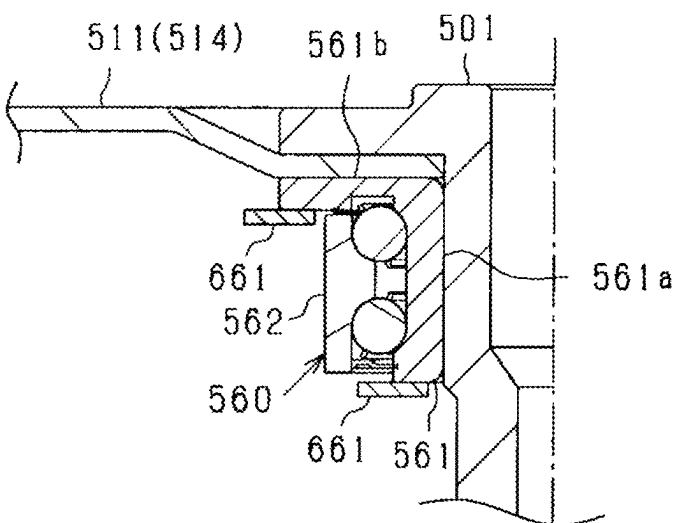


FIG.73B

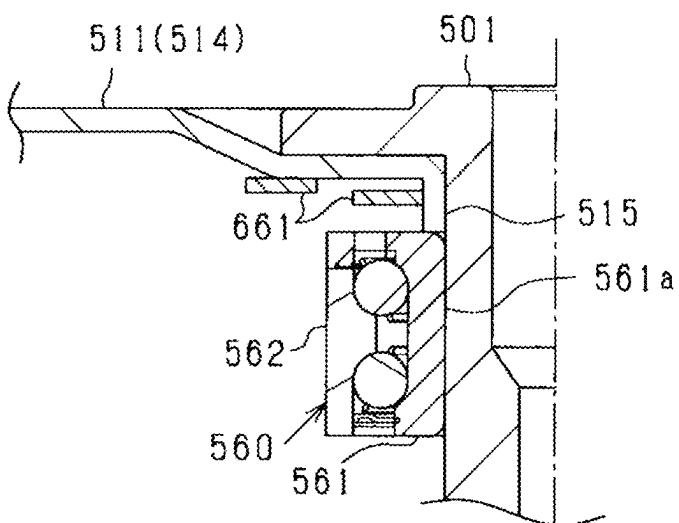


FIG.73C

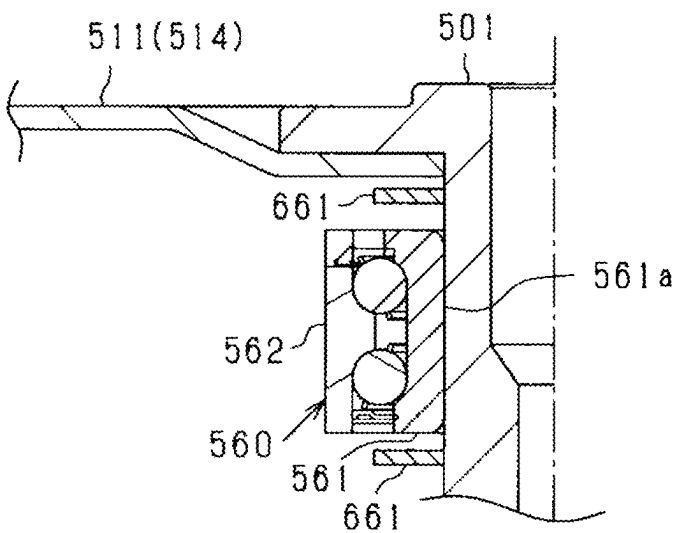


FIG. 74A

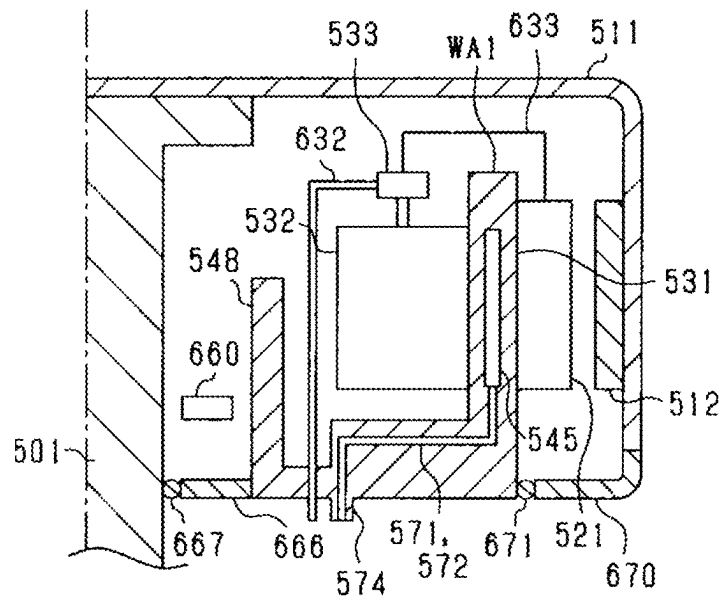


FIG. 74B

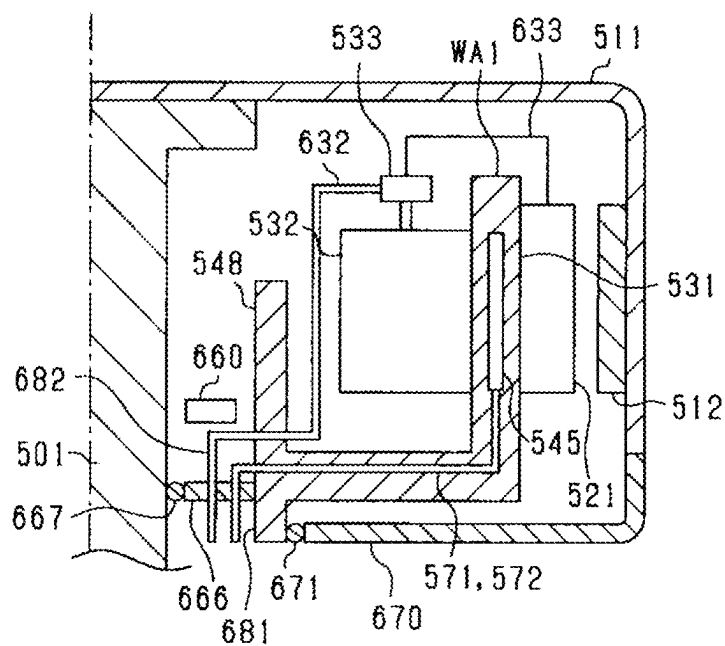


FIG. 75

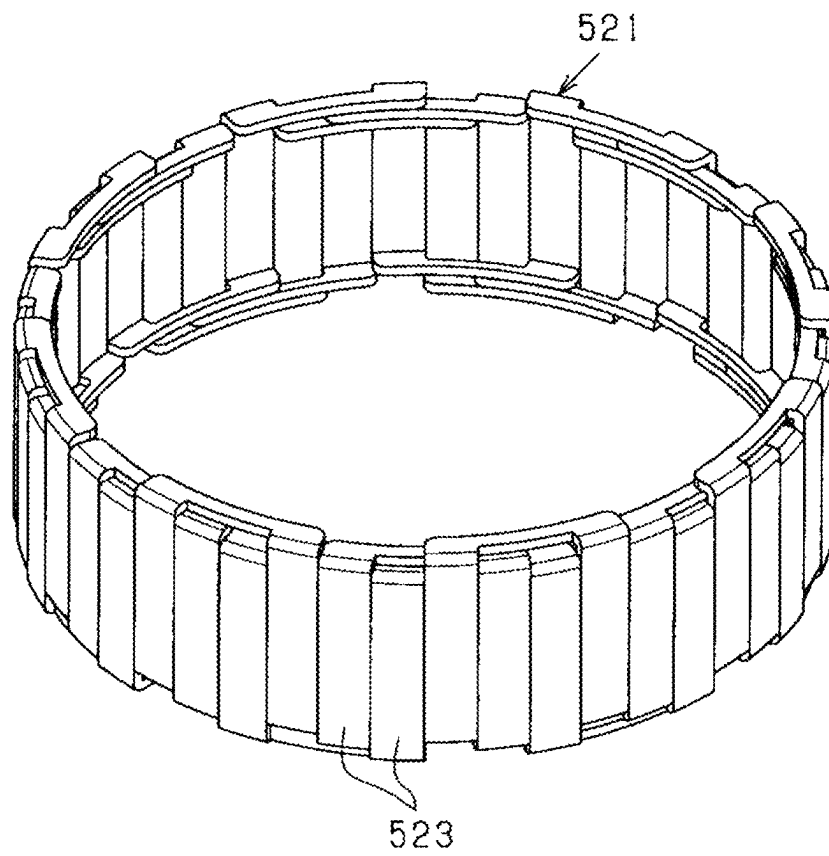


FIG. 76A

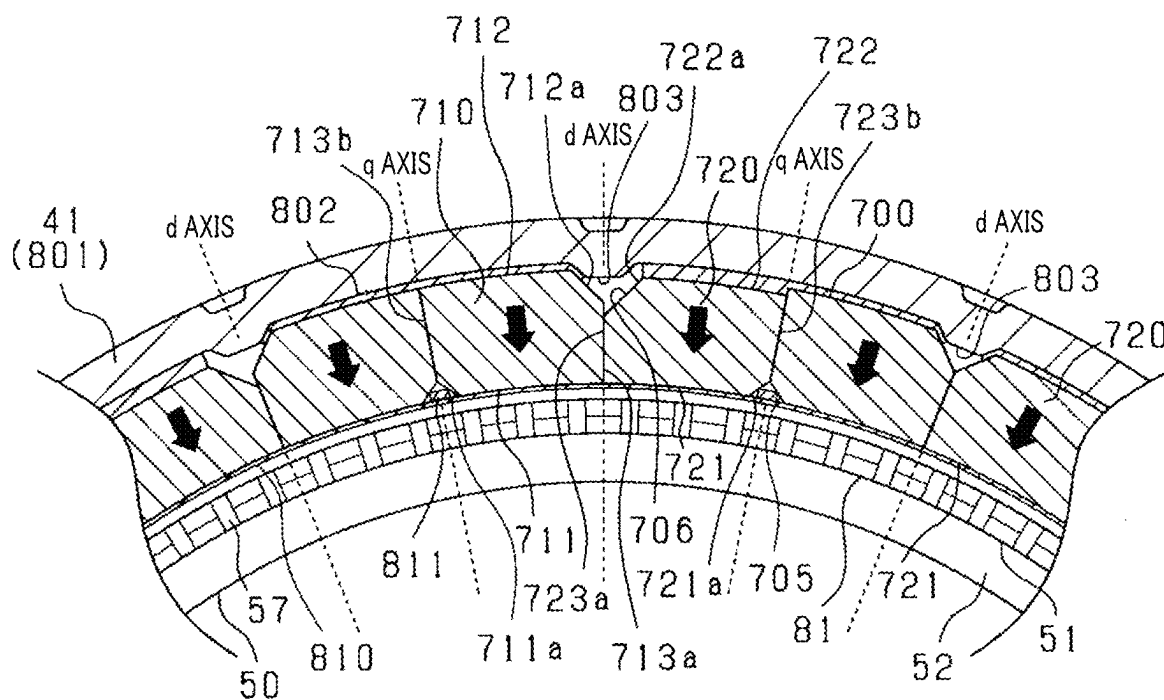


FIG. 76B

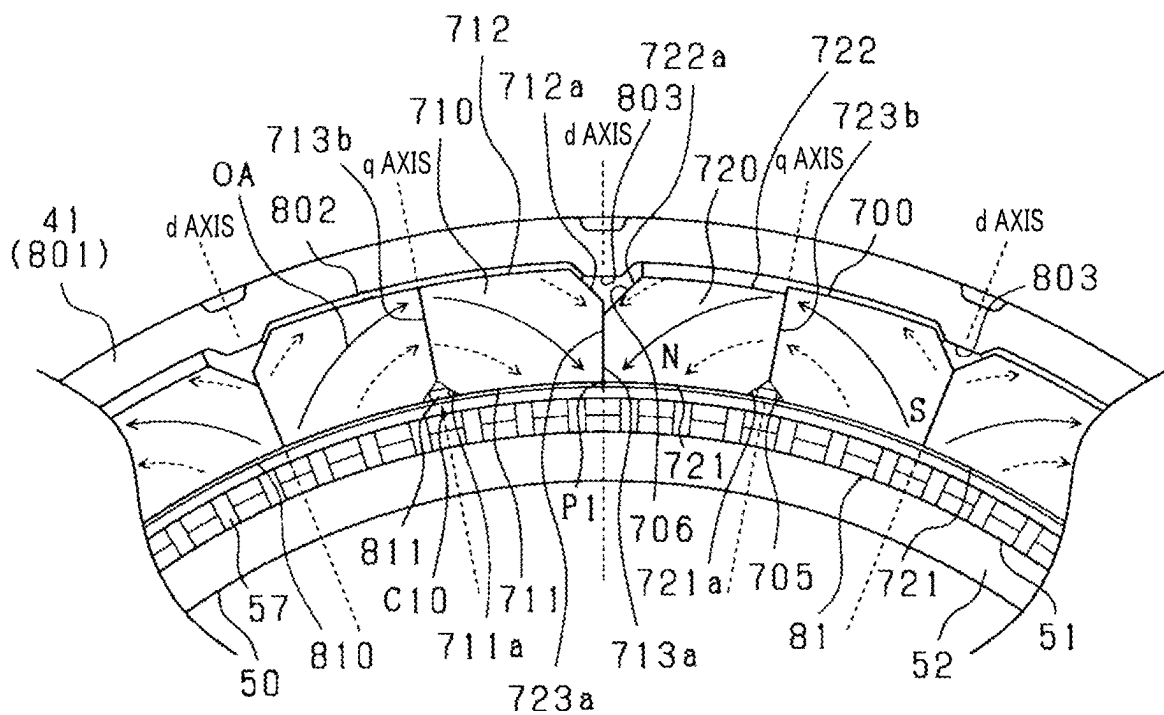


FIG. 77

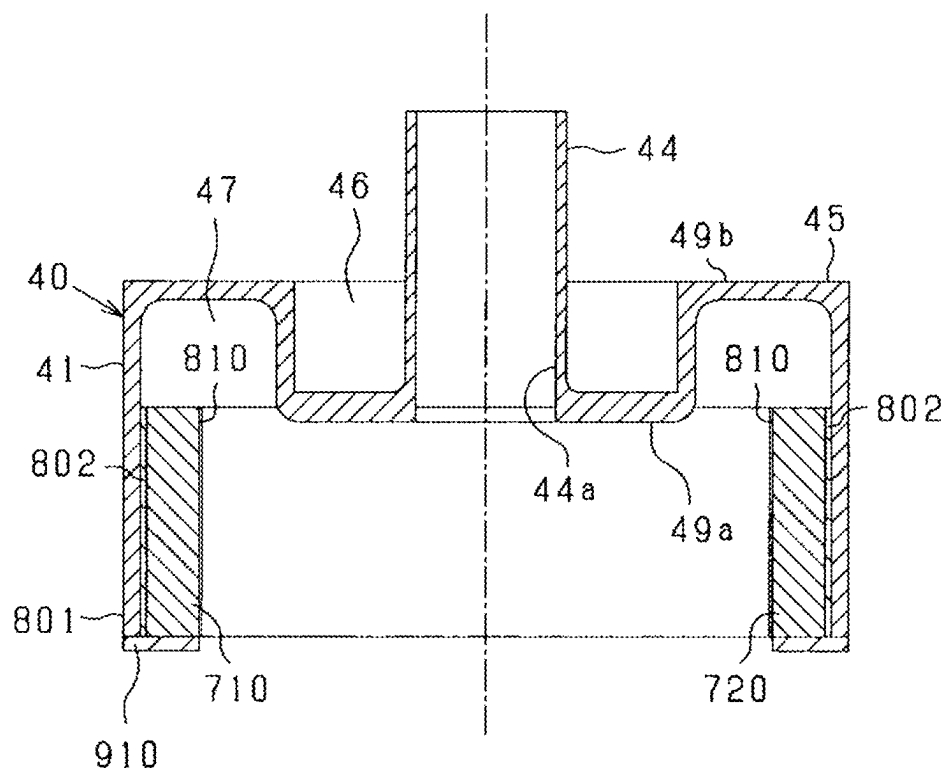


FIG. 78

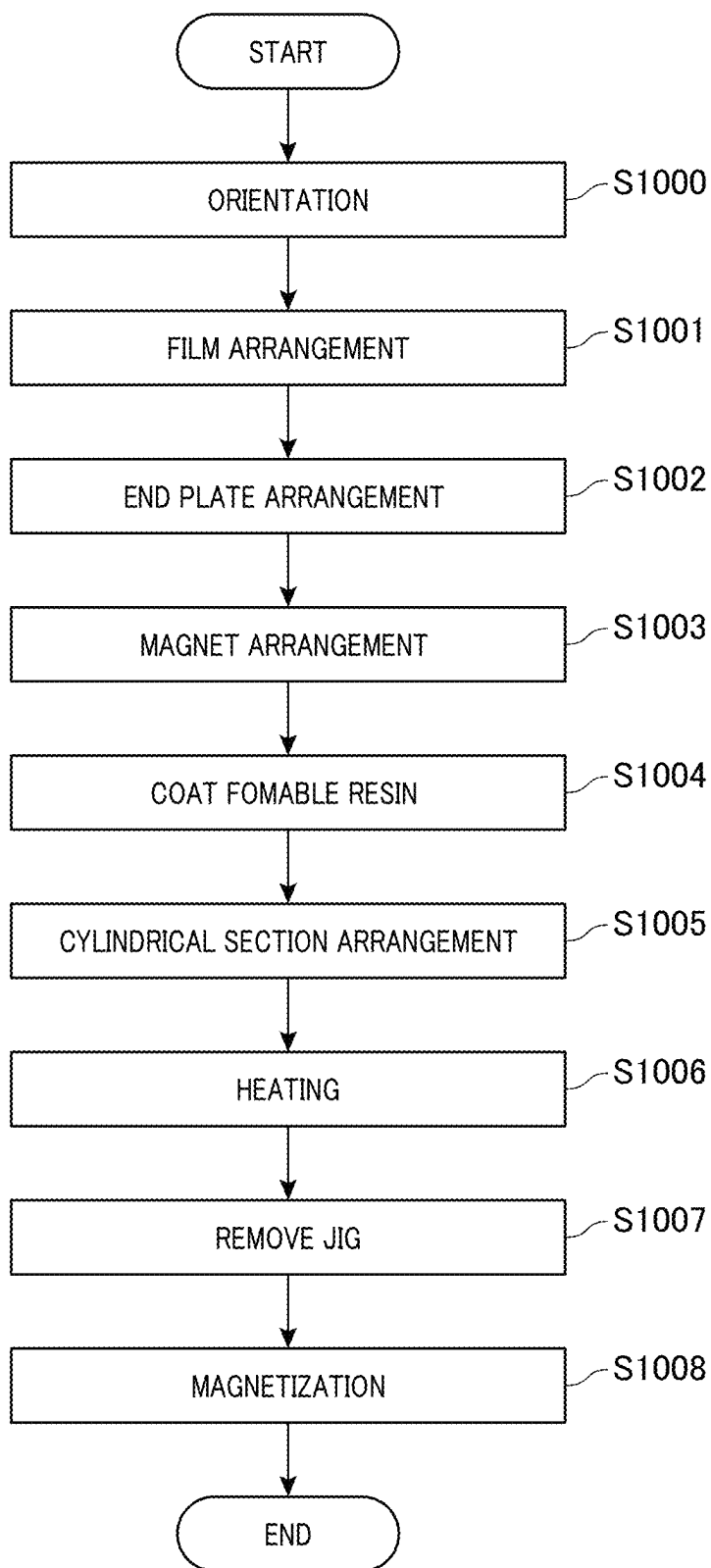


FIG. 79

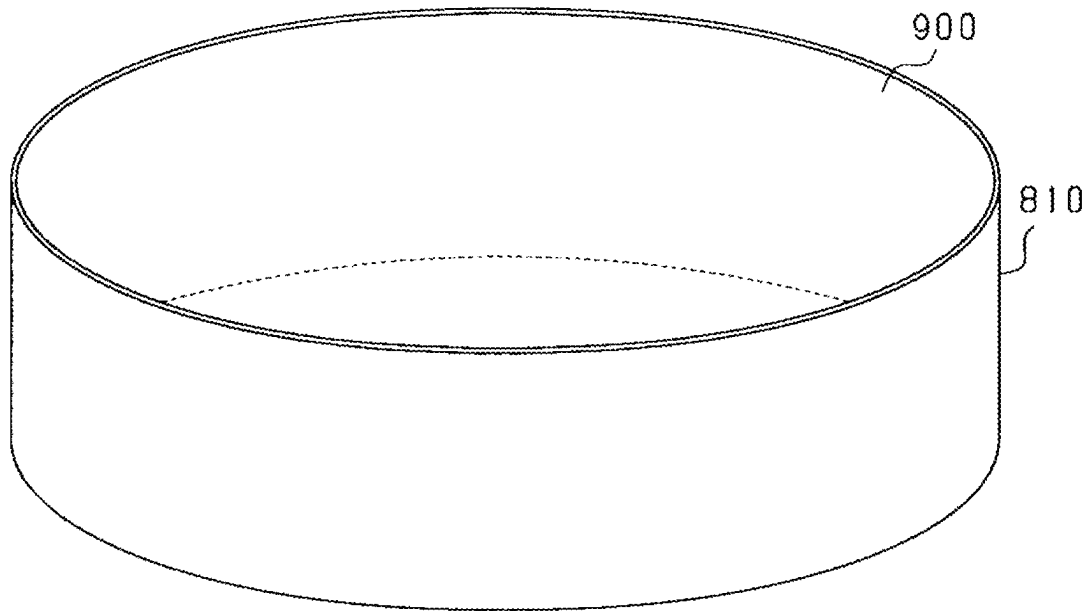


FIG. 80

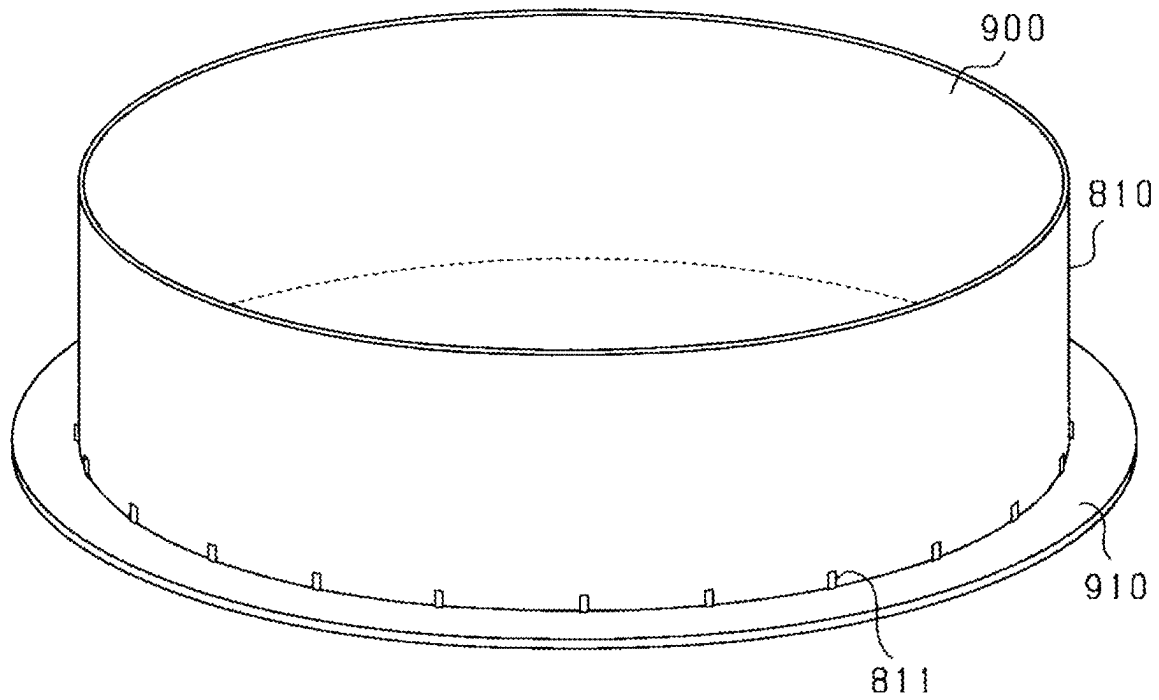


FIG.81

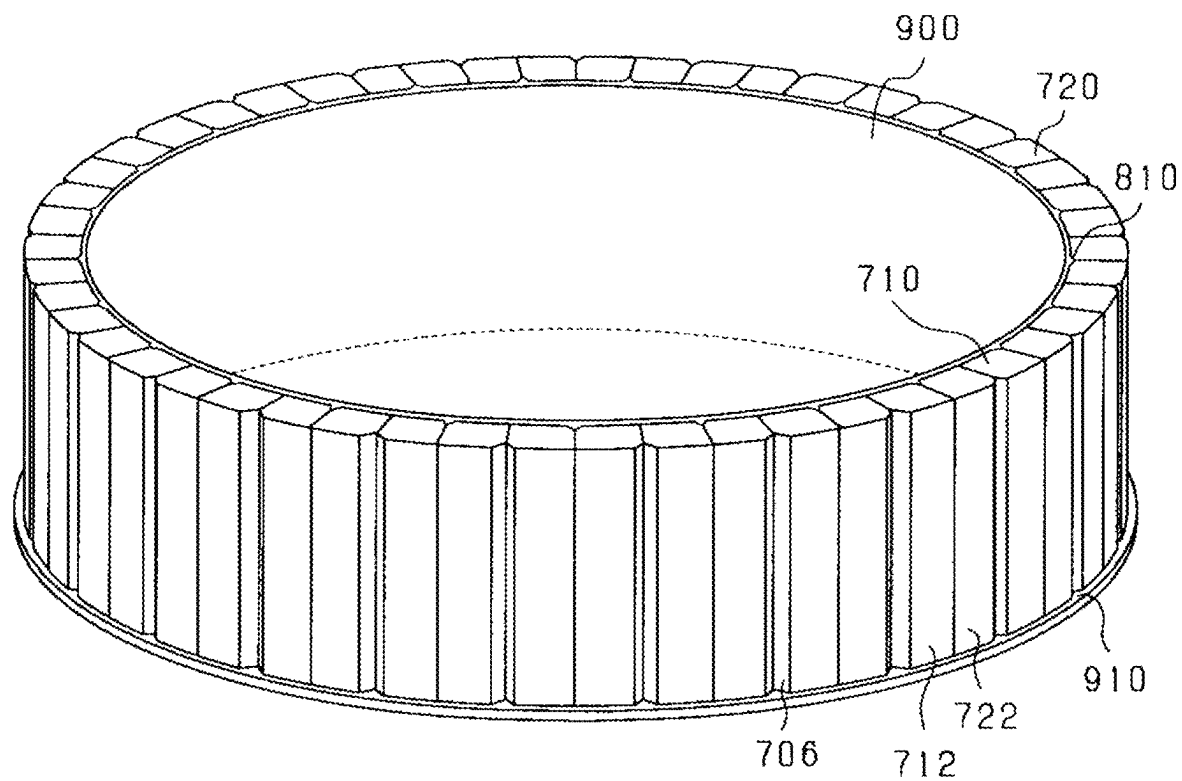


FIG.82

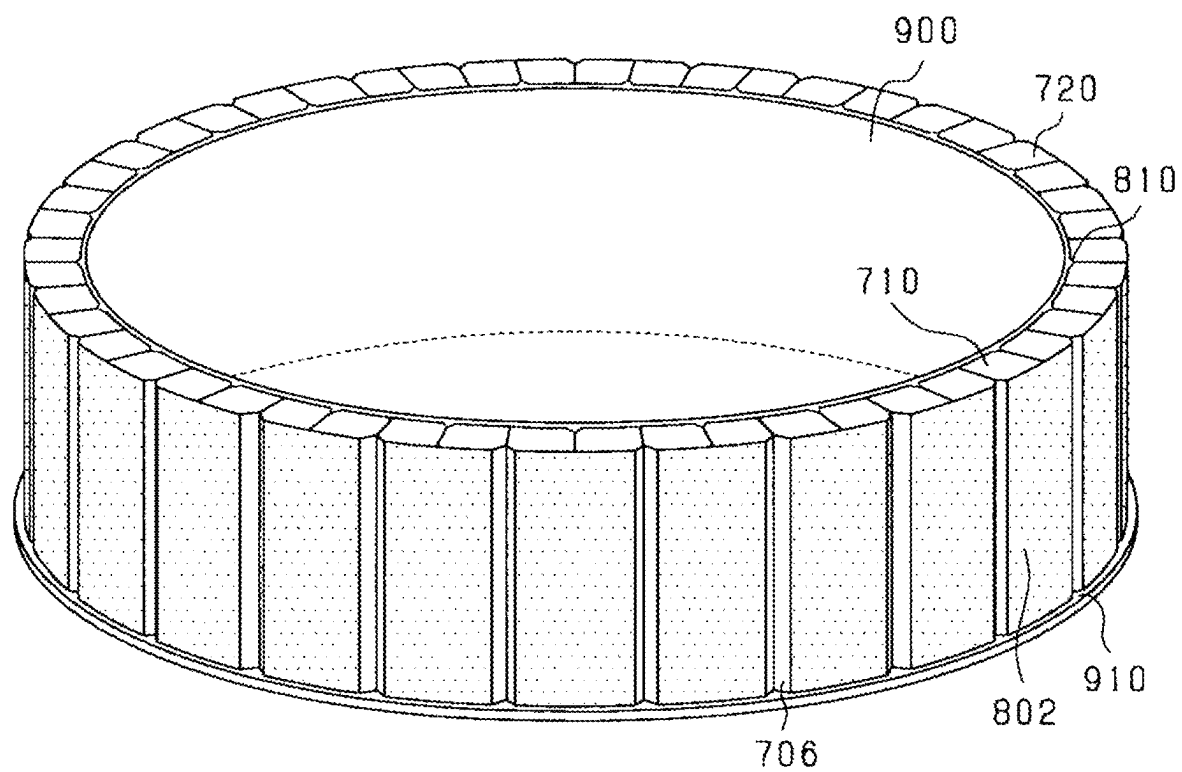
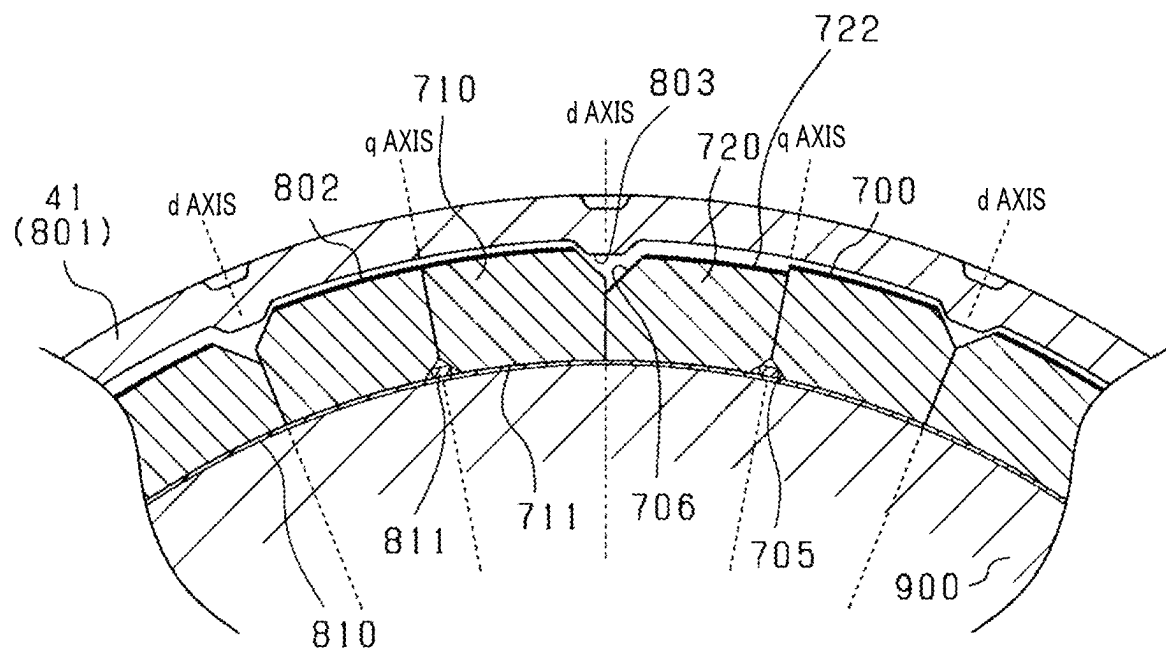


FIG. 83



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**ROTATING ELECTRIC MACHINE AND
MANUFACTURING METHOD THEREOF****CROSS REFERENCE TO RELATED
APPLICATION**

This application is the U.S. bypass application of International Application No. PCT/JP2020/031877 filed on Aug. 24, 2020, which designated the U.S. and claims priority to Japanese Patent Application No. 2019-153901, filed Aug. 26, 2019, the contents of both of these are incorporated herein by reference.

BACKGROUND**Technical Field**

The disclosure of the present specification relates to a rotating electric machine and a manufacturing method thereof.

Conventionally, an outer rotor type motor is known in which a rotor having magnets is disposed. According to the outer rotor type motor, a plurality of magnets are fixed to an inner peripheral surface of a rotor yoke having a cylindrical shape.

SUMMARY

The present disclosure provides a rotating electric machine provided with: a field magnet having a magnet unit including a plurality of magnetic poles having alternating polarities in a circumferential direction and a magnet supporting section having a cylindrical shape to which the magnet unit is fixed; and an armature having a multi-phase armature winding, disposed facing the magnet unit, either one of the field magnet or the armature being configured as a rotor, wherein the magnet unit is configured annularly along the circumferential direction; the magnet unit is fixed to a fixed surface which is either an outer peripheral surface or an inner peripheral surface of the magnet supporting section via a foamable resin; and a thickness dimension of the foamable resin in a radial direction is set to be differentiated depending on a distance from the fixed surface of the magnet supporting section to an anti-armature side circumferential surface of the magnet.

BRIEF DESCRIPTION OF THE DRAWINGS

The above features and advantages of the present disclosure will be clarified further by the following detailed description with reference to the accompanying drawings. The drawings are:

FIG. 1 is a vertical cross-sectional perspective view of a rotating electric machine.

FIG. 2 is a vertical cross-sectional view of the rotating electric machine.

FIG. 3 is a sectional view taken along a line III-III of FIG. 2.

FIG. 4 is a cross-sectional view illustrating a part of FIG. 3 in an enlarged manner.

FIG. 5 is an exploded view of the rotating electric machine.

FIG. 6 is an exploded view of an inverter unit.

FIG. 7 is a torque line diagram illustrating a relation between ampere-turns of a stator winding and torque density.

FIG. 8 is a cross-sectional view of a rotor and a stator.

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FIG. 9 is a view illustrating a part of FIG. 8 in an enlarged manner.

FIG. 10 is a cross-sectional view of the stator.

FIG. 11 is a vertical cross-sectional view of the stator.

FIG. 12 is a perspective view of the stator winding.

FIG. 13 is a perspective view illustrating a configuration of a conductor.

FIG. 14 is a schematic diagram illustrating a configuration of wires.

FIG. 15A is a diagram illustrating a form of each conductor in an nth layer.

FIG. 15B is a diagram illustrating a form of each conductor in an nth layer.

FIG. 16 is a side view illustrating each conductor of the nth layer and an n+1th layer.

FIG. 17 is a diagram illustrating a relation between an electrical angle and a magnetic flux density of a magnet of an embodiment.

FIG. 18 is a diagram illustrating a relation between an electrical angle and a magnetic flux density of a magnet of a comparative example.

FIG. 19 is an electrical circuit diagram of a control system of the rotating electric machine.

FIG. 20 is a functional block diagram illustrating current feedback control processing by a control device.

FIG. 21 is a functional block diagram illustrating torque feedback control processing by the control device.

FIG. 22 is a cross-sectional view of a rotor and a stator in a second embodiment.

FIG. 23 is a view illustrating a part of FIG. 22 in an enlarged manner.

FIG. 24A is a diagram specifically illustrating a flow of magnetic flux in a magnet unit.

FIG. 24B is a diagram specifically illustrating a flow of magnetic flux in a magnet unit.

FIG. 25 is a cross-sectional view of a stator in a first modification.

FIG. 26 is a cross-sectional view of the stator in the first modification.

FIG. 27 is a cross-sectional view of a stator in a second modification.

FIG. 28 is a cross-sectional view of a stator in a third modification.

FIG. 29 is a cross-sectional view of a stator in a fourth modification.

FIG. 30 is a cross-sectional view of a rotor and a stator in a seventh modification.

FIG. 31 is a functional block diagram illustrating a part of the processing of an operation signal generation unit in a eighth modification.

FIG. 32 is a flowchart illustrating a procedure of carrier frequency changing processing.

FIG. 33A is a diagram illustrating a connection form of each conductor constituting a conductor group in a ninth modification.

FIG. 33B is a diagram illustrating a connection form of each conductor constituting a conductor group in a ninth modification.

FIG. 33C is a diagram illustrating a connection form of each conductor constituting a conductor group in a ninth modification.

FIG. 34 is a diagram illustrating a configuration in which four pairs of conductors are laminated in the ninth modification.

FIG. 35 is a cross-sectional view of an inner rotor type rotor and a stator in tenth modification.

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FIG. 36 is a view illustrating a part of FIG. 35 in an enlarged manner.

FIG. 37 is a vertical cross-sectional view of an inner rotor type rotating electric machine.

FIG. 38 is a vertical cross-sectional view illustrating a schematic configuration of the inner rotor type rotating electric machine.

FIG. 39 is a diagram illustrating a configuration of a rotating electric machine having an inner rotor structure in an eleventh modification.

FIG. 40 is a diagram illustrating a configuration of a rotating electric machine having an inner rotor structure in the eleventh modification.

FIG. 41 is a diagram illustrating a configuration of a rotating electric machine having an inner rotor structure in a twelfth modification.

FIG. 42 is a cross-sectional view illustrating a configuration of a conductor in a fourteenth modification.

FIG. 43 is a diagram illustrating a relation between a reluctance torque, a magnet torque, and DM.

FIG. 44 is a diagram illustrating teeth.

FIG. 45 is a perspective view illustrating a wheel having an in-wheel motor structure and its peripheral structure.

FIG. 46 is a vertical cross-sectional view of the wheel and its peripheral structure.

FIG. 47 is an exploded perspective view of the wheel.

FIG. 48 is a side view of a rotating electric machine as viewed from the protruding side of a rotating shaft.

FIG. 49 is a cross-sectional view taken along a line 49-49 of FIG. 48.

FIG. 50 is a cross-sectional view taken along a line 50-50 of FIG. 49.

FIG. 51 is an exploded cross-sectional view of the rotating electric machine.

FIG. 52 is a partial cross-sectional view of the rotor.

FIG. 53 is a perspective view of a stator winding and a star core.

FIG. 54A is a front view illustrating the stator winding developed in a plane.

FIG. 54B is a front view illustrating the stator winding developed in a plane.

FIG. 55 is a diagram illustrating a skew of the conductor.

FIG. 56 is an exploded cross-sectional view of the inverter unit.

FIG. 57 is an exploded cross-sectional view of the inverter unit.

FIG. 58 is a diagram illustrating a state of arrangement of each electric module in an inverter housing.

FIG. 59 is a circuit diagram illustrating an electrical configuration of a power converter.

FIG. 60 is a diagram illustrating an example of a cooling structure of a switch module.

FIG. 61A is a diagram illustrating an example of the cooling structure of the switch module.

FIG. 61B is a diagram illustrating an example of the cooling structure of the switch module.

FIG. 62A is a diagram illustrating an example of the cooling structure of the switch module.

FIG. 62B is a diagram illustrating an example of the cooling structure of the switch module.

FIG. 62C is a diagram illustrating an example of the cooling structure of the switch module.

FIG. 63A is a diagram illustrating an example of the cooling structure of the switch module.

FIG. 63B is a diagram illustrating an example of the cooling structure of the switch module.

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FIG. 64 is a diagram illustrating an example of the cooling structure of the switch module.

FIG. 65 is a diagram illustrating an arrangement order of each electric module with respect to a cooling water passage.

FIG. 66 is a cross-sectional view taken along a line 66-66 of FIG. 49.

FIG. 67 is a cross-sectional view taken along a line 67-67 of FIG. 49.

FIG. 68 is a perspective view illustrating a busbar module alone.

FIG. 69 is a diagram illustrating an electrical connection state between each electric module and the busbar module.

FIG. 70 is a diagram illustrating an electrical connection state between each electric module and the busbar module.

FIG. 71 is a diagram illustrating an electrical connection state between each electric module and the busbar module.

FIG. 72A is a configuration diagram for explaining a first modification in an in-wheel motor.

FIG. 72B is a configuration diagram for explaining a first modification in an in-wheel motor.

FIG. 72C is a configuration diagram for explaining a first modification in an in-wheel motor.

FIG. 72D is a configuration diagram for explaining a first modification in an in-wheel motor.

FIG. 73A is a configuration diagram for explaining a second modification in the in-wheel motor.

FIG. 73B is a configuration diagram for explaining a second modification in the in-wheel motor.

FIG. 73C is a configuration diagram for explaining a second modification in the in-wheel motor.

FIG. 74A is a configuration diagram for explaining a third modification in the in-wheel motor.

FIG. 74B is a configuration diagram for explaining a third modification in the in-wheel motor.

FIG. 75 is a configuration diagram for explaining a fourth modification in the in-wheel motor.

FIG. 76A is a cross-sectional view of a rotor and a stator in a fifteenth modification.

FIG. 76B is a cross-sectional view of a rotor and a stator in the fifteenth modification.

FIG. 77 is a vertical cross-sectional view of the wheel and its peripheral structure.

FIG. 78 is a flowchart illustrating a manufacturing method of a rotor in the fifteenth embodiment.

FIG. 79 is a perspective view illustrating a rotor during assembly.

FIG. 80 is a perspective view illustrating a rotor during assembly.

FIG. 81 is a perspective view illustrating a rotor during assembly.

FIG. 82 is a perspective view illustrating a rotor during assembly.

FIG. 83 is a cross-sectional view illustrating a rotor during assembly.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

Conventionally, for example, JP-2003-259611 discloses an outer rotor type motor in which a rotor having magnets is disposed. According to the above-mentioned patent literature, a plurality of magnets are fixed to an inner peripheral surface of a rotor yoke having a cylindrical shape.

According to the above-mentioned patent literature, a fixing apparatus body is thermally expanded to press a plurality of magnets radially outward, thereby pushing on the inner peripheral surface of the rotor yoke and being fixed

by an adhesive. However, it is difficult to press the respective magnets equally. For example, as disclosed in the above-described patent literature, in the case where the fixing apparatus body is thermally expanded, it is difficult to expand it equally. Further, for manufacturing magnets, since manufacturing tolerance is present, even when pressing respective magnets with respect to the outer peripheral surface of the magnets, positions of the magnets may be shifted on the inner peripheral surface in the radial direction. For this reason, a problem arises in which an air gap between the magnets and stator windings is required to be larger.

Hereinafter, a plurality of embodiments will be described with reference to the drawings. In the plurality of embodiments, functionally and/or structurally corresponding parts and/or associated parts may be designated with the same reference sign or reference signs that are different in the hundreds or higher position. For corresponding and/or associated parts, the description of other embodiments can be referred to.

The rotating electric machine in this embodiment is used, for example, as a vehicle power source. However, the rotating electric machine can be widely used for industrial use, vehicle use, home appliance use, OA equipment use, game machine use, and the like. Note that, in each of the following embodiments, parts that are the same or equivalent to each other are designated by the same reference signs in the drawings, and the description thereof will be incorporated for the parts having the same reference signs.

First Embodiment

A rotating electric machine 10 according to the present embodiment is a synchronous multi-phase AC motor and has an outer rotor structure (outer rotating structure). The outline of the rotating electric machine 10 is illustrated in FIGS. 1 to 5. FIG. 1 is a vertical cross-sectional perspective view of the rotating electric machine 10, FIG. 2 is a vertical cross-sectional view of the rotating electric machine 10 in a direction along a rotating shaft 11, FIG. 3 is a cross-sectional view of the rotating electric machine 10 in a direction orthogonal to the rotating shaft 11 (cross-sectional view taken along a line of FIG. 2), FIG. 4 is a cross-sectional view illustrating a part of FIG. 3 in an enlarged manner, and FIG. 5 is an exploded view of the rotating electric machine 10. Note that, in FIG. 3, for convenience of illustration, hatching indicating a cut surface is omitted except for a rotating shaft 11. In the following description, the direction in which the rotating shaft 11 extends is the axial direction, the direction extending radially from the center of the rotating shaft 11 is the radial direction, and the direction extending circumferentially around the rotating shaft 11 is the circumferential direction.

The rotating electric machine 10 includes, substantially, a bearing unit 20, a housing 30, a rotor 40, a stator 50, and an inverter unit 60. Each of these members is arranged coaxially with the rotating shaft 11 and is assembled in the axial direction in a predetermined order to form the rotating electric machine 10. The rotating electric machine 10 of the present embodiment has a configuration having a rotor 40 as a “field magnet” and a stator 50 as an “armature”, and is embodied as a revolving-field type rotating electric machine.

The bearing unit 20 includes two bearings 21 and 22 arranged apart from each other in the axial direction, and a holding member 23 that holds the bearings 21 and 22. The bearings 21 and 22 are, for example, radial ball bearings, each of which has an outer ring 25, an inner ring 26, and a plurality of balls 27 arranged between the outer ring 25 and

the inner ring 26. The holding member 23 has a cylindrical shape, and the bearings 21 and 22 are assembled radially thereinside. In addition, the rotating shaft 11 and the rotor 40 are rotatably supported radially inside the bearings 21 and 22. The bearings 21 and 22 constitute a pair of bearings that rotatably support the rotating shaft 11.

In the respective bearings 21 and 22, balls 27 are held by retainers (not illustrated), and the pitch between the balls is maintained in that state. The bearings 21 and 22 has sealing members at the upper and lower portions in the axial direction of the retainer, and the inside of the sealing members is filled with nonconductive grease (for example, nonconductive urea grease). Further, the position of the inner ring 26 is mechanically held by a spacer, and a constant pressure preload that rises in the up-down direction from the inside is applied.

A housing 30 has a cylindrical peripheral wall 31. The peripheral wall 31 has a first end and a second end facing each other in the axial direction thereof. The peripheral wall 31 has an end face 32 at the first end and an opening 33 at the second end. The opening 33 is open throughout the second end. A circular hole 34 is formed in the center of the end face 32, and a bearing unit 20 is fixed by a fixture such as a screw or a rivet in a state of being inserted through the hole 34. Further, a hollow cylindrical rotor 40 and a hollow cylindrical stator 50 are housed in the housing 30, that is, in the internal space partitioned by the peripheral wall 31 and the end face 32. In the present embodiment, the rotating electric machine 10 is an outer rotor type, and in the housing 30, the stator 50 is arranged radially inside the cylindrical rotor 40. The rotor 40 is cantilevered and supported by the rotating shaft 11 on the side of the end face 32 in the axial direction.

The rotor 40 has a magnet holder 41 formed in a hollow tubular shape and an annular magnet unit 42 provided radially inside the magnet holder 41. The magnet holder 41 has a substantially cup shape and has a function as a magnet holding member. The magnet holder 41 has a cylindrical section 43 having a cylindrical shape, a fixing section (attachment) 44 having a cylindrical shape and a diameter smaller than that of the cylindrical section 43, and an intermediate section 45 serving as a part connecting the cylindrical section 43 and the fixing section 44. The magnet unit 42 is attached to the inner peripheral surface of the cylindrical section 43.

Moreover, the magnet holder 41 is made of a steel plate cold commercial (SPCC) having sufficient mechanical strength, forging steel, carbon fiber reinforced plastic (CFRP), or the like.

The rotating shaft 11 is inserted through a through hole 44a of the fixing section 44. The fixing section 44 is fixed to the rotating shaft 11 arranged in the through hole 44a. That is, the magnet holder 41 is fixed to the rotating shaft 11 by the fixing section 44. Moreover, the fixing section 44 is preferably fixed to the rotating shaft 11 by spline coupling, key coupling, welding, caulking, or the like using a protrusion and a recess. As a result, the rotor 40 rotates integrally with the rotating shaft 11.

Further, bearings 21 and 22 of the bearing unit 20 are assembled radially inside the fixing section 44. Since the bearing unit 20 is fixed to the end face 32 of the housing 30 as described above, the rotating shaft 11 and the rotor 40 are rotatably supported by the housing 30. As a result, the rotor 40 is rotatable in the housing 30.

The rotor 40 is provided with the fixing section 44 only on one of the two ends facing in the axial direction, whereby the rotor 40 is cantilevered and supported by the rotating shaft

11. Here, the fixing section 44 of the rotor 40 is rotatably supported by the bearings 21 and 22 of the bearing unit 20 at two positions different in the axial direction. In other words, the rotor 40 is rotatably supported by the two bearings 21 and 22 separated in the axial direction at one of the two ends of the magnet holder 41 facing in the axial direction. Therefore, even if the rotor 40 is cantilevered and supported by the rotating shaft 11, stable rotation of the rotor 40 can be achieved. In this case, the rotor 40 is supported by the bearings 21 and 22 at a position displaced to one side with respect to the axial center position of the rotor 40.

Further, in the bearing unit 20, the bearing 22 near the center of the rotor 40 (lower side in the figure) and the bearing 21 on the opposite side (upper side in the figure) have different gap dimensions between the outer ring 25 and the inner ring 26, and the ball 27, and for example, the bearing 22 near the center of the rotor 40 has a larger gap dimension than that of the bearing 21 on the opposite side. In this case, even if shaking of the rotor 40 or vibration due to imbalance caused by component tolerance acts on the bearing unit 20 on the side closer to the center of the rotor 40, the influence of the shake or vibration is well absorbed. Specifically, by increasing the allowance dimension (gap dimension) by preloading the bearing 22 near the center of the rotor 40 (lower side of the figure), the vibration generated in the cantilever structure is absorbed by the allowance portion. The preload may be either a fixed position preload or a constant pressure preload. In the case of fixed position preload, both the outer rings 25 of the bearing 21 and the bearing 22 are joined to the holding member 23 by a method such as press fitting or adhesion. Further, both the inner rings 26 of the bearing 21 and the bearing 22 are joined to the rotating shaft 11 by a method such as press fitting or adhesion. Here, the preload can be generated by arranging the outer ring 25 of the bearing 21 at a different position in the axial direction with respect to the inner ring 26 of the bearing 21. The preload can also be generated by arranging the outer ring 25 of the bearing 22 at a different position in the axial direction with respect to the inner ring 26 of the bearing 22.

Further, in a case where a constant pressure preload is adopted, a preload spring, for example, a waved washer 24 or the like is arranged in the same region sandwiched between the bearing 22 and the bearing 21 in such a manner that preload is generated from the region sandwiched between the bearing 22 and the bearing 21 toward the outer ring 25 of the bearing 22 in the axial direction. Also in this case, both the inner rings 26 of the bearing 21 and the bearing 22 are joined to the rotating shaft 11 by a method such as press fitting or adhesion. The bearing 21 or the outer ring 25 of the bearing 22 is arranged with respect to the holding member 23 via a predetermined clearance. With such a configuration, the spring force of the preload spring acts on the outer ring 25 of the bearing 22 in the direction away from the bearing 21. Then, when this force is transmitted through the rotating shaft 11, a force that presses the inner ring 26 of the bearing 21 in the direction of the bearing 22 acts. As a result, the positions of the outer ring 25 and the inner ring 26 in the axial direction of both the bearings 21 and 22 are displaced, and the two bearings can be preloaded in the same manner as the aforementioned fixed position preload.

Moreover, when generating the constant pressure preload, it is not always necessary to apply the spring force to the outer ring 25 of the bearing 22 as illustrated in FIG. 2. For example, the spring force may be applied to the outer ring 25 of the bearing 21. Further, the inner ring 26 of either of

the bearings 21 and 22 may be arranged with respect to the rotating shaft 11 via a predetermined clearance, and the outer ring 25 of the bearings 21 and 22 may be joined to the holding member 23 by press fitting or adhesion, thereby preloading the two bearings.

Furthermore, in a case where a force is applied in such a manner that the inner ring 26 of the bearing 21 is separated from the bearing 22, it is better to apply a force in such a manner that the inner ring 26 of the bearing 22 is also separated from the bearing 21. On the contrary, in a case where a force is applied in such a manner that the inner ring 26 of the bearing 21 approaches the bearing 22, it is better to apply a force in such a manner that the inner ring 26 of the bearing 22 also approaches the bearing 21.

Moreover, in a case where the rotating electric machine 10 is applied to a vehicle for the purpose of a vehicle power source or the like, there is a possibility that vibration having a component in the preload generation direction is applied to a mechanism that generates the preload, and the direction of gravity applied to an object to which the preload is applied may fluctuate. Therefore, in the case where the rotating electric machine 10 is applied to a vehicle, it is desirable to adopt a fixed position preload.

Further, the intermediate section 45 has an annular inner shoulder section 49a and an annular outer shoulder section 49b. The outer shoulder section 49b is located outside the inner shoulder section 49a in the radial direction of the intermediate section 45. The inner shoulder section 49a and the outer shoulder section 49b are separated from each other in the axial direction of the intermediate section 45. As a result, the cylindrical section 43 and the fixing section 44 partially overlap in the radial direction of the intermediate section 45. That is, the cylindrical section 43 protrudes outward in the axial direction from the base end portion (back side end portion on the lower side in the figure) of the fixing section 44. In this configuration, the rotor 40 can be supported with respect to the rotating shaft 11 at a position near the center of gravity of the rotor 40, as compared with a case where the intermediate section 45 is provided in a flat plate shape without a step, and the operational stability of the rotor 40 can be achieved.

According to the configuration of the intermediate section 45 described above, in the rotor 40, a bearing housing recess 46 that houses a part of the bearing unit 20 is formed in an annular shape at a position that surrounds the fixing section 44 in the radial direction and is inward of the intermediate section 45, and a coil housing recess 47 that houses the coil end 54 of the stator winding 51 of the stator 50 which will be described below is formed at a position that surrounds the bearing housing recess 46 in the radial direction and is outward of the intermediate section 45. In addition, these respective housing recesses 46 and 47 are arranged so as to be adjacent to each other inside and outside in the radial direction. That is, a part of the bearing unit 20 and the coil end 54 of the stator winding 51 are arranged so as to overlap inside and outside in the radial direction. This makes it possible to shorten the axial length dimension in the rotating electric machine 10.

The intermediate section 45 is provided so as to project radially outward from the rotating shaft 11 side. In addition, the intermediate section 45 is provided with a contact avoiding section that extends in the axial direction and avoids contact of the stator winding 51 of the stator 50 with respect to the coil end 54. The intermediate section 45 corresponds to a projecting section.

By bending the coil end 54 inward or outward in the radial direction, the axial dimension of the coil end 54 can be

reduced, and the axial length of the stator 50 can be shortened. The bending direction of the coil end 54 may be in consideration of assembly with the rotor 40. Assuming that the stator 50 is assembled radially inside the rotor 40, the coil end 54 may be preferably bent radially inside on the insertion tip side with respect to the rotor 40. The bending direction of the coil end on the side opposite to the coil end 54 may be arbitrary, but a shape in which the coil end is bent outward with a sufficient space is preferable in manufacturing.

Further, the magnet unit 42 as a magnet section is composed of a plurality of permanent magnets that are arranged on the radial inside of the cylindrical section 43 in such a manner that the polarities alternate along the circumferential direction. As a result, the magnet unit 42 has a plurality of magnetic poles in the circumferential direction. However, the details of the magnet unit 42 will be described below.

The stator 50 is provided radially inside the rotor 40. The stator 50 has a stator winding 51 formed by winding in a substantially tubular shape (annular shape) and a stator core 52 as a base member arranged radially inside the stator winding 51. The stator winding 51 is arranged so as to face the annular magnet unit 42 with a predetermined air gap therebetween. The stator winding 51 is composed of a plurality of phase windings. Each of these phase windings is configured by connecting a plurality of conductors arranged in the circumferential direction to each other at a predetermined pitch. In the present embodiment, a U-phase, V-phase, and W-phase three-phase winding and an X-phase, Y-phase, and Z-phase three-phase winding are used. Two of these three-phase windings are used, and the stator winding 51 is thereby configured as a six-phase winding.

The stator core 52 is formed in an annular shape by laminated steel sheets in which electromagnetic steel sheets which are soft magnetic materials are laminated, and is assembled radially inside the stator winding 51. The electromagnetic steel sheet is, for example, a silicon steel sheet in which approximately several % (for example, 3%) of silicon is added to iron. The stator winding 51 corresponds to an armature winding, and the stator core 52 corresponds to an armature core.

The stator winding 51 is a portion that overlaps the stator core 52 in the radial direction, and has a coil side section 53 that is radially outside the stator core 52, and coil ends 54 and 55 that respectively project to one end side and to the other end side of the stator core 52 in the axial direction. The coil side section 53 faces the stator core 52 and the magnet unit 42 of the rotor 40 in the radial direction, respectively. In a state where the stator 50 is arranged inside the rotor 40, the coil end 54 on the side of the bearing unit 20 (upper side in the figure) of the coil ends 54 and 55 on both sides in the axial direction is housed in the coil housing recess 47 formed by the magnet holder 41 of the rotor 40. Note that, the details of the stator 50 will be described below.

The inverter unit 60 has a unit base 61 fixed to the housing 30 by fasteners such as bolts, and a plurality of electric components 62 assembled to the unit base 61. The unit base 61 is made of, for example, carbon fiber reinforced plastic (CFRP). The unit base 61 has an end plate 63 fixed to the edge of the opening 33 of the housing 30, and a casing 64 integrally provided with the end plate 63 and extending in the axial direction. The end plate 63 has a circular opening 65 at the center thereof, and the casing 64 is formed so as to stand up from the peripheral edge portion of the opening 65.

The stator 50 is assembled on the outer peripheral surface of the casing 64. That is, the outer diameter dimension of the casing 64 is the same as the inner diameter dimension of the

stator core 52, or slightly smaller than the inner diameter dimension of the stator core 52. By assembling the stator core 52 to the outside of the casing 64, the stator 50 and the unit base 61 are integrated. Further, since the unit base 61 is fixed to the housing 30, the stator 50 is integrated with the housing 30 in a state where the stator core 52 is assembled to the casing 64.

Moreover, the stator core 52 is preferably assembled to the unit base 61 by adhesion, shrink fitting, press fitting, or the like. As a result, the displacement of the stator core 52 in the circumferential direction or the axial direction with respect to the unit base 61 side is suppressed.

Further, the radial inside of the casing 64 is a housing space for housing the electric component 62, and the electric component 62 is arranged in the housing space so as to surround the rotating shaft 11. The casing 64 has a role as a housing space forming section. The electric component 62 includes a semiconductor module 66 constituting an inverter circuit, a control board 67, and a capacitor module 68.

Moreover, the unit base 61 is provided radially inside the stator 50 and corresponds to a stator holder (armature holder) that holds the stator 50. The housing 30 and the unit base 61 constitute the motor housing of the rotating electric machine 10. In this motor housing, the holding member 23 is fixed to the housing 30 on one side in the axial direction with the rotor 40 therebetween, and the housing 30 and the unit base 61 are coupled to each other on the other side. For example, in a motor vehicle or the like which is an electric car, the rotating electric machine 10 is mounted to the motor car or the like by attaching a motor housing to the side of the motor car or the like.

Here, the configuration of the inverter unit 60 will be further described with reference to FIG. 6 which is an exploded view of the inverter unit 60, in addition to FIGS. 1 to 5 described above.

In the unit base 61, the casing 64 has a tubular section 71 and an end face 72 provided on one (end on the bearing unit 20 side) of both ends facing each other in the axial direction thereof. Of the both ends of the tubular section 71 in the axial direction, the side opposite to the end face 72 is completely opened through the opening 65 of the end plate 63. A circular hole 73 is formed in the center of the end face 72, and the rotating shaft 11 can be inserted into the hole 73. The hole 73 is provided with a sealing material 171 that seals the space between the hole 73 and the outer peripheral surface of the rotating shaft 11. The sealing material 171 is preferably, for example, a sliding seal made of a resin material.

The tubular section 71 of the casing 64 is a partition section that partitions between the rotor 40 and the stator 50 arranged radially outside and the electric component 62 arranged radially inside. The rotor 40, the stator 50, and the electric component 62 are arranged side by side radially inside and outside with the tubular section 71 therebetween.

Further, the electric component 62 is an electric component constituting an inverter circuit, and has a power running function of passing a current through each phase winding of the stator winding 51 in a predetermined order to rotate the rotor 40 and a power generation function of inputting a three-phase AC current flowing through the stator winding 51 with the rotation of the rotating shaft 11 and outputting same to the outside as generated power. Moreover, the electric component 62 may have only one of the power running function and the power generation function. The power generation function is, for example, a regenerative

function that outputs regenerative power to the outside when the rotating electric machine 10 is used as a power source for a vehicle.

As a specific configuration of the electric component 62, as illustrated in FIG. 4, a hollow cylindrical capacitor module 68 is provided around the rotating shaft 11, and a plurality of semiconductor modules 66 are arranged side by side in the circumferential direction on the outer peripheral surface of the capacitor module 68. The capacitor module 68 includes a plurality of smoothing capacitors 68a connected in parallel to each other. Specifically, the capacitor 68a is a laminated film capacitor in which a plurality of film capacitors are laminated, and has a trapezoidal cross section. The capacitor module 68 is configured by arranging twelve capacitors 68a side by side in an annular shape.

Moreover, in the manufacturing process of the capacitor 68a, for example, a long film having a predetermined width in which a plurality of films are laminated is used, the film width direction is the trapezoid height direction, and the long film is cut into an isosceles trapezoid shape in such a manner that the upper bottom and the lower bottom of the trapezoid alternate, thereby making a capacitor element. Then, the capacitor 68a is manufactured by attaching an electrode or the like to the capacitor element.

The semiconductor module 66 has a semiconductor switching element such as a MOSFET or an IGBT, and is formed in a substantially plate shape. In the present embodiment, since the rotating electric machine 10 includes two sets of three-phase windings and an inverter circuit is provided for each of the three-phase windings, a semiconductor module group 66A formed by arranging a total of 12 semiconductor modules 66 in an annular shape is provided in the electric component 62.

The semiconductor module 66 is arranged in a state of being sandwiched between the tubular section 71 of the casing 64 and the capacitor module 68. The outer peripheral surface of the semiconductor module group 66A is in contact with the inner peripheral surface of the tubular section 71, and the inner peripheral surface of the semiconductor module group 66A is in contact with the outer peripheral surface of the capacitor module 68. In this case, the heat generated in the semiconductor module 66 is transferred to the end plate 63 via the casing 64 and released from the end plate 63.

The semiconductor module group 66A preferably has a spacer 69 between the semiconductor module 66 and the tubular section 71 on the outer peripheral surface side, that is, in the radial direction. In this case, in the capacitor module 68, the cross-sectional shape of the cross section orthogonal to the axial direction is a regular dodecagon, whereas the cross-sectional shape of the inner peripheral surface of the tubular section 71 is circular. Thus, the inner peripheral surface of the spacer 69 is a flat surface and the outer peripheral surface of the spacer 69 is a curved surface. The spacer 69 may be integrally provided so as to be connected in an annular shape on the radial outside of the semiconductor module group 66A. The spacer 69 is a good thermal conductor, and is preferably, for example, a metal such as aluminum, a heat radiating gel sheet, or the like. Moreover, it is also possible to make the cross-sectional shape of the inner peripheral surface of the tubular section 71 the same dodecagon as that of the capacitor module 68. In this case, it is preferable that both the inner peripheral surface and the outer peripheral surface of the spacer 69 are flat surfaces.

Further, in the present embodiment, a cooling water passage 74 for flowing cooling water is formed in the tubular section 71 of the casing 64, and the heat generated in the

semiconductor module 66 is also released to the cooling water flowing through the cooling water passage 74. That is, the casing 64 is provided with a water-cooling mechanism. As illustrated in FIGS. 3 and 4, the cooling water passage 74 is formed in an annular shape so as to surround the electric component 62 (the semiconductor module 66 and the capacitor module 68). The semiconductor module 66 is arranged along the inner peripheral surface of the tubular section 71, and the cooling water passage 74 is provided at a position overlapping the semiconductor module 66 inside and outside in the radial direction.

Since the stator 50 is arranged on the outside of the tubular section 71 and the electric component 62 is arranged on the inside of the tubular section 71, the heat of the stator 50 is transferred to the tubular section 71 from the outside, and the heat of the electric component 62 (for example, the heat of the semiconductor module 66) is transferred from the inside. In this case, the stator 50 and the semiconductor module 66 can be cooled at the same time, and the heat of the heat-generating member in the rotating electric machine 10 can be efficiently released.

Furthermore, at least a part of the semiconductor module 66 constituting a part or the whole of the inverter circuit that operates the rotating electric machine by energizing the stator winding 51 is arranged in a region surrounded by the stator core 52 arranged radially outside the tubular section 71 of the casing 64. Preferably, the entire one semiconductor module 66 is arranged in a region surrounded by the stator core 52. Furthermore, preferably, the whole of all the semiconductor modules 66 is arranged in a region surrounded by the stator core 52.

Further, at least a part of the semiconductor module 66 is arranged in a region surrounded by the cooling water passage 74. Preferably, the whole of all the semiconductor modules 66 is arranged in a region surrounded by a yoke 141.

Further, the electric component 62 includes an insulating sheet 75 provided on one end face of the capacitor module 68 and a wiring module 76 provided on the other end face in the axial direction. In this case, the capacitor module 68 has two end faces facing each other in the axial direction, that is, a first end face and a second end face. The first end face of the capacitor module 68 near the bearing unit 20 faces the end face 72 of the casing 64, and is superimposed on the end face 72 with the insulating sheet 75 sandwiched therebetween. Further, a wiring module 76 is assembled on the second end face of the capacitor module 68 near the opening 65.

The wiring module 76 has a main body section 76a made of a synthetic resin material and having a circular plate shape and a plurality of bus bars 76b, 76c embedded therein, and the bus bars 76b, 76c form an electrical connection with the semiconductor module 66 and the capacitor module 68. Specifically, the semiconductor module 66 has a connecting pin 66a extending from its axial end face, and the connecting pin 66a is connected to the busbar 76b on the radial outside of the main body section 76a. Further, a busbar 76c extends to the side opposite to the capacitor module 68 on the radial outside of the main body section 76a, and is connected to a wiring member 79 at the tip end portion thereof (see FIG. 2).

As described above, according to the configuration in which the insulating sheet 75 is provided on the first end face of the capacitor module 68 facing the axial direction and the wiring module 76 is provided on the second end face of the capacitor module 68, as a heat dissipation path of the capacitor module 68, a path from the first end face and the second end face of the capacitor module 68 to the end face

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72 and the tubular section 71 is formed. In other words, a path from the first end face to the end face 72 and a path from the second end face to the tubular section 71 are formed. As a result, heat can be dissipated from the end face portion of the capacitor module 68 other than the outer peripheral surface on which the semiconductor module 66 is provided. That is, not only heat dissipation in the radial direction but also heat dissipation in the axial direction is possible.

Further, since the capacitor module 68 has a hollow cylindrical shape and the rotating shaft 11 is arranged on the inner peripheral portion thereof with a predetermined gap interposed therebetween, the heat of the capacitor module 68 can be released from the hollow portion as well. In this case, the rotation of the rotating shaft 11 causes an air flow to enhance the cooling effect.

A disk-shaped control board 67 is attached to the wiring module 76. The control board 67 has a printed circuit board (PCB) on which a predetermined wiring pattern is formed, and a control device 77 corresponding to a control unit composed of various ICs and a microcomputer is mounted on the board. The control board 67 is fixed to the wiring module 76 by a fixture such as a screw. The control board 67 has, in the central portion thereof, an insertion hole 67a in which the rotation shaft 11 is inserted.

Moreover, the wiring module 76 has a first surface and a second surface that face each other in the axial direction, that is, face each other in the thickness direction thereof. The first surface faces the capacitor module 68. The wiring module 76 is provided with the control board 67 on the second surface thereof. The busbar 76c of the wiring module 76 extends from one side of both sides of the control board 67 to the other side. In such a configuration, the control board 67 is preferably provided with a notch to avoid interference with the busbar 76c. For example, a part of the outer edge portion of the control board 67 having a circular shape is preferably notched.

As described above, according to the configuration in which the electric component 62 is housed in the space surrounded by the casing 64, and the housing 30, the rotor 40 and the stator 50 are provided in layers on the outside thereof, the electromagnetic noise generated in the inverter circuit is suitably shielded. In other words, in the inverter circuit, switching control is performed in each semiconductor module 66 by utilizing PWM control using a predetermined carrier frequency, and it is conceivable that electromagnetic noise is generated by the switching control. However, the noise can be suitably shielded by the housing 30, the rotor 40, the stator 50, and the like on the radially outside the electric component 62.

Furthermore, at least a part of the semiconductor module 66 is arranged in the region surrounded by the stator core 52 arranged radially outside the tubular section 71 of the casing 64. Thus, compared to a configuration in which the semiconductor module 66 and the stator winding 51 are arranged without the stator core 52, even if magnetic flux is generated from the semiconductor module 66, the stator winding 51 is less likely to be affected. Further, even if the magnetic flux is generated from the stator winding 51, it is unlikely to affect the semiconductor module 66. Moreover, it is more effective if the entire semiconductor module 66 is arranged in a region surrounded by the stator core 52 arranged radially outside the tubular section 71 of the casing 64. Further, in a case where at least a part of the semiconductor module 66 is surrounded by the cooling water passage 74, it is possible to obtain the effect that the heat generated from the stator

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winding 51 and the magnet unit 42 is suppressed from reaching the semiconductor module 66.

In the tubular section 71, a through hole 78 is formed in the vicinity of the end plate 63, through which the wiring member 79 (see FIG. 2) that electrically connects the outer stator 50 and the inner electric component 62 is inserted. As illustrated in FIG. 2, the wiring member 79 is connected to the end of the stator winding 51 and the busbar 76c of the wiring module 76 by crimping, welding, or the like, respectively. The wiring member 79 is, for example, a busbar, and it is desirable that the joint surface thereof be flattened. The through holes 78 are preferably provided at one place or a plurality of places, and in the present embodiment, the through holes 78 are provided at two places. In the configuration in which the through holes 78 are provided at two places, the winding terminals extending from the two sets of three-phase windings can be easily connected by the wiring members 79 respectively, which is suitable for performing multi-phase connection.

As described above, as illustrated in FIG. 4, the rotor 40 and the stator 50 are provided in the housing 30 in this order from the outside in the radial direction, and an inverter unit 60 is provided radially inside the stator 50. Here, when the radius of the inner peripheral surface of the housing 30 is d , the rotor 40 and the stator 50 are arranged radially outside the distance of $d \times 0.705$ from the center of rotation of the rotor 40. In this case, when the region of the rotor 40 and the stator 50, that is radially inside from the inner peripheral surface of the stator 50 that is inside in the radial direction (that is, the inner peripheral surface of the stator core 52) is a first region X1, and the region between the inner peripheral surface of the stator 50 and the housing 30 is a second region X2, the cross-sectional area of the first region X1 is larger than the cross-sectional area of the second region X2. Further, the volume of the first region X1 is larger than the volume of the second region X2 when viewed in the radial direction in the range where the magnet unit 42 of the rotor 40 and the stator winding 51 overlap.

Moreover, when the rotor 40 and the stator 50 are a magnetic circuit component assembly, in the housing 30, the first region X1 radially inside from the inner peripheral surface of the magnetic circuit component assembly has a larger volume than that of the second region X2 between the inner peripheral surface of the magnetic circuit component assembly and the housing 30 in the radial direction.

Next, the configurations of the rotor 40 and the stator 50 will be described in more detail. Generally, as a structure of a stator in a rotating electric machine, a structure is known in which a stator core made of a laminated steel sheet and forming an annular shape is provided with a plurality of slots in the circumferential direction, and a stator winding is wound in the slots. Specifically, the stator core has a plurality of teeth extending in the radial direction from a yoke at predetermined intervals, and slots are formed between the teeth adjacent to each other in the circumferential direction. In addition, for example, a plurality of layers of conductors are housed in the slots in the radial direction, and the stator winding is composed of the conductors.

However, in the above-mentioned stator structure, when the stator winding is energized, magnetic saturation occurs in the teeth portion of the stator core as the magnetomotive force of the stator winding increases, which may limit the torque density of the rotating electric machine. That is, in the stator core, it is considered that magnetic saturation occurs when the rotating magnetic flux generated by the energization of the stator winding is concentrated on the teeth.

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Further, generally, as a configuration of an IPM (Interior Permanent Magnet) rotor in a rotating electric machine, a permanent magnet is arranged on the d-axis in the d-q coordinate system and a rotor core is arranged on the q-axis. In such a case, the stator winding near the d-axis is excited, and thus the exciting magnetic flux flows from the stator to the q-axis of the rotor according to Fleming law. It is considered that this causes a wide range of magnetic saturation in the q-axis core portion of the rotor.

FIG. 7 is a torque line diagram illustrating a relation between an ampere-turn [AT] indicating the magnetomotive force of a stator winding and a torque density [Nm/L]. The broken line indicates the characteristics of a general IPM rotor type rotating electric machine. As illustrated in FIG. 7, in a general rotating electric machine, by increasing the magnetomotive force in the stator, magnetic saturation occurs in two places, the teeth portion between the slots and the q-axis core portion, which limits the increase in torque. As described above, in the general rotating electric machine, the ampere-turn design value is limited by A1.

Accordingly, in the present embodiment, in order to eliminate the limitation caused by magnetic saturation, the rotating electric machine 10 is provided with the following configuration. In other words, as a first measure, in order to eliminate the magnetic saturation that occurs in the teeth of the stator core in the stator, a slotless structure is adopted in the stator 50, and in order to eliminate the magnetic saturation that occurs in the q-axis core portion of the IPM rotor, an SPM (Surface Permanent Magnet) rotor is adopted. According to the first measure, it is possible to eliminate the above-mentioned two parts where magnetic saturation occurs, but it is considered that the torque in the low current region is reduced (see the alternate long and short dash line in FIG. 7). Therefore, as a second measure, in order to recover the torque decrease by increasing the magnetic flux of the SPM rotor, in the magnet unit 42 of the rotor 40, a polar anisotropic structure in which the magnet magnetic path is lengthened to increase the magnetic force is adopted.

Further, as a third measure, in the coil side section 53 of the stator winding 51, a flat conductor structure in which the radial thickness of the stator 50 of the conductor is reduced is adopted to recover the torque decrease. Here, it is conceivable that a larger eddy current is generated in the stator winding 51 facing the magnet unit 42 due to the above-mentioned polar anisotropic structure in which the magnetic force is increased. However, according to the third measure, since the flat conductor structure is thin in the radial direction, it is possible to suppress the generation of eddy current in the radial direction in the stator winding 51. As described above, according to each of these first to third configurations, as illustrated by the solid line in FIG. 7, by adopting a magnet with a high magnetic force, it is expected that the torque characteristics will be significantly improved, and at the same time, the concern about the generation of a large eddy current that may occur due to the magnet with a high magnetic force also can be improved.

Furthermore, as a fourth measure, a magnet unit having a magnetic flux density distribution close to a sine wave is adopted by utilizing a polar anisotropic structure. According to this, the sine wave matching rate can be increased by pulse control or the like described below to increase the torque, and since the magnetic flux changes more slowly than the radial magnet, the eddy current loss (copper loss due to eddy current: eddy current loss) can also be further suppressed.

Hereinafter, the sine wave matching rate will be described. The sine wave matching rate can be obtained by

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comparing the measured waveform of the surface magnetic flux density distribution measured by tracing the surface of the magnet with a magnetic flux probe and the sine wave having the same period and peak value. In addition, the ratio of the amplitude of the primary waveform, which is the fundamental wave of the rotating electric machine, to the amplitude of the actually measured waveform, that is, the amplitude of the fundamental wave plus other higher harmonic components, corresponds to the sine wave matching rate. As the sine wave matching rate increases, the waveform of the surface magnetic flux density distribution approaches a sine wave shape. In addition, when a primary sine wave current is supplied from the inverter to a rotating electric machine equipped with a magnet having an improved sine wave matching rate, the waveform of the surface magnetic flux density distribution of the magnet is close to the sine wave shape, and correlatively, can generate a large torque. Moreover, the surface magnetic flux density distribution may be estimated by a method other than an actual measurement, for example, an electromagnetic field analysis using Maxwell equations.

Further, as a fifth measure, the stator winding 51 has a wire conductor structure in which a plurality of wires are gathered and bundled. According to this, since the wires are connected in parallel, a large current can flow, and the cross-sectional area of each wire is small, the eddy current generated by the conductors that spread in the circumferential direction of the stator 50 in the flat conductor structure can be suppressed more effectively than thinning in the radial direction by the third measure. In addition, by forming a structure in which a plurality of wires are twisted together, for the magnetomotive force from the conductor, it is possible to cancel the eddy current with respect to the magnetic flux generated by the right-handed screw rule in the current energizing direction.

In this way, if the fourth and fifth measures are further added, the torque can be increased while adopting the second measure, a magnet with a high magnetic force, while suppressing the eddy current loss caused by the high magnetic force.

Hereinafter, the slotless structure of the stator 50 described above, the flat conductor structure of the stator winding 51, and the polar anisotropic structure of the magnet unit 42 will be individually described. Here, first, the slotless structure of the stator 50 and the flat conductor structure of the stator winding 51 will be described. FIG. 8 is a cross-sectional view of the rotor 40 and the stator 50, and FIG. 9 is a view illustrating a part of the rotor 40 and the stator 50 illustrated in FIG. 8 in an enlarged manner. FIG. 10 is a cross-sectional view illustrating a cross section of the stator 50 along a line X-X of FIG. 11, and FIG. 11 is a cross-sectional view illustrating a vertical cross section of the stator 50. Further, FIG. 12 is a perspective view of the stator winding 51. Note that FIGS. 8 and 9 illustrate the magnetization direction of the magnet in the magnet unit 42 with arrows.

As illustrated in FIGS. 8 to 11, the stator core 52 has a cylindrical shape in which a plurality of electromagnetic steel sheets are laminated in the axial direction and has a predetermined thickness in the radial direction, and the stator winding 51 is assembled on the radially outside that is the rotor 40 side. In the stator core 52, the outer peripheral surface on the rotor 40 side is a conductor installation section (conductor area). The outer peripheral surface of the stator core 52 has a curved surface without unevenness, and a plurality of conductor groups 81 are arranged at predetermined intervals in the circumferential direction on the outer

peripheral surface thereof. The stator core 52 functions as a back yoke that is a part of a magnetic circuit for rotating the rotor 40. In this case, teeth (that is, an iron core) made of a soft magnetic material are not provided between each of the two conductor groups 81 adjacent to each other in the circumferential direction (that is, a slotless structure). In the present embodiment, the resin material of the sealing member 57 is inserted into a void 56 of each of the conductor groups 81. That is, in the stator 50, the interconductor member provided between the respective conductor groups 81 in the circumferential direction is configured as the sealing member 57 which is a non-magnetic material. In a state before sealing of the sealing member 57, on the outer side of the stator core 52 in the radial direction, the conductor groups 81 are respectively arranged at predetermined intervals in the circumferential direction with the void 56, which is an inter-conductor region, interposed therebetween. As a result, the stator 50 having a slotless structure is constructed. In other words, each conductor group 81 is composed of two conductors (conductor) 82 described below, and only a non-magnetic material occupies a region between the two conductor groups 81 adjacent to each other in the circumferential direction of the stator 50. The non-magnetic material includes a non-magnetic gas such as air, a non-magnetic liquid, and the like in addition to the sealing member 57. Moreover, in the following, the sealing member 57 is also referred to as an inter-conductor member (conductor-to-conductor member).

Moreover, a configuration in which the teeth are provided between the conductor groups 81 arranged in the circumferential direction is considered to be a configuration in which the teeth have a predetermined thickness in the radial direction and a predetermined width in the circumferential direction, and thus a part of a magnetic circuit, that is, a magnet magnetic path is formed between the conductor groups 81. In this respect, it can be said that a configuration in which the teeth are not provided between the respective conductor groups 81 is a configuration in which the above-mentioned magnetic circuit is not formed.

As illustrated in FIG. 10, the stator winding (i.e., armature winding) 51 is formed so as to have a predetermined thickness T2 (hereinafter, also referred to as a first dimension) and a width W2 (hereinafter, also referred to as a second dimension). The thickness T2 is the shortest distance between the outer surface and the inner surface facing each other in the radial direction of the stator winding 51. The width W2 is the circumferential length of the stator winding 51 of a part of the stator winding 51 which functions as one of the polyphase of the stator winding 51 (in the example, three phases: three phases of U phase, V phase and W phase or three phases of X phase, Y phase and Z phase). Specifically, in FIG. 10, in a case where two conductor groups 81 adjacent to each other in the circumferential direction function as one of the three phases, for example, the U phase, the width W2 is from one end to the other of the two conductor groups 81 in the circumferential direction. In addition, the thickness T2 is smaller than the width W2.

Moreover, the thickness T2 is preferably smaller than the total width dimension of the two conductor groups 81 existing in the width W2. Further, if the cross-sectional shape of the stator winding 51 (more specifically, the conductor wire 82) is a perfect circle, an ellipse, or a polygon, in the cross sections of the conductor wire 82 along the radial direction of the stator 50, the maximum length in the radial direction of the stator 50 may be W12, and the maximum length in the circumferential direction of the stator 50 may be W11.

As illustrated in FIGS. 10 and 11, the stator winding 51 is sealed by the sealing member 57 made of a synthetic resin material as a sealing material (molding material). That is, the stator winding 51 is molded together with the stator core 52 by the molding material. The resin can be regarded as a non-magnetic substance or an equivalent of the non-magnetic substance as $B_s=0$.

Looking at the cross section of FIG. 10, the sealing member 57 is provided between the respective conductor groups 81, that is, the void 56 is filled with the synthetic resin material, and with this sealing member 57, an insulating member is interposed between the respective conductor groups 81. That is, the sealing member 57 functions as an insulating member in the void 56. The sealing member 57 is provided on the radial outside of the stator core 52 in a range including all of the conductor groups 81, that is, in a range in which the radial thickness dimension is larger than the radial thickness dimension of each conductor group 81.

Further, when viewed in the vertical cross section of FIG. 11, the sealing member 57 is provided in a range including a turn section 84 of the stator winding 51. Inside the stator winding 51 in the radial direction, the sealing member 57 is provided within a range including at least a part of the end faces of the stator core 52 facing in the axial direction. In this case, the stator winding 51 is resin-sealed almost entirely except for the ends of the phase winding of each phase, that is, the connection terminals with the inverter circuit.

In the configuration in which the sealing member 57 is provided in a range including the end face of the stator core 52, the laminated steel sheet of the stator core 52 can be pressed inward in the axial direction by the sealing member 57. As a result, the laminated state of each steel sheet can be maintained with the use of the sealing member 57. Moreover, in the present embodiment, the inner peripheral surface of the stator core 52 is not resin-sealed, but instead, the entire stator core 52 including the inner peripheral surface of the stator core 52 may be resin-sealed.

In a case where the rotating electric machine 10 is used as a vehicle power source, the sealing member 57 is preferably made of a highly heat-resistant fluororesin, epoxy resin, PPS resin, PEEK resin, LCP resin, silicon resin, PAI resin, PI resin, or the like. Further, considering the linear expansion coefficient from the viewpoint of suppressing cracking due to the expansion difference, it is desirable that the material is the same as that of the outer coating of the conductor of the stator winding 51. In other words, a silicon resin having a linear expansion coefficient that is generally more than double that of other resins is preferably excluded. Moreover, for electric products that do not have an engine that utilizes combustion, such as electric vehicles, PPO resin, phenol resin, and FRP resin that have heat resistance of approximately 180° C. are also candidates. This does not apply in the field where the ambient temperature of the rotating electric machine is considered to be below 100° C.

The torque of the rotating electric machine 10 is proportional to the magnitude of the magnetic flux. Here, in a case where the stator core has teeth, the maximum amount of magnetic flux in the stator is limited depending on the saturation magnetic flux density in the teeth, but in a case where the stator core does not have teeth, the maximum amount of magnetic flux in the stator is not limited. Therefore, the configuration is advantageous in increasing the energization current for the stator winding 51 to increase the torque of the rotating electric machine 10.

In the present embodiment, the inductance of the stator 50 is reduced by using a structure (slotless structure) in which the stator 50 has no teeth. Specifically, in the stator of a

general rotating electric machine in which a conductor is housed in each slot partitioned by a plurality of teeth, the inductance is, for example, around 1 mH, whereas in the stator **50** of the present embodiment, the inductance is reduced to approximately 5 to 60 pH. In the present embodiment, it is possible to reduce a mechanical time constant T_m by reducing the inductance of the stator **50** while using the rotating electric machine **10** having an outer rotor structure. That is, it is possible to reduce the mechanical time constant T_m while increasing the torque. Moreover, when the inertia is J , the inductance is L , the torque constant is K_t , and the counter electromotive force constant is K_e , the mechanical time constant T_m is calculated by the following formula.

$$T_m = (J * L) / (K_t * K_e)$$

In this case, it can be confirmed that the mechanical time constant T_m is reduced by reducing the inductance L .

Each conductor group **81** on the radially outside the stator core **52** is configured by arranging a plurality of conductor wires **82** having a flat rectangular cross section side by side in the radial direction of the stator core **52**. Each conductor wire **82** is arranged in a direction in such a manner that “radial dimension < circumferential dimension” in the cross section. As a result, the thickness of each conductor group **81** is reduced in the radial direction. Further, the thickness in the radial direction is reduced, and the conductor region extends flatly to the region where the teeth have been conventionally, forming a flat conductor region structure. As a result, the increase in the amount of heat generated of the conductor, which is a concern because the cross-sectional area becomes smaller due to the thinning, is suppressed by flattening in the circumferential direction and increasing the cross-sectional area of the conductor. Moreover, even if a plurality of conductors are arranged in the circumferential direction and connected in parallel, the conductor cross-sectional area of the conductor coating is reduced, but the effect by the same reason can be obtained. Moreover, in the following, each conductor group **81** and each conductor wire **82** will also be referred to as a conductive member.

Since there is no slot, in the stator winding **51** in the present embodiment, the conductor region occupied by the stator winding **51** in one circumference in the circumferential direction can be designed to be larger than the conductor non-occupied region which the stator winding **51** does not occupy. Moreover, in a conventional rotating electric machine for vehicles, it is natural that the conductor region/conductor non-occupied region in one circumference in the circumferential direction of the stator winding is one or less. On the other hand, in the present embodiment, each conductor group **81** is provided in such a manner that the conductor region is equal to the conductor non-occupied region or the conductor region is larger than the conductor non-occupied region. Here, as illustrated in FIG. **10**, when the conductor region in which the conductor wire **82** (that is, a straight section **83** described below) is arranged in the circumferential direction is W_A and the region between the adjacent conductor wires **82** is W_B , the region W_A is larger than the interconductor region W_B in the circumferential direction.

As a configuration of the conductor group **81** in the stator winding **51**, the radial thickness dimension of the conductor group **81** is smaller than the circumferential width dimension for one phase in one magnetic pole. In other words, a configuration in which the conductor group **81** is composed of two layers of conductor wires **82** in the radial direction and two conductor groups **81** are provided in the circumferential direction for one phase in one magnetic pole fulfills

“ $T_c * 2 < W_c * 2$ ” when the radial thickness dimension of each conductor wire **82** is T_c , and the circumferential width dimension of each conductor wire **82** is W_c . Moreover, as another configuration, a configuration in which the conductor group **81** is composed of two layers of conductor wires **82** and one conductor group **81** is provided in the circumferential direction for one phase in one magnetic pole preferably fulfills a relation “ $T_c * 2 < W_c$ ”. In short, the conductor section (conductor group **81**) arranged at predetermined intervals in the circumferential direction in the stator winding **51** has a radial thickness dimension that is smaller than the circumferential width dimension for one phase in one magnetic pole.

In other words, it is preferable that the radial thickness dimension T_c of each conductor wire **82** is smaller than the circumferential width dimension W_c . Furthermore, a radial thickness dimension ($2 T_c$) in the radial direction of the conductor group **81** composed of two layers of conductors **82**, that is, a radial thickness dimension ($2 T_c$) in the circumferential direction of the conductor group **81** is preferably smaller than the width dimension W_c .

The torque of the rotating electric machine **10** is substantially inversely proportional to the radial thickness of the stator core **52** of the conductor group **81**. In this regard, the thickness of the conductor group **81** is reduced on the radial outside of the stator core **52**, which is advantageous in increasing the torque of the rotating electric machine **10**. The reason is that the magnetic resistance can be lowered by reducing the distance from the magnet unit **42** of the rotor **40** to the stator core **52** (that is, the distance of the iron-free portion). According to this, the interlinkage magnetic flux of the stator core **52** by the permanent magnet can be increased, and the torque can be increased.

Further, by reducing the thickness of the conductor group **81**, even if the magnetic flux leaks from the conductor group **81**, it is easily collected by the stator core **52**, and the magnetic flux can be prevented from not being effectively used for improving torque and leaking to the outside. That is, it is possible to suppress a decrease in magnetic force due to magnetic flux leakage, and it is possible to increase the interlinkage magnetic flux of the stator core **52** by the permanent magnet to increase the torque.

The conductor wire (conductor) **82** is composed of a coated conductor in which the surface of the conductor (conductor body) **82a** is coated with an insulating coating **82b**, and insulation is ensured between the conductor wires **82** that overlap each other in the radial direction and between the conductor wires **82** and the stator core **52**, respectively. This insulating coating **82b** is composed of a coating if a wire **86** described below is a self-fusion coated wire, or an insulating member laminated separately from the coating of the wire **86**. Moreover, each phase winding composed of the conductor wire **82** retains the insulating property by the insulating coating **82b** except for the exposed portion for connection. The exposed portion is, for example, an input/output terminal portion or a neutral point portion in the case of a star-shaped connection. In the conductor group **81**, the conductor wires **82** adjacent to each other in the radial direction are fixed to each other with the use of resin fixing or a self-fusion coated wire. As a result, dielectric breakdown, vibration, and sound due to the friction of the conductor wires **82** against each other are suppressed.

In the present embodiment, the conductor **82a** is configured as an aggregate of a plurality of wires (wire) **86**. Specifically, as illustrated in FIG. **13**, the conductor **82a** is formed in a twisted state by twisting a plurality of wires **86**. Further, as illustrated in FIG. **14**, the wires **86** are configured

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as a composite in which thin fibrous conductive materials **87** are bundled. For example, the wire **86** is a composite of CNT (carbon nanotube) fibers, and as the CNT fiber, a fiber containing boron-containing fine fibers in which at least a part of carbon is replaced with boron is used. As the carbon-based fine fiber, a vapor-grown carbon fiber (VGCF) or the like can be used in addition to the CNT fiber, but it is preferable to use the CNT fiber. Moreover, the surface of the wire **86** is covered with a polymer insulating layer such as enamel. Further, it is preferable that the surface of the wire **86** is covered with a so-called enamel coating made of a polyimide film or an amide-imide film.

The conductor wire **82** constitutes an n-phase winding in the stator winding **51**. In addition, the respective wires **86** of the conductor wire **82** (that is, the conductor **82a**) are adjacent to each other in contact with each other. In the conductor wire **82**, the winding conductor has a portion formed by twisting the plurality of wires **86** at one or more places in a phase, and the resistance value among the twisted wires **86** is larger than the resistance value of each wire **86** per se. In other words, if each of the two adjacent wires **86** has a first electrical resistivity in its adjacent direction and each of the wires **86** has a second electrical resistivity in its length direction, then the first electrical resistivity is larger than the second electrical resistivity. Moreover, the conductor wire **82** may be formed of the plurality of wires **86**, and may be an aggregate of wires covering the plurality of wires **86** by an insulating member having an extremely high first electrical resistivity. Further, the conductor **82a** of the conductor wire **82** is composed of a plurality of twisted wires **86**.

Since the conductor **82a** is configured by twisting the plurality of wires **86**, it is possible to suppress the generation of the eddy current in the respective wires **86** and reduce the eddy current in the conductor **82a**. Further, since each wire **86** is twisted, portions where the magnetic field application directions are opposite to each other are generated in one wire **86**, and the counter electromotive voltage is canceled out. Therefore, the eddy current can also be reduced. In particular, by forming the wire **86** with the fibrous conductive material **87**, it is possible to make the wire thinner and to significantly increase the number of twists, and it is possible to more preferably reduce the eddy current.

Moreover, the method for insulating the wires **86** from each other here is not limited to the above-mentioned polymer insulating film, and may be a method for making it difficult for current to flow between the twisted wires **86** by utilizing contact resistance. That is, if the resistance value between the twisted wires **86** is larger than the resistance value of the wire **86** per se, the above effect can be obtained by the potential difference generated due to the difference in the resistance values. For example, by using the manufacturing equipment for creating the wire **86** and the manufacturing equipment for making the stator **50** (armature) of the rotating electric machine **10** as separate non-continuous equipment, the wire **86** can be oxidized due to the movement time, work interval, and the like, and the contact resistance can be increased, which is suitable.

As described above, the conductor wires **82** have a flat rectangular cross section and are arranged side by side in the radial direction, and for example, a plurality of wires **86** covered with a self-fusion coated wire having a fusion layer and an insulating layer is assembled in a twisted state, and the fusion layers are fused to maintain the shape of the conductor wires **82**. Moreover, the wires having no fusion layer or the wires with the self-fusion coated wire may be twisted and solidified and molded into a desired shape with

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a synthetic resin or the like. When the thickness of the insulating coating **82b** in the conductor wire **82** is set to, for example, 80 μm to 100 μm and is set to be thicker than the film thickness (5 to 40 μm) of a commonly used conductor, even if an insulating paper or the like is not interposed between the conductor wire **82** and the stator core **52**, the insulating property between the two can be ensured.

Further, it is desirable that the insulating coating **82b** has higher insulating performance than that of the insulating layer of the wire **86** and is configured to be able to insulate between phases. For example, when the thickness of the polymer insulating layer of the wire **86** is set to, for example, approximately 5 μm , it is desirable that the thickness of the insulating coating **82b** of the conductor wire **82** is set to approximately 80 μm to 100 μm , and thus insulation between phases can be preferably performed.

Further, the conductor wire **82** may have a configuration in which a plurality of wires **86** are bundled without being twisted. That is, the conductor wire **82** may have any of a configuration in which a plurality of wires **86** are twisted in the total length, a configuration in which a plurality of wires **86** are twisted in a part of the total length, and a configuration in which a plurality of wires **86** are bundled without being twisted anywhere in the total length. In summary, each conductor wire **82** constituting the conductor section is a wire aggregate in which a plurality of wires **86** are bundled and the resistance value between the bundled wires is larger than the resistance value of the wire **86** per se.

Each conductor wire **82** is bent and formed so as to be arranged in a predetermined arrangement pattern in the circumferential direction of the stator winding **51**, and as a result, a phase winding for each phase is formed as the stator winding **51**. As illustrated in FIG. 12, in the stator winding **51**, the coil side section **53** is formed by the straight section **83** of each conductor wire **82** extending linearly in the axial direction, and the coil ends **54** and **55** are formed by the turn section **84** protruding to both outsides from the coil side section **53** in the axial direction. Each conductor wire **82** is configured as a series of wave winding-shaped conductors by alternately repeating the straight section **83** and the turn section **84**. The straight sections **83** are arranged at positions facing the magnet unit **42** in the radial direction, and in-phase straight sections **83** arranged at positions on the axially outer side of the magnet unit **42** at predetermined intervals are connected to each other by the turn section **84**. Note that the straight section **83** corresponds to a "magnet facing section".

In the present embodiment, the stator winding **51** is wound in an annular shape by distributed winding. In this case, in the coil side section **53**, straight sections **83** are arranged in the circumferential direction at intervals corresponding to one pole pair of the magnet unit **42** for each phase, and in the coil ends **54** and **55**, the respective straight sections **83** for each phase are connected to each other by the turn section **84** formed in a substantially V shape. The directions of the currents of the straight section **83** that are paired corresponding to one-pole pair are opposite to each other. Further, the combination of the pair of straight sections **83** connected by the turn section **84** is different between one coil end **54** and the other coil end **55**, and the connections at the coil ends **54** and **55** are repeated in the circumferential direction, and thus the stator winding **51** is formed in a substantially cylindrical shape.

More specifically, the stator winding **51** constitutes a winding for each phase with the use of two pairs of conductor wires **82** for each phase, and one three-phase winding (U-phase, V-phase, W-phase) and the other three-phase

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winding (X-phase, Y-phase, Z-phase) of the stator winding **51** are provided in two layers inside and outside in the radial direction. In this case, if the number of phases of the stator winding **51** is S (6 in the case of the example) and the number of conductor wires **82** per phase is m , then $2 \cdot S \cdot m = 2 \cdot S \cdot m$ conductor wires **82** will be formed for each pole pair. In the present embodiment, since the number of phases S is 6, the number of m is 4, and an 8-pole pair (16 poles) rotating electric machine is used, $6 \cdot 4 \cdot 8 = 192$ conductor wires **82** are arranged in the circumferential direction of the stator core **52**.

In the stator winding **51** illustrated in FIG. 12, in the coil side section **53**, the straight sections **83** are arranged so as to overlap in two layers adjacent in the radial direction, and in the coil ends **54** and **55**, the turn sections **84** extend in directions opposite to each other in the circumferential direction, from each of the straight sections **83** overlapping in the radial direction. That is, in the respective conductor wires **82** adjacent to each other in the radial direction, the directions of the turn sections **84** are opposite to each other except for the ends of the stator winding **51**.

Here, the winding structure of the conductor wire **82** in the stator winding **51** will be specifically described. In the present embodiment, a plurality of conductor wires **82** formed by wave winding are provided so as to be stacked in a plurality of layers (for example, two layers) adjacent to each other in the radial direction. FIGS. 15A and 15B are diagrams illustrating a form of each conductor wire **82** in an n th layer, FIG. 15A illustrates a shape of the conductor wire **82** seen from the side of the stator winding **51**, and FIG. 15B illustrates a shape of the conductor wire **82** seen from one side in the axial direction of the stator winding **51**. Moreover, in FIG. 15A and FIG. 15B, the positions where the conductor group **81** is arranged are illustrated as D1, D2, D3, . . . , respectively. Further, for convenience of explanation, only three conductor wires **82** are illustrated, which are referred to as a first conductor **82_A**, a second conductor **82_B**, and a third conductor **82_C**.

In each of the conductors **82_A** to **82_C**, the straight sections **83** are all arranged at the n th layer position, that is, at the same position in the radial direction, and the straight sections **83** separated from each other by 6 positions ($3 \cdot m$ pairs) in the circumferential direction are connected to each other by the turn section **84**. In other words, in each of the conductors **82_A** to **82_C**, on the same circle centered on the shaft center of the rotor **40**, both ends of the seven straight sections **83** arranged adjacent to each other in the circumferential direction of the stator winding **51** are connected to each other by one turn section **84**. For example, in the first conductor **82_A**, a pair of straight sections **83** are arranged at D1 and D7, respectively, and the pair of straight sections **83** are connected to each other by an inverted V-shaped turn section **84**. Further, the other conductors **82_B** and **82_C** are arranged in the same n th layer with their circumferential positions shifted by one. In this case, since the conductors **82_A** to **82_C** are all arranged in the same layer, it is conceivable that the turn sections **84** might interfere with each other. Therefore, in the present embodiment, an interference avoidance section is formed in the turn section **84** of each of the conductors **82_A** to **82_C** with a part thereof offset in the radial direction.

Specifically, the turn section **84** of each of the conductors **82_A** to **82_C** has one tilted portion **84a** which is a portion extending in the circumferential direction on the same circle (first circle), a top portion **84b** that shifts from the tilted portion **84a** radially inward (upper side in FIG. 15B) of the same circle and reaches another circle (second circle), an

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tilted portion **84c** that extends in the circumferential direction on the second circle, and a return portion **84d** that returns from the first circle to the second circle. The top portion **84b**, tilted portion **84c**, and return portion **84d** correspond to the interference avoidance section. Moreover, the tilted portion **84c** may be configured to shift outward in the radial direction with respect to the tilted portion **84a**.

That is, the turn sections **84** of the conductors **82_A** to **82_C** have a tilted portion **84a** on one side and a tilted portion **84c** on the other side on both sides of the top portion **84b** which is a central position in the circumferential direction. The radial positions of the tilted portions **84a** and **84c** (the position in the front-rear direction of the plane of FIG. 15A and the position in the up-down direction in FIG. 15B) are different from each other. For example, the turn section **84** of the first conductor **82_A** extends along the circumferential direction with a D1 position of the n th layer as the start point position, bends in the radial direction (for example, inward in the radial direction) at the top portion **84b** which is the central position in the circumferential direction, and then bends again in the circumferential direction, thereby extending along the circumferential direction again, and further bends in the radial direction (for example, outside in the radial direction) again at the return portion **84d**, thereby reaching a D7 position of the n th layer, which is the end point position.

According to the above configuration, in the conductors **82_A** to **82_C**, each tilted portion **84a** on one side is arranged vertically in the order of the first conductor **82_A**, the second conductor **82_B**, and the third conductor **82_C** from the top, and at the top portion **84b**, the top and bottom of each conductor **82_A** to **82_C** are interchanged, and each tilted portion **84c** on the other side is arranged vertically in the order of the third conductor **82_C**, the second conductor **82_B**, and the first conductor **82_A** from the top. Therefore, the respective conductors **82_A** to **82_C** can be arranged in the circumferential direction without interfering with each other.

Here, in a configuration in which a plurality of conductor wires **82** are stacked in the radial direction to form a conductor group **81**, it is preferable that the turn section **84** connected to the straight section **83** on the radially inside and the turn section **84** connected to the straight section **83** on the radial outside of the respective straight sections **83** of the plurality of layers are arranged so as to be radially separated from each of the straight sections **83**. Further, when the conductor wires **82** of a plurality of layers are bent to the same side in the radial direction near the end of the turn section **84**, that is, the boundary portion with the straight section **83**, it is preferable that the insulating property is not impaired due to the interference between the conductor wires **82** of the adjacent layers.

For example, in D7 to D9 of FIG. 15A and FIG. 15B, the respective conductor wires **82** overlapping in the radial direction are respectively bent in the radial direction at the return portion **84d** of the turn section **84**. In this case, as illustrated in FIG. 16, it is preferable that the radius of curvature of the bent portion is different between the conductor wire **82** of the n th layer and the conductor wire **82** of the $n+1$ layer. Specifically, a radius of curvature R1 of the conductor wire **82** on the radially inside (n th layer) is made smaller than a radius of curvature R2 of the conductor wire **82** on the radially outside ($n+1$ th layer).

Further, it is preferable that the radial shift amount is different between the n th layer conductor wire **82** and the $n+1$ th layer conductor wire **82**. Specifically, a shift amount S1 of the conductor wire **82** on the radially inside (n th layer)

is made larger than a shift amount S2 of the conductor wire 82 on the radially outside (n+1th layer).

With the above configuration, mutual interference of the respective conductor wires 82 can be suitably avoided even when the respective conductor wires 82 overlapping in the radial direction are bent in the same direction. As a result, good insulating properties can be obtained.

Next, the structure of the magnet unit 42 in the rotor 40 will be described. In the present embodiment, it is assumed that the magnet unit 42 is made of a permanent magnet, has a residual magnetic flux density $B_r=1.0$ [T], and has an intrinsic coercive force $H_{cj}=400$ [kA/m] or more.

In short, the permanent magnet used in this embodiment is a sintered magnet obtained by sintering and solidifying a granular magnetic material, and the intrinsic coercive force H_{cj} on the J-H curve is 400 [kA/m] or more, and the residual magnetic flux density B_r is 1.0 [T] or more. When 5000 to 10000 [AT] is applied by interphase excitation, if a permanent magnet with a length of 25 [mm] is used in the magnetic length of one pole pair, that is, the N pole and the S pole, in other words, the path of the magnetic flux flowing between the N pole and the S pole, then $H_{cj}=10000$ [A], indicating that demagnetization is not performed.

In other words, the magnet unit 42 has a saturation magnetic flux density J_s of 1.2 [T] or more, a crystal particle size of 10 [μm] or less, and $J_s \cdot \alpha$ of 1.0 [T] or higher when the orientation ratio is a.

The magnet unit 42 will be supplemented below. The magnet unit 42 (magnet) is characterized in that $J_s \geq 1.2$ [T]. In other words, examples of the magnet used in the magnet unit 42 include NdFe11TiN, Nd2Fe14B, Sm2Fe17N3, and FeNi magnets having L10 type crystals. Moreover, configurations such as SmCo5 which is usually called samarium-cobalt, FePt, Dy2Fe14B, and CoPt cannot be used. Note that, like the same type of compounds such as Dy2Fe14B and Nd2Fe14B, in some cases, even a magnet that generally uses dysprosium which is a heavy rare earth to have a high coercive force of Dy while slightly losing the high J_s characteristics of neodymium fulfills $J_s \geq 1.2$ [T], the magnet can be employed in this case as well. In such a case, the magnet is referred to as ([Nd1-xDyx]2Fe14B), for example. Furthermore, the magnet can be achieved by two or more types of magnets having different compositions, for example, magnets made of two or more types of materials such as FeNi plus Sm2Fe17N3, and for example, the magnet can also be achieved by a mixed magnet in which a small amount of $J_s < 1$ [T], for example, Dy2Fe14B is mixed with a magnet of Nd2Fe14B having sufficient value of J_s , i.e. $J_s=1.6$ [T], and the coercive force is increased.

Further, for rotating electric machines that operate at temperatures outside the range of human activity, for example, 60° C. or higher, which exceeds the temperature of a desert, for example, in vehicle motor applications where the temperature inside the vehicle is close to 80° C. in summer, it is desirable to contain the components of FeNi and Sm2Fe17N3, which have a particularly small temperature dependence coefficient. This is because the motor characteristics differ greatly depending on the temperature dependence coefficient in the motor operation from a temperature state close to -40° C. in Northern Europe, which is within the range of human activity, to 60° C. or higher, which exceeds the desert temperature mentioned above, or to a heat resistant temperature of coil enamel coating approximately 180-240° C., and thus it becomes difficult to perform optimum control with the same motor driver. If the aforementioned FeNi having L10 type crystals,

Sm2Fe17N3, or the like is used, the burden on the motor driver can be suitably reduced due to its characteristic of having a temperature dependence coefficient of less than half that of Nd2Fe14B.

Additionally, the magnet unit 42 is characterized in that the size of the particle diameter in the fine powder state before orientation is 10 μm or less and the single magnetic domain particle diameter or more by using the aforementioned magnet blending. In magnets, the coercive force is increased by miniaturizing the powder particles to the order of several hundred nm. Therefore, in recent years, powder as fine as possible has been used. However, if they are made too fine, the BH product of the magnet will drop due to oxidation or the like, and thus a single magnetic domain particle diameter or larger is preferable. It is known that the coercive force increases by the miniaturization when the particle diameter is up to the single magnetic domain particle diameter. Moreover, the size of the particle diameter that has been described here is the size of the particle diameter in the fine powder state at the time of the orientation process in the magnet manufacturing process.

Furthermore, each of the first magnet 91 and the second magnet 92 of the magnet unit 42 is a sintered magnet formed by so-called sintering, in which magnetic powder is baked and hardened at a high temperature. In this sintering, when the saturation magnetization J_s of the magnet unit 42 is 1.2 T or more, the crystal grain diameter of the first magnet 91 and the second magnet 92 is 10 μm or less, and the orientation ratio is a, sintering is performed in such a manner that $J_s \cdot \alpha$ fulfills a condition of 1.0 T (tesla) or more. Further, each of the first magnet 91 and the second magnet 92 is sintered so as to fulfill the following conditions. In addition, the orientation is performed in the orientation process in the manufacturing process, the orientation ratio is different from the definition of the magnetic force direction by the magnetizing process of the isotropic magnet. With the saturation magnetization J_s of the magnet unit 42 of the present embodiment is 1.2 T or more, a high orientation ratio is set in such a manner that the orientation ratio α of the first magnet 91 and the second magnet 92 is $J_s \cdot \alpha \geq 1.0$ [T]. Moreover, the orientation ratio α referred to here is that, in a case where in each of the first magnet 91 and the second magnet 92, for example, there are six axes of easy magnetization, if five of the axes face a direction A10 in the same direction and the remaining one faces a direction B10 tilted 90 degrees with respect to the direction A10, $\alpha=5/6$, and if the remaining one faces the direction B10 tilted 45 degrees with respect to the direction A10, the component for the remaining one facing the direction A10 is $\cos 45^\circ=0.707$, and thus $\alpha=(5+0.707)/6$. In this example, the first magnet 91 and the second magnet 92 are formed by sintering, but if the above conditions are fulfilled, the first magnet 91 and the second magnet 92 may be molded by another method. For example, a method for forming an MQ3 magnet or the like can be employed.

In this embodiment, since a permanent magnet whose axis of easy magnetization is controlled by orientation is used, the magnetic circuit length inside the magnet can be made longer than the magnetic circuit length of a linearly oriented magnet that conventionally produces 1.0 [T] or more. That is, the magnetic circuit length per pole pair can be achieved with a small amount of magnets, and the reversible demagnetization range can be maintained even when exposed to harsh high thermal conditions compared to the conventional design using linearly oriented magnets. Further, the discloser of the present application has found a configuration in

which characteristics similar to those of a polar anisotropic magnet can be obtained even with the use of a magnet of the prior art.

Note that, the axis of easy magnetization refers to a crystal orientation that is easily magnetized in a magnet. The direction of the axis of easy magnetization in the magnet is a direction in which the orientation ratio indicating the degree to which the directions of the axes of easy magnetization are aligned is 50% or more, or a direction in which the orientation of the magnet is average.

As illustrated in FIGS. 8 and 9, the magnet unit 42 has an annular shape and is provided inside the magnet holder 41 (specifically, radially inside the cylindrical section 43). The magnet unit 42 has a first magnet 91 and a second magnet 92, which are respectively polar anisotropic magnets and have different polarities from each other. The first magnet 91 and the second magnet 92 are arranged alternately in the circumferential direction. The first magnet 91 is a magnet that forms an N pole in a portion close to the stator winding 51, and the second magnet 92 is a magnet that forms an S pole in a portion close to the stator winding 51. The first magnet 91 and the second magnet 92 are permanent magnets made of rare earth magnets such as neodymium magnets.

In each of the magnets 91 and 92, as illustrated in FIG. 9, in the known d-q coordinate system, the magnetization direction extends in an arc shape between the d-axis (direct-axis) which is the center of the magnetic pole and the q-axis (quadrature-axis) which is the magnetic pole boundary between the N pole and the S pole (in other words, the magnetic flux density is 0 tesla). In each of the magnets 91 and 92, the magnetization direction is the radial direction of the annular magnet unit 42 on the d-axis side, and the magnetization direction of the annular magnet unit 42 is the circumferential direction on the q-axis side. Hereinafter, it will be described in more detail. As illustrated in FIG. 9, each of the magnets 91 and 92 has a first portion 250 and two second portions 260 located on both sides of the first portion 250 in the circumferential direction of the magnet unit 42. In other words, the first portion 250 is closer to the d-axis than the second portion 260 is, and the second portion 260 is closer to the q-axis than the first portion 250 is. In addition, the magnet unit 42 is configured in such a manner that the direction of an axis of easy magnetization 300 of the first portion 250 is more parallel to the d-axis than the direction of an axis of easy magnetization 310 of the second portion 260. In other words, the magnet unit 42 is configured in such a manner that an angle $\theta 11$ formed by the axis of easy magnetization 300 of the first portion 250 and the d-axis is smaller than an angle $\theta 12$ formed by the axis of easy magnetization 310 of the second portion 260 and the q-axis.

More specifically, the angle $\theta 11$ is an angle formed by the d-axis and the axis of easy magnetization 300 when the direction from the stator 50 (armature) to the magnet unit 42 is positive on the d-axis. The angle $\theta 12$ is an angle formed by the q-axis and the axis of easy magnetization 310 when the direction from the stator 50 (armature) to the magnet unit 42 is positive on the q-axis. Both the angle $\theta 11$ and the angle $\theta 12$ are 90° or less in this embodiment. Each of the axes of easy magnetization 300 and 310 referred to here is defined by the following. If one axis of easy magnetization faces a direction A11 and the other axis of easy magnetization faces a direction B11 in each of the magnets 91 and 92, the absolute value ($|\cos \theta|$) of the cosine of the angle θ formed by the direction A11 and the direction B11 is the axis of easy magnetization 300 or the axis of easy magnetization 310.

That is, each of the magnets 91 and 92 has a different direction of the axis of easy magnetization on the d-axis side

(the portion near the d-axis) and the q-axis side (the portion near the q-axis), and on the d-axis side, the direction of the axis of easy magnetization is close to the direction parallel to the d-axis, and on the q-axis side, the direction of the easy magnetization axis is close to the direction orthogonal to the q-axis. In addition, an arc-shaped magnet magnetic path is formed in accordance with the direction of the axis of easy magnetization. Moreover, in each of the magnets 91 and 92, the axis of easy magnetization may be oriented parallel to the d-axis on the d-axis side, and the axis of easy magnetization may be oriented orthogonal to the q-axis on the q-axis side.

Further, in the magnets 91 and 92, of the peripheral surfaces of the magnets 91 and 92, the outer surface on the stator side on the stator 50 side (lower side in FIG. 9) and the end face on the q-axis side in the circumferential direction are magnetic flux acting surfaces which are inflow and outflow surfaces of magnetic flux, and a magnet magnetic path is formed so as to connect these magnetic flux acting surfaces (the outer surface on the stator side and the end face on the q-axis side).

In the magnet unit 42, magnetic flux flows in an arc shape between adjacent N and S poles due to the magnets 91 and 92, and thus the magnet magnetic path is longer than that of, for example, a radial anisotropic magnet. Therefore, as illustrated in FIG. 17, the magnetic flux density distribution is close to a sine wave. As a result, unlike the magnetic flux density distribution of the radial anisotropic magnet illustrated as a comparative example in FIG. 18, the magnetic flux can be concentrated on the center side of the magnetic poles, and the torque of the rotating electric machine 10 can be increased. Further, it can be confirmed that the magnet unit 42 of the present embodiment has a difference in the magnetic flux density distribution as compared with a conventional Halbach array magnet. Moreover, in FIGS. 17 and 18, the horizontal axis represents an electrical angle and the vertical axis represents a magnetic flux density. Further, in FIGS. 17 and 18, 90° on the horizontal axis indicates the d-axis (that is, the center of the magnetic pole), and 0° and 180° on the horizontal axis indicate the q-axis.

That is, according to the magnets 91 and 92 having the above configuration, the magnet magnetic flux on the d-axis is strengthened and the change in magnetic flux near the q-axis is suppressed. As a result, magnets 91 and 92 in which the change in surface magnetic flux from the q-axis to the d-axis at each magnetic pole is gentle can be preferably achieved.

The sine wave matching rate of the magnetic flux density distribution should be, for example, a value of 40% or more. By doing so, it is possible to reliably improve the amount of magnetic flux in the central portion of the waveform as compared with the case of using a radially oriented magnet or a parallel oriented magnet having a sine wave matching rate of approximately 30%. Further, if the sine wave matching rate is 60% or more, the amount of magnetic flux in the central portion of the waveform can be reliably improved as compared with the magnetic flux concentrated array such as the Halbach array.

In the radial anisotropic magnet illustrated in FIG. 18, the magnetic flux density changes steeply in the vicinity of the q-axis. The steeper the change in magnetic flux density, the greater the eddy current generated in the stator winding 51. Further, the change in magnetic flux on the stator winding 51 side is also steep. On the other hand, in the present embodiment, the magnetic flux density distribution is a magnetic flux waveform close to a sine wave. Therefore, the change in the magnetic flux density in the vicinity of the q-axis is

smaller than the change in the magnetic flux density of the radial anisotropic magnet. As a result, the generation of eddy current can be suppressed.

In the magnet unit 42, a magnetic flux is generated in the vicinity of the d-axis (that is, the center of the magnetic pole) of each of the magnets 91 and 92 in a direction orthogonal to a magnetic flux acting surface 280 on the stator 50 side, and the farther away from the magnetic flux acting surface 280 on the stator 50 side, the magnetic flux forms an arc shape farther away from the d-axis. Further, the magnetic flux more orthogonal to the magnetic flux acting surface becomes stronger. In this respect, in the rotating electric machine 10 of the present embodiment, since each conductor group 81 is thinned in the radial direction as described above, the radial center position of the conductor group 81 approaches the magnetic flux acting surface of the magnet unit 42, and the stator 50 can receive a strong magnet magnetic flux from the rotor 40.

Further, the stator 50 is provided with a cylindrical stator core 52 on the radial inside of the stator winding 51, that is, on the side opposite to the rotor 40 with the stator winding 51 therebetween. Therefore, the magnetic flux extending from the magnetic flux acting surface of each of the magnets 91 and 92 is attracted to the stator core 52 and orbits while using the stator core 52 as a part of the magnetic path. In this case, the direction and path of the magnet magnetic flux can be optimized.

The procedure for assembling the bearing unit 20, the housing 30, the rotor 40, the stator 50, and the inverter unit 60 illustrated in FIG. 5 will be described below as a method for manufacturing the rotating electric machine 10. Moreover, as illustrated in FIG. 6, the inverter unit 60 has a unit base 61 and an electric component 62, and each work process including the assembling process of the unit base 61 and the electric component 62 will be described. In the following description, the assembly including the stator 50 and the inverter unit 60 is referred to as a first unit, and the assembly including the bearing unit 20, the housing 30 and the rotor 40 is referred to as a second unit.

This manufacturing process has

- a first process for mounting the electric component 62 radially inside the unit base 61,
- a second process for mounting the unit base 61 radially inside the stator 50 to manufacture the first unit,
- a third process for inserting the fixing section 44 of the rotor 40 into the bearing unit 20 assembled to the housing 30 to manufacture the second unit,
- a fourth process for mounting the first unit radially inside the second unit, and
- a fifth process for fastening and fixing the housing 30 and the unit base 61.

The execution order of each of these processes is the first process, second process, third process, fourth process, and fifth process.

According to the above manufacturing method, the bearing unit 20, housing 30, rotor 40, stator 50, and inverter unit 60 are assembled as a plurality of assemblies (subassemblies), and then the assemblies are assembled to each other. Therefore, ease of handling and completion of inspection for each unit can be achieved, and a rational assembly line can be constructed. Consequently, it is possible to easily cope with multi-product production.

In the first process, a good thermal conductor having good thermal conductivity is attached to at least one of the radial inside of the unit base 61 and the radial outside of the electric component 62 by coating, adhesion, or the like, and in that state, the electric component 62 is preferably attached to the

unit base 61. This makes it possible to effectively transmit the heat generated by the semiconductor module 66 to the unit base 61.

In the third process, the rotor 40 is preferably inserted while maintaining the coaxiality between the housing 30 and the rotor 40. Specifically, for example, using a jig for determining the position of the outer peripheral surface of the rotor 40 (outer peripheral surface of the magnet holder 41) or the inner peripheral surface of the rotor 40 (inner peripheral surface of the magnet unit 42) with reference to the inner peripheral surface of the housing 30, the housing 30 and the rotor 40 are assembled while sliding either the housing 30 or the rotor 40 along the jig. As a result, heavy parts can be assembled without applying an unbalanced load to the bearing unit 20, and the reliability of the bearing unit 20 is improved.

In the fourth process, it is preferable to assemble the first unit and the second unit while maintaining the coaxiality between both units. Specifically, for example, using a jig for determining the position of the inner peripheral surface of the unit base 61 with reference to the inner peripheral surface of the fixing section 44 of the rotor 40, each of the first unit and the second unit is assembled while sliding either one of them along the jig. As a result, it is possible to assemble the rotor 40 and the stator 50 while preventing mutual interference with each other between the minimum gaps, and therefore it is possible to eliminate defective products caused by assembly, such as damage to the stator winding 51 and chipping of permanent magnets.

The order of each of the above processes may also be the second process, third process, fourth process, fifth process, and first process. In this case, the delicate electric component 62 is assembled last, and the stress on the electric component 62 in the assembling process can be minimized.

Next, the configuration of the control system that controls the rotating electric machine 10 will be described. FIG. 19 is an electrical circuit diagram of the control system of the rotating electric machine 10, and FIG. 20 is a functional block diagram illustrating control processing by a control device 110.

In FIG. 19, two sets of three-phase windings 51a and 51b are illustrated as the stator winding 51. The three-phase winding 51a includes a U-phase winding, a V-phase winding, and a W-phase winding, and the three-phase winding 51b includes an X-phase winding, a Y-phase winding, and a Z-phase winding. A first inverter 101 and a second inverter 102, which correspond to power converters, are provided for each of the three-phase windings 51a and 51b, respectively. The inverters 101 and 102 are composed of a full bridge circuit having the same number of upper and lower arms as the number of phases of the phase windings, and the energization current is adjusted in each phase winding of the stator winding 51 by turning on/off a switch (semiconductor switching element) provided on each arm.

ADC power supply 103 and a smoothing capacitor 104 are connected in parallel to each of the inverters 101 and 102. The DC power supply 103 is composed of, for example, an assembled battery in which a plurality of single batteries are connected in series. Moreover, each switch of the inverters 101 and 102 corresponds to the semiconductor module 66 illustrated in FIG. 1 and the like, and the capacitor 104 corresponds to the capacitor module 68 illustrated in FIG. 1 and the like.

The control device 110 includes a microcomputer composed of a CPU and various memories, and performs energization control by turning on/off each switch in the inverters 101 and 102 on the basis of various detected information

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in the rotating electric machine **10** and requests for power running and power generation. The control device **110** corresponds to the control device **77** illustrated in FIG. 6. The detected information of the rotating electric machine **10** includes, for example, a rotation angle (electrical angle information) of the rotor **40** detected by an angle detector such as a resolver, a power supply voltage (inverter input voltage) detected by a voltage sensor, and an energization current of each phase detected by a current sensor. The control device **110** generates and outputs an operation signal for operating each switch of the inverters **101** and **102**. Moreover, the request for power generation is, for example, a request for regenerative driving when the rotating electric machine **10** is used as a power source for a vehicle.

The first inverter **101** includes a series connection body of an upper arm switch S_p and a lower arm switch S_n in three phases composed of the U phase, V phase, and W phase. The high potential side terminal of the upper arm switch S_p of each phase is connected to the positive electrode terminal of the DC power supply **103**, and the low potential side terminal of the lower arm switch S_n of each phase is connected to the negative electrode terminal (ground) of the DC power supply **103**. One ends of the U-phase winding, V-phase winding, and W-phase winding are connected to the intermediate connection points between the upper arm switch S_p and the lower arm switch S_n of each phase, respectively. These respective phase windings are connected in a star-shape (Y-connected), and the other ends of the respective phase windings are connected to each other at a neutral point.

The second inverter **102** has the same configuration as that of the first inverter **101**, and includes a series connection body of the upper arm switch S_p and the lower arm switch S_n in three phases composed of the U phase, V phase, and W phase. The high potential side terminal of the upper arm switch S_p of each phase is connected to the positive electrode terminal of the DC power supply **103**, and the low potential side terminal of the lower arm switch S_n of each phase is connected to the negative electrode terminal (ground) of the DC power supply **103**. One ends of the X-phase winding, Y-phase winding, and Z-phase winding are connected to the intermediate connection points between the upper arm switch S_p and the lower arm switch S_n of each phase, respectively. These respective phase windings are connected in a star-shape (Y-connected), and the other ends of the respective phase windings are connected to each other at a neutral point.

FIG. 20 illustrates current feedback control processing for controlling each phase current of the U, V, and W phases, and current feedback control processing for controlling each phase current of the X, Y, and Z phases. Here, first, the control processing on the U, V, and W phase side will be described.

In FIG. 20, a current command value setting unit **111** uses a torque-dq map and sets a d-axis current command value and a q-axis current command value on the basis of the power running torque command value or the power generation torque command value for the rotating electric machine **10** and an electric angular velocity ω obtained by time-differentiating an electrical angle θ . Moreover, the current command value setting unit **111** is commonly provided on the U, V, and W phase side and the X, Y, and Z phase side. Note that the power generation torque command value is, for example, a regenerative torque command value when the rotating electric machine **10** is used as a power source for a vehicle.

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A dq conversion unit **112** converts, the current detected values (three phase currents) by the current sensors provided for each phase, into a d-axis current and a q-axis current which are components of an orthogonal two-dimensional rotation coordinate system with the field magnet direction (direction of an axis of a magnetic field or field direction) as the d-axis.

A d-axis current feedback control unit **113** calculates a d-axis command voltage as an operation amount for feedback-controlling the d-axis current to the d-axis current command value. Further, a q-axis current feedback control unit **114** calculates a q-axis command voltage as an operation amount for feedback-controlling the q-axis current to the q-axis current command value. In each of these feedback control units **113** and **114**, the command voltage is calculated with the use of the PI feedback method on the basis of the deviation with respect to the current command values of the d-axis current and the q-axis current.

A three-phase conversion unit **115** converts the d-axis and q-axis command voltages into U-phase, V-phase, and W-phase command voltages. Moreover, each of the above units **111** to **115** is a feedback control unit that performs feedback control of the fundamental wave current according to the dq conversion theory, and the U-phase, V-phase, and W-phase command voltages are feedback control values.

In addition, an operation signal generation unit **116** uses a well-known triangular wave carrier comparison method to generate an operation signal of the first inverter **101** on the basis of the command voltages of the three phases. Specifically, the operation signal generation unit **116** generates a switch operation signal (duty signal) of the upper and lower arms in each phase by PWM control based on the magnitude comparison between the signal obtained by standardizing the command voltage of the three phases with the power supply voltage and the carrier signal such as a triangular wave signal.

Further, the X, Y, and Z phase side also has the same configuration, and a dq conversion unit **122** converts the current detected values (three phase currents) by the current sensor provided for each phase into the d-axis current and q-axis current, which are components of an orthogonal two-dimensional rotation coordinate system with the field direction as the d-axis.

A d-axis current feedback control unit **123** calculates a d-axis command voltage, and a q-axis current feedback control unit **124** calculates a q-axis command voltage. A three-phase conversion unit **125** converts the d-axis and q-axis command voltages into X-phase, Y-phase, and Z-phase command voltages. In addition, an operation signal generation unit **126** generates an operation signal of the second inverter **102** on the basis of the command voltages of the three phases. Specifically, the operation signal generation unit **126** generates a switch operation signal (duty signal) of the upper and lower arms in each phase by PWM control based on the magnitude comparison between the signal obtained by standardizing the command voltage of the three phases with the power supply voltage and the carrier signal such as a triangular wave signal.

A driver **117** turns on/off the switches S_p and S_n of each of the three phases in the inverters **101** and **102** on the basis of the switch operation signals generated by the operation signal generation units **116** and **126**.

Subsequently, the torque feedback control processing will be described. This process is mainly used for the purpose of increasing the output of the rotating electric machine **10** and reducing the loss under operating conditions in which the output voltage of each of the inverters **101** and **102** becomes

large, such as in a high rotation region and a high output region. The control device **110** selects and executes either the torque feedback control processing or the current feedback control processing on the basis of the operating conditions of the rotating electric machine **10**.

FIG. **21** illustrates torque feedback control processing corresponding to the U, V, and W phases and torque feedback control processing corresponding to the X, Y, and Z phases. Moreover, in FIG. **21**, the same configurations as those in FIG. **20** are designated by the same reference signs and the description thereof will be omitted. Here, first, the control processing on the U, V, and W phase side will be described.

A voltage amplitude calculation unit **127** calculates a voltage amplitude command which is a command value of the magnitude of the voltage vector, on the basis of the power running torque command value or the power generation torque command value for the rotating electric machine **10** and the electric angular velocity ω obtained by time-differentiating the electrical angle θ .

A torque estimation unit **128a** calculates a torque estimated value corresponding to the U, V, and W phases on the basis of the d-axis current and the q-axis current converted by the dq conversion unit **112**. Moreover, the torque estimation unit **128a** may calculate the voltage amplitude command on the basis of the map information in which the d-axis current, the q-axis current, and the voltage amplitude command are associated.

A torque feedback control unit **129a** calculates a voltage phase command which is a command value of the phase of the voltage vector, as an operation amount for feedback-controlling the torque estimated value to the power running torque command value or the power generation torque command value. The torque feedback control unit **129a** calculates the voltage phase command with the use of the PI feedback method on the basis of the deviation of the torque estimated value with respect to the power running torque command value or the power generation torque command value.

The operation signal generation unit **130a** generates an operation signal of the first inverter **101** on the basis of the voltage amplitude command, the voltage phase command, and the electrical angle θ . Specifically, the operation signal generation unit **130a** calculates command voltages of three phases on the basis of the voltage amplitude command, the voltage phase command, and the electrical angle θ , and generates the switch operation signal of the upper and lower arms in each phase by PWM control based on the magnitude comparison between the signal obtained by standardizing the calculated command voltages of three phases with the power supply voltage and the carrier signal such as a triangular wave signal.

By the way, the operation signal generation unit **130a** may generate the switch operation signal on the basis of the pulse pattern information which is map information in which the voltage amplitude command, the voltage phase command, the electrical angle θ , and the switch operation signal are associated, the voltage amplitude command, the voltage phase command, and the electrical angle θ .

Further, the X, Y, and Z phase side also has the same configuration, and the torque estimation unit **128b** calculates a torque estimated value corresponding to the X, Y, and Z phases on the basis of the d-axis current and the q-axis current converted by the dq conversion unit **122**.

The torque feedback control unit **129b** calculates a voltage phase command as an operation amount for feedback-controlling the torque estimated value to the power running

torque command value or the power generation torque command value. The torque feedback control unit **129b** calculates the voltage phase command with the use of the PI feedback method on the basis of the deviation of the torque estimated value with respect to the power running torque command value or the power generation torque command value.

The operation signal generation unit **130b** generates an operation signal of the first inverter **102** on the basis of the voltage amplitude command, the voltage phase command, and the electrical angle θ . Specifically, the operation signal generation unit **130b** calculates command voltages of three phases on the basis of the voltage amplitude command, the voltage phase command, and the electrical angle θ , and generates the switch operation signal of the upper and lower arms in each phase by PWM control based on the magnitude comparison between the signal obtained by standardizing the calculated command voltages of three phases with the power supply voltage and the carrier signal such as a triangular wave signal. The driver **117** turns on/off the switches S_p and S_n of each of the three phases in the inverters **101** and **102** on the basis of the switch operation signals generated by the operation signal generation units **130a** and **130b**.

Incidentally, the operation signal generation unit **130b** may generate switch operation signals on the basis of the pulse pattern information which is map information in which the voltage amplitude command, the voltage phase command, the electrical angle θ , and the switch operation signal are associated, the voltage amplitude command, the voltage phase command, and the electrical angle θ .

By the way, in the rotating electric machine **10**, there is a concern that electrolytic corrosion of the bearings **21** and **22** may occur due to the generation of a shaft current. For example, there is a concern that, when the energization of the stator winding **51** is switched by switching, magnetic flux distortion occurs due to a slight deviation in switching timing (switching imbalance), which causes electrolytic corrosion in the bearings **21** and **22** that support the rotating shaft **11**. The distortion of the magnetic flux occurs in accordance with the inductance of the stator **50**, and the electromotive voltage in the axial direction generated by the distortion of the magnetic flux causes dielectric breakdown in the bearings **21** and **22**, and electrolytic corrosion proceeds.

In this regard, in the present embodiment, the following three countermeasures are taken as countermeasures against electrolytic corrosion. A first electrolytic corrosion countermeasure is an electrolytic corrosion suppression countermeasure by reducing the inductance due to the coreless stator **50** and by smoothing the magnet magnetic flux of the magnet unit **42**. A second electrolytic corrosion countermeasure is an electrolytic corrosion suppression countermeasure by adopting a cantilever structure with the bearings **21** and **22** for the rotating shaft. A third electrolytic corrosion countermeasure is an electrolytic corrosion suppression countermeasure by molding the annular stator winding **51** together with the stator core **52** with a molding material. The details of each of these countermeasures will be described below individually.

First, in the first electrolytic corrosion countermeasure, in the stator **50**, the spaces between each conductor group **81** in the circumferential direction are made toothless, and a sealing member **57** made of a non-magnetic material instead of the teeth (iron core) is provided between each conductor group **81** (see FIG. **10**). This makes it possible to reduce the inductance of the stator **50**. By reducing the inductance of

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the stator **50**, even if the switching timing shift occurs when the stator winding **51** is energized, the occurrence of magnetic flux distortion due to the switching timing shift can be suppressed, and thus it is possible to suppress the electrolytic corrosion of the bearings **21** and **22**. Moreover, it is preferable that the inductance of the d-axis is equal to or less than the inductance of the q-axis.

Further, the magnets **91** and **92** are oriented in such a manner that the direction of the axis of easy magnetization is more parallel to the d-axis on the d-axis side as compared with the q-axis side (see FIG. **9**). As a result, the magnet magnetic flux on the d-axis is strengthened, and the change in surface magnetic flux (increase/decrease in magnetic flux) from the q-axis to the d-axis becomes gentle at each magnetic pole. Therefore, the sudden voltage change caused by the switching imbalance is suppressed, and thus a configuration that can contribute to the suppression of electrolytic corrosion is implemented.

In the second electrolytic corrosion countermeasure, in the rotating electric machine **10**, the respective bearings **21** and **22** are arranged unevenly on either side in the axial direction with respect to the axial center of the rotor **40** (see FIG. **2**). As a result, the influence of electrolytic corrosion can be reduced as compared with a configuration in which a plurality of bearings are provided on both sides of the rotor in the axial direction. That is, in a configuration in which the rotor is supported from both sides by a plurality of bearings, a closed circuit that passes through the rotor, the stator, and each bearing (that is, each bearing on both sides in the axial direction with the rotor therebetween) is formed as a high frequency magnetic flux is generated, and there is a concern about electrolytic corrosion of the bearing due to the shaft current. On the other hand, in the configuration in which the rotor **40** is cantilevered and supported by a plurality of bearings **21** and **22**, the closed circuit is not formed and the electrolytic corrosion of the bearings is suppressed.

Further, the rotating electric machine **10** has the following configuration in connection with the configuration for arranging the bearings **21** and **22** on one side. In the magnet holder **41**, the contact avoiding section that extends in the axial direction and avoids contact with the stator **50** is provided at the intermediate section **45** that projects in the radial direction of the rotor **40** (see FIG. **2**). In this case, when a closed circuit of the shaft current is formed via the magnet holder **41**, the closed circuit length can be lengthened to increase the circuit resistance. As a result, the electrolytic corrosion of the bearings **21** and **22** can be suppressed.

The holding member **23** of the bearing unit **20** is fixed to the housing **30** on one side in the axial direction with the rotor **40** therebetween, and the housing **30** and the unit base **61** (stator holder) are coupled to each other on the other side (see FIG. **2**). According to this configuration, it is possible to preferably implement a configuration in which the respective bearings **21** and **22** are unevenly arranged on one side of the rotating shaft **11** in the axial direction. In addition, in this configuration, since the unit base **61** is connected to the rotating shaft **11** via the housing **30**, the unit base **61** can be arranged at a position electrically separated from the rotating shaft **11**. Moreover, if an insulating member such as resin is interposed between the unit base **61** and the housing **30**, the unit base **61** and the rotating shaft **11** are electrically further separated from each other. As a result, the electrolytic corrosion of the bearings **21** and **22** can be appropriately suppressed.

In the rotating electric machine **10** of the present embodiment, the shaft voltage acting on the bearings **21** and **22** is

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reduced by arranging the respective bearings **21** and **22** on one side, and the like. Further, the potential difference between the rotor **40** and the stator **50** is reduced. Therefore, it is possible to reduce the potential difference acting on the bearings **21** and **22** without using conductive grease in the bearings **21** and **22**. Since the conductive grease generally contains fine particles such as carbon, it is considered that noise is generated. In this regard, in the present embodiment, non-conductive grease is used in the bearings **21** and **22**. Therefore, it is possible to suppress the inconvenience of noise in the bearings **21** and **22**. For example, in the application to an electric vehicle such as an electric vehicle, it is considered that a countermeasure against the noise of the rotating electric machine **10** is required, and it is possible to preferably implement the countermeasure against the noise.

In the third electrolytic corrosion countermeasure, the stator winding **51** is molded together with the stator core **52** with a molding material to suppress the displacement of the stator winding **51** in the stator **50** (see FIG. **11**). In particular, since the rotating electric machine **10** of the present embodiment does not have an interconductor member (teeth) between each conductor group **81** in the circumferential direction of the stator winding **51**, there is a concern that the stator winding **51** may be displaced, but by molding the stator winding **51** together with the stator core **52**, the displacement of the conductor position of the stator winding **51** is suppressed. Consequently, it is possible to suppress the distortion of the magnetic flux due to the displacement of the stator winding **51** and the occurrence of electrolytic corrosion of the bearings **21** and **22** due to the distortion.

Moreover, since the unit base **61** as a housing member that fixes the stator core **52** is made of carbon fiber reinforced plastic (CFRP), the electric discharge to the unit base **61** is suppressed as compared with the case where it is made of, for example, aluminum. Thus, a suitable countermeasure against electrolytic corrosion is possible.

In addition to that, as a countermeasure against electrolytic corrosion of the bearings **21** and **22**, it is also possible to use a configuration in which at least one of the outer ring **25** and the inner ring **26** is made of a ceramic material, or in which an insulating sleeve is provided on the outside of the outer ring **25**.

Hereinafter, other embodiments will be described with a focus on differences from the first embodiment.

Second Embodiment

In the present embodiment, the polar anisotropic structure of the magnet unit **42** in the rotor **40** is changed, which will be described in detail below.

As illustrated in FIGS. **22** and **23**, the magnet unit **42** is composed with the use of a magnet array called a Halbach array. That is, the magnet unit **42** has a first magnet **131** in which the magnetization direction (direction of the magnetization vector) is the radial direction and a second magnet **132** in which the magnetization direction (direction of the magnetization vector) is the circumferential direction. The first magnets **131** are arranged at predetermined intervals in the circumferential direction, and the second magnets **132** are arranged at a position between the adjacent first magnets **131** in the circumferential direction. The first magnet **131** and the second magnet **132** are permanent magnets made of rare earth magnets such as neodymium magnets.

The first magnets **131** are arranged so as to be apart from each other in the circumferential direction in such a manner that the poles on the side facing the stator **50** (inside in the

radial direction) are alternately N poles and S poles. Further, the second magnets **132** are arranged next to each first magnet **131** in such a manner that the polarities alternate in the circumferential direction. The cylindrical section **43** provided so as to surround each of the magnets **131** and **132** is preferably a soft magnetic substance core made of a soft magnetic material, and functions as a back core. Moreover, the magnet unit **42** of the second embodiment also has the same relation of the axis of easy magnetization with respect to the d-axis and the q-axis in the d-q coordinate system as in the first embodiment.

Further, a magnetic substance **133** made of a soft magnetic substance is arranged on radially outside the first magnet **131**, that is, on the side of the cylindrical section **43** of the magnet holder **41**. For example, the magnetic substance **133** is preferably made of an electromagnetic steel sheet, soft iron, or a dust core material. In this case, the circumferential length of the magnetic substance **133** is the same as the circumferential length of the first magnet **131** (particularly, the circumferential length of the outer peripheral portion of the first magnet **131**). Further, in a state where the first magnet **131** and the magnetic substance **133** are integrated, the radial thickness of the integrated object is the same as the radial thickness of the second magnet **132**. In other words, the radial thickness of the first magnet **131** is thinner than that of the second magnet **132** by the amount of the magnetic substance **133**. Each of the magnets **131** and **132** and the magnetic substance **133** are fixed to each other by, for example, an adhesive. In the magnet unit **42**, the radial outside of the first magnet **131** is the opposite side to the stator **50**, and the magnetic substance **133** is provided on the side opposite to the stator **50** (opposite-to-stator side) in both sides of the first magnet **131** in the radial direction.

A key **134** is formed on the outer peripheral portion of the magnetic substance **133** as a protrusion that protrudes outward in the radial direction, that is, toward the side of the cylindrical section **43** of the magnet holder **41**. Further, a key groove **135** is formed on the inner peripheral surface of the cylindrical section **43** as a recess for housing the key **134** of the magnetic substance **133**. The protruding shape of the key **134** and the groove shape of the key groove **135** are the same, and the same number of key grooves **135** as the key **134** are formed corresponding to the key **134** formed on each magnetic substance **133**. By engaging the key **134** and the key groove **135**, the displacement of the first magnet **131** and the second magnet **132** and the magnet holder **41** in the circumferential direction (rotation direction) is suppressed. Moreover, it is optional whether the key **134** and the key groove **135** (protrusion and recess) are provided in either of the cylindrical section **43** or the magnetic substance **133** of the magnet holder **41**, and contrary to the above, it is also possible to provide the key groove **135** on the outer peripheral portion of the magnetic substance **133** and to provide the key **134** on the inner peripheral portion of the cylindrical section **43** of the magnet holder **41**.

Here, in the magnet unit **42**, the magnetic flux density in the first magnet **131** can be increased by alternately arranging the first magnet **131** and the second magnet **132**. Therefore, in the magnet unit **42**, the magnetic flux can be concentrated on one side, and the magnetic flux can be strengthened on the side closer to the stator **50**.

Further, by arranging the magnetic substance **133** radially outside the first magnet **131**, that is, on the opposite-to-stator side, it is possible to suppress partial magnetic saturation on the radial outside of the first magnet **131**, and thus demagnetization of the first magnet **131** caused by the magnetic saturation can be suppressed. As a result, it is accordingly

possible to increase the magnetic force of the magnet unit **42**. The magnet unit **42** of the present embodiment has, so to speak, a configuration in which a portion of the first magnet **131** in which demagnetization is likely to occur is replaced with the magnetic substance **133**.

FIGS. **24A** and **24B** are diagrams specifically illustrating the flow of magnetic flux in the magnet unit **42**, FIG. **24A** illustrates a case where a conventional configuration is used in which the magnet unit **42** does not have the magnetic substance **133**, and FIG. **24B** illustrates a case where the configuration of the present embodiment having the magnetic substance **133** in the magnet unit **42** is used. Moreover, in FIGS. **24A** and **24B**, the cylindrical section **43** of the magnet holder **41** and the magnet unit **42** are illustrated in a linearly developed manner, with the lower side of the figure being the stator side and the upper side being the opposite-to-stator side.

In the configuration of FIG. **24A**, the magnetic flux acting surface of the first magnet **131** and the side surface of the second magnet **132** are in contact with the inner peripheral surface of the cylindrical section **43**, respectively. Further, the magnetic flux acting surface of the second magnet **132** is in contact with the side surface of the first magnet **131**. In this case, in the cylindrical section **43**, the combined magnetic flux of a magnetic flux **F1** that enters the contact surface with the first magnet **131** through the outer path of the second magnet **132** and the magnetic flux that is substantially parallel to the cylindrical section **43** and attracts a magnetic flux **F2** of the second magnet **132** is generated. Therefore, there is a concern that magnetic saturation may partially occur in the vicinity of the contact surface between the first magnet **131** and the second magnet **132** in the cylindrical section **43**.

On the other hand, in the configuration of FIG. **24B**, the magnetic substance **133** is formed between the magnetic flux acting surface of the first magnet **131** and the inner peripheral surface of the cylindrical section **43** on the side opposite to the stator **50** of the first magnet **131**, and therefore the magnetic substance **133** allows the passage of magnetic flux. Consequently, magnetic saturation in the cylindrical section **43** can be suppressed, and the proof stress against demagnetization is improved.

Further, in the configuration of FIG. **24B**, unlike FIG. **24A**, the flux **F2** that promotes magnetic saturation can be cancelled. As a result, the permeance of the entire magnetic circuit can be effectively improved. With such a configuration, the magnetic circuit characteristics can be maintained even under severe high heat conditions.

Further, the magnet magnetic path passing through the inside of the magnet becomes longer than that of a radial magnet in a conventional SPM rotor. Therefore, the magnet permeance can rise, the magnetic force can be enhanced, and the torque can be increased. Furthermore, the magnetic flux is concentrated in the center of the d-axis, and thus the sine wave matching rate can be increased. In particular, if the current waveform is made into a sine wave or a trapezoidal wave by PWM control, or if a switching IC energized at 120 degrees is used, the torque can be increased more effectively.

Moreover, in a case where the stator core **52** is made of an electromagnetic steel sheet, the radial thickness of the stator core **52** is preferably $\frac{1}{2}$ or more than $\frac{1}{2}$ of the radial thickness of the magnet unit **42**. For example, the radial thickness of the stator core **52** is preferably $\frac{1}{2}$ or more of the radial thickness of the first magnet **131** provided at the center of the magnetic pole in the magnet unit **42**. Further, the radial thickness of the stator core **52** is preferably smaller than the radial thickness of the magnet unit **42**. In this case,

the magnet magnetic flux is approximately 1 [T], and the saturation magnetic flux density of the stator core 52 is 2 [T]. Therefore, by setting the radial thickness of the stator core 52 to $\frac{1}{2}$ or more of the radial thickness of the magnet unit 42, it is possible to prevent magnetic flux leakage to the inner peripheral side of the stator core 52.

In a magnet having a Halbach structure or a polar anisotropic structure, since the magnetic path has a pseudo arc shape, the magnetic flux can be increased in proportion to the thickness of the magnet that handles the magnetic flux in the circumferential direction. In such a configuration, it is considered that the magnetic flux flowing through the stator core 52 does not exceed the magnetic flux in the circumferential direction. That is, in a case where an iron-based metal having a saturation magnetic flux density of 2 [T] is used with respect to a magnetic flux of the magnet 1 [T], if the thickness of the stator core 52 is set to half or more of the magnet thickness, it is possible to provide a rotating electric machine that is not magnetically saturated and is suitably small and lightweight. Here, since the diamagnetic field from the stator 50 acts on the magnet magnetic flux, the magnet magnetic flux is generally 0.9 [T] or less. Therefore, if the stator core has half the thickness of the magnet, its magnetic permeability can be kept suitably high.

Hereinafter, a modification in which a part of the above-described configuration is modified will be described.
(First Modification)

In the above embodiment, the outer peripheral surface of the stator core 52 has a curved surface without unevenness, and a plurality of conductor groups 81 are arranged side by side at predetermined intervals on the outer peripheral surface, but this may be changed. For example, as illustrated in FIG. 25, the stator core 52 has an annular yoke 141 provided on the side opposite to the rotor 40 (lower side in the figure) on both sides of the stator winding 51 in the radial direction and protruding sections 142 extending from the yoke 141 so as to protrude between the straight sections 83 adjacent to each other in the circumferential direction. The protruding sections 142 are provided at predetermined intervals on the radially outside the yoke 141, that is, on the rotor 40 side. The respective conductor groups 81 of the stator winding 51 are engaged with the protruding sections 142 in the circumferential direction, and are arranged side by side in the circumferential direction while using the protruding section 142 as a positioning section of the conductor group 81. Moreover, the protruding section 142 corresponds to the "interconductor member".

In the protruding section 142, the radial thickness dimension from the yoke 141, in other words, as illustrated in FIG. 25, in the radial direction of the yoke 141, a distance W from the inner side surface 320 adjacent to the yoke 141 of the straight section 83 to the apex of the protruding section 142 is smaller than $\frac{1}{2}$ of the radial thickness dimension of the straight section 83 radially adjacent to the yoke 141 among the plurality of straight sections 38 inside and outside the radial direction (H1 in the figure). In other words, the non-magnetic member (sealing member 57) should occupy a range of three-quarters of a dimension (thickness) T1 of the conductor group 81 (conducting member) in the radial direction of the stator winding 51 (stator core 52) (twice the thickness of the conductor wire 82, in other words, the shortest distance between the surface 320 in contact with the stator core 52 of the conductor group 81 and a surface 330 facing the rotor 40 of the conductor group 81). Due to the thickness limitation of the protruding section 142, the protruding section 142 does not function as teeth between the conductor groups 81 (that is, the straight section 83) adjacent

to each other in the circumferential direction, and the magnetic path is not formed by the teeth. Not all of the protruding sections 142 may not be provided between the conductor groups 81 arranged in the circumferential direction, but should be provided between at least one set of the conductor groups 81 adjacent to each other in the circumferential direction. For example, the protruding sections 142 are preferably provided at equal intervals for each predetermined number of the conductor groups 81 in the circumferential direction. The shape of the protruding section 142 may be any shape such as a rectangular shape or an arc shape.

Further, the straight section 83 may be provided as a single layer on the outer peripheral surface of the stator core 52. Consequently, in a broad sense, the radial thickness dimension of the protruding section 142 from the yoke 141 may be smaller than $\frac{1}{2}$ of the radial thickness dimension of the straight section 83.

Moreover, assuming a virtual circle centered on the shaft center of the rotating shaft 11 and passing through the radial center position of the straight section 83 radially adjacent to the yoke 141, the protruding section 142 preferably has a shape that protrudes from the yoke 141 within the range of the virtual circle, in other words, a shape that does not protrude radially outward of the virtual circle (that is, on the rotor 40 side).

According to the above configuration, the protruding section 142 has a limited radial thickness dimension and does not function as the teeth between the straight sections 83 adjacent to each other in the circumferential direction. Therefore, it is possible to bring the respective adjacent straight sections 83 closer to each other as compared with the case where the teeth are provided between the respective straight sections 83. As a result, the cross-sectional area of the conductor 82a can be increased, and the heat generated by the energization of the stator winding 51 can be reduced. In such a configuration, the absence of teeth makes it possible to eliminate magnetic saturation and increase the energization current to the stator winding 51. In this case, it is possible to preferably cope with the increase in the amount of heat generated as the energization current increases. Further, in the stator winding 51, since the turn section 84 is shifted in the radial direction and has an interference avoidance section for avoiding interference with other turn sections 84, the different turn sections 84 can be separated from each other in the radial direction. As a result, heat dissipation can be improved even in the turn section 84. As described above, it is possible to optimize the heat dissipation performance of the stator 50.

Further, if the yoke 141 of the stator core 52 and the magnet unit 42 of the rotor 40 (that is, the magnets 91 and 92) are separated by a predetermined distance or more, the radial thickness dimension of the protruding section 142 is not limited to H1 in FIG. 25. Specifically, if the yoke 141 and the magnet unit 42 are separated by 2 mm or more, the radial thickness dimension of the protruding section 142 may be H1 or more in FIG. 25. For example, in a case where the radial thickness dimension of the straight section 83 exceeds 2 mm and the conductor group 81 is composed of two layers of conductor wires 82 inside and outside the radial direction, the protruding section 142 may be provided in the straight section 83 not adjacent to the yoke 141, that is, in the range from the yoke 141 to the half position of the second conductor wire 82. In this case, if the radial thickness dimension of the protruding section 142 is up to " $H1 \times \frac{3}{2}$ ", the effect can be obtained not a little by increasing the conductor cross-sectional area in the conductor group 81.

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Further, the stator core 52 may have the configuration illustrated in FIG. 26. Moreover, although the sealing member 57 is omitted in FIG. 26, the sealing member 57 may be provided. In FIG. 26, for convenience, the magnet unit 42 and the stator core 52 are illustrated in a linearly developed manner.

In the configuration of FIG. 26, the stator 50 has the protruding section 142 as an interconductor member between the conductor wires 82 (that is, the straight section 83) adjacent to each other in the circumferential direction. When the stator winding 51 is energized, the stator 50 magnetically functions together with one of the magnetic poles (N pole or S pole) of the magnet unit 42, and has a part 350 extending in the circumferential direction of the stator 50. When the circumferential length of the stator 50 of this part 350 is W_n , the total width of the protruding sections 142 existing in this length range W_n (that is, the total dimension of the stator 50 in the circumferential direction) is W_t , the saturation magnetic flux density of the protruding section 142 is B_s , the width dimension for one pole of the magnet unit 42 in the circumferential direction is W_m , and the residual magnetic flux density of the magnet unit 42 is B_r , the protruding section 142 is made of a magnetic material of a formula (1).

$$W_t \cdot B_s \leq W_m \cdot B_r \quad (1)$$

Moreover, the range W_n is set so as to include a plurality of conductor groups 81 adjacent to each other in the circumferential direction and include a plurality of conductor groups 81 having overlapping excitation times. In doing so, it is preferable to set the center of the void 56 of the conductor group 81 as a reference (boundary) when setting the range W_n . For example, in the case of the configuration illustrated in FIG. 26, the conductor groups 81 up to the fourth in order from the one with the shortest distance from the center of the magnetic pole of the N pole in the circumferential direction correspond to the aforementioned plurality of conductor groups 81. In addition, the range W_n is set so as to include the four conductor groups 81. In doing so, the ends (starting point and ending point) of the range W_n are set as the center of the void 56.

In FIG. 26, since halves of the protruding sections 142 are included at both ends of the range W_n , the range W_n includes a total of four protruding sections 142. Consequently, when the width of the protruding section 142 (that is, the dimension of the protruding section 142 in the circumferential direction of the stator 50, in other words, the interval between the adjacent conductor groups 81) is A , the total width of the protruding sections 142 included in the range W_n is $W_t = \frac{1}{2}A + A + A + A + \frac{1}{2}A = 4A$.

Specifically, in the present embodiment, the three-phase winding of the stator winding 51 is a distributed winding, and in the stator winding 51, the number of protruding sections 142 with respect to one pole of the magnet unit 42, that is, the number of voids 56 between the respective conductor groups 81 is a "number of phases*Q". Here, Q is the number of the one-phase conductor wires 82 that are in contact with the stator core 52. Moreover, in a case where the conductor wires 82 are the conductor group 81 stacked in the radial direction of the rotor 40, it can also be considered to be the number of the conductor wires 82 on the inner peripheral side of the one-phase conductor group 81. In this case, when the three-phase winding of the stator winding 51 is energized in a predetermined order for each phase, the protruding sections 142 for two phases are excited in one pole. Consequently, the total circumferential width dimension W_t of the protruding section 142 excited by the

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energization of the stator winding 51 in the range for one pole of the magnet unit 42 is "the number of excited phases*Q*A=2*2*A" when the circumferential width dimension of the protruding section 142 (that is, the void 56) is A .

In addition, after the total width dimension W_t is defined in this way, in the stator core 52, the protruding section 142 is configured as a magnetic material fulfilling the relation (1) above. Moreover, the total width dimension W_t is also the circumferential dimension of the portion where the relative magnetic permeability can be larger than 1 in one pole. Further, in consideration of a margin, the total width dimension W_t may be set as the circumferential width dimension of the protruding section 142 in one magnetic pole. Specifically, since the number of protruding sections 142 with respect to one pole of the magnet unit 42 is "number of phases*Q", the circumferential width dimension (total width dimension W_t) of the protruding sections 142 in one magnetic pole may be set to "the number of phases*Q*A=3*2*A=6A".

Moreover, the distributed winding referred to here is a one-pole pair period (N-pole and S-pole) of the magnetic pole, and has a one-pole pair of the stator winding 51. The one-pole pair of the stator winding 51 referred to here is composed of two straight sections 83 and a turn section 84 in which currents flow in opposite directions and which are electrically connected at the turn section 84. If the above conditions are met, even a Short Pitch Winding is regarded as an equivalent of a distributed winding of a Full Pitch Winding.

Next, an example in the case of concentrated winding is indicated. The concentrated winding referred to here is that the width of the one-pole pair of magnetic poles and the width of the one-pole pair of the stator winding 51 are different. Example of concentrated winding include a concentrated winding that has relation such as three conductor groups 81 for one magnetic pole pair, three conductor groups 81 for two magnetic pole pairs, nine conductor groups 81 for four magnetic pole pairs, and nine conductor groups 81 for five magnetic pole pairs.

Here, in a case where the stator winding 51 is a concentrated winding, when the three-phase windings of the stator winding 51 are energized in a predetermined order, the stator windings 51 for two phases are excited. As a result, the protruding sections 142 for two phases are excited. Consequently, the circumferential width dimension W_t of the protruding section 142 excited by the energization of the stator winding 51 in the range for one pole of the magnet unit 42 is " $A*2$ ". In addition, after the total width dimension W_t is defined in this way, the protruding section 142 is configured as a magnetic material fulfilling the relation (1) above. Moreover, in the case of the concentrated winding indicated above, the total circumferential width of the protruding sections 142 of the stator 50 in the region surrounded by the conductor groups 81 of the same phase is defined as A . Further, W_m in the concentrated winding corresponds to "the entire circumference of the surface facing the air gap of the magnet unit 42" * "the number of phases" / "the number of dispersions of the conductor group 81".

Incidentally, for magnets with a BH product of 20 [MGOe (kJ/m³)] or more, such as neodymium magnets, samarium-cobalt magnets, and ferrite magnets, B_d = over 1.0 [T], and for iron, B_r = over 2.0 [T]. Therefore, as the high output motor, in the stator core 52, the protruding section 142 may be a magnetic material fulfilling the relation of $W_t < 1/2 * W_m$.

Further, in a case where the conductor wire **82** includes an outer layer coating **182** as described below, the conductor wire **82** may be arranged in the circumferential direction of the stator core **52** in such a manner that the outer layer coatings **182** of the conductor wires **82** come into contact with each other. In this case, W_t can be regarded as 0 or the thickness of the outer layer coating **182** of both conductor wires **82** in contact with each other.

In the configurations of FIGS. **25** and **26**, an interconductor member (protruding section **142**) that is disproportionately small with respect to the magnet magnetic flux on the rotor **40** side is provided. Moreover, the rotor **40** is a flat surface magnet type rotor having a low inductance and does not have saliency in terms of magnetic resistance. In such a configuration, the inductance of the stator **50** can be reduced, the generation of magnetic flux distortion due to the deviation of the switching timing of the stator winding **51** is suppressed, and thus the electrolytic corrosion of the bearings **21** and **22** is suppressed.

(Second Modification)

The following configuration can also be adopted as the stator **50** using the interconductor member fulfilling the relation of the above formula (1). In FIG. **27**, a tooth-shaped section **143** is provided as an interconductor member on the outer peripheral surface side of the stator core **52** (upper surface side in the figure). The tooth-shaped section **143** is provided at predetermined intervals in the circumferential direction so as to protrude from the yoke **141**, and have the same thickness dimension as that of the conductor group **81** in the radial direction. The side surface of the tooth-shaped section **143** is in contact with each conductor wire **82** of the conductor group **81**. However, there may be a gap between the tooth-shaped section **143** and each conductor wire **82**.

The tooth-shaped section **143** is provided with a limitation on the width dimension in the circumferential direction, and is provided with polar teeth (stator teeth) that are disproportionately thin with respect to the amount of magnets. With such a configuration, the tooth-shaped section **143** is surely saturated by the magnetic flux of the magnet at 1.8 T or more, and the inductance can be lowered by lowering the permeance.

Here, in the magnet unit **42**, when the surface area per pole of the magnetic flux acting surface on the stator side is S_m and the residual magnetic flux density of the magnet unit **42** is B_r , the magnetic flux on the magnet unit side is, for example, " $S_m \cdot B_r$ ". Further, when the surface area on the rotor side in each tooth-shaped section **143** is S_t , the number per phase of the conductor wire **82** is m , and the tooth-shaped sections **143** for two phases are excited in one pole by energization of the stator winding **51**, the magnetic flux on the stator side is, for example, " $S_t \cdot m \cdot 2 \cdot B_s$ ". In this case, the inductance is reduced by limiting the dimension of the tooth-shaped section **143** in such a manner that a relation (2) is established.

$$S_t \cdot m \cdot 2 \cdot B_s < S_m \cdot B_r \quad (2)$$

Moreover, in a case where the magnet unit **42** and the tooth-shaped section **143** have the same axial dimension, when the circumferential width dimension for one pole of the magnet unit **42** is W_m , and the circumferential width dimension of the tooth-shaped section **143** is W_{st} , then the above formula (2) is replaced as in a formula (3).

$$W_{st} \cdot m \cdot 2 \cdot B_s < W_m \cdot B_r \quad (3)$$

More specifically, assuming that, for example, $B_s = 2$ T, $B_r = 1$ T, and $m = 2$, the above formula (3) has a relation of " $W_{st} < W_m / 8$ ". In this case, the inductance is reduced by

making the width dimension W_{st} of the tooth-shaped section **143** smaller than $1/8$ of the width dimension W_m for one pole of the magnet unit **42**. Moreover, if the number m is 1, the width dimension W_{st} of the tooth-shaped section **143** is preferably made to be smaller than $1/4$ of the width dimension W_m for one pole of the magnet unit **42**.

Moreover, in the above formula (3), " $W_{st} \cdot m \cdot 2$ " corresponds to the circumferential width dimension of the tooth-shaped section **143** excited by energization of the stator winding **51** in the range for one pole of the magnet unit **42**.

In the configuration of FIG. **27**, similarly to the configurations of FIGS. **25** and **26** described above, the interconductor member (tooth-shaped section **143**) which is disproportionately small with respect to the magnet magnetic flux on the rotor **40** side is provided. In such a configuration, the inductance of the stator **50** can be reduced, the generation of magnetic flux distortion due to the deviation of the switching timing of the stator winding **51** is suppressed, and thus the electrolytic corrosion of the bearings **21** and **22** is suppressed.

(Third Modification)

In the above embodiment, the sealing member **57** covering the stator winding **51** is provided radially outside the stator core **52** in the range including all the conductor groups **81**, that is, in the range in which the radial thickness dimension is larger than the radial thickness dimension of each conductor group **81**, but this may be changed. For example, as illustrated in FIG. **28**, the sealing member **57** is provided in such a manner that a part of the conductor wire **82** protrudes. More specifically, the sealing member **57** is provided in a state where a part of the conductor wire **82** which is the outermost in the radial direction in the conductor group **81** is exposed on the radial outside, that is, on the stator **50** side. In this case, the radial thickness dimension of the sealing member **57** may be the same as or smaller than the radial thickness dimension of each conductor group **81**.

(Fourth Modification)

As illustrated in FIG. **29**, in the stator **50**, each conductor group **81** may not be sealed by the sealing member **57**. That is, the sealing member **57** that covers the stator winding **51** is not used. In this case, no interconductor member is provided between the conductor groups **81** arranged in the circumferential direction, and an airspace is formed. In short, the interconductor member is not provided between the conductor groups **81** arranged in the circumferential direction. Moreover, air may be regarded as a non-magnetic substance or an equivalent of a non-magnetic substance as $B_s = 0$, and air may be arranged in this airspace.

(Fifth Modification)

In a case where the interconductor member in the stator **50** is made of a non-magnetic material, it is possible to use a material other than resin as the non-magnetic material. For example, a metal-based non-magnetic material may be used, such as using SUS304 which is an austenitic stainless steel.

(Sixth Modification)

The stator **50** may be configured not to include the stator core **52**. In this case, the stator **50** is composed of the stator winding **51** illustrated in FIG. **12**. Moreover, in the stator **50** that does not include the stator core **52**, the stator winding **51** may be sealed with a sealing material. Alternatively, the stator **50** may be configured to include an annular winding holding section made of a non-magnetic material such as synthetic resin as an alternative to the stator core **52** made of a soft magnetic material.

(Seventh Modification)

In the above first embodiment, the plurality of magnets **91** and **92** arranged in the circumferential direction are used as

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the magnet unit 42 of the rotor 40, but this may be changed, and an annular magnet which is an annular permanent magnet may be used as the magnet unit 42. Specifically, as illustrated in FIG. 30, an annular magnet 95 is fixed radially inside the cylindrical section 43 of the magnet holder 41. The annular magnet 95 is provided with a plurality of magnetic poles having alternating polarities in the circumferential direction, and the magnet is integrally formed on both the d-axis and the q-axis. The annular magnet 95 is formed with an arc shaped magnet magnetic path such that the direction of orientation is the radial direction on the d-axis of each magnetic pole and the direction of orientation is the circumferential direction on the q-axis between the respective magnetic poles.

Moreover, in the annular magnet 95, the orientation should be made in such a manner that an arc-shaped magnet magnetic path is formed, in which the axis of easy magnetization is parallel to the d-axis or close to parallel to the d-axis in the portion near the d-axis, and the axis of easy magnetization is orthogonal to the q-axis or close to parallel to the q-axis in the portion near the q-axis.

(Eighth Modification)

In this modification, a part of the control method of the control device 110 is changed. In this modification, the difference from the configuration described in the first embodiment will be mainly described.

First, with reference to FIG. 31, the processing in the operation signal generation units 116 and 126 illustrated in FIG. 20 and the operation signal generation units 130a and 130b illustrated in FIG. 21 will be described. Moreover, the processing in each operation signal generation unit 116, 126, 130a, and 130b is basically the same. Therefore, in the following, the processing of the operation signal generation unit 116 will be described as an example.

The operation signal generation unit 116 includes a carrier generation unit 116a and U, V, W phase comparators 116bU, 116bV, and 116bW. In the present embodiment, the carrier generation unit 116a generates and outputs a triangular wave signal as a carrier signal SigC.

The carrier signal SigC generated by the carrier generation unit 116a and the U, V, W phase command voltages calculated by the three-phase conversion unit 115 are input to the U, V, W phase comparators 116bU, 116bV, and 116bW. The U, V, W phase command voltages are, for example, sinusoidal waveforms, and the phases are shifted by 120° depending on the electrical angle.

The U, V, W phase comparators 116bU, 116bV, and 116bW generate the operation signals of the respective switches Sp and Sn of the upper arm and the lower arm of the U, V, W phases in the first inverter 101, by PWM (pulse width modulation) control based on the magnitude comparison between the U, V, W phase command voltages and the carrier signal SigC. Specifically, the operation signal generation unit 116 generates the operation signals of the respective switches Sp and Sn of the U, V, W phases by PWM control based on the magnitude comparison between the signal obtained by standardizing the U, V, W command voltages with the power supply voltage and the carrier signal. The driver 117 turns on/off each of the switches Sp and Sn of the U, V, W phases in the inverter 101 on the basis of the operation signals generated by the operation signal generation unit 116.

The control device 110 performs processing that changes the carrier frequency fc of the carrier signal SigC, that is, the switching frequency of each of the switches Sp and Sn. The carrier frequency fc is set high in the low torque region or high rotation region of the rotating electric machine 10

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and low in the high torque region of the rotating electric machine 10. This setting is made in order to suppress a decrease in controllability of the current flowing through each phase winding.

That is, as the stator 50 becomes coreless, the inductance of the stator 50 can be reduced. Here, when the inductance becomes low, the electrical time constant of the rotating electric machine 10 becomes small. As a result, there is a concern that the ripple of the current flowing through each phase winding increases, the controllability of the current flowing through the winding decreases, and the current control diverges. The effect of this decrease in controllability can be more pronounced when the current flowing through the winding (for example, the effective value of the current) is included in the low current region than in the high current region. In order to deal with this problem, in this modification, the control device 110 changes the carrier frequency fc.

The processing that changes the carrier frequency fc will be described with reference to FIG. 32. This processing is repeatedly executed by the control device 110, for example, at a predetermined control cycle as the processing of the operation signal generation unit 116.

In step S10, it is determined whether the current flowing through a winding 51a of each phase is in the low current region. This processing is processing for determining that the current torque of the rotating electric machine 10 is in the low torque region. Examples of the method for determining whether the current is included in the low current region include the following first and second methods.

<First Method>

The torque estimated value of the rotating electric machine 10 is calculated on the basis of the d-axis current and the q-axis current converted by the dq conversion unit 112. Then, when it is determined that the calculated torque estimated value is less than the torque threshold value, it is determined that the current flowing through the winding 51a is included in the low current region, and when it is determined that the torque estimated value is equal to or more than the torque threshold value, it is determined that the current flowing through the winding 51a is included in the high current region. Here, the torque threshold value should be set to, for example, 1/2 of the starting torque (also referred to as restraint torque) of the rotating electric machine 10.

<Second Method>

When it is determined that the rotation angle of the rotor 40 detected by the angle detector is equal to or greater than a speed threshold value, it is determined that the current flowing through the winding 51a is included in the low current region, that is, in the high rotation region. Here, the speed threshold value should be set to, for example, the rotation speed when the maximum torque of the rotating electric machine 10 becomes the torque threshold value.

If a negative determination is made in step S10, it is determined to be a high current region, and the processing proceeds to step S11. In step S11, the carrier frequency fc is set to a first frequency fL.

If an affirmative determination is made in step S10, the processing proceeds to step S12, and the carrier frequency fc is set to a second frequency fH which is higher than the first frequency fL.

According to this modification described above, the carrier frequency fc is set higher when the current flowing through each phase winding is included in the low current region than when it is included in the high current region. Therefore, in the low current region, the switching frequencies of the switches Sp and Sn can be increased, and the

increase in current ripple can be suppressed. As a result, it is possible to suppress a decrease in current controllability.

On the other hand, when the current flowing through each phase winding is included in the high current region, the carrier frequency f_c is set lower than when it is included in the low current region. In the high current region, the amplitude of the current flowing through the winding is larger than in the low current region, and therefore the increase in current ripple due to the low inductance has a small effect on the current controllability. Therefore, in the high current region, the carrier frequency f_c can be set lower than in the low current region, and the switching loss of the respective inverters **101** and **102** can be reduced.

In this modification, the following embodiments can be implemented.

In a case where the carrier frequency f_c is set to the first frequency f_L , when an affirmative determination is made in step **S10** of FIG. **32**, the carrier frequency f_c may be gradually changed from the first frequency f_L to the second frequency f_H .

Further, in a case where the carrier frequency f_c is set to the second frequency f_H , when a negative determination is made in step **S10**, the carrier frequency f_c may be gradually changed from the second frequency f_H to the first frequency f_L .

A switch operation signal may be generated by space vector modulation (SVM) control instead of PWM control. Even in this case, the above-mentioned change in switching frequency can be applied.

(Ninth Modification)

In each of the above embodiments, two pairs of conductors of each phase constituting the conductor group **81** are connected in parallel as illustrated in FIG. **33A**. FIG. **33A** is a diagram illustrating the electrical connection of first and second conductors **88a** and **88b**, which are two pairs of conductors. Here, as an alternative to the configuration illustrated in FIG. **33A**, as illustrated in FIG. **33B**, the first and second conductors **88a** and **88b** may be connected in series.

Further, three or more pairs of multilayer conductors may be laminated and arranged in the radial direction. FIG. **34** illustrates a configuration in which first to fourth conductors **88a** to **88d**, which are four pairs of conductors, are laminated and arranged. The first to fourth conductors **88a** to **88d** are arranged in the radial direction in the order of the first, second, third, and fourth conductors **88a**, **88b**, **88c**, and **88d** from the side closer to the stator core **52**.

Here, as illustrated in FIG. **33C**, the third and fourth conductors **88c** and **88d** are connected in parallel, the first conductor **88a** may be connected to one end of the parallel connection body, and the second conductor **88b** may be connected to the other end. When connected in parallel, the current density of the conductors connected in parallel can be reduced, and heat generation during energization can be suppressed. Therefore, in the configuration in which the tubular stator winding is assembled to the housing (unit base **61**) in which the cooling water passage **74** is formed, the first and second conductors **88a** and **88b** that are not connected in parallel are arranged on the stator core **52** side that abuts on the unit base **61**, and the third and fourth conductors **88c** and **88d** that are connected in parallel are arranged on the opposite-to-stator core side. As a result, the cooling performance of each of the conductors **88a** to **88d** in the multilayer conductor structure can be equalized.

Moreover, the radial thickness dimension of the conductor group **81** composed of the first to fourth conductors **88a** to

88d should be smaller than the circumferential width dimension for one phase in one magnetic pole.
(Tenth Modification)

The rotating electric machine **10** may have an inner rotor structure (adduction structure). In this case, for example, in the housing **30**, it is preferable that the stator **50** is provided on the radially outside and the rotor **40** is provided on the radially inside. Further, it is preferable that the inverter unit **60** is provided on one side or both sides of both ends of the stator **50** and the rotor **40** in the axial direction. FIG. **35** is a cross-sectional view of the rotor **40** and the stator **50**, and FIG. **36** is a view illustrating a part of the rotor **40** and the stator **50** illustrated in FIG. **35** in an enlarged manner.

The configurations of FIGS. **35** and **36**, which are premised on an inner rotor structure, have the same configurations as those of FIGS. **8** and **9** except that the rotor **40** and stator **50** are reversed in and out of the radial direction. Briefly, the stator **50** has a stator winding **51** having a flat conductor structure and a stator core **52** having no teeth. The stator winding **51** is assembled radially inside the stator core **52**. The stator core **52** has one of the following configurations, as in the case of the outer rotor structure.

A In the stator **50**, an interconductor member is provided between each conductor section in the circumferential direction, and as the interconductor member, a magnetic material having a relation of $W_t \cdot B_s = W_m \cdot B_r$ is used when the circumferential width dimension of the interconductor member at one magnetic pole is W_t , the saturation magnetic flux density of the interconductor member is B_s , the circumferential width dimension of the magnet unit at one magnetic pole is W_m , and the residual magnetic flux density of the magnet unit is B_r .

B In the stator **50**, an interconductor member is provided between each conductor section in the circumferential direction, and a non-magnetic material is used as the interconductor member.

C The stator **50** has a configuration in which no interconductor member is provided between each conductor section in the circumferential direction.

Further, the same applies to the magnets **91** and **92** of the magnet unit **42**. That is, the magnet unit **42** is composed with the use of the magnets **91** and **92** in which, the orientation was made in such a manner that, on the side of the d-axis, which is the center of the magnetic pole, the direction of the axis of easy magnetization is parallel to the d-axis as compared with the side of the q-axis, which is the magnetic pole boundary. Details such as the magnetization directions of the magnets **91** and **92** are as described above. It is also possible to use the annular magnet **95** (see FIG. **30**) in the magnet unit **42**.

FIG. **37** is a vertical cross-sectional view of the rotating electric machine **10** in the case of an inner rotor type, which is a figure corresponding to FIG. **2** described above. Differences from the configuration of FIG. **2** will be briefly described. In FIG. **37**, an annular stator **50** is fixed to the inside of the housing **30**, and a rotor **40** is rotatably provided inside the stator **50** with a predetermined air gap therebetween. Similarly to FIG. **2**, the respective bearings **21** and **22** are arranged unevenly on either side in the axial direction with respect to the axial center of the rotor **40**, whereby the rotor **40** is cantilevered and supported. Further, the inverter unit **60** is provided inside the magnet holder **41** of the rotor **40**.

FIG. **38** illustrates another configuration of the rotating electric machine **10** having an inner rotor structure. In FIG. **38**, the rotating shaft **11** is rotatably supported by the bearings **21** and **22** in the housing **30**, and the rotor **40** is

fixed to the rotating shaft 11. Similarly to the configuration illustrated in FIG. 2 or the like, the respective bearings 21 and 22 are arranged unevenly on either side in the axial direction with respect to the axial center of the rotor 40. The rotor 40 has the magnet holder 41 and the magnet unit 42.

The rotating electric machine 10 of FIG. 38 is different from the rotating electric machine 10 of FIG. 37 in that the inverter unit 60 is not provided radially inside the rotor 40. The magnet holder 41 is connected to the rotating shaft 11 at a position radially inside the magnet unit 42. Further, the stator 50 has the stator winding 51 and the stator core 52, and is attached to the housing 30. (Eleventh Modification)

Another configuration as a rotating electric machine having an inner rotor structure will be described below. FIG. 39 is an exploded perspective view of a rotating electric machine 200, and FIG. 40 is a side sectional view of the rotating electric machine 200. Here, the up-down direction is illustrated with reference to the states of FIGS. 39 and 40.

As illustrated in FIGS. 39 and 40, the rotating electric machine 200 includes a stator 203 having an annular stator core 201 and a multi-phase stator winding 202, and a rotor 204 rotatably arranged inside the stator core 201. The stator 203 corresponds to an armature and the rotor 204 corresponds to a field magnet. The stator core 201 is composed by laminating a large number of silicon steel plates, and a stator winding 202 is attached to the stator core 201. Although not illustrated, the rotor 204 has a rotor core and a plurality of permanent magnets as a magnet unit. The rotor core is provided with a plurality of magnet insertion holes at equal intervals in the circumferential direction. Each of the magnet insertion holes is equipped with a permanent magnet magnetized in such a manner that the magnetization direction changes alternately for each adjacent magnetic pole. Moreover, the permanent magnet of the magnet unit may have a Halbach array as described with reference to FIG. 23 or a similar configuration. Alternatively, it is preferable that the permanent magnet of the magnet unit has polar anisotropy characteristics such as that described with reference to FIG. 9 and FIG. 30, in which the orientation direction (magnetization direction) extends in an arc shape between the d-axis which is the center of the magnetic pole and the q-axis which is the magnetic pole boundary.

Here, the stator 203 preferably has any of the following configurations.

A In the stator 203, an interconductor member is provided between each conductor section in the circumferential direction, and as the interconductor member, a magnetic material having a relation of $Wt \cdot Bs$ $Wm \cdot Br$ is used when the width dimension of the interconductor member in the circumferential direction at one magnetic pole is Wt , the saturation magnetic flux density of the interconductor member is Bs , the width dimension in the circumferential direction of the magnet unit at one magnetic pole is Wm , and the residual magnetic flux density of the magnet unit is Br .

B In the stator 203, an interconductor member is provided between each conductor section in the circumferential direction, and a non-magnetic material is used as the interconductor member.

C The stator 203 has a configuration in which no interconductor member is provided between each conductor section in the circumferential direction.

Further, in the rotor 204, the magnet unit is composed with the use of a plurality of magnets in which, the orientation was made in such a manner that, on the side of the d-axis, which is the center of the magnetic pole, the direction

of the axis of easy magnetization is parallel to the d-axis as compared with the side of the q-axis, which is the magnetic pole boundary.

An annular inverter case 211 is provided on one end side of the rotating electric machine 200 in the axial direction. The inverter case 211 is arranged in such a manner that the lower surface of the case is in contact with the upper surface of the stator core 201. Inside the inverter case 211, a plurality of power modules 212 constituting the inverter circuit, a smoothing capacitor 213 that suppresses voltage/current pulsation (ripple) generated by the switching operation of the semiconductor switching element, a control board 214 having a control unit, a current sensor 215 that detects a phase current, and a resolver stator 216 that is a rotation speed sensor of the rotor 204 are provided. The power module 212 has an IGBT and a diode which are semiconductor switching elements.

On the periphery of the inverter case 211, a power connector 217 connected to the DC circuit of the battery mounted on a vehicle, and a signal connector 218 used for transferring various signals between the rotating electric machine 200 side and the vehicle side control device are provided. The inverter case 211 is covered with a top cover 219. The direct current power from a vehicle-mounted battery is input via the power connector 217, converted into alternate current by switching of the power module 212, and sent to the stator winding 202 of each phase.

On both sides of the stator core 201 in the axial direction, on the side opposite to the inverter case 211, a bearing unit 221 that rotatably holds the rotating shaft of the rotor 204 and an annular rear case 222 that houses the bearing unit 221 are provided. The bearing unit 221 has, for example, a pair of bearings and is arranged unevenly on either side in the axial direction with respect to the axial center of the rotor 204. However, a plurality of bearings in the bearing unit 221 may be provided in a dispersed manner on both sides of the stator core 201 in the axial direction, and the rotating shafts may be supported from both sides by those respective bearings. The rotating electric machine 200 can be mounted on the vehicle side by bolting and fixing the rear case 222 to a mounting section such as a gear case or a transmission of the vehicle.

A cooling flow path 211a for flowing a refrigerant is formed in the inverter case 211. The cooling flow path 211a is formed by closing a space recessed in an annular shape from the lower surface of the inverter case 211 with the upper surface of the stator core 201. The cooling flow path 211a is formed so as to surround the coil end of the stator winding 202. A module case 212a of the power module 212 is inserted in the cooling flow path 211a. Also in the rear case 222, a cooling flow path 222a is formed so as to surround the coil end of the stator winding 202. The cooling flow path 222a is formed by closing a space recessed in an annular shape from the upper surface of the rear case 222 with the lower surface of the stator core 201.

(Twelfth Modification)

So far, the configuration embodied in the revolving-field type rotating electric machine has been described, but it is also possible to change this and embody it in the rotating armature type rotating electric machine. FIG. 41 illustrates the configuration of a rotating armature type rotating electric machine 230.

In the rotating electric machine 230 of FIG. 41, bearings 232 are fixed to housings 231a and 231b, respectively, and a rotating shaft 233 is rotatably supported by the bearings 232. The bearing 232 is, for example, an oil-impregnated bearing made by impregnating a porous metal with oil. A

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rotor **234** as an armature is fixed to the rotating shaft **233**. The rotor **234** has a rotor core **235** and a multi-phase rotor winding **236** fixed to the outer peripheral portion of the rotor core **235**. In the rotor **234**, the rotor core **235** has a slotless structure, and the rotor winding **236** has a flat conductor structure. That is, the rotor winding **236** has a flat structure in which the region for each phase is longer in the circumferential direction than in the radial direction.

Further, a stator **237** as a field magnet is provided radially outside the rotor **234**. The stator **237** has a stator core **238** fixed to the housing **231a** and a magnet unit **239** fixed to the inner peripheral side of the stator core **238**. The magnet unit **239** has a configuration including a plurality of magnetic poles having alternating polarities in the circumferential direction, and is configured similarly to the magnet unit **42** or the like described above, in which, the orientation was made in such a manner that the direction of the axis of easy magnetization is parallel to the d-axis on the d-axis side which is the center of the magnetic pole as compared with the q-axis side which is the magnetic pole boundary. The magnet unit **239** has an oriented sintered neodymium magnet, the intrinsic coercive force thereof is 400 [kA/m] or more, and the residual magnetic flux density is 1.0 [T] or more.

The rotating electric machine **230** of this modification is a 2-pole 3-coil brushed coreless motor, the rotor winding **236** is divided into three, and the magnet unit **239** has two poles. The number of poles and the number of coils of the brushed motor varies depending on the application, such as 2:3, 4:10, 4:21.

A commutator **241** is fixed to the rotating shaft **233**, and a plurality of brushes **242** are arranged on the radially outside thereof. The commutator **241** is electrically connected to the rotor winding **236** via a conductor **243** embedded in the rotating shaft **233**. A direct current flows in and out of the rotor winding **236** through these commutator **241**, brush **242**, and conductor **243**. The commutator **241** is appropriately divided in the circumferential direction in accordance with the number of phases of the rotor winding **236**. The brush **242** may be directly connected to a DC power source such as a storage battery via an electrical wiring, or may be connected to a DC power source via a terminal block or the like.

The rotating shaft **233** is provided with a resin washer **244** as a sealing material between the bearing **232** and the commutator **241**. The resin washer **244** suppresses the oil seeping out from the bearing **232**, which is an oil-impregnated bearing, from flowing out to the commutator **241** side. (Thirteenth Modification)

In the stator winding **51** of the rotating electric machine **10**, each conductor wire **82** may have a plurality of insulating coatings inside and outside. For example, it is preferable to bundle a plurality of conductors (wires) with an insulating coating into one and cover the conductors with an outer layer coating to form the conductor wire **82**. In this case, the insulating coating of the wire constitutes the inner insulating coating, and the outer layer coating constitutes the outer insulating coating. Further, in particular, it is preferable that the insulating capability of the outer insulating coating among the plurality of insulating coatings on the conductor wire **82** is higher than the insulating capability of the inner insulating coating. Specifically, the thickness of the outer insulating coating is made thicker than the thickness of the inner insulating coating. For example, the thickness of the outer insulating coating is 100 μm , and the thickness of the inner insulating coating is 40 μm . Alternatively, a material having a lower dielectric constant than that of the inner

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insulating coating may be used as the outer insulating coating. At least one of these should be applied. Moreover, it is preferable that the wire is configured as an aggregate of a plurality of conductive materials.

By strengthening the insulation of the outermost layer of the conductor wire **82** as described above, it becomes suitable for use in a high voltage vehicle system. Further, the rotating electric machine **10** can be properly driven even in highlands where the atmospheric pressure is low.

(Fourteenth Modification)

In the conductor wire **82** having a plurality of insulating coatings inside and outside, at least one of the linear expansivity (linear expansion coefficient) and the adhesive strength may be different between the outer insulating coating and the inner insulating coating. The configuration of the conductor wire **82** in this modification is illustrated in FIG. **42**.

In FIG. **42**, the conductor wire **82** includes a plurality of (four in the figure) wires **181**, an outer layer coating **182** made of, for example, a resin (outer insulating coating), that surrounds the plurality of wires **181**, and an intermediate layer **183** (intermediate insulating coating) filled around each wire **181** in the outer layer coating **182**. The wire **181** has a conductive part **181a** made of a copper material and a conductor coating **181b** (inner insulating coating) made of an insulating material. When viewed as a stator winding, the outer layer coating **182** insulates the phases. Moreover, it is preferable that the wire **181** is configured as an aggregate of a plurality of conductive materials.

The intermediate layer **183** has a linear expansion coefficient higher than that of the conductor coating **181b** of the wire **181** and a linear expansion coefficient lower than that of the outer layer coating **182**. That is, in the conductor wire **82**, the linear expansion coefficient is higher toward the outside. Generally, the outer layer coating **182** has a linear expansion coefficient higher than that of the conductor coating **181b**, but by providing the intermediate layer **183** having an intermediate linear expansion coefficient between them, the intermediate layer **183** functions as a cushioning material, and simultaneous cracking on the outer layer side and the inner layer side can be prevented.

Further, in the conductor wire **82**, the conductive part **181a** and the conductor coating **181b** are adhered to each other in the wire **181**, and the conductor coating **181b** and the intermediate layer **183** and the intermediate layer **183** and the outer layer coating **182** are adhered to each other, respectively, and the adhesive strength becomes weaker toward the outside of the conductor wire **82** in each of these adhered portions. That is, the adhesive strength of the conductive part **181a** and the conductor coating **181b** is weaker than the adhesive strength of the conductor coating **181b** and the intermediate layer **183** and the adhesive strength of the intermediate layer **183** and the outer layer coating **182**. Further, comparing the adhesive strength of the conductor coating **181b** and the intermediate layer **183** with the adhesive strength of the intermediate layer **183** and the outer layer coating **182**, it is preferable that the latter (outer side) is weaker or equivalent. Moreover, the magnitude of the adhesive strength between the coatings can be grasped from, for example, the tensile strength required when peeling off the two layers of coatings. By setting the adhesive strength of the conductor wire **82** as described above, it is possible to suppress cracking (co-cracking) on both the inner layer side and the outer layer side even if an internal/external temperature difference occurs due to heat generation or cooling.

Here, the heat generation and temperature change of the rotating electric machine mainly occur as copper loss produced from the conductive part **181a** of the wire **181** and iron loss generated from the inside of the iron core, and these two types of losses are transmitted from the conductive part **181a** in the conductor wire **82** or the outside of the conductor wire **82**, and the intermediate layer **183** does not have a heat source. In this case, the intermediate layer **183** has an adhesive force that can serve as a cushion for both of them, and thus simultaneous cracking can be prevented. Consequently, suitable use is possible even when used in fields with high withstand pressure or large temperature changes such as vehicle applications.

This is supplemented below. The wire **181** may be, for example, an enamel wire, and in such a case, it has a resin coating layer (conductor coating **181b**) such as PA, PI, and PAI. Further, it is desirable that the outer layer coating **182** outside the wire **181** is made of the same PA, PI, PAI, or the like, and has a large thickness. As a result, damage to the coating due to a difference in linear expansion coefficient is suppressed. Moreover, as the outer layer coating **182**, it is desirable to also use one with a dielectric constant smaller than PI and PAI, such as PPS, PEEK, fluorine, polycarbonate, silicon, epoxy, polyethylene naphthalate, and LCP, in addition to one made by thickening the aforementioned materials such as PA, PI, and PAI. With these resins, even if they are thinner than the PI and PAI coatings equivalent to the conductor coating **181b** or the thickness equivalent to the conductor coating **181b**, their insulating capability can be enhanced, and it is thereby possible to increase the occupancy ratio of the conductive part. In general, the resin has better insulation than the insulating coating of an enamel wire in terms of dielectric constant. As a matter of course, there are cases where the dielectric constant is deteriorated depending on the molding state and the mixture. Among them, PPS and PEEK are suitable as the outer layer coating of the second layer because their linear expansion coefficient is generally larger than that of the enamel coating but smaller than that of other resins.

Further, it is desirable that the adhesive strength between the two types of coatings (intermediate insulating coating and outer insulating coating) on the outside of the wire **181** and the enamel coating on the wire **181** is weaker than the adhesive strength between the copper wire and the enamel coating on the wire **181**. As a result, the phenomenon that the enamel coating and the aforementioned two types of coatings are destroyed at once is suppressed.

In a case where a water-cooled structure, a liquid-cooled structure, or an air-cooled structure is added to the stator, it is considered that thermal stress or impact stress is basically applied to the outer layer coating **182** first. However, even when the insulating layer of the wire **181** and the resin of the two types of coatings are different, thermal stress and impact stress can be reduced by providing a portion where the coatings are not adhered. That is, the aforementioned insulated structure is formed by providing a wire (enamel wire) and an space and arranging fluorine, polycarbonate, silicon, epoxy, polyethylene naphthalate, and LCP. In this case, it is desirable to adhere the outer layer coating and the inner layer coating with the use of an adhesive material made of epoxy or the like having a low dielectric constant and having a low linear expansion coefficient. By doing so, it is possible to suppress not only the mechanical strength but also the destruction of the coating due to friction caused by the vibration of the conductive part or the destruction of the outer layer coating due to a difference in the linear expansion coefficient.

As the outermost layer fixing which is generally the final process around the stator winding, which is responsible for mechanical strength, fixing, and the like for the conductor wire **82** having the above configuration, resins such as epoxy, PPS, PEEK, and LCP, which have good moldability and have properties such as dielectric constant and linear expansion coefficient similar to those of an enamel coating, are preferable.

Generally, resin potting with urethane or silicon is usually performed, but the linear expansion coefficient of the aforementioned resin is almost double that of other resins, and thermal stress capable of shearing the resin is generated. Therefore, it is not suitable for applications of 60V or higher where strict insulation regulations are used internationally. In this regard, according to the final insulation process for easily making by injection molding or the like using epoxy, PPS, PEEK, LCP or the like, each of the above requirements can be achieved.

Modifications other than the above are listed below.

A distance DM between the surface of the magnet unit **42** on the armature side in the radial direction and the axial center of the rotor in the radial direction may be 50 mm or more. Specifically, a distance DM between, for example, the surface radially inside the magnet unit **42** (specifically, the first and second magnets **91** and **92**) illustrated in FIG. **4** and the axial center of the rotor **40** in the radial direction may be 50 mm or more.

As a rotating electric machine having a slotless structure, a small-scale one whose output is used for a model of several tens of watts to several hundreds of watts is known. In addition, the discloser of the present application does not grasp a case where the slotless structure is adopted in a large industrial rotating electric machine which generally exceeds 10 kW. The discloser of the present application examined the reason.

In recent years, mainstream rotating electric machines are roughly classified into the following four types. These rotating electric machines are a brushed motor, a basket type induction motor, a permanent magnet type synchronous motor, and a reluctance motor.

An exciting current is supplied to the brushed motor via the brush. Therefore, in the case of a brushed motor of a large machine, the brush becomes large and the maintenance becomes complicated. As a result, with the remarkable development of semiconductor technology, it has been replaced by brushless motors such as induction motors. Meanwhile, in the world of small motors, coreless motors are also supplied to the world because of their low inertia and economic advantages.

In the basket type induction motor, the principle is that torque is generated by receiving the magnetic field generated by the stator winding on the primary side by the iron core of the rotor on the secondary side and intensively passing an induced current through the basket type conductor to form a reaction magnetic field. Therefore, from the viewpoint of small size and high efficiency of equipment, it is not always a good idea to eliminate the iron core on both the stator side and the rotor side.

The reluctance motor is a motor that literally utilizes the reluctance change of the iron core, and it is not desirable to eliminate the iron core in principle.

In recent years, IPMs (that is, embedded magnet type rotors) have become the mainstream of permanent magnet type synchronous motors, and in large machines in particular, IPMs are often used unless there are special circumstances.

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The IPM has a characteristic of having both magnet torque and reluctance torque, and is operated while the ratio of these torques is adjusted in a timely manner by inverter control. Therefore, the IPM is a small motor with excellent controllability.

According to the analysis of the discloser of the present application, the torque on the rotor surface that generates magnet torque and reluctance torque is drawn with the horizontal axis of the distance DM in the radial direction between the surface of the magnet unit on the armature side in the radial direction and the axial center of the rotor, that is, the radius of the stator core of a general inner rotor, as illustrated in FIG. 43.

The potential of the magnet torque is determined by the magnetic field strength generated by the permanent magnet as indicated in the following equation (eq1), whereas the potential of the reluctance torque is determined by the inductance, especially, the magnitude of the q-axis inductance as indicated in the following equation (eq2).

$$\text{Magnet torque} = k \cdot \psi \cdot I_q \quad (\text{eq1})$$

$$\text{Reluctance torque} = k \cdot (L_q - L_d) \cdot I_q \cdot I_d \quad (\text{eq2})$$

Here, the magnetic field strength of the permanent magnet and the magnitude of the inductance of the winding were compared by the DM. The magnetic field strength generated by the permanent magnet, that is, the amount of magnetic flux ψ , is proportional to the total area of the permanent magnet on the surface facing the stator. If a cylindrical rotor is used, it will be the surface area of the cylinder. Strictly speaking, since there are N pole and S pole, it is proportional to the occupied area of half of the cylindrical surface. The surface area of a cylinder is proportional to the radius of the cylinder and the length of the cylinder. That is, if the cylinder length is constant, it is proportional to the radius of the cylinder.

On the other hand, an inductance L_q of the winding depends on the shape of the iron core but has low sensitivity, and is rather proportional to the square of the number of turns of the stator winding, and therefore the number of turns is highly dependent. Moreover, when μ is the magnetic permeability of the magnetic circuit, N is the number of turns, S is the cross-sectional area of the magnetic circuit, and δ is the effective length of the magnetic circuit, the inductance $L = \mu \cdot N^2 \cdot S / \delta$. Since the number of turns of the winding depends on the size of the winding space, in the case of a cylindrical motor, it depends on the winding space of the stator, that is, the slot area. As illustrated in FIG. 44, since the slot shape is substantially quadrangular, the slot area is proportional to the product $a \cdot b$ of a length dimension a in the circumferential direction and a length dimension b in the radial direction.

The circumferential length dimension of the slot is proportional to the diameter of the cylinder because it increases as the diameter of the cylinder increases. The radial length dimension of the slot is exactly proportional to the diameter of the cylinder. That is, the slot area is proportional to the square of the diameter of the cylinder. Further, as can be seen from the above equation (eq2), since the reluctance torque is proportional to the square of the stator current, the performance of the rotating electric machine is determined by how large the current can flow, and that performance depends on the slot area of the stator. From the above, if the length of the cylinder is constant, the reluctance torque is proportional to the square of the diameter of the cylinder. Based on this, FIG. 43 is a diagram plotting the relation between the magnet torque and the reluctance torque and DM.

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As illustrated in FIG. 43, the magnet torque increases linearly with respect to the DM, and the reluctance torque increases quadratically with respect to the DM. It can be seen that the magnet torque is dominant when the DM is relatively small, and the reluctance torque is dominant as the stator core radius increases. The discloser of the present application has concluded that the intersection of the magnet torque and the reluctance torque in FIG. 43 is approximately in the vicinity of the stator core radius=50 mm under a predetermined condition. That is, it is difficult to eliminate the iron core in a 10 kW class motor whose stator core radius sufficiently exceeds 50 mm because it is the current mainstream to utilize reluctance torque, and it is presumed that this is one of the reasons why the slotless structure is not adopted in the field of large machines.

In the case of a rotating electric machine in which an iron core is used as a stator, magnetic saturation of the iron core is always an issue. In particular, in a radial gap type rotating electric machine, the vertical cross-sectional shape of the rotating shaft is a fan shape per magnetic pole, the magnetic path width becomes narrower toward the inner peripheral side of the equipment, and the inner circumference side dimension of the teeth portion forming the slot determines the performance limit of the rotating electric machine. No matter how high-performance permanent magnets are used, if magnetic saturation occurs in this portion, the performance of the permanent magnets cannot be fully brought out. In order not to generate magnetic saturation in this portion, the inner circumference must be designed to be large, resulting in an increase in the size of the equipment.

For example, in a distributed winding rotating electric machine, in the case of a three-phase winding, the magnetic flux is shared by three to six teeth per magnetic pole, but the magnetic flux tends to concentrate on the teeth in the front in the circumferential direction, and thus the magnetic flux does not flow evenly to the three to six teeth. In this case, while the magnetic flux flows intensively through some (for example, one or two) teeth, the teeth that are magnetically saturated with the rotation of the rotor also move in the circumferential direction. This also causes slot ripple.

From the above, in a rotating electric machine having a slotless structure in which the DM is 50 mm or more, it is desired to abolish the teeth in order to eliminate magnetic saturation. However, when the teeth are removed, the magnetic resistance of the magnetic circuit in the rotor and the stator increases, and the torque of the rotating electric machine decreases. The reason for the increase in magnetic resistance is, for example, that the air gap between the rotor and the stator becomes large. Therefore, in the above-mentioned rotating electric machine having a slotless structure in which the DM is 50 mm or more, there is room for improvement in increasing the torque. Consequently, there is a great merit of applying the above-mentioned configuration capable of increasing the torque to the above-mentioned rotating electric machine having a slotless structure in which the DM is 50 mm or more.

Moreover, with regard to not only the rotating electric machine having an outer rotor structure but also the rotating electric machine having an inner rotor structure may have a distance DM of 50 mm or more in the radial direction between the surface of the magnet unit on the armature side in the radial direction and the axial center of the rotor.

In the stator winding 51 of the rotating electric machine 10, the straight section 83 of the conductor wire 82 may be provided in a single layer in the radial direction. Further, when the straight section 83 is arranged in a plurality of layers inside and outside the radial direc-

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tion, the number of layers may be arbitrary, and may be provided in three layers, four layers, five layers, six layers, and the like.

For example, in the configuration of FIG. 2, the rotating shaft **11** is provided so as to protrude to both one end side and the other end side of the rotating electric machine **10** in the axial direction, but this may be changed, and the rotating shaft **11** may be configured to protrude only to one end side. In this case, the rotating shaft **11** may be provided so as to extend outward in the axial direction, with a portion that is cantilevered and supported by the bearing unit **20** as an end. In this configuration, since the rotating shaft **11** does not protrude inside the inverter unit **60**, the internal space of the inverter unit **60**, specifically the internal space of the tubular section **71**, can be used more widely.

In the rotating electric machine **10** having the above configuration, in the bearings **21** and **22**, non-conductive grease is used, but this may be changed, and conductive grease may be used in the bearings **21** and **22**. For example, conductive grease containing metal particles, carbon particles, or the like is used.

As a configuration for rotatably supporting the rotating shaft **11**, bearings may be provided at two locations on one end side and the other end side in the axial direction of the rotor **40**. In this case, in the configuration of FIG. 1, it is preferable that bearings are provided at two locations on one end side and the other end side with the inverter unit **60** therebetween.

In the rotating electric machine **10** having the above configuration, in the rotor **40**, the intermediate section **45** of the magnet holder **41** has the inner shoulder section **49a** and the annular outer shoulder section **49b**. However, these shoulder sections **49a** and **49b** may be eliminated to have a flat surface.

In the rotating electric machine **10** having the above configuration, the conductor **82a** is configured as an aggregate of a plurality of wires **86** in the conductor wire **82** of the stator winding **51**, but this may be changed, and a square conductor having a rectangular cross section may be used as the conductor wire **82**. Further, as the conductor wire **82**, a round conductor having a circular cross section or an elliptical cross section may be used.

In the rotating electric machine **10** having the above configuration, the inverter unit **60** is provided radially inside the stator **50**, but instead of this, the inverter unit **60** may not be provided radially inside the stator **50**. In this case, it is possible to set an internal region inside the stator **50** in the radial direction as a space. Further, it is possible to arrange parts different from the inverter unit **60** in the internal region.

The rotating electric machine **10** having the above configuration may not include the housing **30**. In this case, for example, the rotor **40**, the stator **50**, and the like may be held in a part of the wheel or other vehicle parts.

(Embodiment as an In-Wheel Motor for a Vehicle)

Next, an embodiment in which the rotating electric machine is provided integrally with the wheels of a vehicle as an in-wheel motor will be described. FIG. 45 is a perspective view illustrating a wheel **400** having an in-wheel motor structure and its peripheral structure, FIG. 46 is a vertical cross-sectional view of the wheel **400** and its peripheral structure, and FIG. 47 is an exploded perspective view of the wheel **400**. Each of these figures is a perspective view of the wheel **400** as viewed from the inside of the vehicle. Moreover, in the vehicle, the in-wheel motor struc-

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ture of the present embodiment can be applied in various forms. For example, in a vehicle having two wheels in front of and behind the vehicle, it is possible to apply the in-wheel motor structure of the present embodiment to the two wheels on the front side of the vehicle, the two wheels on the rear side of the vehicle, or the four wheels on the front and rear of the vehicle. However, it can also be applied to a vehicle in which at least one of the front and rear of the vehicle has one wheel. Moreover, the in-wheel motor is an application example as a vehicle drive unit.

As illustrated in FIGS. 45 to 47, the wheel **400** includes, for example, a tire **401** which is a well-known pneumatic tire, a wheel **402** fixed to the inner peripheral side of the tire **401**, and a rotating electric machine **500** fixed to the inner peripheral side of the wheel **402**. The rotating electric machine **500** has a fixing section which is a section including a stator and a rotation section which is a section including a rotor. The fixing section is fixed to the vehicle body side, and the rotation section is fixed to the wheel **402**. The rotation of the rotation section causes the tire **401** and the wheel **402** to rotate. Moreover, the detailed configuration of the rotating electric machine **500** including the fixing section and the rotation section will be described below.

Further, as peripheral devices, a suspension device that holds the wheel **400** with respect to a vehicle body (not illustrated), a steering device that changes the direction of the wheels **400**, and a braking device that brakes the wheel **400** are attached to the wheel **400**.

The suspension device is an independent suspension type suspension, and any type such as a trailing arm type, a strut type, a wishbone type, and a multi-link type can be applied. In the present embodiment, as the suspension device, a lower arm **411** is provided so as to extend toward the center side of the vehicle body, and a suspension arm **412** and a spring **413** are provided so as to extend in the up-down direction. The suspension arm **412** may be configured as, for example, a shock absorber. However, detailed illustration is omitted. The lower arm **411** and the suspension arm **412** are respectively connected to the vehicle body side and to a disk-shaped base plate **405** fixed to the fixing section of the rotating electric machine **500**. As illustrated in FIG. 46, the lower arm **411** and the suspension arm **412** are supported on the rotating electric machine **500** side (base plate **405** side) by support shafts **414** and **415** in a coaxial state with each other.

Further, as the steering device, for example, a rack & pinion type structure, a ball & nut type structure, a hydraulic power steering system, and an electric power steering system can be applied. In the present embodiment, a rack device **421** and a tie rod **422** are provided as steering devices, and the rack device **421** is connected to the base plate **405** on the rotating electric machine **500** side via the tie rod **422**. In this case, when the rack device **421** operates with the rotation of a steering shaft (not illustrated), the tie rod **422** moves in the right-left direction of the vehicle. As a result, the wheel **400** rotates about the support shafts **414** and **415** of the lower arm **411** and the suspension arm **412**, and the wheel direction is changed.

As the braking device, it is preferable to apply a disc brake or a drum brake. In the present embodiment, as the braking device, a disc rotor **431** fixed to a rotating shaft **501** of the rotating electric machine **500** and a brake caliper **432** fixed to the base plate **405** on the rotating electric machine **500** side are provided. In the brake caliper **432**, brake pads are operated by oil pressure or the like, and when the brake pads

are pressed against the disc rotor **431**, a braking force due to friction is generated and the rotation of the wheel **400** is stopped.

Further, the wheel **400** is attached with a housing duct **440** that houses an electric wiring H1 extending from the rotating electric machine **500** and a cooling pipe H2. The housing duct **440** extends from the end of the rotating electric machine **500** on the fixing section side along the end face of the rotating electric machine **500** and is provided so as to avoid the suspension arm **412**, and is fixed to the suspension arm **412** in that state. As a result, the connection portion of the housing duct **440** in the suspension arm **412** has a fixed positional relation with the base plate **405**. Therefore, it is possible to suppress the stress caused by the vibration of the vehicle in the electric wiring H1 and the cooling pipe H2. Moreover, the electrical wiring H1 is connected to an in-vehicle power supply unit and an in-vehicle ECU (not illustrated), and the cooling pipe H2 is connected to a radiator (not illustrated).

Next, the configuration of the rotating electric machine **500** used as an in-wheel motor will be described in detail. In the present embodiment, an example in which the rotating electric machine **500** is applied to an in-wheel motor is indicated. The rotating electric machine **500** has excellent operating efficiency and output as compared with the motor of a vehicle drive unit having a speed reducer as in the prior art. That is, if the rotating electric machine **500** is adopted in an application in which a practical price can be achieved by reducing the cost as compared with the prior art, it may be used as a motor for applications other than the vehicle drive unit. Even in such a case, it exhibits excellent performance as when applied to an in-wheel motor. Moreover, the operating efficiency refers to an index used during a test in a driving mode that derives the fuel efficiency of a vehicle.

The outline of the rotating electric machine **500** is illustrated in FIGS. **48** to **51**. FIG. **48** is a side view of the rotating electric machine **500** as viewed from the protruding side (inside the vehicle) of the rotating shaft **501**, FIG. **49** is a vertical cross-sectional view of the rotating electric machine **500** (cross-sectional view taken along a line **49-49** of FIG. **48**), FIG. **50** is a cross-sectional view of the rotating electric machine **500** (a cross-sectional view taken along a line **50-50** of FIG. **49**), and FIG. **51** is an exploded cross-sectional view of the components of the rotating electric machine **500**. In the following description, in FIG. **51**, the direction in which the rotating shaft **501** extends outward of the vehicle body is the axial direction, the direction extending radially from the rotating shaft **501** is the radial direction, and in FIG. **48**, both of the two directions extending in a circumferential shape from any point other than the center of rotation of the rotating portion on the center line drawn to form a cross section **49** passing through the center of the rotating shaft **501**, in other words, the center of rotation of the rotating portion are defined as the circumferential directions. In other words, the circumferential direction may be either a clockwise direction starting from an arbitrary point on the cross section **49** or a counterclockwise direction. Further, in the vehicle-mounted state, the right side is the outside of the vehicle and the left side is the inside of the vehicle in FIG. **49**. In other words, in the vehicle-mounted state, a rotor **510** which will be described below is arranged outward of the vehicle body with respect to a rotor cover **670**.

The rotating electric machine **500** according to the present embodiment is an outer rotor type surface magnet type rotating electric machine. The rotating electric machine **500** includes, roughly, a rotor **510**, a stator **520**, an inverter unit

530, a bearing **560**, and the rotor cover **670**. Each of these members is arranged coaxially with the rotating shaft **501** integrally provided on the rotor **510** and is assembled in the axial direction in a predetermined order to form the rotating electric machine **500**.

In the rotating electric machine **500**, the rotor **510** and the stator **520** each have a cylindrical shape, and are arranged so as to face each other with an air gap therebetween. As the rotor **510** rotates integrally with the rotating shaft **501**, the rotor **510** rotates on the radial outside of the stator **520**. The rotor **510** corresponds to a “field magnet” and the stator **520** corresponds to an “armature”.

The rotor **510** has a substantially cylindrical rotor carrier **511** and an annular magnet unit **512** fixed to the rotor carrier **511**. The rotating shaft **501** is fixed to the rotor carrier **511**.

The rotor carrier **511** has a cylindrical section **513**. A magnet unit **512** is attached to the inner peripheral surface of the cylindrical section **513**. That is, the magnet unit **512** is provided in a state of being surrounded by the cylindrical section **513** of the rotor carrier **511** from the outside in the radial direction. Further, the cylindrical section **513** has a first end and a second end facing each other in the axial direction thereof. The first end is located in the direction outside the vehicle body, and the second end is located in the direction in which the base plate **405** is present. In the rotor carrier **511**, an end plate **514** is continuously provided at the first end of the cylindrical section **513**. That is, the cylindrical section **513** and the end plate **514** have an integral structure. The second end of the cylindrical section **513** is open. The rotor carrier **511** is formed of, for example, a steel plate cold commercial having sufficient mechanical strength (SPCC or SPHC thicker than SPCC), forging steel, carbon fiber reinforced plastic (CFRP), or the like.

The axial length of the rotating shaft **501** is longer than the axial dimension of the rotor carrier **511**. In other words, the rotating shaft **501** protrudes toward the open end side (inward direction of the vehicle) of the rotor carrier **511**, and the above-mentioned brake device or the like is attached to the protruding side end.

A through hole **514a** is formed in the central portion of the end plate **514** of the rotor carrier **511**. The rotating shaft **501** is fixed to the rotor carrier **511** in a state of being inserted into the through hole **514a** of the end plate **514**. The rotating shaft **501** has a flange **502** extending in a direction intersecting with (orthogonal to) the axial direction at a portion where the rotor carrier **511** is fixed, and in a state where the flange and the surface of the end plate **514** outside the vehicle are surface-joined, the rotating shaft **501** is fixed to the rotor carrier **511**. Moreover, in the wheel **400**, the wheel **402** is fixed with the use of a fastener such as a bolt erected from the flange **502** of the rotating shaft **501** toward the outside of the vehicle.

Further, the magnet unit **512** is composed of a plurality of permanent magnets arranged in such a manner that the polarities alternate along the circumferential direction of the rotor **510**. As a result, the magnet unit **512** has a plurality of magnetic poles in the circumferential direction. The permanent magnet is fixed to the rotor carrier **511** by, for example, adhesion. The magnet unit **512** has the configuration described as the magnet unit **42** with reference to FIGS. **8** and **9** of the first embodiment, and as a permanent magnet, has an intrinsic coercive force of **400** [kA/m] or more and is composed with a sintered neodymium magnet having a residual magnetic flux Br of **1.0** [T] or more.

Similarly to the magnet unit **42** in FIG. **9** or the like, the magnet unit **512** has a first magnet **91** and a second magnet **92**, which are respectively polar anisotropic magnets and

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have different polarities from each other. As described with reference to FIGS. 8 and 9, each of the magnets 91 and 92 has a different direction of the axis of easy magnetization on the d-axis side (the portion located near the d-axis) and the q-axis side (the portion located near the q-axis), and on the d-axis side, the direction of the axis of easy magnetization is close to the direction parallel to the d-axis, and on the q-axis side, the direction of the easy magnetization axis is close to the direction orthogonal to the q-axis. In addition, an arc-shaped magnet magnetic path is formed by the orientation according to the direction of the axis of easy magnetization. Moreover, in each of the magnets 91 and 92, the axis of easy magnetization may be oriented parallel to the d-axis on the d-axis side, and the axis of easy magnetization may be oriented orthogonal to the q-axis on the q-axis side. In short, the magnet unit 512 is configured, in which the orientation was made in such a manner that the direction of the axis of easy magnetization is parallel to the d-axis on the d-axis side which is the center of the magnetic pole as compared with the q-axis side which is the magnetic pole boundary.

According to the magnets 91 and 92, the magnet magnetic flux on the d-axis is strengthened and the change in magnetic flux near the q-axis is suppressed. As a result, the magnets 91 and 92 in which the change in surface magnetic flux from the q-axis to the d-axis at each magnetic pole is gentle can be preferably achieved. As the magnet unit 512, the configurations of the magnet unit 42 illustrated in FIGS. 22 and 23 and the configuration of the magnet unit 42 illustrated in FIG. 30 can also be used.

Moreover, the magnet unit 512 may have a rotor core (back yoke) formed by laminating a plurality of electromagnetic steel sheets in the axial direction on the side of the cylindrical section 513 of the rotor carrier 511, that is, on the outer peripheral surface side. That is, it is also possible to provide a rotor core on the radial inside of the cylindrical section 513 of the rotor carrier 511 and to provide the permanent magnets (magnets 91 and 92) on the radial inside of the rotor core.

As illustrated in FIG. 47, the cylindrical section 513 of the rotor carrier 511 is formed with a recess 513a in a direction extending in the axial direction at predetermined intervals in the circumferential direction. The recess 513a is formed by, for example, press working, and as illustrated in FIG. 52, a protrusion 513b is formed on the inner peripheral surface side of the cylindrical section 513 at a position on the back side of the recess 513a. On the other hand, on the outer peripheral surface side of the magnet unit 512, a recess 512a is formed in accordance with the protrusion 513b of the cylindrical section 513, and the protrusion 513b of the cylindrical section 513 enters the recess 512a, whereby the displacement in the circumferential direction of the magnet unit 512 is suppressed. That is, the protrusion 513b on the rotor carrier 511 side functions as the rotation stop section of the magnet unit 512. Moreover, the method for forming the protrusion 513b may be any method other than press working.

In FIG. 52, the direction of the magnet magnetic path in the magnet unit 512 is indicated by an arrow. The magnet magnetic path extends in an arc shape so as to straddle the q-axis which is the magnetic pole boundary, and is in a direction parallel to or close to parallel to the d-axis on the d-axis which is the center of the magnetic pole. The magnet unit 512 is formed with a recess 512b at positions corresponding to the q-axis on the inner peripheral surface side thereof. In this case, in the magnet unit 512, the magnet magnetic path length differs between the side closer to the

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stator 520 (lower side in the figure) and the side farther from the stator 520 (upper side in the figure), the magnet magnetic path length is shorter on the side closer to the stator 520, and the recess 512b is formed at a position where the magnet magnetic path length is the shortest. That is, in consideration of the fact that it is difficult for the magnet unit 512 to generate a sufficient magnet magnetic flux in a place where the magnet magnetic path length is short, the magnet is omitted in a place where the magnet magnetic flux is weak.

Here, an effective magnetic flux density Bd of the magnet becomes higher as the length of the magnetic circuit passing through the inside of the magnet becomes longer. Further, a permeance coefficient Pc and the effective magnetic flux density Bd of the magnet are in a relation that the higher one is, the higher the other is. According to the configuration of FIG. 52, the amount of magnets can be reduced while suppressing a decrease in the permeance coefficient Pc which is an index of the height of the effective magnetic flux density Bd of the magnet. Moreover, in a B-H coordinates, the intersection of the permeance straight line and the demagnetization curve according to the shape of the magnet is the operating point, and the magnetic flux density at the operating point is the effective magnetic flux density Bd of the magnet. The rotating electric machine 500 of the present embodiment has a configuration in which the amount of iron in the stator 520 is reduced, and in such a configuration, a method for setting a magnetic circuit straddling the q-axis is extremely effective.

Further, the recess 512b of the magnet unit 512 can be used as an air passage extending in the axial direction. Therefore, it is also possible to improve the air cooling performance.

Next, the configuration of the stator 520 will be described. The stator 520 has a stator winding 521 and a stator core 522. FIG. 53 is a perspective view illustrating the stator winding 521 and the stator core 522 in an exploded manner.

The stator winding 521 is composed of a plurality of phase windings formed by winding in a substantially tubular shape (annular shape), and the stator core 522 as a base member is assembled radially inside the stator winding 521. In the present embodiment, a U-phase, V-phase, and W-phase windings are used, and the stator winding 521 is thereby configured as a three-phase winding. Each phase winding is composed of two inner and outer layers of conductors 523 in the radial direction. Similarly to the stator 50 described above, the stator 520 is characterized by having a slotless structure and a flat conductor structure of the stator winding 521, and has the same configuration as or a configuration similar to that of the stator 50 illustrated in FIGS. 8 to 16.

The configuration of the stator core 522 will be described. Similarly to the stator core 52 described above, the stator core 522 has a cylindrical shape in which a plurality of electromagnetic steel sheets are laminated in the axial direction and has a predetermined thickness in the radial direction, and the stator winding 521 is assembled on the radially outside that is the rotor 510 side in the stator core 522. The outer peripheral surface of the stator core 522 has a curved shape without unevenness, and in a state where the stator winding 521 is assembled, the conductor 523 constituting the stator winding 521 are arranged side by side in the circumferential direction on the outer peripheral surface of the stator core 522. The stator core 522 functions as a back core.

The stator 520 may use any of the following A to C.

A In the stator 520, an interconductor member is provided between each conductor 523 in the circumferential direction,

and as the interconductor member, a magnetic material having a relation of $W_t \cdot B_s$ $W_m \cdot B_r$ is used when the width dimension of the interconductor member in the circumferential direction at one magnetic pole is W_t , the saturation magnetic flux density of the interconductor member is B_s , the width dimension in the circumferential direction of the magnet unit **512** at one magnetic pole is W_m , and the residual magnetic flux density of the magnet unit **512** is B_r .

B In the stator **520**, an interconductor member is provided between each conductor **523** in the circumferential direction, and a non-magnetic material is used as the interconductor member.

C The stator **520** has a configuration in which no interconductor member is provided between each conductor **523** in the circumferential direction.

According to such configuration of the stator **520**, the inductance can be reduced as compared with a rotating electric machine having a general teeth structure in which teeth (iron core) for establishing a magnetic path is provided between respective conductor sections as a stator winding. Specifically, the inductance can be reduced to $1/10$ or less. In this case, since the impedance decreases as the inductance decreases, the output power with respect to the input power of the rotating electric machine **500** can be increased, which thus can contribute to the increase in torque. Further, it is possible to provide a rotating electric machine with a higher output than a rotating electric machine using an embedded magnet type rotor that outputs torque utilizing the voltage of the impedance component (in other words, utilizing reluctance torque).

In the present embodiment, the stator winding **521** is integrally molded together with the stator core **522** by a molding material (insulating member) made of resin or the like, and the molding material is interposed between the respective conductors **523** arranged in the circumferential direction. According to such a configuration, the stator **520** of the present embodiment corresponds to the configuration of B among the above A to C. Further, the respective conductors **523** adjacent to each other in the circumferential direction are arranged in such a manner that the end faces in the circumferential direction are in contact with each other or are arranged close to each other at a minute interval, and the configuration of the above C may be adopted in view of this configuration. Moreover, in a case where the configuration A is adopted, it is preferable that a protrusion is provided on the outer peripheral surface of the stator core **522**, in accordance with the direction of the conductor **523** in the axial direction, that is, in accordance with the skew angle of the stator winding **521** having a skew structure, for example.

Next, the configuration of the stator winding **521** will be described with reference to FIG. **54**. FIG. **54** is a front view illustrating the stator winding **521** developed in a plane, FIG. **54A** illustrates each conductor **523** located in the outer layer in the radial direction, and FIG. **54B** illustrates each conductor **523** located in the inner layer in the radial direction.

The stator winding **521** is formed being wound in an annular shape by distributed winding. In the stator winding **521**, the conductor material is wound around the inner and outer two layers in the radial direction, and the respective conductors **523** on the inner layer side and the outer layer side are skewed in different directions (See FIGS. **54A** and **54B**). The respective conductors **523** are insulated from each other. It is preferable that the conductor **523** is configured as an aggregate of a plurality of wires **86** (see FIG. **13**). Further, for example, two conductors **523** having the same phase and the same energizing direction are provided side by side in

the circumferential direction. In the stator winding **521**, one conductor section having the same phase is composed of each conductor **523** having two layers in the radial direction and two conductors in the circumferential direction (that is, a total of four conductors), and the conductor section is provided one per magnetic pole.

It is desirable that the radial thickness dimension of the conductor section be smaller than the circumferential width dimension for one phase in one magnetic pole, whereby the stator winding **521** has a flat conductor structure. Specifically, for example, in the stator winding **521**, one conductor section having the same phase is preferably composed of each conductor **523** having two layers in the radial direction and four conductors in the circumferential direction (that is, a total of eight conductors). Alternatively, in the conductor cross section of the stator winding **521** illustrated in FIG. **50**, the circumferential width dimension is preferably larger than the radial thickness dimension. As the stator winding **521**, the stator winding **51** illustrated in FIG. **12** can also be used. However, in this case, it is necessary to secure a space in the rotor carrier **511** for housing the coil end of the stator winding.

In the stator winding **521**, the conductors **523** are arranged side by side in the circumferential direction, being tilted at a predetermined angle on a coil side **525** that overlaps the stator core **522** inside and outside the radial direction, and coil ends **526** on both sides, which are axially outer than the stator core **522**, are inverted (folded back) inward in the axial direction to form a continuous connection. FIG. **54A** illustrates a range of the coil side **525** and a range of the coil end **526**, respectively. The inner layer side conductor **523** and the outer layer side conductor **523** are connected to each other at the coil end **526**, and as a result, each time the conductor **523** is inverted in the axial direction at the coil end **526** (each time it is folded back), the conductor **523** is switched alternately between the inner layer side and the outer layer side. In short, the stator winding **521** has a configuration in which the inner and outer layers are switched in accordance with the reversal of the direction of the current in the respective conductors **523** that are continuous in the circumferential direction.

Further, in the stator winding **521**, two types of skews are applied in which the skew angles are different between the end regions that are both ends in the axial direction and the central region sandwiched between the end regions. That is, as illustrated in FIG. **55**, in the conductor **523**, a skew angle θ_{s1} in the central region and a skew angle θ_{s2} in the end region are different, and the skew angle θ_{s1} is smaller than the skew angle θ_{s2} . In the axial direction, the end region is defined to include the coil side **525**. The skew angle θ_{s1} and the skew angle θ_{s2} are tilt angles at which each conductor **523** is tilted with respect to the axial direction. The skew angle θ_{s1} in the central region may be set in an angle range appropriate for reducing the harmonic component of the magnetic flux generated by the energization of the stator winding **521**.

The skew angle of each conductor **523** in the stator winding **521** is made different between the central region and the end region, and the skew angle θ_{s1} in the central region is made smaller than the skew angle θ_{s2} in the end region, whereby the winding coefficient of the stator winding **521** can be increased while reducing the coil end **526**. In other words, the length of the coil end **526**, that is, the conductor length of the portion protruding in the axial direction from the stator core **522**, can be shortened while

ensuring a desired winding coefficient. As a result, it is possible to improve the torque while downsizing the rotating electric machine 500.

Here, an appropriate range as the skew angle θ_{s1} in the central region will be described. When X conductors 523 are arranged in one magnetic pole in the stator winding 521, it is conceivable that the Xth order harmonic component is generated by energization of the stator winding 521. When the number of phases is S and the logarithm is m, $X=2*S*m$. The discloser of the present application focused on the fact that the Xth order harmonic component is a component that constitutes a composite wave of the X-1th order harmonic component and the X+1th order harmonic component, and therefore at least one of the X-1th order harmonic component or the X+1th order harmonic component is reduced, whereby the Xth order harmonic component can be reduced. Based on this focus, the discloser of the present application found that the skew angle θ_{s1} is set within the angle range of " $360^\circ/(X+1)$ to $360^\circ/(X-1)$ " in terms of the electrical angle, whereby the Xth harmonic component can be reduced.

For example, when $S=3$ and $m=2$, the skew angle θ_{s1} is set within the angle range of " $360^\circ/13$ to $360^\circ/11$ " in order to reduce the harmonic component of the $X=12$ th order. That is, the skew angle θ_{s1} is preferably set at an angle within the range of 27.7° to 32.7° .

By setting the skew angle θ_{s1} in the central region as described above, the magnet magnetic fluxes alternated at N and S poles can be positively interlinked in the central region, and the winding coefficient of the stator winding 521 can be increased.

The skew angle θ_{s2} in the end region is larger than the skew angle θ_{s1} in the central region described above. In this case, the angle range of the skew angle θ_{s2} is " $\theta_{s1} < \theta_{s2} < 90^\circ$ ".

Further, in the stator winding 521, the inner layer side conductor 523 and the outer layer side conductor 523 are preferably connected by welding or adhesion between the ends of the respective conductors 523, or are preferably connected by bending. In the stator winding 521, the end of each phase winding is electrically connected to a power converter (inverter) via a bus bar or the like on one side (that is, one end side in the axial direction) of each coil end 526 on both sides in the axial direction. Therefore, here, the configuration in which the respective conductors are connected to each other at the coil end 526 will be described while distinguishing between the coil end 526 on the bus bar connection side and the coil end 526 on the opposite side.

The first configuration is such that each conductor 523 is connected at the coil end 526 on the bus bar connection side by welding, and each conductor 523 is connected at the coil end 526 on the opposite side by means other than welding. As the means other than welding, for example, a connection by bending a conductor material is conceivable. At the coil end 526 on the bus bar connection side, it is assumed that the bus bar is connected to the end of each phase winding by welding. Therefore, by connecting each conductor 523 at the same coil end 526 by welding, each welded portion can be handled in a series of processes, and work efficiency can be improved.

The second configuration is such that each conductor 523 is connected at the coil end 526 on the bus bar connection side by means other than welding, and each conductor 523 is connected at the coil end 526 on the opposite side by welding. In this case, if each conductor 523 is connected at the coil end 526 on the bus bar connection side by welding, the separation distance between the bus bar and the coil end

526 needs to be sufficient to avoid contact between the welded portion and the bus bar. However, with this configuration, the separation distance between the bus bar and the coil end 526 can be reduced. As a result, the regulation regarding the length of the stator winding 521 in the axial direction or the bus bar can be relaxed.

As the third configuration, each conductor 523 is connected at the coil ends 526 on both sides in the axial direction by welding. In this case, all of the conductor materials prepared before welding may have a short wire length, and the work efficiency can be improved by reducing the bending process.

As the fourth configuration, each conductor 523 is connected at the coil ends 526 on both sides in the axial direction by means other than welding. In this case, the portion of the stator winding 521 to be welded can be reduced as much as possible, and the concern that insulation peeling may occur in the welding process can be reduced.

Further, in the process of manufacturing the annular stator winding 521, it is preferable to manufacture the strip-shaped windings arranged in a plane shape, and then to form the strip-shaped windings in an annular shape. In this case, it is preferable to weld the conductors at the coil end 526 in a state where the winding is a flat strip. When forming a flat strip-shaped winding in an annular shape, it is preferable to use a cylindrical jig having the same diameter as that of the stator core 522 and wind the strip-shaped winding around the cylindrical jig to form the strip-shaped winding in an annular shape. Alternatively, the strip-shaped winding may be wound directly around the stator core 522.

Moreover, the configuration of the stator winding 521 can also be changed as follows.

For example, in the stator winding 521 illustrated in FIGS. 54A and 54B, the skew angles of the central region and the end region may be the same.

Further, in the stator windings 521 illustrated in FIGS. 54A and 54B, the ends of the in-phase conductors 523 adjacent to each other in the circumferential direction may be connected by a crossover section extending in a direction orthogonal to the axial direction.

The number of layers of the stator winding 521 may be $2*n$ layers (n is a natural number), and the stator winding 521 may have 4 layers, 6 layers, or the like in addition to the 2 layers.

Next, the inverter unit 530, which is a power conversion unit, will be described. Here, the configuration of the inverter unit 530 will be described with reference to FIGS. 56 and 57, which are exploded cross-sectional views of the inverter unit 530. Moreover, in FIG. 57, each member illustrated in FIG. 56 is illustrated as two subassemblies.

The inverter unit 530 includes an inverter housing 531, a plurality of electric modules 532 assembled to the inverter housing 531, and a bus bar module 533 that electrically connects each of the electric modules 532.

The inverter housing 531 has a cylindrical outer wall member 541, an inner wall member 542 having a cylindrical outer peripheral diameter smaller than that of the outer wall member 541 and arranged radially inside the outer wall member 541, and a boss forming member 543 fixed to one end side in the axial direction of the inner wall member 542. Each of these members 541 to 543 is preferably made of a conductive material, for example, made of carbon fiber reinforced plastic (CFRP). The inverter housing 531 is configured by combining the outer wall member 541 and the inner wall member 542 inside and outside the radial direction, and assembling the boss forming member 543 on one

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end side in the axial direction of the inner wall member **542**. The assembled state is the state illustrated in FIG. **57**.

The stator core **522** is fixed to the radial outside of the outer wall member **541** of the inverter housing **531**. As a result, the stator **520** and the inverter unit **530** are integrated. As illustrated in FIG. **56**, the outer wall member **541** is formed with a plurality of recesses **541a**, **541b**, and **541c** on the inner peripheral surface thereof, and the inner wall member **542** is formed with a plurality of recesses **542a**, **542b**, and **542c** on the outer peripheral surface thereof. In addition, by the outer wall member **541** and the inner wall member **542** being assembled to each other, three hollow portions **544a**, **544b**, and **544c** are formed between them (see FIG. **57**). Of these, the central hollow portion **544b** is used as a cooling water passage **545** through which cooling water as a refrigerant flows. Further, a sealing material **546** is housed in the hollow portions **544a** and **544c** on both sides of the hollow portion **544b** (cooling water passage **545**). The hollow portion **544b** (cooling water passage **545**) is sealed by the sealing material **546**. The cooling water passage **545** will be described in detail below.

Further, the boss forming member **543** is provided with a disc ring-shaped end plate **547** and a boss section **548** protruding from the end plate **547** toward the inside of the housing. The boss section **548** is provided in a hollow tubular shape. For example, as illustrated in FIG. **51**, the boss forming member **543** is fixed to the second end, of the first end of the inner wall member **542** in the axial direction and the second end on the protruding side (that is, inside the vehicle) of the rotating shaft **501** facing the first end. Moreover, in the wheels **400** illustrated in FIGS. **45** to **47**, the base plate **405** is fixed to the inverter housing **531** (more specifically, the end plate **547** of the boss forming member **543**).

The inverter housing **531** has a configuration having a double peripheral wall in the radial direction about the shaft center, and the outer peripheral wall of the double peripheral wall is formed by the outer wall member **541** and the inner wall member **542**, and the inner peripheral wall is formed by the boss section **548**. Moreover, in the following description, the outer peripheral wall formed by the outer wall member **541** and the inner wall member **542** is also referred to as an "outer peripheral wall WA1", and the inner peripheral wall formed by the boss section **548** is also referred to as an "inner peripheral wall WA2".

An annular space is formed in the inverter housing **531** between the outer peripheral wall WA1 and the inner peripheral wall WA2, and a plurality of electric modules **532** are arranged side by side in the circumferential direction in the annular space. The electric module **532** is fixed to the inner peripheral surface of the inner wall member **542** by adhesion, screw tightening, or the like. In the present embodiment, the inverter housing **531** corresponds to a "housing member" and the electric module **532** corresponds to an "electric component".

A bearing **560** is housed inside the inner peripheral wall WA2 (boss section **548**), and the rotating shaft **501** is rotatably supported by the bearing **560**. The bearing **560** is a hub bearing that rotatably supports the wheel **400** at the center of the wheel. The bearing **560** is provided at a position overlapping with the rotor **510**, the stator **520**, and the inverter unit **530** in the axial direction. In the rotating electric machine **500** of the present embodiment, the magnet unit **512** can be made thinner in accordance with the orientation of the rotor **510**, and a slotless structure or a flat conductor structure is adopted in the stator **520**. Thus, it is possible to expand the hollow space radially inside the

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magnetic circuit section by reducing the radial thickness dimension of the magnetic circuit section. This makes it possible to arrange the magnetic circuit section, the inverter unit **530**, and the bearing **560** in a state of being stacked in the radial direction. The boss section **548** is a bearing holding section that holds the bearing **560** thereinside.

The bearing **560** is, for example, a radial ball bearing, and has a tubular inner ring **561**, an outer ring **562** having a diameter larger than that of the inner ring **561** and arranged radially outside the inner ring **561**, and a plurality of balls **563** arranged between the inner ring **561** and the outer ring **562**. The bearing **560** is fixed to the inverter housing **531** by assembling the outer ring **562** to the boss forming member **543**, and the inner ring **561** is fixed to the rotating shaft **501**. The inner ring **561**, outer ring **562**, and ball **563** are all made of a metallic material such as carbon steel.

Further, the inner ring **561** of the bearing **560** has a tubular section **561a** that houses the rotating shaft **501** and a flange **561b** that extends in a direction intersecting with (orthogonal to) the axis direction from one end in the axial direction of the tubular section **561a**. The flange **561b** is a portion that comes into contact with the end plate **514** of the rotor carrier **511** from the inside, and in a state where the bearing **560** is assembled to the rotating shaft **501**, the rotor carrier **511** is held in a state of being sandwiched between the flange **502** of the rotating shaft **501** and the flange **561b** of the inner ring **561**. In this case, the flange **502** of the rotating shaft **501** and the flange **561b** of the inner ring **561** have the same angle of intersection with respect to the axial direction (both are right angles in the present embodiment), and the rotor carrier **511** is held in a state of being sandwiched between these respective flanges **502** and **561b**.

According to the configuration in which the rotor carrier **511** is supported from the inside by the inner ring **561** of the bearing **560**, the angle of the rotor carrier **511** with respect to the rotating shaft **501** can be maintained at an appropriate angle, and thus the parallelism of the magnet unit **512** with respect to the rotating shaft **501** can be kept good. As a result, even if the rotor carrier **511** is expanded in the radial direction, the resistance to vibration and the like can be enhanced.

Next, the electric module **532** housed in the inverter housing **531** will be described.

The plurality of electric modules **532** are obtained by dividing electric components such as semiconductor switching elements and smoothing capacitors constituting a power converter into a plurality of individual modules. The electric module **532** includes a switch module **532A** having a semiconductor switching element that is a power element and a capacitor module **532B** having a smoothing capacitor.

As illustrated in FIGS. **49** and **50**, a plurality of spacers **549** having a flat surface for attaching the electric module **532** are fixed to the inner peripheral surface of the inner wall member **542**, and the electric module **532** is attached to the spacer **549**. That is, the inner peripheral surface of the inner wall member **542** is a curved surface, whereas the mounting surface of the electric module **532** is a flat surface, and thus the spacer **549** forms a flat surface on the inner peripheral surface side of the inner wall member **542**, and the electric module **532** is fixed to the flat surface.

Note that the configuration in which the spacer **549** is interposed between the inner wall member **542** and the electric module **532** is not essential, and it is also possible to attach the electric module **532** directly to the inner wall member **542** by flattening the inner peripheral surface of the inner wall member **542** or by making the mounting surface of the electric module **532** curved. Further, it is also possible

to fix the electric module **532** to the inverter housing **531** in a state of non-contact with the inner peripheral surface of the inner wall member **542**. For example, the electric module **532** is fixed to the end plate **547** of the boss forming member **543**. It is also possible to fix the switch module **532A** to the inner peripheral surface of the inner wall member **542** in a contact state, and to fix the capacitor module **532B** to the inner peripheral surface of the inner wall member **542** in a non-contact state.

Moreover, in a case where the spacer **549** is provided on the inner peripheral surface of the inner wall member **542**, the outer peripheral wall **WA1** and the spacer **549** correspond to the "tubular section". Further, in a case where the spacer **549** is not used, the outer peripheral wall **WA1** corresponds to the "tubular section".

As described above, the outer peripheral wall **WA1** of the inverter housing **531** is formed with the cooling water passage **545** through which cooling water as a refrigerant flows, and each electric module **532** is cooled by the cooling water flowing through the cooling water passage **545**. Moreover, it is also possible to use cooling oil as the refrigerant as an alternative to the cooling water. The cooling water passage **545** is provided in an annular shape along the outer peripheral wall **WA1**, and the cooling water flowing in the cooling water passage **545** flows from the upstream side to the downstream side while passing through each electric module **532**. In the present embodiment, the cooling water passage **545** is provided in an annular shape so as to overlap each electric module **532** inside and outside the radial direction and to surround these respective electric modules **532**.

The inner wall member **542** is provided with an inlet passage **571** for flowing the cooling water into the cooling water passage **545** and an outlet passage **572** for discharging the cooling water from the cooling water passage **545**. As described above, the plurality of electric modules **532** are fixed to the inner peripheral surface of the inner wall member **542**, and in such a configuration, the interval between the electric modules adjacent to each other in the circumferential direction is expanded in only one place, and a part of the inner wall member **542** is protruded inward in the radial direction to form a protruding section **573** in the expanded portion. In addition, the protruding section **573** is provided with the inlet passage **571** and the outlet passage **572** in a side-by-side manner along the radial direction.

FIG. **58** illustrates a state of arrangement of each electric module **532** in the inverter housing **531**. Note that FIG. **58** is the same vertical cross-sectional view as FIG. **50**.

As illustrated in FIG. **58**, the respective electric modules **532** are arranged side by side in the circumferential direction with the interval between the electric modules in the circumferential direction as a first interval **INT1** or a second interval **INT2**. The second interval **INT2** is a wider interval than the first interval **INT1**. The respective intervals **INT1** and **INT2** are, for example, the distance between the center positions of two electric modules **532** adjacent to each other in the circumferential direction. In this case, the interval between the electric modules adjacent to each other in the circumferential direction without sandwiching the protruding section **573** is the first interval **INT1**, and the interval between the electric modules adjacent to each other in the circumferential direction sandwiching the protruding section **573** is the second interval **INT2**. That is, the interval between the electric modules adjacent to each other in the circumferential direction is partially widened, and the protruding section **573** is provided at, for example, the central portion of the widened interval (second interval **INT2**).

The respective intervals **INT1** and **INT2** may be, for example, the distance of an arc between the center positions of two electric modules **532** adjacent to each other in the circumferential direction, on the same circle centered on the rotating shaft **501**. Alternatively, the interval between the electric modules in the circumferential direction may be defined by angular distances $\theta i1$ and $\theta i2$ centered on the rotating shaft **501** ($\theta i1 < \theta i2$).

Moreover, in the configuration illustrated in FIG. **58**, the respective electric modules **532** arranged at the first interval **INT1** are arranged in a state of being separated from each other in the circumferential direction (non-contact state), but instead of this configuration, the respective electric modules **532** may be arranged in the circumferential direction in a state of being in contact with each other.

As illustrated in FIG. **48**, the end plate **547** of the boss forming member **543** is provided with a water channel port **574** in which the passage ends of the inlet passage **571** and the outlet passage **572** are formed. A circulation path **575** for circulating cooling water is connected to the inlet passage **571** and the outlet passage **572**. The circulation path **575** is composed of a cooling water pipe. A pump **576** and a heat radiating device **577** are provided in the circulation path **575**, and the cooling water circulates through the cooling water passage **545** and the circulation path **575** as the pump **576** is driven. The pump **576** is an electric pump. The heat radiating device **577** is, for example, a radiator that releases the heat of the cooling water to the atmosphere.

As illustrated in FIG. **50**, since the stator **520** is arranged on the outside of the outer peripheral wall **WA1** and the electric module **532** is arranged on the inside of the outer peripheral wall **WA1**, the heat of the stator **520** is transferred to the outer peripheral wall **WA1** from the outside, and the heat of the electric module **532** is transferred from the inside. In this case, the stator **520** and the electric module **532** can be cooled at the same time by the cooling water flowing the cooling water passage **545**, and the heat of the heat-generating component in the rotating electric machine **500** can be efficiently released.

Here, the electrical configuration of the power converter will be described with reference to FIG. **59**.

As illustrated in FIG. **59**, the stator winding **521** is composed of a U-phase winding, a V-phase winding, and a W-phase winding, and an inverter **600** is connected to the stator winding **521**. The inverter **600** is composed of a full bridge circuit having the same number of upper and lower arms as the number of phases, and a series connection body composed of an upper arm switch **601** and a lower arm switch **602** is provided for each phase. These respective switches **601** and **602** are turned on/off by the drive circuit **603**, and the winding of each phase is energized by the on/off. The respective switches **601** and **602** are composed of a semiconductor switching element such as a MOSFET or an IGBT. Further, in the upper and lower arms of each phase, a charge supply capacitor **604** that supplies the charge required for switching to the respective switches **601** and **602** is connected in parallel to the series connection body of the switches **601** and **602**.

A control device **607** includes a microcomputer composed of a CPU and various memories, and performs energization control by turning on/off the respective switches **601** and **602** on the basis of various detected information in the rotating electric machine **500** and requests for power running and power generation. The control device **607** performs on/off control of the respective switches **601** and **602** by, for example, PWM control at a predetermined switching frequency (carrier frequency) and rectangular wave control.

The control device 607 may be a built-in control device built in the rotating electric machine 500, or an external control device provided outside the rotating electric machine 500.

Incidentally, in the rotating electric machine 500 of the present embodiment, the electric time constant is small because the inductance of the stator 520 is reduced, and in a situation where the electrical time constant is small, it is desirable to increase the switching frequency (carrier frequency) and increase the switching speed. In this respect, the charge supply capacitor 604 is connected in parallel to the series connection body of the switches 601 and 602 of each phase, and thus the wiring inductance becomes low, and appropriate surge countermeasures are possible even with a configuration in which the switching speed is increased.

The high potential side terminal of the inverter 600 is connected to the positive electrode terminal of a DC power supply 605, and the low potential side terminal is connected to the negative electrode terminal (ground) of the DC power supply 605. Further, a smoothing capacitor 606 is connected to the high potential side terminal and the low potential side terminal of the inverter 600, in parallel to the DC power supply 605.

The switch module 532A has the respective switches 601 and 602 (semiconductor switching elements), the drive circuit 603 (specifically, an electric element constituting the drive circuit 603), and the charge supply capacitor 604, as heat-generating components. Further, the capacitor module 532B has the smoothing capacitor 606 as a heat-generating component. A specific configuration example of the switch module 532A is illustrated in FIG. 60.

As illustrated in FIG. 60, the switch module 532A has a module case 611 as a housing case, and the switches 601 and 602 for one phase housed in the module case 611, the drive circuit 603, and the charge supply capacitor 604. The drive circuit 603 is configured as a dedicated IC or a circuit board and is provided in the switch module 532A.

The module case 611 is made of an insulating material such as resin, and is fixed to the outer peripheral wall WA1 with its side surface in contact with the inner peripheral surface of the inner wall member 542 of the inverter unit 530. The module case 611 is filled with a molding material such as resin. In the module case 611, the switches 601 and 602 and the drive circuit 603, and the switches 601 and 602 and the capacitor 604 are electrically connected by a wiring 612, respectively. More specifically, the switch module 532A is attached to the outer peripheral wall WA1 via the spacer 549, but the spacer 549 is not illustrated.

In a state where the switch module 532A is fixed to the outer peripheral wall WA1, the side of the switch module 532A closer to the outer peripheral wall WA1, that is, the side closer to the cooling water passage 545 has higher cooling performance. Therefore, the order of the arrangement of the switches 601 and 602, the drive circuit 603, and the capacitor 604 is determined in accordance with the cooling performance. Specifically, when comparing the amount of heat generated, the switches 601 and 602, the capacitor 604, and the drive circuit 603 are in the order from the largest, and therefore the switches 601 and 602, the capacitor 604, and the drive circuit 603 are arranged in this order from the side closer to the outer peripheral wall WA1 in accordance with the magnitude order of the amount of heat generated. Moreover, the contact surface of the switch module 532A is preferably smaller than the contactable surface on the inner peripheral surface of the inner wall member 542.

Furthermore, although detailed illustration of the capacitor module 532B is omitted, the capacitor module 532B is

configured such that the capacitor 606 is housed in a module case having the same shape and size as that of the switch module 532A. Similarly to the switch module 532A, the capacitor module 532B is fixed to the outer peripheral wall WA1 in a state where the side surface of the module case 611 is in contact with the inner peripheral surface of the inner wall member 542 of the inverter housing 531.

The switch module 532A and the capacitor module 532B do not necessarily have to be arranged concentrically on the radial inside of the outer peripheral wall WA1 of the inverter housing 531. For example, the switch module 532A may be arranged radially inside the capacitor module 532B, or vice versa.

When the rotating electric machine 500 is driven, heat exchange is performed between the switch module 532A and the capacitor module 532B and the cooling water passage 545 via the inner wall member 542 of the outer peripheral wall WA1. As a result, the switch module 532A and the capacitor module 532B are cooled.

Each electric module 532 may have a configuration in which cooling water is drawn into the module and the cooling water is used to cool the inside of the module. Here, the water-cooled structure of the switch module 532A will be described with reference to FIGS. 61A and 61B. FIG. 61A is a vertical cross-sectional view illustrating a cross-sectional structure of the switch module 532A in a direction crossing the outer peripheral wall WA1, and FIG. 61B is a cross-sectional view taken along a line 61B-61B of FIG. 61A.

As illustrated in FIGS. 61A and 61B, as is the case with FIG. 60, the switch module 532A has the module case 611, the switches 601 and 602 for one phase, the drive circuit 603, and the capacitor 604, as in FIG. 60. In addition, the switch module 532A has a cooling device composed of a pair of piping sections 621 and 622 and a cooler 623. In the cooling device, the pair of piping sections 621 and 622 are composed of an inflow side piping section 621 that allows cooling water to flow in from the cooling water passage 545 of the outer peripheral wall WA1 to the cooler 623 and an outflow side piping section 622 that allows cooling water to flow out from the cooler 623 to the cooling water passage 545. The cooler 623 is provided in accordance with an object to be cooled, and a one-stage or multiple-stage cooler 623 is used in the cooling device. In the configurations of FIGS. 61A and 61B, two-stage coolers 623 are provided in a direction away from the cooling water passage 545, that is, in the radial direction of the inverter unit 530, in a state of being separated from each other, and cooling water is supplied to those respective coolers 623 via the pair of piping sections 621 and 622. The cooler 623 has, for example, a hollow inside. However, an inner fin may be provided inside the cooler 623.

In the configuration including the two-stage coolers 623, (1) the outer peripheral wall WA1 side of the first-stage cooler 623, (2) between the first-stage and second-stage coolers 623, and (3) the opposite-to-outer peripheral wall side of the second stage cooler 623 is the place where the electric components to be cooled are placed, and each of these places is (2), (1), and (3) in order from the one with the highest cooling performance. That is, the place sandwiched between the two coolers 623 has the highest cooling performance, and in the place adjacent to any one of the coolers 623, the place closer to the outer peripheral wall WA1 (cooling water passage 545) has higher cooling performance. Taking this into consideration, in the configurations illustrated in FIGS. 61A and 61B, the switches 601 and 602 are arranged (2) between the first-stage and second-stage

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coolers 623, the condenser 604 is arranged on (1) the outer peripheral wall WA1 side of the first-stage cooler 623, and the drive circuit 603 is arranged on (3) the opposite-to-outer peripheral wall side of the second-stage cooler 623. Moreover, although not illustrated, the drive circuit 603 and the capacitor 604 may be arranged in reverse.

In either case, in the module case 611, the switches 601 and 602 and the drive circuit 603, and the switches 601 and 602 and the capacitor 604 are electrically connected by a wiring 612, respectively. Further, since the switches 601 and 602 are located between the drive circuit 603 and the capacitor 604, the wiring 612 extending from the switches 601 and 602 toward the drive circuit 603 and the wiring 612 extending from the switches 601 and 602 toward the capacitor 604 are in a relation in which they extend in opposite directions.

As illustrated in FIG. 61B, the pair of piping sections 621 and 622 are arranged side by side in the circumferential direction, that is, on the upstream side and the downstream side of the cooling water passage 545, and cooling water flows into the cooler 623 from the inflow side piping section 621 located on the upstream side, and then the cooling water flows out from the outflow side piping section 622 located on the downstream side. Moreover, in order to promote the inflow of the cooling water into the cooling device, the cooling water passage 545 is preferably provided with a regulating section 624 that regulates the flow of cooling water, at a position between the inflow side piping section 621 and the outflow side piping section 622 when viewed in the circumferential direction. The regulating section 624 may be a blocking section that blocks the cooling water passage 545, or a throttle section that reduces the passage area of the cooling water passage 545.

FIG. 62 illustrates another cooling structure of the switch module 532A. FIG. 62A is a vertical cross-sectional view illustrating a cross-sectional structure of the switch module 532A in a direction crossing the outer peripheral wall WA1, and FIG. 62B is a cross-sectional view taken along a line 62B—62B of FIG. 62A.

The configurations of FIGS. 62A and 62B differ from the configurations of FIGS. 61A and 61B described above in that the pair of piping sections 621 and 622 in the cooling device are arranged differently, and a pair of piping sections 621 and 622 are arranged side by side in the axial direction. Further, as illustrated in FIG. 62C, in the cooling water passage 545, the passage portion communicating with the inflow side piping section 621 and the passage portion communicating with the outflow side piping section 622 are separated in the axial direction. Each of these passage portions is communicated with each other through the respective piping sections 621 and 622 and each cooler 623.

In addition, the following configuration can be used as the switch module 532A.

In the configuration illustrated in FIG. 63A, the cooler 623 is changed from two stages to one stage as compared with the configuration illustrated in FIG. 61A. In this case, the place where the cooling performance is highest in the module case 611 is different from that in FIG. 61A, and the cooling performance is highest in the place on the outer peripheral wall WA1 side of both sides in the radial direction (both sides in the right-left direction in the figure) of the cooler 623, and then the cooling performance is lowered in the order of the place on the opposite-to-outer peripheral wall side of the cooler 623 and the place away from the cooler 623. Taking this into consideration, in the configuration illustrated in FIG. 63A, the switches 601 and 602 are arranged on the outer peripheral wall WA1 side of both sides

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in the radial direction (both sides in the right-left direction in the figure) of the cooler 623, the capacitor 604 is arranged on the opposite-to-outer peripheral wall side of the cooler 623, and the drive circuit 603 is arranged in the place away from the cooler 623.

Further, in the switch module 532A, it is possible to change the configuration in which the switches 601 and 602 for one phase, the drive circuit 603, and the capacitor 604 are housed in the module case 611. For example, the module case 611 may house either one of the switches 601 and 602 for one phase and the drive circuit 603 and the capacitor 604.

In FIG. 63B, the pair of piping sections 621 and 622 and the two-stage coolers 623 are provided in the module case 611, the switches 601 and 602 are arranged between the first-stage and second-stage coolers 623, and the capacitor 604 or the drive circuit 603 is arranged on the outer peripheral wall WA1 side of the first stage cooler 623. Further, it is also possible to integrate the switches 601 and 602 and the drive circuit 603 into a semiconductor module, and to house the semiconductor module and the capacitor 604 in the module case 611.

Moreover, in FIG. 63B, in the switch module 532A, a capacitor is preferably arranged on the side opposite to the switches 601 and 602 in at least one of the coolers 623 arranged on both sides with the switches 601 and 602 therebetween. That is, there may be a configuration in which the capacitor 604 is arranged only on one of the outer peripheral wall WA1 side of the first-stage cooler 623 and the opposite-to-peripheral wall side of the second-stage cooler 623, or a configuration in which the capacitor 604 is arranged on the both sides.

In the present embodiment, of the switch module 532A and the capacitor module 532B, only the switch module 532A is configured to draw cooling water from the cooling water passage 545 into the module. However, the configuration may be changed in such a manner that cooling water is drawn into both modules 532A and 532B from the cooling water passage 545.

Further, it is also possible to cool each electric module 532 by directly applying cooling water to the outer surface of each electric module 532. For example, as illustrated in FIG. 64, by embedding the electric module 532 in the outer peripheral wall WA1, the cooling water is applied to the outer surface of the electric module 532. In this case, it is conceivable to immerse a part of the electric module 532 in the cooling water passage 545, or to expand the cooling water passage 545 further in the radial direction than in the configuration illustrated in FIG. 58 or the like to immerse all the electric modules 532 in the cooling water passage 545. In the case where the electric module 532 is immersed in the cooling water passage 545, the cooling performance can be further improved by providing a fin in the module case 611 (the immersed portion of the module case 611) to be immersed.

Further, the electric module 532 includes a switch module 532A and a capacitor module 532B, and there is a difference in the amount of heat generated when both of them are compared. In consideration of this point, it is also possible to devise the arrangement of each electric module 532 in the inverter housing 531.

For example, as illustrated in FIG. 65, a plurality of switch modules 532A are arranged in the circumferential direction without being dispersed, and arranged on the upstream side of the cooling water passage 545, that is, on the side close to the inlet passage 571. In this case, the cooling water flowing in from the inlet passage 571 is first used for cooling the three switch modules 532A, and then

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used for cooling each capacitor module 532B. Moreover, in FIG. 65, the pair of piping sections 621 and 622 are arranged side by side in the axial direction as in the preceding FIGS. 62A and 62B, but the present invention is not limited to this, and the pair of piping sections 621 and 622 may be arranged side by side in the circumferential direction as in the preceding FIGS. 61A and 61B.

Next, the configuration related to the electrical connection in each electric module 532 and the bus bar module 533 will be described. FIG. 66 is a cross-sectional view taken along a line 66-66 of FIG. 49, and FIG. 67 is a cross-sectional view taken along a line 67-67 of FIG. 49. FIG. 68 is a perspective view illustrating the busbar module 533 alone. Here, the configuration related to the electrical connection in each electric module 532 and the bus bar module 533 will be described with reference to each of these figures.

As illustrated in FIG. 66, in the inverter housing 531, at positions adjacent to each other in the circumferential direction of the protruding section 573 provided on the inner wall member 542 (that is, the protruding section 573 provided with the inlet passage 571 and the outlet passage 572 communicating with the cooling water passage 545), three switch modules 532A are arranged side by side in the circumferential direction, and next to them, six capacitor modules 532B are arranged side by side in the circumferential direction. As the outline, in the inverter housing 531, the inside of the outer peripheral wall WA1 is equally divided into 10 regions (that is, the number of modules+1) in the circumferential direction, and one electric module 532 is arranged in each of the nine regions, and the protruding section 573 is provided in the remaining one region. The three switch modules 532A are a U-phase module, a V-phase module, and a W-phase module.

As illustrated in FIG. 66 and the above-mentioned FIGS. 56 and 57, each electric module 532 (switch module 532A and capacitor module 532B) has a plurality of module terminals 615 extending from the module case 611. The module terminal 615 is a module input/output terminal for performing electrical input/output in each electric module 532. The module terminal 615 is provided so as to extend in the axial direction, and more specifically, the module terminal 615 is provided so as to extend from the module case 611 toward the back side of the rotor carrier 511 (outside the vehicle) (see FIG. 51).

The module terminals 615 of each electric module 532 are connected to the bus bar module 533, respectively. The number of module terminals 615 differs between the switch module 532A and the capacitor module 532B. The switch module 532A is provided with four module terminals 615, and the capacitor module 532B is provided with two module terminals 615.

Further, as illustrated in FIG. 68, the bus bar module 533 has an annular section 631 forming an annular shape, three external connection terminals 632 extending from the annular section 631 and enabling connection with external devices such as a power supply device and an ECU (electronic control unit), and a winding connection terminal 633 connected to the winding end of each phase in the stator winding 521. The bus bar module 533 corresponds to a "terminal module".

The annular section 631 is arranged in the inverter housing 531 at a position on the radial inside of the outer peripheral wall WA1 and on one side in the axial direction of each electric module 532. The annular section 631 has an annular main body formed of, for example, an insulating member such as resin, and a plurality of bus bars embedded therein. The plurality of bus bars are connected to the

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module terminal 615 of each electric module 532, each external connection terminal 632, and each phase winding of the stator winding 521. The details will be described below.

The external connection terminal 632 is composed of a high potential side power terminal 632A and a low potential side power terminal 632B connected to the power supply device, and one signal terminal 632C connected to an external ECU. Each of these external connection terminals 632 (632A to 632C) is provided so as to be arranged in a line in the circumferential direction and to extend in the axial direction on the radial inside of the annular section 631. As illustrated in FIG. 51, in a state where the bus bar module 533 is assembled to the inverter housing 531 together with each electric module 532, one end of the external connection terminal 632 is configured to protrude from the end plate 547 of the boss forming member 543. Specifically, as illustrated in FIGS. 56 and 57, the end plate 547 of the boss forming member 543 is provided with an insertion hole 547a, a cylindrical grommet 635 is attached to the insertion hole 547a, and the external connection terminal 632 is provided with the grommet 635 inserted. The grommet 635 also functions as a sealed connector.

The winding connection terminal 633 is a terminal connected to the winding end of each phase of the stator winding 521, and is provided so as to extend radially outward from the annular section 631. The winding connection terminal 633 has a winding connection terminal 633U connected to the end of the U-phase winding in the stator winding 521, a winding connection terminal 633V connected to the end of the V-phase winding, and a winding connection terminal 633W connected to each connection at the end of the W-phase winding. It is preferable to provide a current sensor 634 that detects the current (U-phase current, V-phase current, W-phase current) flowing through each of these winding connection terminals 633 and each phase winding (see FIG. 70).

Moreover, the current sensor 634 may be arranged outside the electric module 532 and around each winding connection terminal 633, or may be arranged inside the electric module 532.

Here, the connection between each electric module 532 and the bus bar module 533 will be described more specifically with reference to FIGS. 69 and 70. FIG. 69 is a diagram illustrating each electric module 532 developed in a plane and schematically illustrating an electrical connection state between each electric module 532 and the bus bar module 533. FIG. 70 is a diagram schematically illustrating the connection between each electric module 532 and the bus bar module 533 in a state where each electric module 532 is arranged in an annular shape. Moreover, in FIG. 69, the path for power transmission is illustrated by a solid line, and the path of the signal transmission system is illustrated by a dashed line. FIG. 70 illustrates only the path for power transmission.

The bus bar module 533 has a first bus bar 641, a second bus bar 642, and a third bus bar 643 as bus bars for power transmission. Of these, the first bus bar 641 is connected to the high potential side power terminal 632A, and the second bus bar 642 is connected to the low potential side power terminal 632B. Further, three third bus bars 643 are connected to the U-phase winding connection terminal 633U, the V-phase winding connection terminal 633V, and the W-phase winding connection terminal 633W, respectively.

Further, the winding connection terminal 633 and the third bus bar 643 are portions that easily generate heat due to the operation of the rotating electric machine 10. Therefore, a terminal block (not illustrated) may be interposed

between the winding connection terminal **633** and the third bus bar **643**, and the terminal block may be brought into contact with the inverter housing **531** having the cooling water passage **545**. Alternatively, the winding connection terminal **633** or the third bus bar **643** may be bent into a crank shape to bring the winding connection terminal **633** or the third bus bar **643** into contact with the inverter housing **531** having the cooling water passage **545**.

With such a configuration, the heat generated at the winding connection terminal **633** and the third bus bar **643** can be dissipated to the cooling water in the cooling water passage **545**.

Moreover, in FIG. **70**, the first bus bar **641** and the second bus bar **642** are illustrated as bus bars having an annular shape, but each of these bus bars **641** and **642** does not necessarily have to be connected in an annular shape and may have a substantially C-shape with a part discontinuous in the circumferential direction. Further, each winding connection terminal **633U**, **633V**, and **633W** may be individually connected to the switch module **532A** corresponding to each phase, and therefore may be directly connected to each switch module **532A** (actually, the module terminal **615**) without going through the bus bar module **533**.

Meanwhile, each switch module **532A** has four module terminals **615** composed of a positive electrode side terminal, a negative electrode side terminal, a winding terminal, and a signal terminal. Of these, the positive electrode side terminal is connected to the first bus bar **641**, the negative electrode side terminal is connected to the second bus bar **642**, and the winding terminal is connected to the third bus bar **643**.

Further, the bus bar module **533** has a fourth bus bar **644** as a bus bar of the signal transmission system. The signal terminal of each switch module **532A** is connected to the fourth bus bar **644**, and the fourth bus bar **644** is connected to the signal terminal **632C**.

In the present embodiment, the control signal for each switch module **532A** is input from the external ECU via the signal terminal **632C**. That is, the respective switches **601** and **602** in each switch module **532A** are turned on/off by a control signal input via the signal terminal **632C**. Therefore, each switch module **532A** is connected to the signal terminal **632C** without going through a control device disposed in the rotating electric machine on the way. However, it is also possible to change this configuration in such a manner that a rotating electric machine has a built-in control device and the control signal from the control device is input to each switch module **532A**. Such a configuration is illustrated in FIG. **71**.

In the configuration of FIG. **71**, a control board **651** on which a control device **652** is mounted is provided, and the control device **652** is connected to each switch module **532A**. Further, the signal terminal **632C** is connected to the control device **652**. In this case, the control device **652** inputs a command signal related to power running or power generation from, for example, an external ECU which is a higher-level control device, and appropriately turns on/off the switches **601** and **602** of each switch module **532A** on the basis of the command signal.

In the inverter unit **530**, the control board **651** is preferably arranged on the further outside of the vehicle than the bus bar module **533** (back side of the rotor carrier **511**). Alternatively, the control board **651** may be arranged between each electric module **532** and the end plate **547** of the boss forming member **543**. The control board **651** is

preferably arranged in such a manner that at least a part thereof overlaps with each electric module **532** in the axial direction.

Further, each capacitor module **532B** has two module terminals **615** composed of a positive electrode side terminal and a negative electrode side terminal, the positive electrode side terminal is connected to the first bus bar **641**, and the negative electrode side terminal is connected to the second bus bar **642**.

As illustrated in FIGS. **49** and **50**, the protruding section **573** having the inlet passage **571** and the outlet passage **572** for cooling water is provided in the inverter housing **531** at a position aligned with each electric module **532** in the circumferential direction, and the external connection terminals **632** is provided so as to be adjacent to the protruding section **573** in the radial direction. In other words, the protruding section **573** and the external connection terminal **632** are provided at the same angular position in the circumferential direction. In the present embodiment, the external connection terminal **632** is provided at a position on the radial inside of the protruding section **573**. Further, when viewed from the inside of the vehicle of the inverter housing **531**, the end plate **547** of the boss forming member **543** is provided with the water channel port **574** and the external connection terminal **632** arranged side by side in the radial direction (see FIG. **48**).

In this case, by arranging the protruding section **573** and the external connection terminal **632** side by side in the circumferential direction together with the plurality of electric modules **532**, the inverter unit **530** can be downsized, and thus the rotating electric machine **500** can be downsized.

Referring to FIGS. **45** and **47** illustrating the structure of the wheel **400**, the cooling pipe **H2** is connected to the water channel port **574**, the electric wiring **H1** is connected to the external connection terminal **632**, and in that state, the electric wiring **H1** and the cooling pipe **H2** are housed in the housing duct **440**.

Moreover, in the above configuration, the three switch modules **532A** are arranged side by side in the circumferential direction next to the external connection terminal **632** in the inverter housing **531**, and next to them, the six capacitor modules **532B** are arranged side by side in the circumferential direction. However, this may be changed. For example, the three switch modules **532A** may be arranged side by side at a position farthest from the external connection terminal **632**, that is, a position opposite to the external connection terminal **632** with the rotating shaft **501** therebetween. Further, it is also possible to disperse each switch module **532A** in such a manner that the capacitor modules **532B** are arranged on both sides of each switch module **532A**.

If each switch module **532A** is arranged at the position farthest from the external connection terminal **632**, that is, a position opposite to the external connection terminal **632** with the rotating shaft **501** therebetween, a malfunction or the like caused by mutual inductance between the external connection terminal **632** and each switch module **532A** can be suppressed.

Next, the configuration of a resolver **660** provided as a rotation angle sensor will be described.

As illustrated in FIGS. **49** to **51**, the inverter housing **531** is provided with a resolver **660** that detects the electrical angle θ of the rotating electric machine **500**. The resolver **660** is an electromagnetic induction type sensor, and includes a resolver rotor **661** fixed to the rotating shaft **501** and a resolver stator **662** arranged so as to face the radial outside of the resolver rotor **661**. The resolver rotor **661** has

a disc ring shape, and is provided coaxially with the rotating shaft **501** with the rotating shaft **501** inserted. The resolver stator **662** includes an annular stator core **663** and a stator coil **664** wound around a plurality of teeth formed on the stator core **663**. The stator coil **664** includes a one-phase excitation coil and a two-phase output coil.

The exciting coil of the stator coil **664** is excited by a sinusoidal excitation signal, and the magnetic flux generated in the exciting coil by the excitation signal interlinks a pair of output coils. In doing so, since the relative arrangement relation between the exciting coil and the pair of output coils changes periodically in accordance with the rotation angle of the resolver rotor **661** (that is, the rotation angle of the rotation shaft **501**), the amount of magnetic flux interlinking the pair of output coils changes periodically. In the present embodiment, the pair of output coils and the exciting coil are arranged in such a manner that the phases of the voltages generated in the pair of output coils are shifted by $\pi/2$ from each other. As a result, the output voltage of each of the pair of output coils becomes a modulated wave in which the excitation signal is modulated by the modulated waves $\sin \theta$, and $\cos \theta$, respectively. More specifically, when the excitation signal is " $\sin \Omega t$ ", the modulated waves are " $\sin \theta \cdot \sin \Omega t$ " and " $\cos \theta \cdot \sin \Omega t$ ", respectively.

The resolver **660** has a resolver digital converter. The resolver digital converter calculates the electrical angle θ by detection based on the generated modulated wave and the excitation signal. For example, the resolver **660** is connected to the signal terminal **632C**, and the calculation result of the resolver digital converter is output to an external device via the signal terminal **632C**. Further, in a case where the rotating electric machine **500** has a built-in control device, the calculation result of the resolver digital converter is input to the control device.

Here, the assembly structure of the resolver **660** in the inverter housing **531** will be described.

As illustrated in FIGS. **49** and **51**, the boss section **548** of the boss forming member **543** constituting the inverter housing **531** has a hollow tubular shape, and on the inner peripheral side of the boss section **548**, a protruding portion **548a** extending in a direction orthogonal to the axial direction is formed. Then, the resolver stator **662** is fixed by a screw or the like in a state of being in contact with the protruding section **548a** in the axial direction. In the boss section **548**, the bearing **560** is provided on one side in the axial direction with the protruding section **548a** therebetween, and the resolver **660** is coaxially provided on the other side.

Further, in the hollow portion of the boss section **548**, a protruding section **548a** is provided on one side of the resolver **660** in the axial direction, and a disc ring-shaped housing cover **666** that closes the housing space of the resolver **660** is attached on the other side. The housing cover **666** is made of a conductive material such as carbon fiber reinforced plastic (CFRP). A hole **666a** through which the rotating shaft **501** is inserted is formed in the central portion of the housing cover **666**. In the hole **666a**, a sealing material **667** that seals the airspace therebetween with the outer peripheral surface of the rotating shaft **501**. The resolver housing space is sealed by the sealing material **667**. The sealing material **667** is preferably, for example, a sliding seal made of a resin material.

The space in which the resolver **660** is housed is a space surrounded by the boss section **548** forming an annular shape in the boss forming member **543** and sandwiched between the bearing **560** and the housing cover **666** in the axial direction, and the circumference of the resolver **660** is

surrounded by a conductive material. This makes it possible to suppress the influence of electromagnetic noise on the resolver **660**.

Further, as described above, the inverter housing **531** has the outer peripheral wall **WA1** and the inner peripheral wall **WA2** that together form a double wall (see FIG. **57**), the stator **520** is arranged on the outside of the double peripheral walls (outside the outer peripheral wall **WA1**), the electric module **532** is arranged between the double peripheral walls (between **WA1** and **WA2**), and the resolver **660** is arranged inside the double peripheral walls (inside the inner peripheral wall **WA2**). Since the inverter housing **531** is a conductive member, the stator **520** and the resolver **660** are arranged so as to be separated from each other by a conductive partition wall (particularly a double conductive partition wall in the present embodiment), and the occurrence of mutual magnetic interference between the stator **520** side (magnetic circuit side) and the resolver **660** can be suitably suppressed.

Next, the rotor cover **670** provided on the open end side of the rotor carrier **511** will be described.

As illustrated in FIGS. **49** and **51**, one side of the rotor carrier **511** in the axial direction is open, and the substantially disc ring-shaped rotor cover **670** is attached to the open end. The rotor cover **670** is preferably fixed to the rotor carrier **511** by any joining method such as welding, adhesion, or screwing. It is more preferable that the rotor cover **670** has a portion whose dimension is set smaller than the inner circumference of the rotor carrier **511** in such a manner that the movement of the magnet unit **512** in the axial direction can be suppressed. The outer diameter dimension of the rotor cover **670** matches the outer diameter dimension of the rotor carrier **511**, and the inner diameter dimension of the rotor cover **670** is slightly larger than the outer diameter dimension of the inverter housing **531**. The outer diameter dimension of the inverter housing **531** and the inner diameter dimension of the stator **520** are the same.

As described above, the stator **520** is fixed to the radial outside of the inverter housing **531**. At the joint portion where the stator **520** and the inverter housing **531** are joined to each other, the inverter housing **531** protrudes axially with respect to the stator **520**. In addition, the rotor cover **670** is attached so as to surround the protruding portion of the inverter housing **531**. In this case, a sealing material **671** that seals the gap between the end face of the rotor cover **670** on the inner peripheral side and the outer peripheral surface of the inverter housing **531** is provided. The housing space of the magnet unit **512** and the stator **520** is sealed by the sealing material **671**. The sealing material **671** is preferably, for example, a sliding seal made of a resin material.

According to the present embodiment described in detail above, the following excellent effects can be obtained.

In the rotating electric machine **500**, the outer peripheral wall **WA1** of the inverter housing **531** is arranged radially inside the magnetic circuit section composed of the magnet unit **512** and the stator winding **521**, and the cooling water passage **545** is formed on the outer peripheral wall **WA1**. Further, a plurality of electric modules **532** are arranged in the circumferential direction along the outer peripheral wall **WA1**, on the radial inside of the outer peripheral wall **WA1**. As a result, the magnetic circuit section, the cooling water passage **545**, and the power converter can be arranged so as to be stacked in the radial direction of the rotating electric machine **500**, and an efficient component arrangement is possible while reducing the dimensions in the axial direction. Further, the plurality of electric modules **532** constituting the power converter can be efficiently cooled. As a

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result, the high efficiency and downsizing of the rotating electric machine 500 can be achieved.

The electric module 532 (switch module 532A, capacitor module 532B) having heat-generating components such as a semiconductor switching element and a capacitor is provided in contact with the inner peripheral surface of the outer peripheral wall WA1. As a result, the heat in each electric module 532 is transferred to the outer peripheral wall WA1, and the electric module 532 is suitably cooled by the heat exchange in the outer peripheral wall WA1.

In the switch module 532A, the coolers 623 are arranged on both sides of the switches 601 and 602, respectively, and in at least one of the coolers 623 on both sides of the switches 601 and 602, the capacitor 604 is arranged on the side opposite to the switches 601 and 602. As a result, the cooling performance for the switches 601 and 602 can be improved, and the cooling performance of the capacitor 604 can also be improved.

In the switch module 532A, the coolers 623 are arranged on both sides of the switches 601 and 602, respectively, and in one of the coolers 623 on both sides of the switches 601 and 602, the drive circuit 603 is arranged on the side opposite to the switches 601 and 602, and in the other of the coolers 623, the capacitor 604 is arranged on the side opposite to the switches 601 and 602. As a result, the cooling performance for the switches 601 and 602 can be improved, and the cooling performance of the drive circuit 603 and the capacitor 604 can also be improved.

For example, in the switch module 532A, cooling water flows into the module from the cooling water passage 545, and the semiconductor switching element or the like is cooled by the cooling water. In this case, the switch module 532A is cooled by heat exchange by the cooling water inside the module in addition to heat exchange by the cooling water on the outer peripheral wall WA1. As a result, the cooling effect of the switch module 532A can be enhanced.

In the cooling system in which the cooling water flows into the cooling water passage 545 from the external circulation path 575, the switch module 532A is arranged on the upstream side near the inlet passage 571 of the cooling water passage 545, and the capacitor module 532B is arranged on the downstream side of the switch module 532A. In this case, assuming that the cooling water flowing through the cooling water passage 545 is lower in temperature toward the upstream side, it is possible to implement a configuration in which the switch module 532A is preferentially cooled.

The interval between the electric modules adjacent to each other in the circumferential direction is partially widened, and the protruding section 573 having the inlet passage 571 and the outlet passage 572 is provided in the portion where the interval is widened (second interval INT2). As a result, the inlet passage 571 and the outlet passage 572 of the cooling water passage 545 can be suitably formed in the portion that is radially inside of the outer peripheral wall WA1. That is, in order to improve the cooling performance, it is necessary to secure the flow amount of the refrigerant, and for that purpose, it is conceivable to increase the opening areas of the inlet passage 571 and the outlet passage 572. In this regard, as described above, by partially widening the interval between the electric modules and providing the protruding section 573, the inlet passage 571 and the outlet passage 572 having a desired size can be suitably formed.

The external connection terminal 632 of the bus bar module 533 is arranged at a position radially aligned with the protruding section 573 on the radial inside of the outer peripheral wall WA1. That is, the external connection terminal 632 is arranged together with the protruding section

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573 in the portion where the interval between the electric modules adjacent to each other in the circumferential direction is widened (the portion corresponding to the second interval INT2). As a result, the external connection terminal 632 can be suitably arranged while avoiding interference with each electric module 532.

In the outer rotor type rotating electric machine 500, the stator 520 is fixed to the radial outside of the outer peripheral wall WA1, and a plurality of electric modules 532 are arranged on the radial inside. As a result, the heat of the stator 520 is transferred to the outer peripheral wall WA1 from the radial outside, and the heat of the electric module 532 is transferred from the radial inside. In this case, the stator 520 and the electric module 532 can be cooled at the same time by the cooling water flowing the cooling water passage 545, and the heat of the heat-generating member in the rotating electric machine 500 can be efficiently released.

The electric module 532 on the radial inside and the stator winding 521 on the radial outside are electrically connected by the winding connection terminal 633 of the bus bar module 533 with the outer peripheral wall WA1 therebetween. Further, in this case, the winding connection terminal 633 is provided at a position axially separated from the cooling water passage 545. As a result, even in a configuration in which the cooling water passage 545 is formed in an annular shape on the outer peripheral wall WA1, that is, the inside and outside of the outer peripheral wall WA1 are separated by the cooling water passage 545, the electric module 532 and the stator winding 521 can be suitably connected.

In the rotating electric machine 500 of the present embodiment, by reducing or eliminating the teeth (iron core) between the respective conductors 523 arranged in the circumferential direction in the stator 520, the torque limitation caused by the magnetic saturation between the respective conductors 523 is suppressed, and the torque decrease is suppressed by making the conductor 523 flat and thin. In this case, even if the outer diameter dimension of the rotating electric machine 500 is the same, the region on the radial inside of the magnetic circuit section can be expanded by reducing the thickness of the stator 520, and with the use of the inner region, the outer peripheral wall WA1 having the cooling water passage 545 and the plurality of electric modules 532 provided radially inside the outer peripheral wall WA1 can be suitably arranged.

In the rotating electric machine 500 of the present embodiment, the magnet magnetic flux in the magnet unit 512 is collected on the d-axis side, and thus the magnet magnetic flux on the d-axis is strengthened, and the torque can be increased accordingly. In this case, as the radial thickness dimension can be reduced (thinned) in the magnet unit 512, the region on the radial inside of the magnetic circuit section can be expanded by reducing the thickness of the stator 520, and with the use of the inner region, the outer peripheral wall WA1 having the cooling water passage 545 and the plurality of electric modules 532 provided radially inside the outer peripheral wall WA1 can be suitably arranged.

Further, not only the magnetic circuit section, the outer peripheral wall WA1, and the plurality of electric modules 532, but also the bearing 560 and the resolver 660 can be suitably arranged in the radial direction in the same manner.

The wheel 400 using the rotating electric machine 500 as an in-wheel motor is mounted on a vehicle body via the base plate 405 fixed to the inverter housing 531 and a mounting mechanism such as a suspension device. Here, since the rotating electric machine 500 has been downsized, it is

possible to save space even if it is assumed to be assembled to a vehicle body. Therefore, it is possible to implement an advantageous configuration in expanding the installation region of the power supply device such as a battery in the vehicle and expanding the vehicle interior space.

A modification on an in-wheel motor will be described below.

(First Modification in an In-Wheel Motor)

In the rotating electric machine 500, the electric module 532 and the bus bar module 533 are arranged radially inside the outer peripheral wall WA1 of the inverter unit 530, and the electric module 532 and the bus bar module 533 and the stator 520 are arranged radially inside and outside so as to be separated from each other by the outer peripheral wall WA1, respectively. In such a configuration, the position of the bus bar module 533 with respect to the electric module 532 can be arbitrarily set. Further, when connecting each phase winding of the stator winding 521 and the bus bar module 533 across the outer peripheral wall WA1 in the radial direction, a winding connection wire (for example, the winding connection terminal 633) used for the connection can be arbitrarily set.

That is, as the position of the bus bar module 533 with respect to the electric module 532, a configuration ($\alpha 1$) in which the bus bar module 533 is located further outside of the vehicle in the axial direction than the electric module 532, that is, on the back side in the rotor carrier 511 side and a configuration ($\alpha 2$) in which the bus bar module 533 is located further inside of the vehicle in the axial direction than the electric module 532, that is, on the front side in the rotor carrier 511 side are conceivable.

Further, as a position to guide the winding connection winding connection wire, a configuration ($\beta 1$) in which the winding connection wire is guided in the axial direction on the outside of the vehicle, that is, on the back side in the rotor carrier 511 side and a configuration ($\beta 2$) in which the winding connection wire is guided in the axial direction on the inside of the vehicle, that is, on the front side in the rotor carrier 511 side are conceivable.

Hereinafter, four configuration examples relating to the arrangement of the electric module 532, the bus bar module 533, and the winding connection wire will be described with reference to FIGS. 72A to 72D. FIGS. 72A to 72D are vertical cross-sectional views illustrating a simplified configuration of the rotating electric machine 500, in which the same reference signs are given to the configurations already described. The winding connection wire 637 is an electric wiring that connects each phase winding of the stator winding 521 and the bus bar module 533, and for example, the winding connection terminal 633 described above corresponds to this.

In the configuration of FIG. 72A, the above ($\alpha 1$) is adopted as the position of the bus bar module 533 with respect to the electric module 532, and the above ($\beta 1$) is adopted as the position for guiding the winding connection wire 637. That is, the electric module 532, the bus bar module 533, the stator winding 521, and the bus bar module 533 are all connected on the outside of the vehicle (the back side of the rotor carrier 511). This corresponds to the configuration illustrated in FIG. 49.

According to this configuration, the cooling water passage 545 can be provided on the outer peripheral wall WA1 without fear of interference with the winding connection wire 637. Further, the winding connection wire 637 that connects the stator winding 521 and the bus bar module 533 can be easily achieved.

In the configuration of FIG. 72B, the above ($\alpha 1$) is adopted as the position of the bus bar module 533 with respect to the electric module 532, and the above ($\beta 2$) is adopted as the position for guiding the winding connection wire 637. That is, the electric module 532 and the bus bar module 533 are connected on the outside of the vehicle (the back side of the rotor carrier 511), and the stator winding 521 and the bus bar module 533 are connected on the inside of the vehicle (the front side of the rotor carrier 511).

According to this configuration, the cooling water passage 545 can be provided on the outer peripheral wall WA1 without fear of interference with the winding connection wire 637.

In the configuration of FIG. 72C, the above ($\alpha 2$) is adopted as the position of the bus bar module 533 with respect to the electric module 532, and the above ($\beta 1$) is adopted as the position for guiding the winding connection wire 637. That is, the electric module 532 and the bus bar module 533 are connected on the inside of the vehicle (the front side of the rotor carrier 511), and the stator winding 521 and the bus bar module 533 are connected on the outside of the vehicle (the back side of the rotor carrier 511).

In the configuration of FIG. 72D, the above ($\alpha 2$) is adopted as the position of the bus bar module 533 with respect to the electric module 532, and the above ($\beta 2$) is adopted as the position for guiding the winding connection wire 637. That is, the electric module 532, the bus bar module 533, the stator winding 521, and the bus bar module 533 are all connected on the inside of the vehicle (the front side of the rotor carrier 511).

According to the configurations of FIGS. 72C and 72D, the bus bar module 533 is arranged inside the vehicle (on the front side of the rotor carrier 511), and thus it is considered the wiring becomes easy when adding an electric component such as a fan motor. Further, it is possible that the bus bar module 533 can be brought closer to the resolver 660 arranged further inside the vehicle than the bearing, and it is considered that wiring to the resolver 660 becomes easier.

(Second Modification in an In-Wheel Motor)

A modification of the mounting structure of the resolver rotor 661 will be described below. That is, the rotating shaft 501, the rotor carrier 511, and the inner ring 561 of the bearing 560 are a rotating body that rotates integrally, and a modification of the mounting structure of the resolver rotor 661 with respect to the rotating body will be described below.

FIGS. 73A to 73C are block diagrams illustrating an example of a mounting structure of the resolver rotor 661 to the rotating body. In any of the configurations, the resolver 660 is provided in a closed space surrounded by the rotor carrier 511, the inverter housing 531 and the like, and protected from external water, mud, and the like. Of FIGS. 73A to 73C, in FIG. 73A, the bearing 560 has the same configuration as that in FIG. 49. Further, in FIGS. 73B and 73C, the bearing 560 has a configuration different from that of FIG. 49, and is arranged at a position away from the end plate 514 of the rotor carrier 511. In each of these figures, two locations are illustrated as mounting locations for the resolver rotor 661. Moreover, although the resolver stator 662 is not illustrated, for example, the boss section 548 of the boss forming member 543 should be extended to the outer peripheral side of the resolver rotor 661 or its vicinity, and the resolver stator 662 should be fixed to the boss section 548.

In the configuration of FIG. 73A, the resolver rotor 661 is attached to the inner ring 561 of the bearing 560. Specifically, the resolver rotor 661 is provided on the axial end face

of the flange **561b** of the inner ring **561**, or is provided on the axial end face of the tubular section **561a** of the inner ring **561**.

In the configuration of FIG. **73B**, the resolver rotor **661** is attached to the rotor carrier **511**. Specifically, the resolver rotor **661** is provided on the inner surface of the end plate **514** in the rotor carrier **511**. Alternatively, in a configuration in which the rotor carrier **511** has a tubular section **515** extending from the inner peripheral edge portion of the end plate **514** along the rotating shaft **501**, the resolver rotor **661** is provided on the outer peripheral surface of the tubular section **515** of the rotor carrier **511**. In the latter case, the resolver rotor **661** is arranged between the end plate **514** of the rotor carrier **511** and the bearing **560**.

In the configuration of FIG. **73C**, the resolver rotor **661** is attached to the rotating shaft **501**. Specifically, on the rotating shaft **501**, the resolver rotor **661** is provided between the end plate **514** of the rotor carrier **511** and the bearing **560**. Alternatively, on the rotating shaft **501**, the resolver rotor **661** is arranged on the side opposite to the rotor carrier **511** with the bearing **560** therebetween. (Third Modification in an In-Wheel Motor)

A modification of the inverter housing **531** and the rotor cover **670** will be described below with reference to FIG. **74**. FIGS. **74A** and **74B** are vertical cross-sectional views illustrating a simplified configuration of the rotating electric machine **500**, in which the same reference signs are given to the configurations already described. Moreover, the configuration illustrated in FIG. **74A** substantially corresponds to the configuration described with reference to FIG. **49** and the like, and the configuration illustrated in FIG. **74B** corresponds to the configuration in which a part of the configuration of FIG. **74A** is modified.

In the configuration illustrated in FIG. **74A**, the rotor cover **670** fixed to the open end of the rotor carrier **511** is provided so as to surround the outer peripheral wall **WA1** of the inverter housing **531**. That is, the end face on the inner diameter side of the rotor cover **670** faces the outer peripheral surface of the outer peripheral wall **WA1**, and the sealing material **671** is provided between them. Further, the housing cover **666** is attached to the hollow portion of the boss section **548** of the inverter housing **531**, and the sealing material **667** is provided between the housing cover **666** and the rotating shaft **501**. The external connection terminal **632** constituting the bus bar module **533** penetrates the inverter housing **531** and extends to the inside of the vehicle (lower side in the figure).

Further, in the inverter housing **531**, the inlet passage **571** and the outlet passage **572** communicating with the cooling water passage **545** are formed, and the water channel port **574** including the passage ends of the inlet passage **571** and the outlet passage **572** is formed.

On the other hand, in the configuration illustrated in FIG. **74B**, the inverter housing **531** (specifically, the boss forming member **543**) is formed with an annular protrusion **681** extending toward the protruding side (inside the vehicle) of the rotating shaft **501**. The rotor cover **670** is provided so as to surround the protrusion **681** of the inverter housing **531**. That is, the end face on the inner diameter side of the rotor cover **670** faces the outer peripheral surface of the protrusion **681**, and the sealing material **671** is provided between them. Further, the external connection terminal **632** constituting the bus bar module **533** penetrates the boss section **548** of the inverter housing **531** and extends into the hollow region of the boss section **548**, and also penetrates the housing cover **666** and extends to the inside of the vehicle (lower side of the figure).

Further, the inverter housing **531** is formed with the inlet passage **571** and the outlet passage **572** communicating with the cooling water passage **545**, those inlet passage **571** and outlet passage **572** extend into the hollow region of the boss section **548** and extend to the further inside of the vehicle (lower side of the figure) than the housing cover **666** via a relay pipe **682**. In this configuration, the piping portion extending from the housing cover **666** to the inside of the vehicle is the water channel port **574**.

According to the configurations of FIGS. **74A** and **74B**, the rotor carrier **511** and the rotor cover **670** can be suitably rotated with respect to the housing **531** while maintaining airtightness of the internal space of the rotor carrier **511** and the rotor cover **670**.

Moreover, in particular, according to the configuration of FIG. **74B**, the inner diameter of the rotor cover **670** is smaller than that of the configuration of FIG. **74A**. Therefore, the inverter housing **531** and the rotor cover **670** can be provided double in the axial direction at a position further inside the vehicle than the electric module **532**, and the inconvenience caused by electromagnetic noise, which is a concern in the electric module **532**, is suppressed. Further, by reducing the inner diameter of the rotor cover **670**, the sliding diameter of the sealing material **671** can be reduced, and mechanical loss in the rotating sliding portion can be suppressed.

(Fourth Modification in an In-Wheel Motor)

A modification of the stator winding **521** will be described below. FIG. **75** illustrates a modification on the stator winding **521**.

As illustrated in FIG. **75**, in the stator winding **521**, a conductor material having a rectangular cross section is used, and is wound by a wave winding with the long side of the conductor material extending in the circumferential direction. In this case, the conductors **523** of each phase on the coil side of the stator winding **521** are arranged at predetermined pitch intervals for each phase and are connected to each other at the coil ends. The conductors **523** adjacent to each other in the circumferential direction on the coil side are in contact with each other at the end faces in the circumferential direction, or are arranged close to each other at a minute interval.

Further, in the stator winding **521**, the conductor material is bent in the radial direction for each phase at the coil end. More specifically, the stator winding **521** (conductor material) is bent inward in the radial direction at a different position for each phase in the axial direction, whereby interference with each other in the respective U-phase, V-phase, and W-phase windings is avoided. In the illustrated configuration, the conductors are bent at a right angle inward in the radial direction for each phase, with each phase winding being different by the thickness of the conductor material. In each of the conductors **523** arranged in the circumferential direction, the length dimension between both ends in the axial direction is preferably the same for each of the conductors **523**.

Moreover, in a case where the stator core **522** is assembled to the stator winding **521** to manufacture the stator **520**, a part of the annular shape of the stator winding **521** is preferably opened as a non-connection part (that is, the stator winding **521** is preferably made to be substantially C-shaped), and after assembling the stator core **522** on the inner peripheral side of the stator winding **521**, the disconnecting portions are preferably connected to each other to form the stator winding **521** in an annular shape.

In addition to the above, it is also possible to divide the stator core **522** into a plurality of parts (for example, three

or more) in the circumferential direction, and assemble the core pieces divided into a plurality of pieces onto the inner peripheral side of the stator winding 521 formed in an annular shape.

(Other Modifications)

For example, as shown in FIG. 50, in the rotating electric machine 500, the inlet passage 571 and the outlet passage 572 of the cooling water passage 545 are provided at one place. However, this configuration may be changed such that the inlet passage 571 and the outlet passage 572 may be provided at different positions in the circumferential direction. For example, the inlet passage 571 and the outlet passage 572 may be provided at different positions by 180 degrees, or at least either the inlet passage 571 or the outlet passage 572 may be provided in plural number.

According to the wheel 400 of the present disclosure, the rotating shaft 501 is configured to protrude towards one side in the axial direction of the rotating electric machine 500. However, this configuration may be modified such that the rotating shaft 501 may protrude towards both sides in the axial direction. This configuration is preferably applied to a vehicle provided with a single wheel at at least one of the front side or rear side thereof.

As the rotating electric machine used for the wheel 400, an inner rotor type rotating electric machine can be utilized.

(Fifteenth Modification)

In the above-described embodiments or the above-described modifications, the configuration of the magnet unit may be modified in the following manners.

Hereinafter, a configuration of mainly the magnet unit 700 of this modification will be described. According to the present modification, portions different from the configurations described in the above-described embodiments and modifications will be mainly described. Further, according to the present modification, as a basic configuration of the rotating electric machine, the configuration of the first embodiment will be described as an example.

As shown in FIG. 76, the magnet unit 700 includes a plurality of magnets 710 and 720 arranged in the circumferential direction and a cylindrical section 801 that supports these magnets 710 and 720. The magnets 710 and 720 are fixed to the cylindrical section 801 via a foamable resin 802 as a foamable adhesive. According to the present modification, the cylindrical section 801 corresponds to magnet supporting section. The magnets 710 and 720 are provided in a plural number and arranged alternately in the circumferential direction.

Firstly, the shapes of respective magnets 710 and 720 will be described. The magnets 710 and 720 are each formed such that the cross-sectional shape is substantially an arc shape, showing an annular shape when being arranged in the circumferential direction.

Specifically, the respective magnets 710 have a stator side circumferential surface 711 (armature side circumferential surface) in a radially inside (stator side) and an anti-stator side circumferential surface 712 (anti-armature side circumferential surface) in a radially outside (cylindrical section side). Also, each magnet 710 has a circumferential end face at both ends in the circumferential direction as a plane along the radial direction. Note that in each magnet 710, the circumferential end face in the d-axis side may be indicated by d-axis side end face 713a and the circumferential end face in the q-axis side may be indicated by q-axis side end face 713b.

Similarly, each magnet 720 has a stator side circumferential surface 721 (armature side circumferential surface) in a radially inside (stator side) and an anti-stator side circumferential surface 722 (anti-armature side circumferential surface) in a radially outside (cylindrical section side). Also, each magnet 720 has a circumferential end face at both ends in the circumferential direction as a plane along the radial direction. Note that in each magnet 720, the circumferential end face in the d-axis side may be indicated by d-axis side end face 723a and the circumferential end face in the q-axis side may be indicated by q-axis side end face 723b.

The magnets 710 and 720 are each provided to have a predetermined height dimension in the axial direction. The magnets 710 and 720 are each provided between the d-axis which is the center of the pole and the q-axis which is the magnetic pole boundary adjacently positioned in the circumferential direction with respect to the d-axis. In other words, the magnets 710 and 720 are divided by the d-axis and the q-axis. Moreover, the magnet 710 and the magnet 720 are laterally symmetric in the circumferential direction of which the center is the q-axis (or d-axis).

The stator side circumferential surfaces 711 and 721 of the respective magnets 710 and 720 are formed in substantially an arc shape along the circumferential direction. In the stator side circumferential surfaces 711 and 721 of the respective magnets 710 and 720, q-axis side inclined surfaces 711a and 721a which are inclined with respect to the stator side circumferential surfaces 711 and 721 are provided. With these q-axis side inclined surfaces 711a and 721a, when the magnets 710 and 720 are arranged in an annular shape, a first recess 705 recessed in the radial direction is provided in the q-axis side in the stator side circumferential surfaces 711 and 721.

Specifically, a corner portion between the stator side circumferential surfaces 711 and 721 of the respective magnets 710 and 720 and the q-axis side end faces 713b and 723b is chamfered. That is, the q-axis side inclined surfaces 711a and 721a, which are inclined towards the anti-stator side with respect to the stator side circumferential surfaces 711 and 721, are provided at the corner portion in the q-axis side between the stator side circumferential surfaces 711 and 721 of the respective magnets 710 and 720 and the q-axis side end faces 713b and 723b. These q-axis side inclined surfaces 711a and 721a are configured as inclined surfaces inclined radially outside.

In other words, a dimension in the radial direction from the stator 50 to the q-axis side inclined surfaces 711a and 721a is larger than a dimension in the radial direction from the stator 50 to the stator side circumferential surfaces 711 and 721. Also, a dimension in the radial direction from the stator 50 to the q-axis side inclined surfaces 711a and 721a are set such that the closer to the q-axis, the larger the dimension is.

In the case where unevenness is present on the outer peripheral surface of the stator 50, the dimension in the radial direction from the stator 50 to the stator side circumferential surfaces 711 and 721 (or q-axis side inclined surfaces 711a and 721a) refers to a dimension from an outer peripheral surface positioned at an outer most side (rotor side) of the stator 50. Further, although the dimension is defined from the stator 50 to the stator side circumferential surfaces 711 and 721 (or q-axis side inclined surfaces 711a and 721a), the dimension may be defined from the center of rotation to the stator side circumferential surfaces 711 and 721 (or q-axis side inclined surfaces 711a and 721a).

The anti-stator side circumferential surfaces 712 and 722 of the respective magnets 710 and 720 are formed in a

substantially arc shape. Further, for the anti-stator side circumferential surfaces **712** and **722** of the respective magnets **710** and **720**, d-axis side inclined surfaces **712a** and **722a**, which are each inclined with respect to the anti-stator side circumferential surfaces **712** and **722**, are provided at d-axis side end faces. With these d-axis side inclined surfaces **712a** and **722a**, when the magnets **710** and **720** are arranged in an annular shape, a second recess **706** recessed in the radial direction is provided in the d-axis side in the anti-stator side circumferential surfaces **712** and **722**.

Specifically, a corner portion between the anti-stator side circumferential surfaces **712** and **722** of the respective magnets **710** and **720** and the d-axis side end faces **713a** and **723a** is chamfered. That is, the d-axis side inclined surfaces **712a** and **722a**, which are inclined towards the stator side with respect to the anti-stator side circumferential surfaces **712** and **722**, are provided at the corner portion in the d-axis side between the anti-stator side circumferential surfaces **712** and **722** of the respective magnets **710** and **720** and the d-axis side end faces **713a** and **723a**. These d-axis side inclined surfaces **712a** and **722a** are configured as inclined surfaces inclined radially inside.

In other words, a dimension in the radial direction from the stator **50** to the d-axis side inclined surfaces **712a** and **722a** is smaller than a dimension in the radial direction from the stator **50** to the stator side circumferential surfaces **712** and **722**. Also, a dimension in the radial direction from the stator **50** to the d-axis side inclined surfaces **712a** and **722a** are set such that the closer to the d-axis, the smaller the dimension is.

In the case where unevenness is present on the outer peripheral surface of the stator **50**, the dimension in the radial direction from the stator **50** to the anti-stator side circumferential surfaces **712** and **722** (or d-axis side inclined surfaces **712a** and **722a**) refers to a dimension from an outer peripheral surface positioned at an outer most side (rotor side) of the stator **50**. Further, although the dimension is defined from the stator **50** to the anti-stator side circumferential surfaces **712** and **722** (or d-axis side inclined surfaces **712a** and **722a**), the dimension may be defined from the center of rotation to the anti-stator side circumferential surfaces **712** and **722** (or d-axis side inclined surfaces **712a** and **722a**).

Next, the magnet magnetic path of the magnets **710** and **720** will be described with reference to FIG. **76B**. These magnets **710** and **720** are each arranged such that orientation of the axis of easy magnetization in the d-axis side as the center of the magnetic pole is set to be more parallel to the d-axis compared to locations on the q-axis side as the magnetic pole boundary, and the magnet magnetic path is formed along the axis of easy magnetization. Specifically, the axis of easy magnetization of the magnets **710** and **720** is arranged to be substantially parallel to the d-axis in a portion located close to the d-axis and to be orthogonal to the q-axis or substantially orthogonal to the q-axis in a portion located close to the q-axis.

More specifically, as shown in FIG. **76B**, in the magnets **710** and **720**, a plurality of axes of easy magnetization are arranged in an arc shape with respect to an orientation center point **C10** as the center thereof set on the q-axis. These plurality of axes of easy magnetization having arc shape include an axis of easy magnetization on the arc **OA** of which the center is the orientation center point **C10** set on the q-axis, passing through the first cross point **P1** between the d-axis side end face and stator side circumferential surfaces **711** and **721**. Note that the shape of the magnet magnetic path may be an arc shape as a part of circle or an

arc shape as a part of an ellipse. Further, although the orientation center point **C10** is set on the q-axis, but may not be set on the q-axis. However, the orientation center point **C10** may preferably be set to be located closer to the q-axis side than the d-axis.

As shown in FIG. **76B**, the arc **OA** is set such that a tangential line at the first cross point **P1** on the arc **OA** is formed having a predetermined orientation inclined angle with respect to the d-axis. In more detail, the axes of easy magnetization are arranged on the magnets **710** and **720** to have an inclined angle within a predetermined angle range (e.g. 15 to 45 [deg]) with respect to the d-axis at at least first cross point **P1** between the d-axis side end face and the stator side circumferential surfaces **711** and **712**. That is, the axes of easy magnetization are arranged such that the orientation inclined angle is within the predetermined angle range.

It is found that, compared to a case where the axes of easy magnetization are set to be parallel to the d-axis, by setting the plurality of axes of easy magnetization to be inclined for a certain degree, vectors of magnetic flux flowing from the stator side circumferential surfaces **711** and **721** (flux outflow surface) are concentrated on the d-axis, thereby improving the magnetic flux density of the d-axis.

However, when setting the axes of easy magnetization to be inclined by more than certain degree, vector components of the magnetic flux in the radial direction which flows out from the stator side circumferential surfaces **711** and **721** (flux outflow surface) becomes significantly small, thereby lowering the magnetic flux. Hence, for the orientation inclined angle, the predetermined angle range is set, considering the shapes of magnets **710** and **720** and the size of the air gap, such that the magnetic density in the d-axis is larger than that of a case where the axes of easy magnetization are set to be parallel to the d-axis.

Further, the orientation center point **C10** is set to be positioned closer to a stator side than the position of the stator side circumferential surfaces **711** and **712** in the radial direction in order to set the orientation inclined angle to be within the predetermined angle range. Thus, thickness dimension of the magnets **710** and **720** in the radial direction can be reduced.

According to the above-described configuration, among the axes of easy magnetization (and the magnet magnetic path) of the magnets **710** and **720**, axes of easy magnetization (and the magnet magnetic path) along the arc **OA** are likely to be longer, and axes of easy magnetization (and the magnet magnetic path) are likely to be shorter further from the arc **OA**. For example, in a portion close to the q-axis side among magnet magnetic paths of the magnets **710** and **720**, a magnet magnetic path (indicated by a dotted line) that passes through a portion in the stator side rather than the anti-stator side is likely to be shorter. Also, for example, in a portion close to the d-axis side among magnet magnetic paths of the magnets **710** and **720**, a magnet magnetic path (indicated by dotted line) that passes through a portion in the anti-stator side rather than the stator side is likely to be shorter.

When the magnet magnetic path is shorter, demagnetization is likely to occur due to an external magnetic field, making it useless for improving the magnetic flux in the d-axis. Therefore, even when the first recess **705** is provided by the q-axis side inclined surfaces **711a** and **721a**, the magnetic density in the d-axis hardly decreases. Similarly, even when the second recess **706** is provided by the d-axis side inclined surfaces **712a** and **722a**, the magnetic density

in the d-axis hardly decreases. That is, the number of magnets can be reduced without affecting the magnetic flux density in the d-axis.

The magnet magnetic path of the magnet **710** is formed laterally symmetrical to the magnet magnetic path of the magnet **720** with respect to the d-axis in the circumferential direction, where the magnet **720** is positioned adjacently to the magnet **710** with respect to the d-axis in the circumferential direction. Note that the magnet **710** and the magnet **720** adjacently positioned to the magnet **710** with respect to the d-axis in the circumferential direction are referred to as a pair of magnets **710** and **720**.

In the magnet unit **700**, in order to make the polarities of adjacent d-axes in the circumferential direction to be different, magnetization directions (magnetizing direction) of the magnets **710** and **720** are made opposite at each pair of magnets **710** and **720**. Specifically, as shown in FIG. 76B, the magnetization direction (magnetizing direction) of the pair of magnets **710** and **720** which are paired with respect to the d-axis of which the polarity is positive (N pole) are set such that lines of magnetic flux proceed towards the d-axis. On the other hand, the magnetization direction (magnetizing direction) of the pair of magnets **710** and **720** which are paired with respect to the d-axis of which the polarity is negative (S pole) are set such that lines of magnetic flux proceed away from the d-axis.

Next, an overall manufacturing method of the magnets **710** and **720** will be described. Each of the magnets **710** and **720** is a sintered magnet manufactured by a sintering process. Specifically, materials such as neodymium, boron and iron are dissolved to be alloyed (first process). Next, the alloy produced at the first process is pulverized to be in particulate form (second process). Then, a powder obtained at the second process is introduced in a metal mold and press-molded in a magnetic field (third process). After the press-molding, the molding is sintered (fourth process) and heat treated (fifth process) after completing the sintering. During the heat treatment process, heating and cooling processes are repeated for a several times. Then, machining such as polishing and surface treatment are performed (sixth process), and thereafter, a magnetizing process is applied (seventh process), thereby producing the respective magnets **710** and **720**.

Next, the cylindrical section **801** of the magnet holder **41** according to the fifteenth modification will be described. The cylindrical section **801** is formed in a cylindrical shape. The magnets **710** and **720** are fixed to the inner peripheral surface of the cylindrical section **801** via the foamable resin **802**. Hence, the cylindrical section **801** serves as a magnet supporting section that supports the magnets **710** and **720**. Further, the inner peripheral surface of the cylindrical section **801** serves as a fixed surface of the magnets **710** and **720**.

In the magnets **710** and **720** which are adjacently positioned in the circumferential direction, their faces **713a**, **723a**, **713b**, and **723b** are in contact with each other. In other words, a gap between the magnets **710** and **720** in the circumferential direction is set to be shorter than the thickness dimension of the foamable resin **802**.

Further, a second protrusion **803** that protrudes radially inside (stator side) along the d-axis is provided on the inner peripheral surface of the cylindrical section **801**. The second protrusion **803** is formed to be accommodated in the second recess **706**.

The second protrusion **803** is provided extending over the entire region of the magnets **710** and **720** in the axial direction. Moreover, the second protrusion **803** has a side

surface capable of engaging with the d-axis side inclined surfaces **712a** and **722a** in the circumferential direction. The side surface of the second protrusion **803** is formed depending on the inclination angle of the d-axis side inclined surfaces **712a** and **722a**.

The second protrusion **803** is formed such that the cross section has a trapezoidal shape and the height dimension in the radial direction is set to be shorter than the depth dimension of the second recess **706**. Thus, for the second protrusion **803**, when being accommodated in the second recess **706**, a gap is formed between a radially-inside end section of the second protrusion **803** and the bottom section of the second recess **706**. The gap allows excess foamable resin **802** to flow into the gap.

A film **810** is provided as an annular wall member in the stator side with respect to the magnet unit **700**. The film **810** is provided for the entire region of the magnets **710** and **720** in the axial direction such that the length thereof is the same as that of the magnets **710** and **720** or more. Further, the stator side circumferential surfaces **711** and **721** of the magnets **710** and **720** are provided to come into contact with the outer peripheral surface of the film **810** (that is, circumferential surface of the magnet unit **700** side).

Also, a first protrusion **811** protruding towards radially outside along the q-axis (i.e. magnet unit **700** side) is provided on the outer peripheral surface of the film **810**. The first protrusion **811** is provided to be accommodated in the first recess **705**. For the first protrusion **811**, when being accommodated in the first recess **705**, the side surface of the first protrusion **811** is configured to be capable of engaging in the circumferential direction with the q-axis side inclined surfaces **711a** and **721a** which constitute the first recess **705**.

A predetermined air gap is provided between the inner peripheral surface of the film **810** and the stator **50**. Note that the film **810** is made of non-magnetic material such as resin. The thickness dimension of the film **810** in the radial direction is formed to be significantly thinner than the thickness dimension of the magnets **710** and **720**.

Further, as shown in FIG. 77, an annular sheet-shaped end plate **910** is disposed in the end portion of the magnet holder **41** and the magnets **710** and **720** (opening end of the magnet holder **41**). The end plate **910** regulates a movement of the magnets **710** and **720** in the axial direction, thereby preventing the magnets **710** and **720** from being detached.

In the case where a sintered magnet is utilized, since press-fitting, sintering and the like are required, a manufacturing tolerance occurs to cause a difficulty in setting the thickness dimensions of the magnets **710** and **720** to be the same. Hence, in the case where an adhesive such as an adhesive sheet having the same thickness is used for fixing the magnets **710** and **720** on the inner peripheral surface of the cylindrical section **801**, unevenness occurs on the stator side circumferential surfaces **711** and **712** of the magnets **711** and **712** because of difference in the thickness between the magnets **710** and **720**. In this case, the air gap between the rotor **40** and the stator **50** is required to be larger to avoid contact with the stator **50**, which lowers the torque.

In this respect, according to the fifteenth embodiment, as shown in FIGS. 76A and 76B, the thickness dimension of the foamable resin **802** in the radial direction is changed depending on the distance from the inner peripheral surface of the cylindrical section **801** to the anti-stator side circumferential surfaces **712** and **722** of the magnets **710** and **720**. Specifically, the thickness dimension of the foamable resin **802** in the radial direction is changed such that the stator side circumferential surfaces **711** and **721** of the magnets **710** and **720** are on the same circumferential surface of which the

center thereof is the center of rotation. In FIGS. 76A and 76B, for convenience of explanation, although the thickness dimension of the magnets 710 and 720 are intentionally shown to be different for emphasis, the difference in the dimension between the magnets 710 and 720 may preferably be as small as possible.

In a state where the respective magnets 710 and 720 are arranged in the circumferential direction, the thickness dimension of the foamable resin 802 is adjusted such that the foamable resin 802 applies pressing force from the anti-stator side circumferential surfaces 712 and 722 of the magnets 710 and 720 to the radially inside (stator side) thereof. That is, the foamable resin 802 is configured to be capable of being elastically deformed. For this reason, the thickness dimension of the foamable resin 802 is set to be slightly larger than the distance from the cylindrical section 801 to the magnets 710 and 720, and the foamable resin 802 is elastically deformed in the radial direction by the cylindrical section 801 and the magnets 710 and 720. Thus, as shown with an arrow in FIG. 76A, it is configured such that a pressing force (elastic force) is applied to the respective magnets 710 and 720.

Next, a manufacturing method of the rotor 40 will be described. According to the present modification example, after the predetermined axes of easy magnetization are oriented in the magnets 710 and 720 (after a sixth process of the manufacturing method of the magnets 710 and 720), the magnets 710 and 720 before being magnetized are fixed to the cylindrical section 801, thereafter they are magnetized (subsequent magnetization). Hereinafter, detailed description will be provided with reference to FIG. 78.

Firstly, as described above, the manufacturing step proceeds to the sixth process, predetermined axes of easy magnetization are oriented in the magnets 710 and 720. Next as shown in FIG. 79, a film 810 is disposed on a jig 900 having a cylindrical shape to cover its surface (step S1001). Thus, an annular shaped film 810 is provided. Next, as shown in FIG. 80, an annular shaped end plate 910 is disposed (S1002) in a lower end in the axial direction (vertical direction). Further, at this time, the first protrusion 811 is provided on the outer peripheral surface of the film 810.

Then, as shown in FIG. 81, the magnets 710 and 720 which are oriented with the predetermined axes of easy magnetization and not being magnetized are arranged in the circumferential direction (step S1003) so as to surround the circumference of the jig 900 covered by the film 810. That is, the film 810 is interposed between the magnets 710 and 720, and the jig 900. At this moment, the magnets 710 and 720 are arranged such that the stator side circumferential surfaces 711 and 721 of the magnets 710 and 720 come into contact with the jig 900 covered by the film 810. Further, the magnets 710 and 720 are arranged such that the first protrusion 811 of the film 810 is accommodated in the first recess 705 of the magnet unit 700. Note that the position of the lower end in the axial direction (vertical direction) is aligned with respect to the end plate 910.

Subsequently, as shown in FIG. 82, the foamable resin 802 is coated on the anti-stator side circumferential surfaces 712 and 722 of the magnets 710 and 720 (step S1004). Next, as shown in FIG. 83, the cylindrical section 801 is disposed on the jig 900 in a state where the center thereof are aligned (step S1005). At this moment, the magnets 710 and 720 are arranged on the inner peripheral surface of the cylindrical section 801. At that time, as shown in FIG. 83, the second protrusion 803 of the cylindrical section 801 is positionally-aligned in the circumferential direction such that the second

protrusion 803 of the cylindrical section 801 is accommodated in the second recess 706 provided in the magnet unit 700. The step 1005 corresponds to arrangement process. Note that for convenience of explanation, difference in the thickness dimension between the magnets 710 and 720 caused by manufacturing tolerance are intentionally shown emphasized. As shown in FIG. 83, before the heating process, the thickness dimension of the foamable resin 802 may be smaller than the distance between the cylindrical section 801 and the magnets 710 and 720. That is, a gap may be provided.

The foamable resin 802 may be coated on the inner peripheral surface of the cylindrical section 801. Further, the foamable resin 802 may be coated on both of the cylindrical section 801 and the magnets 710 and 720.

Then, the cylindrical section 801 and the magnet unit 700 are heated for each jig 900, and the foamable resin 802 is foamed, thereby fixing the magnets 710 and 720 on the cylindrical section 801 via the foamable resin 802 (step S1006). The step S1006 corresponds to heating process.

The foamable resin 802 is foamed and becomes larger by heating. Thus, by heating, the thickness dimension of the foamable resin 802 in the radial direction can be differentiated depending on the distance from the inner peripheral surface of the cylindrical section 801 to the anti-stator side circumferential surfaces 712 and 722 of the magnets 712 and 722 so as to occupy the gap between the cylindrical section 801 and the magnets 710 and 720 (see FIGS. 76A and 76B). Specifically, due to the manufacturing tolerance, the thickness dimensions of the magnets 710 and 720 are different. Hence, although the distance between the cylindrical section 801 and the magnets 710 and 720 is changed, the thickness dimension of the foamable resin 802 in the radial direction can be differentiated so as to occupy the gap.

Then, a heating process is continuously applied until the thickness dimension of the foamable resin 802 becomes slightly larger than the distance from the inner peripheral surface of the cylindrical section 801 to the anti-stator side circumferential surfaces 712 and 722. Thus, pressing force is applied to the magnets 710 and 720 from radially outside by the foamable resin 802.

At this moment, the stator side circumferential surfaces 711 and 721 of the magnets 710 and 720 are in contact with the cylindrical shaped jig 900 via the film 810. Hence, when the magnets 710 and 720 are applied with the pressing force from radially outside by the foamable resin 802, the stator side circumferential direction 711 and 721 of the magnets 710 and 720 are pressed on the outer peripheral surface and serve as the same circumferential surface.

Note that the foamable resin 802 has fluidity when being heated. Hence, excess foamable resin 802 flows into a gap formed between the second recess 706 and the second protrusion 803. Thus, the magnets 710 and 720 can be prevented from being applied with a predetermined pressing force or a pressing force larger than the predetermined pressing force.

The magnets 710 and 720 are heated, fixed to the cylindrical section 801, and thereafter cooled and the jig 900 is removed (step S1007). For the magnets 710 and 720, end faces 713a, 723a, 713b and 723b are engaged with each other, and respective magnets 710 and 720 are in contact with the outer peripheral surface of the film 810. Hence, in a state of a pressing force being applied, even when the jig 900 is removed, the stator side circumferential surfaces 711 and 721 of the magnets 710 and 720 are maintained at the same circumferential surface. Thereafter, the magnets 710

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and 720 are disposed in a magnetizing apparatus and magnetized (step S1008). Thus, the rotor 40 is completed.

According to the present modification example, the rotor 40 is assembled from the components in the radially inside portion. However, the assembly order may be arbitrarily changed. For example, components may be disposed from the radially outside portion. In other words, the cylindrical section 801 may be disposed first and then magnets 710 and 720 may be disposed. Further, after fixing the magnets 710 and 720 and removing the jig 900, the film 810 may be assembled. Also, the magnets 710 and 720 after being magnetized may be fixed to the cylindrical section 801.

Advantageous effects in the modification example thus configured will be described.

The magnet unit 700 is fixed to the inner peripheral surface of the cylindrical section 801 via the foamable resin 802. For the foamable resin 802, the thickness dimension thereof can be changed by heating. Therefore, the thickness dimension of the foamable resin 802 in the radial direction can be differentiated depending on the distance in the radial direction from the inner peripheral surface of the cylindrical section 801 to the anti-stator side circumferential surfaces 712 and 722 of the magnet unit 700. In other words, the thickness dimension of the foamable resin 802 can be differentiated so as to occupy the gap between the inner peripheral surface of the cylindrical section 801 and the anti-stator side circumferential surfaces 712 and 722. Therefore, considering the manufacturing tolerance of the magnets 710 and 720, the thickness dimension of the foamable resin 802 is differentiated, whereby the stator side circumferential surfaces 711 and 721 of the magnet unit 700 can be close to the same circumferential surface. Thus, the air gap can be small.

Also, according to the present modification example, the thickness dimension of the foamable resin 802 in the radial direction is differentiated such that the stator side circumferential surfaces 711 and 721 of the magnet unit 700 are on the same circumferential surface. Thus, the air gap can be small.

The foamable resin 802 is configured to be applied with a pressing force from the anti-stator side circumferential surfaces 712 and 722 of the magnet unit 700 to the stator side circumferential surfaces 711 and 721. Hence, the stator side circumferential surfaces 711 and 721 of the magnet unit 700 can reliably come into contact with the annular shaped film 810. Thus, the stator side circumferential surfaces 711 and 721 become the same circumferential surface and the air gap can be small. Further, the stator side circumferential surfaces 711 and 721 can be protected by the film 810.

The first protrusion 811 protruding towards magnet unit 700 side in the radial direction is provided on the outer peripheral surface (magnet section side circumferential surface) of the film 810. On the other hand, the first recess 705 recessed in the radial direction is provided on the stator side circumferential surfaces 711 and 721. The first protrusion 811 is accommodated in the first recess to be capable of being engaged with the first recess 705 in the circumferential direction. Thus, the magnet unit 700 and the film 810 can be prevented from being shifted in the circumferential direction. Moreover, positional alignment can readily be performed.

The second protrusion 803 protruding towards magnet unit 700 side in the radial direction is provided on the inner peripheral surface of the cylindrical section 801. On the other hand, the second recess 706 recessed in the radial direction is provided on the anti-stator side circumferential surfaces 712 and 722. The second protrusion 803 is accom-

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modated in the second recess 706 to be capable of being engaged with the second recess 706 in the circumferential direction. Thus, the magnet unit 700 and the cylindrical section 801 can be prevented from being shifted in the circumferential direction. Moreover, positional alignment can readily be performed.

The second recess 706 is formed to be larger than the second protrusion and is configured such that the foamable resin 802 is capable of flowing into a gap formed between the second recess 706 and the second protrusion 803. Thus, excess foamable resin 802 can be made to flow into the gap between the second recess 706 and the second protrusion 803. Hence, the thickness dimension of the foamable resin 802 can readily be adjusted.

In the magnet unit 700, the axes of easy magnetization in the d-axis as the center of the magnetic pole are oriented to be parallel to the d-axis compared to the axes of easy magnetization in the q-axis as the magnetic pole boundary, and the magnet magnetic path is formed along the axes of easy magnetization. Thus, flux leakage from the anti-stator side of the magnet unit 700 can be reduced in the rotor 40, the magnetic flux density of the d-axis is enhanced and surface magnetic flux waveform can be close to a sinusoidal wave. Further, the cylindrical section 801 can be thinner.

A gap formed between magnets 710 and 720 adjacently positioned in the circumferential direction is configured to be shorter than the thickness dimension of the foamable resin 802 in the radial direction. Hence, the magnet magnetic path is likely to be connected between the magnets 710 and 720 adjacently positioned in the circumferential direction so that the magnetic flux leakage can be suppressed.

Since a magnetization is performed after the magnets 710 and 720 are fixed to the cylindrical section, compared to a case where magnetized magnets 710 and 720 are fixed, the magnets 710 and 720 can readily be arranged because no influence of magnetic force is present. Further, the first recess 705 and the second recess 706 are provided at a portion where the magnet magnetic path is short and demagnetization is likely to occur. Therefore, the number of magnets can be reduced while suppressing influence on the magnetic density.

(Other Examples of Fifteenth Modification)

The rotor 40 in the above-described fifteenth modification may be modified as follows.

In the magnets 710 and 720 according to the above-described fifteenth modification, the d-axis side inclined surfaces 712a and 722a may not be provided. In other words, the second protrusion 803 may not be provided. Similarly, q-axis side inclined surfaces 711a and 721a may not be provided. That is, the first recess 705 may not be provided. In this case, the first protrusion 811 may not be provided.

In the magnets 710 and 720 according to the above-described fifteenth modification, the axis of easy magnetization at the first cross point P1 may be set to be parallel to the d-axis. Also, an arc-shaped magnet magnetic path is formed in the magnets 710 and 720, but a linear shaped magnet magnetic path may be provided.

In the magnets 710 and 720 according to the above-described fifteenth modification, the q-axis side inclined surfaces 711a and 721a and the d-axis side inclined surfaces 712a and 722a may be formed as curved surface. Further, the first recess 705 and the second recess 706 may be formed in a stepped shape.

According to the above-described fifteenth modification, an outer rotor type rotor is employed. However, an inner rotor type rotor may be employed. Note that the outer

circumferential surface of the magnet supporting section serves as a fixed surface in the case where the inner rotor type rotor is employed.

In the above-described fifteenth modification, the film **810** may not be provided. In other words, the stator side circumferential surfaces **711** and **721** may be exposed. Moreover, instead of using the film **810**, annular shaped wall member may be provided by spirally winding a non-magnetic yarn.

In the above-described fifteenth modification, the end plate **910** may not be provided.

In the above-described fifteenth modification, a ring magnet having an annular shape may be employed.

The disclosure herein is not limited to the illustrated embodiments. The disclosure includes exemplary embodiments and modifications by persons skilled in the art based on the exemplary embodiments. For example, the disclosure is not limited to the parts and/or element combinations indicated in the embodiments. The disclosure can be carried out in various combinations. The disclosure can have additional parts that can be added to the embodiments. The disclosure includes those in which the parts and/or elements of the embodiments are omitted. The disclosure includes the replacement or combination of parts and/or elements between one embodiment and another. The technical scope disclosed is not limited to the description of the embodiments. Some technical scopes disclosed are indicated by the statement of the claims and should be understood to include all modifications within the meaning and scope equivalent to the claims statement.

While the present disclosure has been described in accordance with the examples, the present disclosure should be understood such that the present disclosure is not limited to the examples and structures. The present disclosure also includes various modifications and modifications within an equivalent range. Additionally, various combinations and forms, as well as other combinations and forms further including only one element, more, or less, also fall within the category and scope of the present disclosure.

CONCLUSION

The present disclosure has been made in view of the above circumstances and provides a rotating electric machine and a manufacturing method thereof capable of reducing an air gap.

A first aspect of the present disclosure is a rotating electric machine provided with: a field magnet having a magnet unit including a plurality of magnetic poles having alternating polarities in a circumferential direction and a magnet supporting section having a cylindrical shape to which the magnet unit is fixed; and an armature having a multi-phase armature winding, disposed facing the magnet unit, either one of the field magnet or the armature being configured as a rotor, wherein the magnet unit is configured annularly along the circumferential direction; the magnet unit is fixed to a fixed surface which is either an outer peripheral surface or an inner peripheral surface of the magnet supporting section via a foamable resin; and a thickness dimension of the foamable resin in a radial direction is set to be differentiated depending on a distance from the fixed surface of the magnet supporting section to an anti-armature side circumferential surface of the magnet.

The magnet unit is fixed to the fixed surface of the magnet supporting section via the foamable resin. When fixing, the foamable resin is heated to allow the thickness dimension of the foamable resin to be changed. Hence, the thickness dimension of the foamable resin in a radial direction is

capable of being differentiated depending on a distance from the fixed surface of the magnet supporting section to an anti-armature side circumferential surface of the magnet. That is, considering a manufacturing tolerance of the magnet unit, the thickness dimension of the foamable resin is made to be differentiated, whereby the armature side circumferential surface of the magnet unit can be substantially the same circumferential surface. Thus, the air gap can be small.

According to a second aspect, in the first aspect, the thickness dimension of the foamable resin in the radial direction is set to be differentiated such that armature side circumferential surfaces of the magnet unit are on the same circumferential surface, of which the center thereof is a rotating shaft of the rotor.

Considering a manufacturing tolerance of the magnet unit, the thickness dimension of the foamable resin is made to be differentiated, whereby the armature side circumferential surface of the magnet unit is set to be the same circumferential surface. Thus, the air gap can be small.

According to a third aspect, in the first aspect or the second aspect, an annular shaped wall member is provided in an armature side of the magnet unit, the annular shaped wall member coming into contact with the armature side circumferential surfaces of respective magnets; and the foamable resin is configured to apply pressing force to an armature side circumferential surface of the magnet unit from an anti-armature side circumferential surface of the magnet unit.

Since the foamable resin is configured to apply pressing force to an armature side circumferential surface of the magnet unit from an anti-armature side circumferential surface of the magnet unit, the armature side circumferential surfaces of respective magnets can reliably come into contact with the annular shaped wall member. Thus, the armature side circumferential surface becomes the same circumferential surface and the air gap can be small. Further, the armature side circumferential surface of the magnet unit can be protected by the wall member.

According to a fourth aspect, in the third aspect a first protrusion is additionally provided on a magnet unit side circumferential surface of the wall member, the first protrusion protruding towards a magnet unit side in the radial direction; a first recess is provided on an armature side circumferential surface of the magnet unit; and the first protrusion is accommodated in the first recess to be capable of being engaged with the first recess in the circumferential direction.

Thus, the magnet unit and the wall member can be prevented from being positionally-shifted in the circumferential direction.

According to a fifth aspect, in the first to fourth aspects, a second protrusion is provided on fixed surface of the magnet supporting section, the second protrusion protruding towards a magnet unit side in the radial direction; a second recess is provided on an anti-armature side circumferential surface of the magnet unit, the second recess being recessed in the radial direction; and the second protrusion is accommodated in the second recess to be capable of being engaged with the second recess in the circumferential direction.

Thus, the magnet unit and the wall member can be prevented from being positionally-shifted in the circumferential direction.

According to a sixth aspect, in the fifth aspect, the second recess is formed to be larger than the second protrusion and configured such that the foamable resin is capable of flowing into a gap formed between the second recess and the second protrusion.

Thus, excess foamable resin can be made to flow into the gap between the second recess and the second protrusion. Hence, the thickness dimension of the foamable resin can readily be adjusted.

According to a seventh aspect, in the first to sixth aspects, in the magnetic unit, axes of easy magnetization in a d-axis as the center of the magnetic pole are oriented to be parallel to the d-axis compared to axes of easy magnetization in a q-axis as a magnetic pole boundary, and a magnet magnetic path is formed along the axes of easy magnetization.

Thus, leakage flux from the anti-stator side can be reduced, the magnetic flux density of the d-axis is enhanced and the surface magnetic flux waveform can be close to sinusoidal wave. Further, the magnet supporting section can be thinner.

According to an eighth aspect, in the first to seventh aspects, an end plate that regulates movement of the magnet unit is provided in one end side of the magnet unit in the axial direction.

Thus, the magnet unit can be prevented from being positionally-shifted in the axial direction and the magnet unit can be prevented from becoming detached.

According to a ninth aspect, in the first to eighth aspects, the magnet unit is configured to include a plurality of magnets each having an arc shape, the plurality of magnets are arranged in the circumferential direction, whereby the magnet unit is formed in an annular shape; and a gap formed between the magnets in the circumferential direction is shorter than a thickness dimension of the foamable resin in the radial direction. Thus, the magnet magnetic path is likely to be completed between the magnets adjacently positioned in the circumferential direction so that the magnetic flux leakage can be suppressed.

According to a tenth aspect, a manufacturing method of a rotating electric machine configured annularly along a circumferential direction is provided. The rotating electric machine includes: a field magnet having a magnet unit including a plurality of magnetic poles having alternating polarities in a circumferential direction and a magnet supporting section having a cylindrical shape to which the magnet unit is fixed; and an armature having a multi-phase armature winding, disposed facing the magnet unit, either one of the field magnet or the armature being configured as a rotor. The manufacturing method of the rotating electric machine includes: an arrangement process that arranges the magnet unit to face a fixed surface via a foamable resin, the fixed surface being either an outer peripheral surface or an inner peripheral surface of the magnet supporting section; and a heating process that heats the foamable resin to differentiate a thickness dimension of the foamable resin in a radial direction depending on a distance from the fixed surface of the magnet supporting section to an anti-armature side circumferential surface of the magnet.

The magnet unit faces the fixed surface of the magnet supporting section via the foamable resin. When fixing, the foamable resin is heated to allow the thickness dimension of the foamable resin to be changed. Hence, the thickness dimension of the foamable resin in a radial direction is capable of being differentiated depending on a distance from the fixed surface of the magnet supporting section to an anti-armature side circumferential surface of the magnet. That is, considering a manufacturing tolerance of the magnet unit, the thickness dimension of the foamable resin is made to be differentiated, whereby the armature side circumferential surface of the magnet unit can be substantially the same circumferential surface. Thus, the air gap can be small.

What is claimed is:

1. A rotating electric machine comprising:

a field magnet having (i) a magnet unit including a plurality of magnetic poles having alternating polarities in a circumferential direction and (ii) a magnet supporting section having a cylindrical shape to which the magnet unit is fixed; and

an armature having a multi-phase armature winding that faces the magnet unit, wherein:

one of the field magnet and the armature is configured as a rotor;

the magnet unit is configured annularly along the circumferential direction;

the magnet unit is fixed to a fixed surface which is either an outer peripheral surface or an inner peripheral surface of the magnet supporting section via a foamable resin;

a thickness dimension of the foamable resin in a radial direction is differentiated depending on a distance from the fixed surface of the magnet supporting section to an anti-armature side circumferential surface of the magnet unit;

a second protrusion is provided on the fixed surface of the magnet supporting section, the second protrusion protruding toward a magnet unit side in the radial direction;

a second recess is provided on the anti-armature side circumferential surface of the magnet unit, the second recess being recessed in the radial direction;

the second protrusion is accommodated in the second recess so as to be capable of being engaged with the second recess in the circumferential direction;

in the magnet unit, axes of easy magnetization at a d-axis, which is at a center of a magnetic pole, are oriented to be more parallel to the d-axis than axes of easy magnetization at a q-axis, which is at a magnetic pole boundary, and a magnet magnetic path is formed along the axes of easy magnetization;

the second recess is provided on the d-axis in the circumferential direction; and

the magnet unit comprises a plurality of magnets, an adjacent two of the plurality of magnets being divided at the d-axis but closely arranged in the circumferential direction so as to contact each other.

2. The rotating electric machine according to claim 1, wherein

the thickness dimension of the foamable resin in the radial direction is differentiated such that armature side circumferential surfaces of the plurality of magnets constituting the magnet unit form a same circumferential surface of the magnet unit, a center of which is a rotating shaft of the rotor.

3. The rotating electric machine according to claim 1, wherein

an annular shaped wall member is provided at an armature side of the magnet unit, the annular shaped wall member coming into contact with armature side circumferential surfaces of the plurality of magnets constituting the magnet unit; and

the foamable resin is configured to apply pressing force to an armature side circumferential surface of the magnet unit from the anti-armature side circumferential surface of the magnet unit.

4. The rotating electric machine according to claim 3, wherein

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- a first protrusion is provided on a magnet unit side circumferential surface of the wall member, the first protrusion protruding towards a magnet unit side in the radial direction;
- a first recess is provided on the armature side circumferential surface of the magnet unit; and
- the first protrusion is accommodated in the first recess so as to be capable of being engaged with the first recess in the circumferential direction.
5. The rotating electric machine according to claim 1, wherein
- an end plate that regulates movement of the magnet unit is provided at one end side of the magnet unit in an axial direction.
6. The rotating electric machine according to claim 1, wherein
- the plurality of magnets each have an arc shape and are arranged in the circumferential direction, whereby the magnet unit is formed in an annular shape; and
- a gap formed between ones of the magnets in the circumferential direction is shorter than the thickness dimension of the foamable resin in the radial direction.
7. The rotating electric machine according to claim 1, wherein
- a dimension of the second protrusion in the radial direction is shorter than a depth dimension of the second recess such that a gap is formed between the second recess and the second protrusion at the d-axis to allow the foamable resin to flow into the gap when the second protrusion is accommodated in the second recess.
8. A manufacturing method of a rotating electric machine configured annularly along a circumferential direction, the rotating electric machine comprising:
- a field magnet having (i) a magnet unit including a plurality of magnetic poles having alternating polarities in the circumferential direction and (ii) a magnet supporting section having a cylindrical shape to which the magnet unit is fixed; and

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- an armature having a multi-phase armature winding that faces the magnet unit, one of the field magnet and the armature being configured as a rotor, and
- the manufacturing method of the rotating electric machine comprising:
- an arrangement process that disposes the magnet unit to face a fixed surface via a foamable resin, the fixed surface being either an outer peripheral surface or an inner peripheral surface of the magnet supporting section; and
- a heating process that heats the foamable resin to differentiate a thickness dimension of the foamable resin in a radial direction depending on a distance from the fixed surface of the magnet supporting section to an anti-armature side circumferential surface of the magnet unit, wherein:
- a second protrusion is provided on the fixed surface of the magnet supporting section, the second protrusion protruding toward a magnet unit side in the radial direction;
- a second recess is provided on the anti-armature side circumferential surface of the magnet unit, the second recess being recessed in the radial direction;
- the second protrusion is accommodated in the second recess so as to be capable of being engaged with the second recess in the circumferential direction;
- in the magnet unit, axes of easy magnetization at a d-axis, which is at a center of a magnetic pole, are oriented to be more parallel to the d-axis than axes of easy magnetization at a q-axis, which is at a magnetic pole boundary, and a magnet magnetic path is formed along the axes of easy magnetization;
- the second recess is provided on the d-axis in the circumferential direction; and
- the magnet unit comprises a plurality of magnets, an adjacent two of the plurality of magnets being divided at the d-axis but closely arranged in the circumferential direction so as to contact each other.

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